FRM Part II

Recall + Priority Map Workbook

Volume 2: Tiers B to D

Map topics with Telegram or sheet evidence beyond Tier A. Revision notes for every section; questions where a specific was reported.

44 sections · 63 questions

Built from the GARP 2026 source readings and SchweserNotes Book 5 where no GARP text exists. Retake: 25 November 2026.

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Priority table

Every section in this volume in reading order. **Aug** = you saw it in the August 2026 sitting. **Map** = Cross-Source Priority Map entry and how many of 16 sources name it. **TG** = a Telegram recall cites it.

ID	Section	Aug	Map	TG	Qs	Page
B1	Credit VaR and Granularity		#10, 6/16	✓	2	15
B2	Expected Shortfall and Coherence		#12, 6/16	✓	3	19
B3	Default Correlation, Joint PD and Copulas		#16, 6/16	✓	3	24
B4	Factor Theory, Fama-French and Style		#18, 5/16	✓	2	29
B5	Climate Risk (not in 2026 LOs: notes only)		#19, 5/16	✓	notes	34
C1	Regression Hedging, DV01 and PCA		#23, 5/16	✓	2	37
C2	Backtesting VaR: Exceedance vs PIT		#21, 4/16	✓	3	41
C3	VaR Mapping		#22, 4/16	✓	2	46
C4	Policy-Mix VaR and Risk Budgeting		#35, 4/16	✓	3	50
C5	Basel III Post-Crisis Reforms and Stress-Test Regimes		#30, 4/16	✓	2	56
C6	Model Risk Management		#32, 3/16	✓	2	61
C7	CMT Swap Valuation and OAS on a Rate Tree		#33, 3/16		1	66
C8	Counterparty Exposure Profiles: EE, PFE, EPE		#37, 3/16	✓	2	69
C9	Stress-Testing Practices and Scenario Design		#38, 3/16	✓	2	74
D1	CIR and Lognormal Short-Rate Models		#39, 4/16	✓	2	80
D2	Outsourcing and Third-Party Risk		#51, 3/16	✓	2	84
D3	Regulatory vs Economic Capital and Capital Adequacy		#61, 3/16	✓	2	89
D4	Stressed VaR and Market Risk Capital		#73, 3/16	✓	2	93
D5	Credit Spread and PD from Spreads		#42, 2/16		2	97
D6	Repurchase Agreements		#43, 2/16		2	101
D7	Early Warning Indicators and PIT vs TTC		#45, 2/16	✓	2	106
D8	US Dollar Shortage and the Cross-Currency Basis		#46, 2/16	✓	2	110
D9	Illiquid Assets		#48, 2/16		2	115
D10	BHC Capital Planning and VaR Model Validation		#53, 2/16	✓	2	119
D11	Empirical Properties of Correlation		#54, 2/16		2	124
D12	Convexity and Jensen's Inequality		#55, 2/16		1	128
D13	Retail Credit Risk		#57, 2/16	✓	1	131
D14	Credit Derivatives and Credit-Linked Notes		#58, 2/16		1	134
D15	Structured Credit and Implied Correlation		#60, 2/16	✓	1	137
D16	Monetary and Fiscal Policy		#67, 2/16	✓	1	140
D17	2023 Bank Failures and Credit Suisse (not in 2026 LOs: notes only)		#69, 2/16	✓	notes	143
D18	Inflation and Market Stress Across Bond Maturities		#72, 2/16	✓	1	146
D19	LDA for Operational Risk		#74, 2/16		1	149
D20	Credit Model Comparison: Vasicek, CreditRisk+, CreditMetrics, KMV		#41, 2/16		notes	152
D21	Investor Protection, MiFID and the Volcker Rule		#52, 2/16		notes	155
D22	Operational Risk Identification, Mitigation and Reporting		#75, 2/16		notes	158
D23	Sovereign Default Risk and CDS		#56, 1/16		1	162
D24	Central Clearing and the CCP Loss Waterfall		#59, 1/16		1	165
D25	Contingent Convertible Bonds		#70, 1/16	✓	1	168
D26	Contingency Funding Planning		#62, 1/16		notes	171
D27	Intraday Liquidity and Dealer Bank Failure Mechanics		#65, 1/16		notes	174
D28	Portfolio Construction and the No-Trade Region		#66, 1/16		notes	178
D29	Crypto, Tokenisation and Digital Resilience		#68, 1/16		notes	181
D30	Geopolitical Risk and Financial Stability		not in map		2	184

Specifics coverage index

Every reported shape (S-number) and the questions that drill it. Notes-only sections list their shapes for revision.

ID	S	Reported shape	Questions
B1	S1	Credit VaR by direct formula: worst-case loss at the confidence level minus expected loss	B1.1
B1	S2	formula-card form and the dropped-EL trap	B1.1
B1	S3	granularity effect 1: Credit VaR falls as the portfolio becomes more granular	B1.2
B1	S4	granularity effect 2: it does not fall to zero when defaults are correlated	B1.2
B2	S1	ES from a sample and its bootstrap resamples: exactly which losses enter the tail average, 4, 5 or 6	B2.1
B2	S2	ES as the average of tail VaRs read off a confidence-level table, against the 99% VaR it replaces under FRTB	B2.2
B2	S3	ES within filtered historical simulation	B2.3
B2	S4	coherence: ES is subadditive, VaR is not (Q3)	B2.3
B3	S1	default correlation of two firms	B3.1
B3	S2	joint probability of default, direct calculation	B3.2
B3	S3	default correlation between credits with unequal default probabilities	B3.1, B3.2
B3	S4	Gaussian copula and implied correlation	B3.3
B3	S5	dropped-term trap: omitting $\pi_1 \pi_2$	B3.1, B3.2
B3	S6	conditional default probability given the common factor	B3.3
B3	S7	LO 40.i impacts table for tranches (D15.1)	D15.1
B4	S1	SMB (small minus big) vs SML (security market line), with HML, in a Fama-French portfolio composition	B4.1
B4	S2	investing style (style, investment or dynamic factors) as a theory item	B4.2
B4	S3	value vs momentum: return and risk profiles	B4.2
B4	S4	the low-risk anomaly: volatility anomaly vs beta anomaly and their explanations (Q2)	B4.2
B5	S1	climate risk drivers: physical vs transition	notes
B5	S2	climate risk and inflation	notes
B5	S3	catastrophe bonds and risk transfer	notes
B5	S4	where climate still appears in 2026 readings (operational risk identification, reporting, ICT outages, AI energy demand, fiscal pressure)	notes
C1	S1	hedge ratio from a regression output that lists alpha, beta and both DV01s, with alpha as the tempting extra input	C1.1
C1	S2	face amount of the offsetting position in a regression hedge, paired with the hedged P&L volatility	C1.1
C1	S3	DV01 hedging: the DV01-neutral amount as the planted distractor against the regression amount	C1.1, C1.2
C1	S4	drawbacks of a DV01-neutral hedge and why the regression hedge improves it	C1.2
C1	S5	PCA as the construction of a hedging portfolio: how many PCs two hedging securities can neutralize, and the risk-weight sum (Q2)	C1.2
C2	S1	backtesting VaR: counting exceptions against the model	C2.1
C2	S2	PIT-based versus exceedance-based backtesting	C2.2
C2	S3	Kupiec POF (unconditional coverage) versus Christoffersen (independence, conditional coverage)	C2.3
C2	S4	Type I versus Type II errors	C2.3
C2	S5	Basel penalty zones and the multiplier they set (Q1)	C2.1
C3	S1	difference among principal, duration and cash-flow mapping	C3.2
C3	S2	VaR relative to a benchmark: tracking error VaR and the interpretation of variance improvement	C3.1
C3	S3	ranking the three methods by how faithfully they capture risk (Q2)	C3.2
C4	S1	VaR calculated by the policy-mix method: policy weights on benchmark returns, diversified	C4.1
C4	S2	the six-way sibling set: absolute, relative, policy-mix, active-management, funding, sponsor risk	C4.1, C4.3
C4	S3	portfolio return computation: expected excess return of a pool of managers	C4.2
C4	S4	IR and active risk numerical: IR from expected excess return and TEV, portfolio IR	C4.2
C4	S5	risk budgets across asset classes and across active managers	C4.1, C4.2
C5	S1	pre-SCAP versus post-SCAP features of stress testing	C5.2
C5	S2	the standardised versus internal-models split across accords: who uses which, and where the December 2017 reforms removed models	C5.1
C5	S3	output floors: the Basel II floor on Basel I and the Basel III floor on the revised standardised approaches	C5.1
C5	S4	the Basel I / 1996 / II / 2.5 / III progression, each change attached to the right accord	C5.1
C5	S5	the SCAP / CEBS / EBA / CCAR sibling swap: which banks failed, base versus stress disclosure, which risk weights (Q2)	C5.2
C6	S1	the friction between investor and rating agency labelled model error, tied to the GARP Gaussian copula case	C6.1
C6	S2	the three components of model risk management: development, implementation and use; validation; governance, policies and controls	C6.2
C6	S3	the independence requirement for validation, with effective challenge (incentives, competence, influence)	C6.2
C7	S1	value a CMT swap backward through a two-period rate tree with risk-neutral probabilities, inside an OAS wrapper	C7.1
C7	S2	OAS: shift the discount rates only, then read cheap versus rich from the sign	C7.1
C7	S3	true versus risk-neutral probabilities and the drift they imply, with the real-world probabilities supplied as a temptation (Q1)	C7.1

ID	S	Reported shape	Questions
C8	S1	the six-way metric family: expected MtM (EFV), expected exposure / EPE, PFE, ENE, effective EE/EPE and average EPE, maximum PFE	C8.1, C8.2
C8	S2	the netting row: aggregation, EFV additive but EPE not, netting factor	C8.1, C8.2
C8	S3	sold option red herring: zero exposure whatever Black-Scholes data the stem supplies (Q1)	C8.1
C9	S1	appropriate stress-testing practices as a best-practice audit: what to stress, how to aggregate, what to avoid	C9.1
C9	S2	market-driven scenarios: plausible scenario construction, z-scores and covariance choice	C9.2
C9	S3	the SCAP / EBA / CCAR sibling swap on which regime did what (Q2; regime comparison also in C5.2)	C9.2
D1	S1	CIR model solving: one step of the short-rate change with drift, volatility and a random draw	D1.1
D1	S2	CIR basis-point volatility $\sigma\sqrt{r}$ versus the constant parameter σ	D1.1, D1.2
D1	S3	non-negativity: CIR and lognormal versus normal models	D1.2
D1	S4	lognormal models: basis-point volatility proportional to r, with deterministic drift or with mean reversion	D1.2
D1	S5	the sibling set against Vasicek and Ho-Lee (Q2)	D1.2
D2	S1	outsource the activity, never the accountability, applied to a vendor and subcontractor failure	D2.1
D2	S2	concentration risk in a critical service provider and its resilience implications	D2.2
D3	S1	regulatory capital as a formula-driven minimum versus economic capital as the bank's own estimate	D3.2
D3	S2	unexpected loss into economic capital: EC $=$ UL$_p$ $\times$ CM, loan level ULC$_i$ $\times$ CM	D3.1, D3.2
D3	S3	risk-weighted assets (standardized weights plus IRB $K \times 12.5 \times$ EAD) and the capital adequacy ratio against a minimum (Q1)	D3.1
D4	S1	stressed market VaR calculation: the SVaR term with its own max and multiplier	D4.1, D4.2
D4	S2	market risk capital charge calculation: max$(VaR_{t-1}, mVaR_{avg})$, $\sqrt{10}$ scaling, multiplier from backtesting	D4.1, D4.2
D4	S3	mixed vignette: market stressed VaR charge plus credit capital given the 99.9th percentile default rate, one error per option (Q2)	D4.2
D5	S1	credit spread from the market value of debt, with the card's inverted sign as the distractor	D5.1
D5	S2	deriving PD from credit spreads	D5.2
D5	S3	one-period PD $= (1/\text{LGD})\big(1-(1+r_f)/(1+y)\big)$, where recovery decides the ranking (Q2)	D5.2
D6	S1	repo settlement amount from the closed-book template, with accrued interest and haircut, and the repurchase price at the unwind	D6.1
D6	S2	which party is more vulnerable when a bank will not roll the repo over (Q2)	D6.2
D7	S1	characteristics of sound EWIs: leading, sharp, calibrated thresholds	D7.1
D7	S2	applications of EWIs in the liquidity risk process: escalation to the right authority	D7.1
D7	S3	PIT versus TTC definitions and users	D7.2
D7	S4	the procyclicality polarity: PIT is the volatile, procyclical one (Q2)	D7.2
D8	S1	why non-US (chiefly European) banks faced a US dollar shortage in the GFC	D8.1
D8	S2	covered interest parity violations and the cross-currency basis, the adjacent reading (Q2)	D8.2
D9	S1	biases in reported illiquid asset returns: survivorship, selection, infrequent trading, unsmoothing	D9.1
D9	S2	illiquidity risk premiums across versus within asset classes, and the four ways to harvest them	D9.2
D9	S3	portfolio allocation with illiquid assets: optimal holdings, hurdle rates, rebalancing point	D9.2
D10	S1	capital planning governance: board versus senior management, reporting of overlays	D10.1
D10	S2	BHC scenario design and estimation practices that the Federal Reserve calls stronger or weaker	D10.1
D10	S3	validation standards for BHC VaR models: conceptual soundness, sensitivity analysis, confidence intervals, benchmarking (Q2)	D10.2
D11	S1	best-fit distribution for equity, bond and default probability correlations	D11.2
D11	S2	mean reversion rate from regressing the correlation change on the lagged level	D11.1
D11	S3	the corresponding autocorrelation (Q1)	D11.1
D12	S1	Jensen's inequality applied to bond prices: the expected price exceeds the price at the expected rate	D12.1
D12	S2	the value of convexity increases with volatility	D12.1
D12	S3	the value of convexity increases with time to maturity, so forward rates bend down more at longer terms (Q1)	D12.1
D13	S1	risk-based pricing and adverse selection under a "one price fits all" policy; tiered pricing direction	D13.1
D13	S2	the dark side of retail credit: a systematic factor that drives many credits at once	D13.1
D13	S3	retail versus corporate credit risk: predictable EL, small but non-zero capital (Q1)	D13.1
D14	S1	CLN structure: funded, the note investor bears the credit risk	D14.1
D14	S2	total return swaps: who owns the bond, who earns the spread	D14.1
D14	S3	CDS mechanics: obligations of buyer and seller, settlement	D14.1
D14	S4	synthetic CDO valuation and the two implied correlations, compound and base (Q1, option C)	D14.1
D15	S1	LO 40.i impacts table: how default probability and correlation move each tranche	D15.1
D15	S2	the one possible calculation: reading tranche values and writedowns off the simulation grid	D15.1
D15	S3	implied correlation backed out of a tranche price	D15.1
D15	S4	Gaussian copula and implied correlation	D15.1
D16	S1	fiscal and monetary policy: how the two interact, including the tension that high debt creates (fiscal dominance)	D16.1
D16	S2	LO 104.c: the interest-growth differential and the risks of high public debt (Q1)	D16.1
D17	S1	SVB and the March 2023 US failures: duration mismatch plus an uninsured-deposit run	notes
D17	S2	Credit Suisse: a confidence-and-governance collapse, resolved by takeover and an AT1 write-down	notes
D17	S3	case↔lesson matching against the liquidity cases in the 2026 readings	notes
D18	S1	higher expected inflation as higher expected short rates: effect on forward rates and bond values grows with duration	D18.1
D18	S2	market stress as a higher risk premium and higher rate volatility: premium term grows with duration, convexity term pulls the other way	D18.1
D18	S3	a climate-related shock framed as an inflation and stress scenario for short versus long bonds (Q1)	D18.1

ID	S	Reported shape	Questions
D19	S1	loss frequency × severity convolution into an aggregated annual loss distribution	D19.1
D19	S2	operational risk capital as the 99.9th percentile of the aggregated distribution, with aggregation across units of measure	D19.1
D20	S1	CreditRisk+ as the actuarial, default-mode model (notes; distractors 1, 5)	notes
D20	S2	CreditMetrics as the ratings-migration, mark-to-market model (notes; distractor 2)	notes
D20	S3	Vasicek as the model under the Basel IRB formula (notes; distractors 3, 6)	notes
D20	S4	Moody's KMV as the equity-market based Merton implementation (notes; distractor 4)	notes
D20	S5	four-way comparison on loss recognition and correlation modelling (comparison table)	notes
D21	S1	suitability and appropriateness obligations	notes
D21	S2	best execution, with the Deutsche Bank Securities case	notes
D21	S3	conflicts-of-interest management under MiFID II and MiFIR	notes
D21	S4	the Volcker Rule's proprietary-trading prohibition and its market-making and hedging exemptions, with scope/condition errors	notes
D22	S1	the control sibling set: preventive, detective, corrective, directive, with key controls, automation and testing	notes
D22	S2	risk and control self-assessment (RCSA): inherent and residual risk, scales, heatmap, ownership of outcomes	notes
D22	S3	key risk indicators, against KPIs and KCIs	notes
D22	S4	loss-data collection: regulatory criteria, fields, dates, boundary events, completeness	notes
D23	S1	sovereign CDS spreads as a measure of country default risk, compared with ratings and bond default spreads (CR-8 g) (Q1)	D23.1
D24	S1	the order of the loss waterfall, with sequence errors as distractors (Q1)	D24.1
D25	S1	trigger mechanics: Core Tier 1 to RWA threshold or regulator's nonviability declaration, and the 5.125% line between Additional Tier 1 and Tier 2	D25.1
D25	S2	equity conversion versus principal write-down: both increase equity	D25.1
D25	S3	AT1 ranking and the inversion controversy: CoCo holders bear losses not so different from equity holders (Q1)	D25.1
D26	S1	governance of the CFP: who designs, invokes, oversees and approves (notes: governance and roles)	notes
D26	S2	trigger design and escalation: EWIs, liquidity health measures, escalation levels (notes: monitoring and escalation)	notes
D26	S3	the menu of contingent actions and their reliability under stress, including signalling effects (notes: contingent actions, market signals)	notes
D26	S4	testing requirements and operational readiness (notes: testing)	notes
D27	S1	intraday liquidity monitoring tools: measures of flows versus measures of risk levels (notes: measurement bullets)	notes
D27	S2	the payment-system angle: uses and sources of intraday liquidity, daylight overdraft policy, FMU defences and tools (notes: LTR-6 bullets)	notes
D27	S3	dealer-bank runs on repo, on prime-brokerage balances and by OTC derivatives counterparties, ending with the clearing bank (notes: LTR-8 bullets)	notes
D27	S4	case↔lesson matching against Northern Rock (notes: case table, LTR-1 b)	notes
D28	S1	alpha refinement: scaling, trimming, and neutralisation (benchmark, cash, risk-factor, industry)	notes
D28	S2	transaction costs: amortisation over the holding period	notes
D28	S3	the optimal no-trade region for MCVA and for alpha, and Leland's band around a target allocation	notes
D28	S4	the four techniques as a four-statement audit: screens, stratification, LP, QP	notes
D29	S1	CI-6: crypto asset categories, ecosystem entities and their regulation, national approaches, BCBS treatment, Bali Fintech Agenda	notes
D29	S2	CI-7: tokenisation, trust, sharedness and programmability, lifecycle frictions, externalities, retail and broker effects, ledger models and trust failures	notes
D29	S3	CI-8: cyber versus ICT risk, cloud concentration, contagion to financial risk, macroprudential tools and their challenges	notes
D30	S1	channels from geopolitical risk to asset prices and financial stability, and policy responses	D30.1
D30	S2	effects on banks and nonbank financial institutions: valuations, lending, fund flows	D30.1
D30	S3	pricing in equity and option markets (GPR betas, downside and tail-risk premiums)	D30.2
D30	S4	sovereign risk premiums, and commodity importers vs exporters (Q2)	D30.2

Syllabus map (2026 LOs)

Each section's official 2026 reading ID, reading title, source text and chapter, and the LO letters it covers, per GARP's 2026 FRM Part II Learning Objectives.

ID	Topic	Reading	Official 2026 title	Source and chapter	LOs
B1	Credit VaR and Granularity	**CR-10**	Credit Value at Risk	*Hull, RMFI 6e, Chapter 19*	a to c, f
		CR-11	Portfolio Credit Risk	*Malz, Financial Risk Management: Models, History, and Institutions, Chapter 8 (Sections 8.1, 8.2, 8.3 only)*	c, g, h
		CR-12	Credit Risk	*Hull, Options, Futures, and Other Derivatives 11e, Chapter 24*	g
B2	Expected Shortfall and Coherence	**MR-1**	Estimating Market Risk Measures: An Introduction and Overview	*Dowd, Measuring Market Risk 2e, Chapter 3*	c to e
		MR-2	Non-Parametric Approaches	*Dowd, Chapter 4*	a, c, d
		MR-18	Fundamental Review of the Trading Book	*Hull, Risk Management and Financial Institutions 6e, Chapter 18*	a, b
		ORR-19	Range of Practices and Issues in Economic Capital Frameworks	*Basel Committee on Banking Supervision, March 2009*	a
		CR-14	Derivatives	*Gregory, The xVA Challenge, Chapter 2*	j
B3	Default Correlation, Joint PD and Copulas	**CR-11**	Portfolio Credit Risk	*Malz, Financial Risk Management: Models, History, and Institutions, Chapter 8 (Sections 8.1, 8.2, 8.3 only)*	a to f
		CR-12	Credit Risk	*Hull, Options, Futures, and Other Derivatives 11e, Chapter 24*	e, f
		MR-10	Financial Correlation Modeling: Bottom-Up Approaches	*Meissner, Chapter 5 (pp. 126-134 only)*	a to c
		CR-22	Structured Credit Risk	*Malz, Chapter 9*	l
B4	Factor Theory, Fama-French and Style	**IM-1**	Factor Theory	*Ang, Asset Management: A Systematic Approach to Factor Investing, Chapter 6*	a to f
		IM-2	Factors	*Ang, Chapter 7*	a to e
		IM-3	Alpha (and the Low-Risk Anomaly)	*Ang, Chapter 10*	a, c, e to i
B5	Climate Risk (not in 2026 LOs: notes only)	**CI-4**	Global Financial Stability Report, Chapter 2	*IMF, April 2025*	cross-reference
		ORR-3	Risk Identification	*GARP, Chapter 3*	a
		ORR-5	Risk Mitigation	*GARP, Chapter 5*	g
		ORR-6	Risk Reporting	*GARP, Chapter 6*	c
		CI-2	The Financial Stability Implications of Artificial Intelligence	*FSB, November 2024*	c
		CI-5	Monetary and Fiscal Policy: Safeguarding Stability and Trust	*Annual Economic Report 2023, Section 2, BIS, June 2023*	c
		CI-8	Digital Resilience and Financial Stability: The Quest for Policy Tools in the Financial Sector	*Jose Ramon Martinez, Banco de España, April 2023*	a
		Map evidence relates to an earlier GFSR edition; not in the 2026 LOs (notes only).			
C1	Regression Hedging, DV01 and PCA	**MR-11**	Regression Hedging and Principal Component Analysis	*Tuckman & Serrat, Fixed Income Securities 4e, Chapter 6*	a to h
C2	Backtesting VaR: Exceedance vs PIT	**MR-4**	Backtesting VaR	*Jorion, Value at Risk 3e, Chapter 6*	a to f
		MR-7	Beyond Exceedance-Based Backtesting of Value-at-Risk Models: Methods for Backtesting the Entire Forecasting Distribution Using Probability Integral Transform	*Chapter 4*	a to d
		MR-18	Fundamental Review of the Trading Book	*Hull, Risk Management and Financial Institutions 6e, Chapter 18*	c
		ORR-21	Capital Regulation Before the Global Financial Crisis	*Mark Carey, GARP Risk Institute, April 2019*	c
C3	VaR Mapping	**MR-5**	VaR Mapping	*Jorion, Chapter 11*	a to g
C4	Policy-Mix VaR and Risk Budgeting	**IM-6**	VaR and Risk Budgeting in Investment Management	*Jorion, Chapter 17*	a to h

ID	Topic	Reading	Official 2026 title	Source and chapter	LOs
C5	Basel III Post-Crisis Reforms and Stress-Test Regimes	**ORR-21**	Capital Regulation Before the Global Financial Crisis	*Mark Carey, GARP Risk Institute, April 2019*	a to d, g, h
		ORR-22	Solvency, Liquidity, and Other Regulation After the Global Financial Crisis	*Mark Carey, GARP Risk Institute, April 2019*	a, d, e, h
		ORR-23	High-Level Summary of Basel III Reforms	*Basel Committee on Banking Supervision, December 2017*	a to c
		ORR-24	Basel III: Finalising Post-Crisis Reforms	*Basel Committee on Banking Supervision, December 2017, pp. 128-136*	a to c
		ORR-17	Stress Testing Banks	*Til Schuermann, International Journal of Forecasting 30:3, pp. 717-728, 2014*	a
C6	Model Risk Management	**ORR-15**	Supervisory Guidance on Model Risk Management	*Federal Deposit Insurance Corporation, June 2017*	a to d
		ORR-16	Case Study: Model Risk and Model Validation	*GARP, Chapter 16*	a to c
		ORR-3	Risk Identification	*GARP, Chapter 3*	c
C7	CMT Swap Valuation and OAS on a Rate Tree	**MR-12**	Arbitrage Pricing with Term Structure Models	*Tuckman & Serrat 4e, Chapter 7*	a to j
C8	Counterparty Exposure Profiles: EE, PFE, EPE	**CR-19**	Future Value and Exposure	*Gregory, Chapter 11*	a to i
		CR-12	Credit Risk	*Hull, Options, Futures, and Other Derivatives 11e, Chapter 24*	a
C9	Stress-Testing Practices and Scenario Design	**CR-21**	The Evolution of Stress Testing Counterparty Exposures	*Siddique & Hasan (eds.), Stress Testing: Approaches, Methods, and Applications, Chapter 4 (David Lynch, Federal Reserve Board)*	a to h
		ORR-17	Stress Testing Banks	*Til Schuermann, International Journal of Forecasting 30:3, pp. 717-728, 2014*	a to c
		IM-15	Market-Driven Scenarios: An Approach for Plausible Scenario Construction	*Golub (ed.), BlackRock's Guide to Fixed-Income Risk Management, Chapter 5*	a to d
D1	CIR and Lognormal Short-Rate Models	**MR-15**	The Art of Term Structure Models: Volatility and Distribution	*Tuckman & Serrat 3e, Chapter 10*	d to f
		MR-14	The Art of Term Structure Models: Drift	*Tuckman & Serrat 3e, Chapter 9*	c, d, f
D2	Outsourcing and Third-Party Risk	**ORR-12**	Guidance on Managing Outsourcing Risk	*Board of Governors of the Federal Reserve System, December 2013*	a to c
		ORR-13	Case Study: Third-Party Risk Management	*GARP, Chapter 13*	a, b
		ORR-5	Risk Mitigation	*GARP, Chapter 5*	g
		ORR-4	Risk Measurement and Assessment	*GARP, Chapter 4*	g
		ORR-3	Risk Identification	*GARP, Chapter 3*	d
D3	Regulatory vs Economic Capital and Capital Adequacy	**CR-4**	Capital Structure in Banks	*Schroeck, Risk Management and Value Creation in Financial Institutions, Chapter 5 (pp. 170-186 only)*	a to d, g to i
		CR-5	Introduction to Credit Risk Modeling and Assessment	*Doumpos, Lemonakis, Niklis & Zopounidis, Analytical Techniques in the Assessment of Credit Risk, Chapter 1*	a to c
		ORR-18	Risk Capital Attribution and Risk-Adjusted Performance Measurement	*Crouhy, Galai & Mark, The Essentials of Risk Management 2e, Chapter 17*	a
D4	Stressed VaR and Market Risk Capital	**ORR-22**	Solvency, Liquidity, and Other Regulation After the Global Financial Crisis	*Mark Carey, GARP Risk Institute, April 2019*	a to c
		ORR-21	Capital Regulation Before the Global Financial Crisis	*Mark Carey, GARP Risk Institute, April 2019*	c, f
D5	Credit Spread and PD from Spreads	**CR-10**	Credit Value at Risk	*Hull, RMFI 6e, Chapter 19*	f
		CR-12	Credit Risk	*Hull, Options, Futures, and Other Derivatives 11e, Chapter 24*	c
		CR-9	Estimating Default Probabilities	*Hull, Risk Management and Financial Institutions 6e, Chapter 17*	h, j, l

ID	Topic	Reading	Official 2026 title	Source and chapter	LOs
D6	Repurchase Agreements	**LTR-13**	Repurchase Agreements and Financing	*Tuckman & Serrat, Fixed Income Securities 3e, Chapter 12*	a to g
D7	Early Warning Indicators and PIT vs TTC	**LTR-3**	Early Warning Indicators	*Venkat & Baird, Liquidity Risk Management: A Practitioner's Perspective, Chapter 6*	a to c
		CR-6	Credit Scoring and Rating	*Doumpos et al., Chapter 2*	a, b, d
		ORR-18	Risk Capital Attribution and Risk-Adjusted Performance Measurement	*Crouhy, Galai & Mark, The Essentials of Risk Management 2e, Chapter 17*	d
D8	US Dollar Shortage and the Cross-Currency Basis	**LTR-15**	The US Dollar Shortage in Global Banking and the International Policy Response	*McGuire & von Peter, BIS Working Papers, 2009*	a to c
		LTR-16	Covered Interest Parity Lost: Understanding the Cross-Currency Basis	*Borio, McCauley, McGuire & Sushko, BIS Quarterly Review, Q3 2016*	a to c
D9	Illiquid Assets	**IM-17**	Illiquid Assets	*Ang, Asset Management, Chapter 13*	a to f
D10	BHC Capital Planning and VaR Model Validation	**ORR-20**	Capital Planning at Large Bank Holding Companies: Supervisory Expectations and Range of Current Practice	*Board of Governors of the Federal Reserve System, August 2013*	a, b
		MR-6	Validating Bank Holding Companies' Value-at-Risk Models for Market Risk	*Lynch, Hasan & Siddique (eds.), Validation of Risk Management Models, Chapter 2*	a to d
D11	Empirical Properties of Correlation	**MR-9**	Empirical Properties of Correlation: How Do Correlations Behave in the Real World?	*Meissner, Chapter 2*	a to c
D12	Convexity and Jensen's Inequality	**MR-13**	Expectations, Risk Premium, Convexity, and the Shape of the Term Structure	*Tuckman & Serrat 4e, Chapter 8*	a to d
		MR-14	The Art of Term Structure Models: Drift	*Tuckman & Serrat 3e, Chapter 9*	cross-reference
D13	Retail Credit Risk	**CR-7**	Credit Scoring and Retail Credit Risk Management	*Crouhy, Galai & Mark, The Essentials of Risk Management 2e, Chapter 9*	a to h
		CR-6	Credit Scoring and Rating	*Doumpos et al., Chapter 2*	a, b
D14	Credit Derivatives and Credit-Linked Notes	**CR-13**	Credit Derivatives	*Hull, OFOD 11e, Chapter 25*	a to h
		CR-9	Estimating Default Probabilities	*Hull, Risk Management and Financial Institutions 6e, Chapter 17*	g
		CR-23	An Introduction to Securitisation	*Choudhry, Structured Credit Products 2e, Chapter 12*	cross-reference
D15	Structured Credit and Implied Correlation	**CR-22**	Structured Credit Risk	*Malz, Chapter 9*	h to l
D16	Monetary and Fiscal Policy	**CI-5**	Monetary and Fiscal Policy: Safeguarding Stability and Trust	*Annual Economic Report 2023, Section 2, BIS, June 2023*	a to c
D17	2023 Bank Failures and Credit Suisse (not in 2026 LOs: notes only)	**CI-4**	Global Financial Stability Report, Chapter 2	*IMF, April 2025*	cross-reference
		LTR-1	Liquidity Risk	*Hull, Risk Management and Financial Institutions 5e, Chapter 24*	b
		LTR-2	Liquidity and Leverage	*Malz, Financial Risk Management: Models, History, and Institutions, Chapter 12*	b
		LTR-3	Early Warning Indicators	*Venkat & Baird, Liquidity Risk Management: A Practitioner's Perspective, Chapter 6*	a
		LTR-9	Liquidity Stress Testing	*Venkat & Baird, Chapter 3*	c
		LTR-17	Risk Management for Changing Interest Rates: Asset-Liability Management and Duration Techniques	*Rose & Hudgins, Chapter 7*	c
		ORR-6	Risk Reporting	*GARP, Chapter 6*	d
		ORR-22	Solvency, Liquidity, and Other Regulation After the Global Financial Crisis	*Mark Carey, GARP Risk Institute, April 2019*	g
		IM-9	Risk, Regulation and Organizational Structure	*Stowell & Stowell, Chapter 14*	b

ID	Topic	Reading	Official 2026 title	Source and chapter	LOs
		CI-1	Advances in Artificial Intelligence: Implications for Capital Markets Activities	*IMF, October 2024*	c
		CI-2	The Financial Stability Implications of Artificial Intelligence	*FSB, November 2024*	b
		CI-8	Digital Resilience and Financial Stability: The Quest for Policy Tools in the Financial Sector	*Jose Ramon Martinez, Banco de España, April 2023*	b, c
		Map evidence relates to an earlier GFSR edition; not in the 2026 LOs (notes only).			
D18	Inflation and Market Stress Across Bond Maturities	**MR-13**	Expectations, Risk Premium, Convexity, and the Shape of the Term Structure	*Tuckman & Serrat 4e, Chapter 8*	a to d
		MR-15	The Art of Term Structure Models: Volatility and Distribution	*Tuckman & Serrat 3e, Chapter 10*	d
		IM-2	Factors	*Ang, Chapter 7*	b
D19	LDA for Operational Risk	**ORR-4**	Risk Measurement and Assessment	*GARP, Chapter 4*	a, f
		ORR-6	Risk Reporting	*GARP, Chapter 6*	c
D20	Credit Model Comparison: Vasicek, CreditRisk+, CreditMetrics, KMV	**CR-10**	Credit Value at Risk	*Hull, RMFI 6e, Chapter 19*	d, e
		CR-5	Introduction to Credit Risk Modeling and Assessment	*Doumpos, Lemonakis, Niklis & Zopounidis, Analytical Techniques in the Assessment of Credit Risk, Chapter 1*	f
		CR-12	Credit Risk	*Hull, Options, Futures, and Other Derivatives 11e, Chapter 24*	g
D21	Investor Protection, MiFID and the Volcker Rule	**ORR-14**	Case Study: Investor Protection and Compliance Risks in Investment Activities	*GARP, Chapter 14*	a, b
		ORR-12	Guidance on Managing Outsourcing Risk	*Board of Governors of the Federal Reserve System, December 2013*	a
		ORR-5	Risk Mitigation	*GARP, Chapter 5*	d
		ORR-3	Risk Identification	*GARP, Chapter 3*	d
D22	Operational Risk Identification, Mitigation and Reporting	**ORR-3**	Risk Identification	*GARP, Chapter 3*	a to d
		ORR-4	Risk Measurement and Assessment	*GARP, Chapter 4*	a to g
		ORR-5	Risk Mitigation	*GARP, Chapter 5*	a to g
		ORR-6	Risk Reporting	*GARP, Chapter 6*	a to d
D23	Sovereign Default Risk and CDS	**CR-8**	Country Risk: Determinants, Measures, and Implications	*Damodaran, Country Risk — The 2022 Edition*	a to g
D24	Central Clearing and the CCP Loss Waterfall	**CR-18**	Central Clearing	*Gregory, Chapter 8*	a to j
D25	Contingent Convertible Bonds	**ORR-22**	Solvency, Liquidity, and Other Regulation After the Global Financial Crisis	*Mark Carey, GARP Risk Institute, April 2019*	g
D26	Contingency Funding Planning	**LTR-11**	Contingency Funding Planning	*Venkat & Baird, Chapter 7*	a to c
D27	Intraday Liquidity and Dealer Bank Failure Mechanics	**LTR-6**	Intraday Liquidity Risk Management	*Venkat & Baird, Chapter 4*	a to c
		LTR-8	The Failure Mechanics of Dealer Banks	*Darrell Duffie, Journal of Economic Perspectives 24:1, pp. 51-72, 2010*	a to c
		LTR-1	Liquidity Risk	*Hull, Risk Management and Financial Institutions 5e, Chapter 24*	b
D28	Portfolio Construction and the No-Trade Region	**IM-4**	Portfolio Construction	*Grinold & Kahn, Active Portfolio Management 2e, Chapter 14*	a to i
D29	Crypto, Tokenisation and Digital Resilience	**CI-6**	Regulating the Crypto Ecosystem: The Case of Unbacked Crypto Assets	*IMF Fintech Notes, September 2022*	a to d
		CI-7	Tokenization and Financial Market Inefficiencies	*IMF Fintech Notes, January 2025*	a to e
		CI-8	Digital Resilience and Financial Stability: The Quest for Policy Tools in the Financial Sector	*Jose Ramon Martinez, Banco de España, April 2023*	a to c
D30	Geopolitical Risk and Financial Stability	**CI-4**	Global Financial Stability Report, Chapter 2	*IMF, April 2025*	a to f

Chapters by book

Every 2026 reading this volume teaches, grouped by book, with the sections where it appears.

Book 1 · Market Risk Measurement and Management		14 readings
MR-1	Estimating Market Risk Measures: An Introduction and Overview	B2
MR-2	Non-Parametric Approaches	B2
MR-4	Backtesting VaR	C2
MR-5	VaR Mapping	C3
MR-6	Validating Bank Holding Companies' Value-at-Risk Models for Market Risk	D10
MR-7	Beyond Exceedance-Based Backtesting of Value-at-Risk Models: Methods for Backtesting the Entire Forecasting Distribution Using Probability Integral Transform	C2
MR-9	Empirical Properties of Correlation: How Do Correlations Behave in the Real World?	D11
MR-10	Financial Correlation Modeling: Bottom-Up Approaches	B3
MR-11	Regression Hedging and Principal Component Analysis	C1
MR-12	Arbitrage Pricing with Term Structure Models	C7
MR-13	Expectations, Risk Premium, Convexity, and the Shape of the Term Structure	D12, D18
MR-14	The Art of Term Structure Models: Drift	D1
MR-15	The Art of Term Structure Models: Volatility and Distribution	D1, D18
MR-18	Fundamental Review of the Trading Book	B2, C2

Book 2 · Credit Risk Measurement and Management		15 readings
CR-4	Capital Structure in Banks	D3
CR-5	Introduction to Credit Risk Modeling and Assessment	D3, D20
CR-6	Credit Scoring and Rating	D7, D13
CR-7	Credit Scoring and Retail Credit Risk Management	D13
CR-8	Country Risk: Determinants, Measures, and Implications	D23
CR-9	Estimating Default Probabilities	D5, D14
CR-10	Credit Value at Risk	B1, D5, D20
CR-11	Portfolio Credit Risk	B1, B3
CR-12	Credit Risk	B1, B3, C8, D5, D20
CR-13	Credit Derivatives	D14
CR-14	Derivatives	B2
CR-18	Central Clearing	D24
CR-19	Future Value and Exposure	C8
CR-21	The Evolution of Stress Testing Counterparty Exposures	C9
CR-22	Structured Credit Risk	B3, D15

Book 3 · Operational Risk and Resilience		17 readings
ORR-3	Risk Identification	B5, C6, D2, D21, D22
ORR-4	Risk Measurement and Assessment	D2, D19, D22
ORR-5	Risk Mitigation	B5, D2, D21, D22
ORR-6	Risk Reporting	B5, D17, D19, D22
ORR-12	Guidance on Managing Outsourcing Risk	D2, D21
ORR-13	Case Study: Third-Party Risk Management	D2
ORR-14	Case Study: Investor Protection and Compliance Risks in Investment Activities	D21
ORR-15	Supervisory Guidance on Model Risk Management	C6
ORR-16	Case Study: Model Risk and Model Validation	C6
ORR-17	Stress Testing Banks	C5, C9
ORR-18	Risk Capital Attribution and Risk-Adjusted Performance Measurement	D3, D7
ORR-19	Range of Practices and Issues in Economic Capital Frameworks	B2
ORR-20	Capital Planning at Large Bank Holding Companies: Supervisory Expectations and Range of Current Practice	D10
ORR-21	Capital Regulation Before the Global Financial Crisis	C2, C5, D4
ORR-22	Solvency, Liquidity, and Other Regulation After the Global Financial Crisis	C5, D4, D17, D25
ORR-23	High-Level Summary of Basel III Reforms	C5
ORR-24	Basel III: Finalising Post-Crisis Reforms	C5

Book 4 · Liquidity and Treasury Risk Measurement and Management — 11 readings

LTR-1	Liquidity Risk	D17, D27
LTR-2	Liquidity and Leverage	D17
LTR-3	Early Warning Indicators	D7, D17
LTR-6	Intraday Liquidity Risk Management	D27
LTR-8	The Failure Mechanics of Dealer Banks	D27
LTR-9	Liquidity Stress Testing	D17
LTR-11	Contingency Funding Planning	D26
LTR-13	Repurchase Agreements and Financing	D6
LTR-15	The US Dollar Shortage in Global Banking and the International Policy Response	D8
LTR-16	Covered Interest Parity Lost: Understanding the Cross-Currency Basis	D8
LTR-17	Risk Management for Changing Interest Rates: Asset-Liability Management and Duration Techniques	D17

Book 5 · Risk Management and Investment Management — 8 readings

IM-1	Factor Theory	B4
IM-2	Factors	B4, D18
IM-3	Alpha (and the Low-Risk Anomaly)	B4
IM-4	Portfolio Construction	D28
IM-6	VaR and Risk Budgeting in Investment Management	C4
IM-9	Risk, Regulation and Organizational Structure	D17
IM-15	Market-Driven Scenarios: An Approach for Plausible Scenario Construction	C9
IM-17	Illiquid Assets	D9

Book 5 · Current Issues in Financial Markets — 7 readings

CI-1	Advances in Artificial Intelligence: Implications for Capital Markets Activities	D17
CI-2	The Financial Stability Implications of Artificial Intelligence	B5, D17
CI-4	Global Financial Stability Report, Chapter 2	D30
CI-5	Monetary and Fiscal Policy: Safeguarding Stability and Trust	B5, D16
CI-6	Regulating the Crypto Ecosystem: The Case of Unbacked Crypto Assets	D29
CI-7	Tokenization and Financial Market Inefficiencies	D29
CI-8	Digital Resilience and Financial Stability: The Quest for Policy Tools in the Financial Sector	B5, D17, D29

Tier B: High Corroboration

Map Tier B topics not already in Tier A.

B1 Credit VaR and Granularity

Readings: CR-10 a, b, c, f · Credit Value at Risk (Hull, RMFI 6e, Ch.19); CR-11 c, g, h · Portfolio Credit Risk (Malz, Ch.8, 8.1 to 8.3); CR-12 g · Credit Risk (Hull, OFOD 11e, Ch.24, 24.3)

MAP TIER B · #10 · 6/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #10: "Reported as 'CVaR calculation, direct formula application' and 'CVaR (1) FC'. Credit VaR = worst-case loss at the confidence level minus expected loss: the classic error is forgetting to subtract EL. Two granularity effects are separately examinable (LO 29.h): Credit VaR per unit of exposure falls as the portfolio becomes more granular, but does not fall to zero because of systematic correlation."
Shapes covered here:
S1 Credit VaR by direct formula: worst-case loss at the confidence level minus expected loss (Q1)
S2 formula-card form and the dropped-EL trap (Q1)
S3 granularity effect 1: Credit VaR falls as the portfolio becomes more granular (Q2)
S4 granularity effect 2: it does not fall to zero when defaults are correlated (Q2)

Revision notes

- **The specific: Credit VaR as a quantile minus EL** (CR-11 c, S1). Malz: portfolio Credit VaR "is a quantile of the credit loss, minus the expected loss of the portfolio". Because EL is always subtracted, Credit VaR can be *negative*: a single credit with $\pi = 2\%$ has a 95% Credit VaR of $0 - 20{,}000{,}000 = -20{,}000{,}000$ (a gain), since the 95th percentile loss is zero.
- **Hull's wording** (CR-10 b; CR-12 g). Hull defines credit risk VaR "similarly to market risk VaR": "the credit risk loss over a certain time period that will not be exceeded with a certain confidence level". His Vasicek estimate $L(1-R)V(X,T)$ is a loss quantile with no EL deduction (Example 12.5: USD 100 million retail book, PD 2%, recovery 60%, $\rho = 0.1$: WCDR 12.8%, 99.9% credit VaR USD 5.13 million). When a question says Credit VaR and gives EL or PD, apply the Malz definition (quantile minus EL); when it reproduces Hull's $L(1-R)V(X,T)$, it is asking for the loss quantile.
- **Market VaR versus credit VaR** (CR-10 a). Definition: same concept (loss not exceeded at a confidence level); some credit models count only default losses, others also downgrades and spread changes. Horizon: market VaR often one day; credit VaR for instruments not held for trading usually one year (Basel: one year, 99.9%). Tools: historical simulation is the main market-risk tool; credit VaR needs "a more elaborate model". Scenario PDs should be real-world; risk-neutral estimates are then used to value the portfolio on the horizon date.
- **Credit correlation** (CR-10 a; CR-11 c). Defaults are not independent: in a downturn most companies become more likely to default. Credit correlation rises in stressed conditions and so does the risk of a credit portfolio. Malz: correlation "affects the volatility and extreme quantiles of loss rather than the expected loss".
- **Two extremes of correlation** (CR-11 c). Default correlation 1: the portfolio behaves as if $n = 1$ (all-or-nothing), no diversification. Default correlation 0: the number of defaults is binomial(n, π) and significant diversification is possible. Malz's USD 1 billion book, $\pi = 2\%$, zero recovery, $n = 50$: 95th percentile = 3 defaults, loss USD 60 million, Credit VaR $60 - 20 =$ USD 40 million; $n = 1{,}000$: 28 defaults, Credit VaR USD 8 million.
- **Granularity, effect 1** (CR-11 h, S3). For a given default probability, Credit VaR "decreases as the credit portfolio becomes more granular". The convergence is more drastic with a high default probability; conversely "it is harder to reduce VaR by making the portfolio more granular, if the default probability is low". Expected loss is unchanged by granularity ($\pi\times$ portfolio value in every panel of Malz's Table 11.1).
- **Granularity, effect 2** (CR-11 d, h, S4). With independent credits the loss converges to EL "almost certainly" and Credit VaR goes to zero. In the single-factor model the granular portfolio's loss rate converges to the conditional default probability $p(m)$, so "portfolio returns are driven by the market factor": idiosyncratic risk diversifies away, systematic risk does not, and Credit VaR stays positive. Malz Ch.9 adds that a more granular collateral pool lets a securitisation carry a larger senior tranche with no increase in Credit VaR (see D15).

- **Simulation of joint defaults** (CR-11 g). Draw correlated standard normals (copula correlation), map each to a default time or rating outcome, compute the loss on each thread, repeat, and read Credit VaR off the sorted simulated losses (Malz, Ch.9, 22.3: the 10th largest of 1,000 losses for 99%, minus EL). Hull's CreditRisk+ Monte Carlo: sample an overall default rate, link each category's rate to it, sample defaults and LGDs, total the loss, repeat, read VaR.
- **Rating transition matrices** (CR-10 c). Historical one-year matrices (S&P 1981 to 2020: A stays A with 92.61%, defaults with 0.05%). Assuming independence across periods, the n-year matrix is the nth power; six months needs the square root, three months the fourth root. Independence is not exactly true: recently downgraded companies are more likely to be downgraded again (ratings momentum).
- **CreditMetrics mechanics** (CR-10 c; CR-12 g). Sample correlated normals x_A, x_B (copula correlation set to the equity-return correlation, e.g. 0.2), convert transition probabilities into thresholds with N^{-1} of cumulative probabilities, revalue every counterparty at the horizon; upgrades count as negative losses. See D20 for the model comparison.
- **Credit spread risk** (CR-10 f). Trading-book credit products depend critically on spreads. Approach 1: historical simulation of 500 days of spread changes for a one-day 99% VaR, scaled by $\sqrt{10}$. Problems: a company alive today did not default in the past, so no future default is assumed (regulators add jump-to-default risk); spread data are not updated daily for all names. Approach 2: a 10-day transition matrix plus a distribution of spread changes per rating category, simulated; correlation through a Gaussian copula on ratings and near-perfectly correlated spread moves across categories.
- **Example 10.1** (CR-10 f). Two-year zero, USD 1,000 principal, yield 5%, price 907.03; worst outcome default (0.1%, value 400, loss 507.03); next worst B at 500 bp (0.267%, loss 44.17). VaR is 507.03 above 99.9% confidence, 44.17 between 99.633% and 99.9%, and 2.92 at 99%.
- **Constant level of risk** (CR-10 f). Rebalancing monthly back to BBB: default probability $12 \times 0.0001 \times 0.9999^{11} = 0.0012$ versus 0.0017 buy-and-hold; one downgrade to BB about 0.040 versus 0.0378. Fewer big losses, more small ones: VaR and ES are generally *smaller* under the constant level of risk strategy.

Formulas

Credit VaR (Malz). $\text{Credit VaR}_\alpha = q_\alpha(\text{credit loss}) - \text{EL}, \qquad \text{EL} = \sum_i \text{EAD}_i \times \text{LGD}_i \times \pi_i$

Worst-case default rate (Vasicek; Hull 10.1, 12.10).

$$\text{WCDR}(T,X) = V(X,T) = N\left(\frac{N^{-1}(\text{PD}) + \sqrt{\rho}\,N^{-1}(X)}{\sqrt{1-\rho}}\right)$$

Loss quantile (Hull; Gordy, eq. 10.2). $\sum_i \text{WCDR}_i(T,X) \times \text{EAD}_i \times \text{LGD}_i$; for a homogeneous book $L(1-R)V(X,T)$; loan contribution $L_i(1-R)V(X,T)$

Transition matrices. $M_{n\text{ years}} = M^n, \qquad M_{6\text{ months}} = M^{1/2}, \qquad M_{3\text{ months}} = M^{1/4}$

Constant level of risk. $P(\text{one event in 12 months}) = 12\,q_m(1-q_m)^{11}$ with q_m the one-month probability

Credit spread VaR (historical simulation). $\text{VaR}_{\text{10-day}} = \sqrt{10} \times \text{VaR}_{\text{1-day}}$

Exam traps

- **Dropped term (EL):** reporting the worst-case loss as Credit VaR. In Malz's framework the EL is always subtracted, which is also why Credit VaR can be negative.
- **Quantile threshold:** the 99% quantile is the *smallest* number of defaults whose cumulative probability is at least 99%, not the last row below 99%.
- **Wrong-input twin:** recovery rate in place of LGD in both the worst-case loss and the EL.
- **Sibling swap:** granularity changes the *dispersion* of losses, not the expected loss.
- **Polarity flip:** granularity helps *most* when the default probability is high; with a low PD it is harder to reduce Credit VaR.
- **Absolute claim:** "a granular portfolio has zero Credit VaR" is true only for independent defaults; with correlation the common factor remains.

Question B1.1

CR-11 c, h; CR-10 b · S1, S2 · [Map #10]

A credit portfolio manager at a Canadian bank is computing the one-year Credit VaR of a CAD 400 million book that the bank's economic capital model treats in Malz's portfolio credit risk framework. Data:

- 40 loans of CAD 10 million each, to 40 different obligors
- One-year probability of default of each loan: 3%; recovery rate: 40%
- Defaults are assumed independent; the cumulative probability of the number of defaults is:

Number of defaults	0	1	2	3	4	5	6
Cumulative probability	0.2957	0.6615	0.8822	0.9686	0.9933	0.9988	0.9998

Which of the following is closest to the 99% Credit VaR of the book?

A. CAD 10.8 million

B. CAD 11.2 million

C. CAD 16.8 million

D. CAD 24.0 million

Solution

Answer: C

Step 1 – 99th percentile of the number of defaults

The quantile is the smallest count whose cumulative probability is at least 0.99: $P(\leq 3) = 0.9686 < 0.99$, $P(\leq 4) = 0.9933 \geq 0.99$, so the 99th percentile is 4 defaults.

$\Rightarrow$ *In 99% of years the book suffers 4 or fewer defaults.*

Step 2 – Worst-case loss at 99%

$$\text{Loss}_{99\%} = 4 \times 10 \times (1 - 0.40) = 4 \times 6.0 = 24.0$$

$\Rightarrow$ *Each default costs CAD 6.0 million (LGD 60%), so the 99% loss quantile is CAD 24.0 million.*

Step 3 – Expected loss

$$\text{EL} = 400 \times 0.03 \times 0.60 = 7.2$$

$\Rightarrow$ *On average the book loses CAD 7.2 million a year, which is priced into the loans and reserved for.*

Step 4 – Credit VaR

$$\text{Credit VaR}_{99\%} = \text{Loss}_{99\%} - \text{EL} = 24.0 - 7.2 = 16.8$$

Interpretation

CAD 16.8 million is the unexpected loss at 99%: the economic capital buffer beyond the provisioned EL. Making the book more granular (more, smaller independent loans) would shrink this number while leaving the CAD 7.2 million EL unchanged.

Why the other options are wrong

A is incorrect. Took the last row below 99% as the quantile: $3 \times 6.0 - 7.2 = 10.8$.

B is incorrect. Used the 40% recovery rate as the LGD in both terms: $4 \times 10 \times 0.40 - 400 \times 0.03 \times 0.40 = 16.0 - 4.8 = 11.2$.

D is incorrect. Forgot to subtract the expected loss and reported the 99% loss quantile, 24.0.

LO: CR-11 c: Assess the impact of correlation on a credit portfolio and its Credit VaR. CR-11 h: Assess the effect of granularity on Credit VaR. CR-10 b: Define and calculate Credit VaR. · Reference: Malz, Ch.8, 8.2 Granularity and Portfolio Credit Value-at-Risk, Table 11.1; Hull, RMFI 6e, Ch.19

Question B1.2

CR-11 c, d, h · S3, S4 · [Map #10]

The CRO of a Japanese regional bank wants to cut the Credit VaR of its SME loan book. She proposes replacing 20 loans of JPY 5 billion each with 2,000 loans of JPY 50 million each to similar borrowers. The default probability, the LGD and the pairwise default correlation (0.05, estimated with a single-factor model) would all be unchanged. A risk analyst is drafting the memo to the risk committee. Which of the following statements should the analyst

include?

A. The move lowers Credit VaR, because spreading the book across more borrowers lowers the portfolio's expected loss, and the expected loss is subtracted from the loss quantile.

B. The move lowers Credit VaR, and the proportional reduction from granularity is largest when the default probability of the loans is low.

C. The move lowers Credit VaR, but with a positive default correlation it cannot bring Credit VaR to zero, because losses on the granular book still move with the common factor.

D. The move lowers Credit VaR and, with enough borrowers, drives it to zero whatever the default correlation, because the loss rate converges to the default probability.

Solution **Answer: C**

Step 1 – What granularity changes

With the total exposure fixed, more and smaller positions reduce the variance of portfolio losses, so the loss quantile moves toward the expected loss and Credit VaR falls (Malz Table 11.1: at $\pi = 2\%$ and 99%, Credit VaR is USD 980 million for $n = 1$, 60 million for $n = 50$, 11 million for $n = 1{,}000$).

Step 2 – What it cannot change

In the single-factor model the loss rate of a granular book converges to the conditional default probability $p(m) = \Phi\big((k - \beta m)/\sqrt{1-\beta^2}\big)$, not to π. Idiosyncratic risk is diversified away; the market factor m still moves the loss rate, so Credit VaR stays positive.

$\Rightarrow$ *The CRO's plan removes concentration risk but leaves the systematic part of Credit VaR in place.*

Interpretation

C is correct. The memo should present the reduction as a cut in idiosyncratic risk only; the residual Credit VaR is set by the default probability and the correlation to the common factor.

Why the other options are wrong

A is incorrect. Sibling swap. Granularity changes the dispersion and extreme quantiles of loss, not the expected loss, which stays $\pi\times$ portfolio value for every n.

B is incorrect. Polarity flip. Malz: the convergence "is more drastic with a high default probability" and "it is harder to reduce VaR by making the portfolio more granular, if the default probability is low".

D is incorrect. Absolute claim / scope error. Convergence of the loss to EL and Credit VaR to zero holds for a very large number of *independent* positions; with correlation 0.05 the common factor remains.

LO: CR-11 h: Assess the effect of granularity on Credit VaR. CR-11 c: Assess the impact of correlation on a credit portfolio and its Credit VaR. CR-11 d: Describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation. · Reference: Malz, Ch.8, 8.2 and 8.3

B2 Expected Shortfall and Coherence

Readings: MR-1 c, d, e · Estimating Market Risk Measures: An Introduction and Overview (Dowd, Ch.3); MR-2 a, c, d · Non-Parametric Approaches (Dowd, Ch.4); MR-18 a, b · Fundamental Review of the Trading Book (Hull, RMFI Ch.18); ORR-19 a · Range of Practices and Issues in Economic Capital Frameworks (BCBS 2009); CR-14 j · Derivatives (Gregory, Ch.2)

MAP TIER B · #12 · 6/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #12: "One recall captures the exact trap: 'ES numerical given the sample and resampling values: I took the average of all 5 samples, it could be average of 4 losses or 5.' That off-by-one on tail membership is the intended distractor. A second reported form is ES as the average of tail VaRs read off a confidence-level table, and a third is ES within filtered historical simulation. Also the coherence argument: ES is subadditive, VaR is not."
Shapes covered here:
S1 ES from a sample and its bootstrap resamples: exactly which losses enter the tail average, 4, 5 or 6 (Q1)
S2 ES as the average of tail VaRs read off a confidence-level table, against the 99% VaR it replaces under FRTB (Q2)
S3 ES within filtered historical simulation (Q3)
S4 coherence: ES is subadditive, VaR is not (Q3).

Revision notes

- **Expected shortfall** (MR-1 c): "the probability-weighted average of tail losses". For standard normal losses the 95% VaR is 1.645 and the 95% ES is 2.063. ES is "the easiest" coherent risk measure to estimate.
- **Which observations enter the tail** (MR-1 a, c; S1): with n observations and confidence level α, the tail holds $(1-\alpha)n$ losses and the VaR is the $\big((1-\alpha)n+1\big)$th highest loss, "the highest loss observation outside the tail". With 1,000 observations the 95% VaR is the 51st highest loss (1.704) and the ES is "the average of the 50 highest losses" (2.196). With 100 observations the 95% VaR is the 6th largest loss and the 96% VaR the 5th largest, so the 95% ES averages the 5 largest losses.
- **Dowd's footnote on the boundary:** taking the VaR as the 50th, the 51st or a mix of the two is "just an issue of approximation"; the convention adopted throughout is the highest loss *outside* the tail. Gregory (CR-14) words ES as the average loss "equal to or above" VaR; for the MR-1 numerical, follow Dowd and leave the VaR observation out.
- **Average-of-tail-VaRs method** (MR-1 c, d; S2): slice the tail into n equal-probability slices, estimate the $n-1$ tail VaRs at confidence levels *above* α, and average them. With $n = 10$ at 95%: nine VaRs from 95.5% to 99.5%, average 2.0250 against a true 2.063. The estimate rises with n and converges (2.0513 at $n = 50$, 2.0626 at $n = 10{,}000$): coarse slicing *understates* ES.
- **Coherent risk measures** (MR-1 d): $M_\phi = \int_0^1 \phi(p) q_p \, dp$, a weighted average of loss quantiles. ES sets $\phi(p) = 1/(1-\alpha)$ for $p \geq \alpha$ and 0 otherwise; spectral measures (e.g. exponential, $\gamma = 0.05$) put rising weights on higher quantiles and need far more slices to converge (judged by the halving error). An existing VaR engine needs only "the last few lines of code" changed to produce them.
- **Standard errors** (MR-1 e): quantile standard errors rise as the quantile moves into the tail and fall with n. Yamai and Yoshiba found ES estimators have much bigger standard errors than VaR for heavy-tailed distributions ("prone to the effects of large but infrequent losses"); Acerbi found the two roughly comparable in empirically realistic ranges.
- **Bootstrapped HS** (MR-2 a; S1): resample with replacement; each resample gives a VaR and an ES ("the average of the losses in each resample exceeding the resample VaR"); the best estimate is the *mean* of the resample estimates (1.669 and 2.114 against basic HS 1.704 and 2.196). The bootstrapped ES histogram is better behaved than the VaR histogram because ES is an average of tail VaRs; 90% bootstrap interval for ES [1.986, 2.271].
- **Age-weighted HS** (BRW, MR-2 c): $w(i) = \lambda^{i-1}(1-\lambda)/(1-\lambda^n)$; traditional HS is the special case $\lambda \to 1$; more responsive to recent large losses, reduces ghost effects, but shrinks the effective sample size and is ad hoc.

- **Volatility-weighted HS** (Hull and White, MR-2 c): $r^*_{t,i} = (\sigma_{T,i}/\sigma_{t,i})\, r_{t,i}$; returns are scaled up when current volatility exceeds the volatility at the time; VaR and ES can exceed the largest historical loss. **Correlation-weighted HS** adjusts returns through Choleski matrices of the historical and current correlation matrices.
- **Filtered historical simulation** (MR-2 c; S3): a semi-parametric bootstrap. (1) Fit an asymmetric GARCH (AGARCH, captures the leverage effect); (2) divide realised returns by the model's volatility forecasts to get standardised, iid returns; (3) bootstrap these and multiply each by the AGARCH forecast of *tomorrow's* volatility; (4) read VaR off the simulated losses. It can exceed the maximum historical loss, keeps the correlation structure for multi-asset portfolios, and "can also be extended easily to allow for the estimation of ES". Pritsker's concern that FHS VaR may not pay enough attention to extremes has the partial response: use ES instead of VaR.
- **ES as the more informative measure** (MR-2 c): after a crash, an equal-weight HS VaR may not move except at very high confidence, but "the increase in risk would show up in ES estimates just after the first shock".
- **Non-parametric pros and cons** (MR-2 d): no distributional assumption (fat tails, skewness accommodated), easy to implement and explain, no covariance matrices; but results depend completely on the historical data set, are constrained by the largest loss, slow to reflect shifts, and subject to ghost effects. Rule of thumb: at least 500 daily observations.
- **Coherence and subadditivity** (ORR-19 a, CR-14 j; S4): the four conditions are monotonicity, positive homogeneity, translation invariance and subadditivity ("the risk measure of two portfolios, if combined, is always smaller or equal to the sum of the risk measures of the two individual portfolios"). VaR is subadditive for elliptical distributions such as the normal, but not for very skewed losses, asymmetric copulas, or independent but very heavy-tailed risk factors; the gap matters more for credit and operational risk. ES is coherent, so capital allocation and limits for sub-portfolios stay consistent with the portfolio measure.
- **FRTB** (MR-18 a, b): stressed ES at 97.5% replaces 99% VaR and stressed VaR. For normal losses, 99% VaR $= \mu + 2.326\sigma$ and 97.5% ES $= \mu + 2.338\sigma$; with a heavier tail the 97.5% ES "can be considerably greater" than the 99% VaR. Liquidity horizons 10, 20, 40, 60, 120 days; overlapping 10-day changes; cascade formula across categories; partial (desk) ES sum always exceeds the whole-portfolio ES; WES $= 0.5\,\text{ES}_T + 0.5\sum \text{ES}_{P,j}$. See also C2 for FRTB backtesting.

Formulas

HS from n losses at confidence α. Tail size $(1-\alpha)n$; VaR = the $\big((1-\alpha)n+1\big)$th highest loss; ES $= \dfrac{1}{(1-\alpha)n}\sum_{i=1}^{(1-\alpha)n} L_{(i)}$ (the losses *beyond* the VaR).

Bootstrap point estimate. $\widehat{\text{ES}}_B = \dfrac{1}{B}\sum_{b=1}^{B} \text{ES}^{(b)}$

Average tail VaR (n slices). $\text{ES}_\alpha \approx \dfrac{1}{n-1}\sum_{j=1}^{n-1} \text{VaR}_{\alpha+j(1-\alpha)/n}$ (the VaR at α itself is excluded)

Coherent risk measure. $M_\phi = \int_0^1 \phi(p)\, q_p\, dp$; ES: $\phi(p) = \dfrac{1}{1-\alpha}$ for $p \geq \alpha$, else 0.

Weighted HS. Age: $w(i) = \dfrac{\lambda^{i-1}(1-\lambda)}{1-\lambda^n}$. Volatility: $r^*_{t,i} = \dfrac{\sigma_{T,i}}{\sigma_{t,i}}\, r_{t,i}$. FHS: $z_t = r_t/\hat{\sigma}_t$, simulated return $= z^{(b)} \times \hat{\sigma}_{T+1}$.

Subadditivity. $\rho(X+Y) \leq \rho(X) + \rho(Y)$

FRTB. 99% VaR $= \mu + 2.326\sigma$; 97.5% ES $= \mu + 2.338\sigma$ (normal). Cascade: ES $= \sqrt{\text{ES}_1^2 + \sum_{j\geq 2}\left(\text{ES}_j\sqrt{\frac{LH_j - LH_{j-1}}{10}}\right)^2}$; WES $= 0.5\,\text{ES}_T + 0.5\sum_j \text{ES}_{P,j}$.

Book checks. 1,000 obs: 95% VaR 1.704, ES 2.196; bootstrap 1.669, 2.114. Ten slices: ES 2.0250 vs true 2.063.

Exam traps

- **Off-by-one on tail membership (S1):** at 95% with 100 losses the tail is 5 losses. Averaging 4 gives the 96% ES; averaging 6 drags the VaR observation into the tail.
- **Including the VaR row (S2):** in a tail-VaR table the ES at α averages only the VaRs at confidence levels *above* α; and divide by the number of tail VaRs $(n-1)$, not by the number of slices n.
- **Bootstrap intermediate result:** the best bootstrap ES is the mean across resamples; one resample's ES is only an intermediate number.

- **Normal shortcut misapplied (sibling swap):** "97.5% ES ≈ 99% VaR" holds for the normal only; for a heavy-tailed book the ES is larger.
- **Scope error on subadditivity:** VaR is subadditive for elliptical distributions; claiming it for skewed or heavy-tailed losses, or claiming ES fails subadditivity, are the two corruptions.
- **FHS sibling swap:** FHS (like Hull and White) can produce VaR and ES beyond the largest historical loss; only basic HS is capped by it. FHS scales by *tomorrow's* forecast volatility, not the volatility of the day drawn.

Question B2.1

MR-1 c; MR-2 a · S1 · [Map #12]

A market risk analyst at a Canadian asset manager estimates the 1-day 95% expected shortfall of an equity portfolio using bootstrapped historical simulation. Each resample contains 100 daily P/L observations drawn with replacement from the historical sample. For illustration, only three resamples are used:

- ES computed from resample 1: CAD 7.10 million; ES computed from resample 2: CAD 7.32 million
- Ten largest losses in resample 3, in CAD million, ranked: 9.8, 8.1, 7.4, 6.9, 6.9, 6.2, 5.6, 5.3, 5.0, 4.8

Which of the following is closest to the bootstrapped estimate of the 1-day 95% ES?

A. CAD 7.32 million

B. CAD 7.41 million

C. CAD 7.49 million

D. CAD 7.82 million

Solution

Answer: B

Step 1 – Size of the tail

$$(1-\alpha)\,n = (1-0.95)\times 100 = 5 \text{ losses}$$

⇒ *Five of the 100 resampled losses lie in the 5% tail. The 95% VaR is the 6th largest loss, CAD 6.2 million, the highest loss outside the tail.*

Step 2 – ES of resample 3

$$\text{ES}^{(3)} = \frac{9.8+8.1+7.4+6.9+6.9}{5} = \frac{39.1}{5} = 7.82$$

⇒ *Average of the losses exceeding the resample VaR. The repeated 6.9 is legitimate: resampling with replacement can draw the same day twice.*

Step 3 – Bootstrap point estimate

$$\widehat{\text{ES}}_B = \frac{7.10+7.32+7.82}{3} = \frac{22.24}{3} = 7.41$$

Interpretation

The bootstrapped 95% ES is about CAD 7.41 million: on the worst 5% of days the portfolio is expected to lose that amount on average. Averaging across resamples is what makes the bootstrap estimate more accurate than a single raw sample estimate (Dowd: 2.114 against 2.196 for basic HS).

Why the other options are wrong

A is incorrect. Averaged six losses, pulling the VaR observation into the tail: $(39.1+6.2)/6 = 7.55$, then $(7.10+7.32+7.55)/3 = 7.32$.

C is incorrect. Averaged only four losses, treating the 5th largest loss as the VaR (that is the 96% VaR): $32.2/4 = 8.05$, then $(7.10+7.32+8.05)/3 = 7.49$. This is the reported "4 or 5" trap.

D is incorrect. Stopped at the intermediate result: the ES of resample 3 alone, 7.82, instead of the mean over all resamples.

LO: MR-1 c: Estimate the expected shortfall given profit and loss (P&L) or return data. MR-2 a: Apply the bootstrap historical simulation approach to estimate coherent risk measures. · Reference: Dowd, Measuring Market Risk 2e, Ch.3, 1.2 (footnote 2) and 1.4; Ch.4, 2.2 (Basic and Bootstrapped Historical Simulation)

Question B2.2

MR-1 c, d; MR-18 a · S2 · [Map #12]

A market risk manager at an Australian bank is preparing a trading desk for the switch from a 99% VaR to a 97.5% ES under FRTB. The desk's P/L distribution is heavy-tailed. The desk's model slices the tail beyond the 97.5% confidence level into 10 equal-probability slices and reports the VaR at each confidence level (AUD million):

Confidence level	97.50%	97.75%	98.00%	98.25%	98.50%	98.75%	99.00%	99.25%	99.50%	99.75%
VaR	4.29	4.50	4.75	5.04	5.39	5.84	6.42	7.26	8.60	11.43

Which of the following is closest to the desk's 97.5% ES and the change in its headline risk number when the 97.5% ES replaces the 99% VaR?

A. ES AUD 5.92 million; change −AUD 0.50 million

B. ES AUD 6.35 million; change −AUD 0.07 million

C. ES AUD 6.42 million; change AUD 0.00 million

D. ES AUD 6.58 million; change +AUD 0.16 million

Solution

Answer: D

Step 1 – Identify the tail VaRs

With $n = 10$ slices beyond 97.5% there are $n - 1 = 9$ tail VaRs, at 97.75% through 99.75%. The 97.5% VaR (4.29) marks the start of the tail and is excluded.

Step 2 – Average the tail VaRs

$$\text{ES}_{97.5\%} \approx \frac{4.50 + 4.75 + 5.04 + 5.39 + 5.84 + 6.42 + 7.26 + 8.60 + 11.43}{9} = \frac{59.23}{9} = 6.58$$

⇒ *On the worst 2.5% of days the desk loses about AUD 6.58 million on average.*

Step 3 – Compare with the 99% VaR

$$\Delta = \text{ES}_{97.5\%} - \text{VaR}_{99\%} = 6.58 - 6.42 = +0.16$$

Interpretation

For a normal distribution the two numbers would almost coincide (2.338σ against 2.326σ); because this desk's tail is heavy, the 97.5% ES exceeds the 99% VaR, which is the FRTB motivation for capturing tail risk. With only 10 slices the estimate is, if anything, too low: Dowd's Table 1.2 shows the average-tail-VaR estimate rising towards the true ES as n grows.

Why the other options are wrong

A is incorrect. Divided the nine tail VaRs by the number of slices (10) instead of the number of tail VaRs (9): $59.23/10 = 5.92$, change $5.92 - 6.42 = -0.50$.

B is incorrect. Included the 97.5% VaR in the average (the off-by-one on tail membership): $(59.23 + 4.29)/10 = 6.35$, change -0.07.

C is incorrect. Applied the normal-distribution equivalence to a heavy-tailed desk and set the 97.5% ES equal to the 99% VaR, so the change is zero.

LO: MR-1 c: Estimate the expected shortfall given profit and loss (P&L) or return data. MR-1 d: Estimate risk measures by estimating quantiles. MR-18 a: Describe the changes to the Basel framework for calculating market risk capital under the Fundamental Review of the Trading Book (FRTB) and the motivations for these changes. · Reference: Dowd, Measuring Market Risk 2e, Ch.3, 1.4, Tables 1.1 and 1.2; Hull, RMFI 6e, Ch.18, 18.1

Question B2.3

MR-2 c; ORR-19 a; CR-14 j · S3, S4 · [Map #12]

The CRO of a European bank is merging a credit trading desk, whose losses are heavily skewed, with a rates desk. A junior risk analyst is preparing a report recommending that the bank replace its basic historical simulation VaR with an expected shortfall estimated by filtered historical simulation (FHS). Which of the following statements is correct for the analyst to include in the report?

A. Because VaR is subadditive for skewed loss distributions, the combined 99% VaR of the two desks cannot exceed the sum of their stand-alone VaRs.

B. Because FHS bootstraps from the historical data set, its VaR and ES estimates, like those of basic HS, cannot

exceed the largest historical loss.

C. Because ES averages all tail losses, its estimators have smaller standard errors than VaR estimators, and the advantage grows as tails become heavier.

D. Because ES is subadditive, the combined ES of the two desks cannot exceed the sum of their stand-alone ESs, and FHS can be extended to estimate ES.

Solution **Answer: D**

Step 1 – Coherence

A coherent risk measure satisfies monotonicity, positive homogeneity, translation invariance and subadditivity, $\rho(X+Y) \leq \rho(X) + \rho(Y)$. ES is coherent; VaR "may not always satisfy the subadditivity condition" (ORR-19), and "may not behave in a sub-additive fashion" (Gregory, CR-14).

Step 2 – FHS and ES

FHS bootstraps GARCH-standardised returns and rescales them by tomorrow's volatility forecast; Dowd notes that it "can also be extended easily to allow for the estimation of ES as well as VaR". Using ES is also the suggested response to Pritsker's finding that FHS VaR may not pay enough attention to extreme observations.

Interpretation

D is correct. For a merger involving a skewed credit book, ES is the measure that guarantees the combined number cannot exceed the sum of the parts, so desk limits and capital allocation stay consistent with the firm-wide figure.

Why the other options are wrong

A is incorrect. Scope/condition error. VaR is subadditive for elliptical distributions such as the normal; non-subadditivity "can occur when assets in portfolios have very skewed loss distributions", exactly the credit desk here.

B is incorrect. Sibling swap. Being capped by the largest historical loss is a property of basic HS; FHS, like the Hull and White approach, scales returns by current volatility and "allows us to get VaR and ES estimates that can exceed the maximum historical loss".

C is incorrect. Polarity flip. Yamai and Yoshiba found ES estimators had *much bigger* standard errors than VaR for particularly heavy-tailed distributions, because ES is more prone to large but infrequent losses.

LO: MR-2 c: Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches. ORR-19 a: Within the economic capital implementation framework, describe the challenges that appear in: defining and calculating risk measures; risk aggregation; validation of models; dependency modeling in credit risk; evaluating counterparty credit risk; assessing interest rate risk in the banking book. CR-14 j: Describe the approaches used and the challenges faced in modeling derivatives risk. · Reference: Dowd, Measuring Market Risk 2e, Ch.4, 2.4 and footnotes 13 and 15; Ch.3, 1.5; BCBS (2009), Range of Practices, risk measures section and footnote 8; Gregory, The xVA Challenge 4e, Ch.2

B3 Default Correlation, Joint PD and Copulas

Readings: CR-11 a, b, c, d, e, f · Portfolio Credit Risk (Malz, Ch.8, 8.1 to 8.3); CR-12 e, f · Credit Risk (Hull, OFOD 11e, Ch.24, 24.2); MR-10 a, b, c · Financial Correlation Modeling: Bottom-Up Approaches (Meissner, Ch.5, pp. 126 to 134); CR-22 l · Structured Credit Risk (Malz, Ch.9, 22.4)

MAP TIER B · #16 · 6/16 SOURCES | MAP TIER D · #60 · 2/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Map #16: "Reported: 'default correlation of 2 firms'; 'joint prob of default (direct calculation)'; 'default correlation between credits'; 'Gaussian copula, implied correlation'. Your formula card carries the full expression: $\rho = [P(\text{both default}) - \pi_1\pi_2]/\sqrt{\pi_1(1-\pi_1)\pi_2(1-\pi_2)}$. The dropped-term trap is omitting the $\pi_1\pi_2$ subtraction in the numerator. Conditional default probability is separately flagged as examinable."
Map #60: "Named: LO 40.i, 'just remember the impacts via the table', with one possible calculation; and the concept of implied correlation, described as 'likely to be testable.' Reported independently as 'Gaussian copula, implied correlation.' "
Shapes covered here:
S1 default correlation of two firms (Q1)
S2 joint probability of default, direct calculation (Q2)
S3 default correlation between credits with unequal default probabilities (Q1, Q2)
S4 Gaussian copula and implied correlation (Q3)
S5 dropped-term trap: omitting $\pi_1\pi_2$ (Q1 option C; Q2 option B)
S6 conditional default probability given the common factor (Q3)
S7 LO 40.i impacts table for tranches (D15.1)

Revision notes

- **Default correlation** (CR-11 a, S1). Two firms, default probabilities π_1, π_2 over a horizon T, joint default probability π_{12}. Defaults are Bernoulli variables: no default $1-\pi_1-\pi_2+\pi_{12}$; firm 1 only $\pi_1-\pi_{12}$; firm 2 only $\pi_2-\pi_{12}$; both π_{12}. π_{12} is "a genuinely new parameter, a primitive": it cannot be computed from π_1 and π_2 unless defaults are independent.
- **Moments** (CR-11 a). Means π_i; variances $\pi_i(1-\pi_i)$; covariance $\pi_{12}-\pi_1\pi_2$; correlation = covariance over the product of standard deviations (eq. 11.1). Independence gives $\pi_{12}=\pi_1\pi_2$ and $\rho_{12}=0$; the larger the excess of π_{12} over $\pi_1\pi_2$, the higher the correlation. $P(\text{at least one}) = \pi_1+\pi_2-\pi_{12}$.
- **Example 11.1.** $\pi_1 = 0.0025$, $\pi_2 = 0.0125$: independent $\pi_{12} = 0.000031$ (under a third of a basis point); with $\rho = 5\%$, $\pi_{12} = 0.000309$, "nearly 10 times as great".
- **Order of magnitude** (CR-11 a). Default is rare, so default correlation is hard to estimate from default data; most studies find one-year correlations around 0.05, varying widely by period, industry and domicile, and often negative. Because PDs are small, an "optically" small correlation can have a large impact.
- **Drawbacks of the correlation framework** (CR-11 b). With $n > 2$ credits there are 2^n events but only $n+1+n(n-1)/2$ conditions (3 credits: 8 versus 7; 10 credits: 1,024 versus 56), so pairwise correlations cannot specify the full distribution. Many pairwise parameters are needed, and a single common parameter must be non-negative to keep the correlation matrix positive-definite. Guarantees and revolvers (option-like), CDS basis trades (technical factors) and convertibles (equity and vega risk) fit the framework poorly.
- **Single-factor model** (CR-11 d, e). $a_i = \beta_i m + \sqrt{1-\beta_i^2}\,\epsilon_i$ with m, ϵ_i independent standard normals; each a_i is standard normal, beta to the factor is β_i, and the asset return correlation of any pair is $\beta_i\beta_j$ (Example 11.3: cyclical 0.5, defensive 0.1, correlation 0.05). Firm i defaults if $a_i \le k_i$, with $\pi_i = \Phi(k_i)$.
- **Conditional default probability** (CR-11 d, S6). Given $m = \bar{m}$, defaults are independent (conditional independence), the conditional mean is $\beta_i\bar{m}$ and the conditional standard deviation $\sqrt{1-\beta_i^2}$: $p(m) = \Phi\left((k_i-\beta_i m)/\sqrt{1-\beta_i^2}\right)$. It exceeds or falls short of π_i unless $m=0$ or $\beta_i=0$. Example 11.4: $\beta=0.4$, $\pi=1\%$: conditional PD 1.78% at $m=-1$ and 6.4% at $m=-2.33$.
- **Joint default in the single-factor model** (CR-11 f). The pair's returns are bivariate standard normal with correlation $\beta_i\beta_j$; $\pi_{12} = P[a_i \le k_i,\, a_j \le k_j]$. Homogeneous case: default correlation $= (\pi_{12}-\pi^2)/(\pi(1-$

π)). Example 11.5: $\pi = 1\%$, default correlation 0.05 needs $\beta = 0.561$, asset correlation 0.315, joint PD 6 bp; $\beta = 0.50$ gives 4.3 bp and 0.034. Asset correlation is far larger than default correlation.

- **Impact of correlation** (CR-11 c). Correlation leaves EL unchanged and moves the tails; see B1 for Credit VaR and granularity.
- **Reduced form versus structural correlation models** (CR-12 e). Reduced form: correlated stochastic hazard rates linked to macro variables; mathematically attractive, but the achievable range of default correlation is limited (fix: jumps in the hazard rate). Structural: correlated asset-value processes in a Merton-type model; correlation can be made as high as desired, but computation is slow. Default correlation is why credit risk cannot be fully diversified and the major reason risk-neutral PDs exceed real-world PDs.
- **Gaussian copula for time to default** (CR-12 f; MR-10 b). Times to default are not even approximately normal, so each t_i is mapped percentile-to-percentile to $x_i = N^{-1}[Q_i(t_i)]$ (5th percentile to -1.645), and the x_i are assumed multivariate normal. The joint distribution is then fully defined by the marginals and the copula correlation, often proxied by equity return correlation. Works with real-world or risk-neutral PDs.
- **One-factor Gaussian copula** (CR-12 f). $x_i = a_i F + \sqrt{1-a_i^2}\, Z_i$; conditional PD $Q_i(T|F) = N\big((N^{-1}[Q_i(T)] - a_i F)/\sqrt{1-a_i^2}\big)$; homogeneous case $a_i = \sqrt{\rho}$. Vasicek's WCDR follows by setting $F = N^{-1}(1-X)$ (see B1, A31).
- **Purpose of copulas** (MR-10 a). A copula joins univariate marginal distributions into one multivariate distribution, $C : [0,1]^n \to [0,1]$ (Sklar 1959). One-parameter: Gaussian (strictly, only the bivariate one) and Archimedean (Gumbel, Clayton, Frank); two-parameter: Student's t, Fréchet, Marshall-Olkin. Vasicek brought one-factor copulas to finance (1987), Li the multivariate default-time copula (2000); embraced, then "fell into disgrace" in the 2007 crisis.
- **Joint PD from the bivariate normal** (MR-10 b). Example 10.1: $Q_B(1) = 6.51\% \to -1.5133$, $Q_{Caa}(1) = 23.83\% \to -0.7118$, $\rho = 0.4$: $M_2 = 3.44\%$. B by year 3 and Caa by year 5: 16.93%. Hence "default-time copula".
- **Default time correlated to all other assets** (MR-10 c). Draw $M_n(\cdot) \in [0,1]$ from the n-variate normal via Cholesky decomposition (it carries the correlation matrix), set $M_n(\cdot) = Q_i(\tau_i)$ and solve $\tau_i = Q_i^{-1}(M_n(\cdot))$ by Newton-Raphson or lookup (no closed form). Example: a draw of 35% gives 5.5 years. Repeat (e.g. 100,000 times) and average.
- **Implied correlation** (CR-22 l). Backed out of standard tranche prices once risk-neutral PDs from single-name CDS are in: the unique copula correlation that makes the modelled tranche value match the market price. Associated with a tranche's attachment point (base correlation); it varies by tranche (correlation skew). Risk-neutral; a market rather than credit risk concept. See D15 and D14 (compound versus base correlation).
- **Six correlation concepts** (CR-22). Default correlation; asset return correlation (Merton-type models, driven by factor loadings); equity return correlation (proxy for unobservable asset correlation); copula correlation (no direct economic interpretation; the Gaussian copula correlation equals the Gaussian single-factor correlation); spread correlation (mark-to-market); implied credit correlation (copula correlation inferred from prices).

Formulas

Default correlation (Malz 11.1). $\rho_{12} = \dfrac{\pi_{12} - \pi_1\pi_2}{\sqrt{\pi_1(1-\pi_1)}\sqrt{\pi_2(1-\pi_2)}}$

Joint PD. $\pi_{12} = \rho_{12}\sqrt{\pi_1(1-\pi_1)}\sqrt{\pi_2(1-\pi_2)} + \pi_1\pi_2$

Outcomes. $P(\text{neither}) = 1-\pi_1-\pi_2+\pi_{12}$, $P(\text{at least one}) = \pi_1+\pi_2-\pi_{12}$, $P(\text{exactly one}) = \pi_1+\pi_2-2\pi_{12}$

Single-factor model. $a_i = \beta_i m + \sqrt{1-\beta_i^2}\,\epsilon_i$, $\text{corr}(a_i, a_j) = \beta_i\beta_j$, $p(m) = \Phi\left(\dfrac{k_i - \beta_i m}{\sqrt{1-\beta_i^2}}\right)$, $k_i = \Phi^{-1}(\pi_i)$

Homogeneous single-factor default correlation. $\rho = \dfrac{\Phi_2(k,k;\beta^2) - \pi^2}{\pi(1-\pi)}$

Gaussian copula (Hull 24; Meissner 10.4). $x_i = N^{-1}[Q_i(t_i)]$; $Q(t_1 \le t, t_2 \le t) = M_2\big(N^{-1}(Q_1(t)), N^{-1}(Q_2(t)); \rho\big)$; $Q(T|F) = N\left(\dfrac{N^{-1}[Q(T)] - \sqrt{\rho}F}{\sqrt{1-\rho}}\right)$

Conditions versus events. 2^n events against $n + 1 + n(n-1)/2$ conditions

Exam traps

- **Dropped term:** omitting $\pi_1\pi_2$ from the numerator of ρ_{12} (overstates it), or omitting $+\pi_1\pi_2$ when rebuilding π_{12} (understates it).
- **Denominator:** the standard deviations are $\sqrt{\pi_i(1-\pi_i)}$, not $\sqrt{\pi_i}$; the gap matters for high-yield PDs.
- **Sibling swap, correlations:** asset (or equity) return correlation $\beta_i\beta_j$ is not default correlation; for $\pi = 1\%$, 0.315 asset correlation is only 0.05 default correlation.
- **Sibling swap, outcomes:** "at least one" is $\pi_1 + \pi_2 - \pi_{12}$; "exactly one" subtracts π_{12} twice.
- **Polarity flip, conditional PD:** in a downturn ($m < 0$) a *higher* beta means a *higher* conditional PD.
- **Scope:** the Gaussian copula makes the transformed variables x_i normal, not the default times; implied correlation is risk-neutral and market-based, not a historical estimate.

Question B3.1

CR-11 a · S1, S3, S5 · [Map #16]

A credit risk analyst at a Brazilian bank is assessing concentration in two high-yield borrowers in the same region. The bank's one-year default model gives:

- Probability of default: borrower 1, 6%; borrower 2, 10%
- Probability that neither borrower defaults over the year: 86.5%
- Correlation of the two borrowers' equity returns, used by the desk as an asset-correlation proxy: 0.55

Which of the following is closest to the one-year default correlation between the two borrowers?

A. 0.019

B. 0.245

C. 0.351

D. 0.267

Solution

Answer: D

Step 1 – Joint default probability from the no-default outcome

$$P(\text{neither}) = 1 - \pi_1 - \pi_2 + \pi_{12} \;\Rightarrow\; \pi_{12} = 0.865 - 1 + 0.06 + 0.10 = 0.025$$

$\Rightarrow$ *A 2.5% chance that both borrowers default in the same year, against* $0.06 \times 0.10 = 0.6\%$ *under independence.*

Step 2 – Covariance of the default indicators

$$\pi_{12} - \pi_1\pi_2 = 0.025 - 0.006 = 0.019$$

Step 3 – Standard deviations

$$\sqrt{\pi_1(1-\pi_1)}\sqrt{\pi_2(1-\pi_2)} = \sqrt{0.06(0.94)}\sqrt{0.10(0.90)} = 0.23749 \times 0.30000 = 0.071246$$

Step 4 – Default correlation

$$\rho_{12} = \frac{0.019}{0.071246} = 0.267$$

$\Rightarrow$ *A default correlation of 0.27 is very high by Malz's yardstick (studies find about 0.05): joint default is over four times as likely as under independence.*

Interpretation

The 0.55 equity correlation is a red herring: it proxies the asset return correlation, a sibling concept that is much larger than default correlation for the same pair. For limits on joint exposure the bank needs the default correlation, 0.267.

Why the other options are wrong

A is incorrect. Stopped at the covariance, $\pi_{12} - \pi_1\pi_2 = 0.019$, without dividing by the standard deviations (intermediate result).

B is incorrect. Used $\sqrt{\pi_1\pi_2} = 0.07746$ as the denominator, dropping the $(1-\pi_i)$ factors: $0.019/0.07746 = 0.245$.
C is incorrect. Omitted the $\pi_1\pi_2$ subtraction in the numerator: $0.025/0.071246 = 0.351$ (the formula-card dropped-term trap).

LO: CR-11 a: Define and calculate default correlation for credit portfolios. · Reference: Malz, Ch.8, 8.1 Defining Default Correlation, eq. 11.1, Example 11.1

Question B3.2

CR-11 a, f · S2, S3, S5 · [Map #16]

A treasury analyst at a UK bank is sizing a stress scenario for two corporate exposures. The bank's limit framework is breached if *at least one* of the two borrowers defaults within a year, and the stress report also requires the probability that *both* default. Inputs:

- One-year probability of default: borrower 1, 1.5%; borrower 2, 4.0%
- Default correlation between the two borrowers: 0.08

Which of the following is closest to the joint default probability and the probability that the limit is breached?

A. Joint default 0.06%; limit breached 5.44%

B. Joint default 0.19%; limit breached 5.31%

C. Joint default 0.25%; limit breached 5.00%

D. Joint default 0.25%; limit breached 5.25%

Solution

Answer: D

Step 1 – Product of the standard deviations

$$\sqrt{\pi_1(1-\pi_1)}\sqrt{\pi_2(1-\pi_2)} = \sqrt{0.015(0.985)}\sqrt{0.04(0.96)} = 0.12155 \times 0.19596 = 0.023819$$

Step 2 – Joint default probability

$$\begin{aligned}\pi_{12} &= \rho_{12}\sqrt{\pi_1(1-\pi_1)}\sqrt{\pi_2(1-\pi_2)} + \pi_1\pi_2 \\ &= 0.08(0.023819) + 0.015(0.04) = 0.001906 + 0.000600 = 0.002506\end{aligned}$$

$\Rightarrow$ *A 0.25% chance both borrowers default, about four times the 0.06% independence figure.*

Step 3 – Probability that at least one defaults

$$P(\text{at least one}) = \pi_1 + \pi_2 - \pi_{12} = 0.015 + 0.040 - 0.0025 = 0.0525$$

$\Rightarrow$ *The limit is breached with probability 5.25%. Positive correlation* lowers *this number relative to independence, because defaults cluster in the same states.*

Interpretation

Correlation pushes probability into the "both" and "neither" cells and out of the "one only" cells: the joint figure rises from 0.06% to 0.25%, and the breach probability falls from 5.44% to 5.25%.

Why the other options are wrong

A is incorrect. Ignored the default correlation: $\pi_{12} = \pi_1\pi_2 = 0.06\%$, breach $0.055 - 0.0006 = 5.44\%$.
B is incorrect. Dropped the $+\pi_1\pi_2$ term: $\pi_{12} = 0.19\%$, breach $0.055 - 0.0019 = 5.31\%$.
C is incorrect. Correct joint PD, but computed the probability that *exactly* one defaults, $\pi_1 + \pi_2 - 2\pi_{12} = 5.00\%$, which misses the joint-default state that also breaches the limit (sibling swap).

LO: CR-11 a: Define and calculate default correlation for credit portfolios. CR-11 f: Estimate the probability of a joint default of any pair of credits and the default correlation between any pair of credits using the single factor model. · Reference: Malz, Ch.8, 8.1 Defining Default Correlation

Question B3.3 CR-11 d; CR-12 f; MR-10 b; CR-22 l · S4, S6 · [Map #16] [Map #60]

A quantitative analyst at a Canadian bank is reviewing the documentation of the bank's portfolio credit model. The model joins each obligor's default-time distribution with a one-factor Gaussian copula, and its correlation parameter is calibrated so that modelled values of standard index tranches match their market prices. Which of the following statements in the documentation is correct?

A. The model assumes that the obligors' times to default are jointly normally distributed, so each obligor's marginal default-time distribution must itself be normal.

B. Conditional on a negative realisation of the common factor, an obligor with a higher factor loading has a lower conditional default probability than an otherwise identical obligor.

C. The correlation calibrated to the tranche prices is a real-world parameter, estimated from the historical default experience of the obligors in the index.

D. Once the common factor is realised, the obligors' defaults are independent, so all default dependence in the model enters through the obligors' loadings on that factor.

Solution **Answer: D**

Step 1 – Conditional independence

In the single-factor model $a_i = \beta_i m + \sqrt{1-\beta_i^2}\,\epsilon_i$ (Hull: $x_i = a_i F + \sqrt{1-a_i^2}\,Z_i$) the idiosyncratic shocks are independent of one another and of the factor. Fixing m leaves only ϵ_i random, so defaults are conditionally independent and the pairwise correlation $\beta_i\beta_j$ comes entirely from the loadings.

Step 2 – Conditional default probability check

$p(m) = \Phi\big((k_i - \beta_i m)/\sqrt{1-\beta_i^2}\big)$. With $\pi = 1\%$ ($k = -2.33$) and $m = -1$: $\beta = 0.4$ gives 1.78% (Malz Example 11.4); a higher β raises the numerator and shrinks the denominator, so the conditional PD is *higher*.

Interpretation

D is correct. Conditional independence is what makes the factor model tractable and underlies the Vasicek large-portfolio result used for Credit VaR (B1) and IRB capital (A31).

Why the other options are wrong

A is incorrect. Sibling swap (t_i for x_i). Hull: a company's time to default "is not even approximately normal"; the copula maps each t_i percentile-to-percentile into $x_i = N^{-1}[Q_i(t_i)]$ and assumes only the x_i are multivariate normal, so the marginals can take any shape.

B is incorrect. Polarity flip. When $m < 0$, a higher loading pushes the conditional mean $\beta_i m$ further down and the conditional PD up: cyclical firms suffer more in a downturn.

C is incorrect. Sibling swap. Implied correlation is inferred from market prices using risk-neutral default probabilities from single-name CDS spreads (Malz 22.4); it is a risk-neutral, market rather than credit risk, parameter, not a historical estimate.

LO: CR-11 d: Describe the use of a single factor model to measure portfolio credit risk, including the impact of correlation. CR-12 f: Describe the Gaussian copula model for time to default and calculate the probability of default using the one-factor Gaussian copula model. MR-10 b: Describe the Gaussian copula and explain how to use it to derive the joint probability of default of two assets. CR-22 l: Define implied correlation and describe how it can be measured. · Reference: Malz, Ch.8, 8.3 Conditional Default Distributions; Hull, OFOD 11e, Ch.24, 24.2; Meissner, Ch.5, The Gaussian Copula; Malz, Ch.9, 22.4 Implied Correlation

B4 Factor Theory, Fama-French and Style

Readings: IM-1 a, b, c, d, e, f · Factor Theory (Ang, Asset Management, Ch.6); IM-2 a, b, c, d, e · Factors (Ang, Ch.7); IM-3 a, c, e, f, g, h, i · Alpha (and the Low-Risk Anomaly) (Ang, Ch.10)

MAP TIER B · #18 · 5/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #18: 'One handwritten source writes 'Portfolio Composition (SML, HML)'. This is almost certainly a transcription error for SMB, the Fama-French size factor, not SML, the security market line. Studying the SML here would be studying the wrong object. Also reported: 'investing style theory question'. Value and momentum comparison (LO 84.e) and the low-risk / volatility anomaly (Reading 85) sit adjacent.'
Shapes covered here: S1 SMB (small minus big) vs SML (security market line), with HML, in a Fama-French portfolio composition (Q1); S2 investing style (style, investment or dynamic factors) as a theory item (Q2); S3 value vs momentum: return and risk profiles (Q2); S4 the low-risk anomaly: volatility anomaly vs beta anomaly and their explanations (Q2).

Revision notes

The specifics first: SMB, HML, style, value vs momentum, low risk (IM-2 d, e; IM-3 a, f, h, i)

- **SMB, not SML (S1).** Fama-French (1993) adds two factors to the CAPM market factor: **SMB**, the returns of **small** stocks minus **big** stocks (market capitalization), and **HML**, high book-to-market (value) minus low book-to-market (growth) stocks. The **security market line** is a different object: the CAPM beta pricing relationship $E(r_i) - r_f = \beta_i(E(r_m) - r_f)$. The SML is a pricing line; SMB is a tradeable long-short factor.
- **Factor-mimicking, long-short portfolios.** SMB = USD 1 in small caps minus USD 1 in large caps; HML = USD 1 in value minus USD 1 in growth. Loadings s and h: $s = 0$ for a medium-size stock, $s > 0$ moves with small stocks, $s < 0$ with large; $h > 0$ value orientation, $h < 0$ growth. The **market is size neutral and value neutral** (zero s and h): the average stock has only market exposure, just as the average investor holds the market. Fama and French did not discover size and value; they gave a parsimonious model to capture them.
- **Berkshire Hathaway as a factor benchmark.** CAPM: alpha 0.72% per month, beta 0.51, adjusted R^2 14%. Fama-French: alpha 0.65%, market 0.67, $s = -0.50$, $h = 0.38$ (large, value), adjusted R^2 27%. Benchmark: USD 0.33 T-bills + 0.67 market − 0.50 small + 0.50 large + 0.38 value − 0.38 growth. Adding UMD: loading -0.04, insignificant, R^2 unchanged, alpha 0.68%.
- **Size has disappeared.** Banz (1981) and Reinganum (1981): small stocks beat large after adjusting for beta. Since the mid-1980s there has been **no significant size effect**, internationally too. Two responses: **data mining** (Fischer Black, 1993; the "file drawer problem") or rational investors **bid up small caps** until the effect vanished (Grossman-Stiglitz near-efficiency), so size should be removed from the model. Small stocks still have higher raw returns, are less liquid, and value and momentum are stronger in them; the unconstrained, investment-only case for a small-cap tilt is not compelling.
- **Value is robust (IM-2 a).** Value stocks have low prices relative to book value; value has produced gains for fifty years but lost in the early-1990s recession, the late-1990s internet bull market and 2007 to 2008. *Rational*: value stocks co-move and the common risk cannot be diversified away; firm investment risk (Berk, Green and Naik: options exercised when market returns are low); Zhang/Cochrane: value firms are stuck with unproductive capital (high, asymmetric adjustment costs). *Behavioral*: overextrapolation of past growth (Lakonishok, Shleifer and Vishny, 1994); loss aversion plus mental accounting (Barberis and Huang, 2001). The book-to-market effect is a three- to six-month effect, unlike Buffett's five- to ten-year "deep value".
- **Value in other asset classes.** Buying high yields (low prices) and selling low yields: **riding the yield curve** in fixed income (a duration premium), the **roll return** in commodities, **carry** in FX (long high-rate currencies, short low-rate).
- **Momentum (S3).** WML (winners minus losers) or UMD: buy stocks with the highest returns over the past six months or so, short the lowest. Cross-sectional, like size and value; returns an **order of magnitude larger** than size and value; observed in every asset class ("trend" investing). Momentum is **not** the opposite of value: correlation of HML with WML only -16%. Value is a **negative feedback**, stabilizing strategy;

momentum is **positive feedback**, destabilizing and prone to crashes (skewness -1.43). Of the eleven largest crashes, seven were in the Great Depression, one in 2001, three in 2008 (bailouts lifted the loser financials the strategy was short). Explanations mainly behavioral: overreaction (conservatism bias; overconfidence and biased self-attribution) or underreaction (Hong and Stein's "news watchers").

- **Investing style (S2).** "Style factors", "investment factors" and "dynamic factors" are used interchangeably (practitioners: "smart beta", "alternative beta"). **Static** factors (the market) are simply held long; **dynamic** factors require constant trading. Many hedge funds and private equity funds are bundles of dynamic factors. For the asset owner: are these bad times bad for *me*? The volatility premium has a rational story, momentum a behavioral one, value both.
- **Style analysis (IM-3 f).** Sharpe (1992): a factor benchmark whose exposures **evolve through time**, built from **tradeable** low-cost funds (SPY, SPYV, SPYG), weights summing to one, estimated over the previous 60 months, with or without a no-shorting constraint. It fixes two shortcomings of Fama-French regressions: the factor portfolios are not tradeable, and loadings vary over time. Buffett: strongly value in 2006, growth in the crisis, value again after. Allowing shorting reduces alphas; time-varying styles have larger standard errors.
- **The low-risk anomaly (S4; IM-3 a, h).** Three effects: volatility is negatively related to future returns; realized beta is negatively related to future returns; minimum variance portfolios beat the market. **Volatility anomaly** (Ang et al., 2006): the highest idiosyncratic-volatility quintile returned 0.1%; raw Sharpe ratios fall from 0.8 to 0.0. **Beta anomaly**: average returns are roughly **flat** across beta quintiles (about 15%, 12.7% for the top quintile), but Sharpe ratios fall from 0.9 to 0.4; contemporaneous beta and returns are positively related; the real mystery is that past betas predict future betas poorly.
- **BAB and VOL factors.** BAB (Frazzini and Pedersen) goes long low-beta and short high-beta, each scaled to unit beta (so the long side is levered); VOL scales to a 15% target volatility and can be traded without the risk-free asset. Correlation BAB/VOL -9%: the anomalies are **distinct**. Alphas: BAB 0.33% per month (t 1.89, borderline), VOL 0.42% (t 4.37). Both tilt to value and momentum; BAB loads $+0.29$ on SMB, VOL -0.63: the beta anomaly shows more in small stocks, the volatility anomaly in large, liquid stocks. Do both.
- **Explanations of the risk anomaly (IM-3 i).** Data mining (but the effect appears across countries, bonds, credit, commodities, FX, options); **leverage constraints** (explain overpricing of high-beta stocks, not the underpricing of low-beta stocks; and institutions actually underweight high-risk stocks); **agency problems** (market-weighted benchmarks with tracking error limits stop managers buying low-risk stock A or shorting high-risk stock B); **preferences** ("hopes and dreams", lottery preferences explain close to half); disagreement with short-sale limits (Hong and Sraer). Buffett's alpha: 12.5%, 11.1% after BAB, 7.0% after a quality factor.

The surrounding material: factor theory and CAPM (IM-1)

- **Factor theory (IM-1 a).** Factors are to assets what nutrients are to food: factors matter, not assets; assets are bundles of factors; different investors need different factors. Unlike nutrients, factor risks are *bad*: each factor defines a set of bad times, and investors are paid premiums for bearing losses in them. 2008 did not kill diversification: asset-class labels misled investors about common factor exposures.
- **CAPM lessons (IM-1 b, c).** Hold the factor (the market diversifies away unrewarded idiosyncratic risk); each investor has his own optimal factor exposure; the average investor holds the market; the market premium has an economic story, $E(r_m) - r_f = \bar{\gamma}\sigma_m^2$; risk is factor exposure (beta $= \rho_{i,m}\sigma_i/\sigma_m$: high beta means **low diversification benefit**); assets paying off in bad times have low, even negative, risk premiums. Equilibrium: demand equals supply, so the MVE portfolio becomes the market. The CAPM is "a spectacular failure" empirically but remains the workhorse.
- **CAPM assumptions and failures.** Only financial wealth (human capital, Jagannathan and Wang); mean-variance utility (downside risk earns a premium); single period (minor; Merton's dynamic CAPM); homogeneous expectations (matters with short-sale limits: Miller, 1977); no taxes or transaction costs; price takers; costless information. Failures should be most visible in illiquid, inefficient markets.
- **Multifactor models and the SDF (IM-1 d, e).** APT (Ross, 1976): factors cannot be arbitraged or diversified away. The pricing kernel (SDF) m is an index of bad times; the CAPM is the special case $m = a + b\, r_m$ with $b < 0$. $P_i = E[m \times \text{payoff}_i]$, $1 = E[m(1 + r_i)]$, $1/(1 + r_f) = E[m]$. Higher covariance with m (paying off in bad times) means a *lower* risk premium. Bad times for the representative agent: high marginal utility, low consumption or growth.
- **Efficient markets (IM-1 f).** Grossman-Stiglitz (1980): markets cannot be perfectly efficient if information is costly; active managers earn returns in pockets of inefficiency (illiquid, poor information). Deviations are rational (premiums persist, scalable) or behavioral (persist only with barriers to entry of capital).

Macro factors, volatility, benchmarks (IM-2 b, c; IM-3 c, e, g)

- **Macro factors.** Shocks matter more than levels. Growth: large stocks 5.6% in recessions vs 12.4% in expansions; government bonds 12.3% vs 5.9%; high yield shows no discernible difference. Inflation: high inflation hurts both stocks (14.7% low vs 8.0% high) and bonds. Volatility: VIX changes correlate -0.39 with stocks (leverage effect) and only 0.12 with bonds; large stocks -4.6% in volatile vs 24.9% in stable times. Also productivity (correlation 48%), demographic and political risk.
- **Managing volatility risk.** Volatility has a **negative price of risk**: buy protection (puts, volatility swaps) or hold more bonds; collect the premium by selling out-of-the-money puts (VIX implied about 2% to 3% above realized). Selling volatility had skewness -8.26 (only -0.37 before 2008) and losses near 70% in September to November 2008. Rebalancing is a short-volatility strategy. Challenge: estimates of γ in the return-volatility trade-off are positive, negative and zero.
- **Alpha is about the benchmark.** Martingale: alpha 1.50% (IR 0.24) against the naive Russell 1000, 3.44% (IR 0.78) against the beta-adjusted 0.27 T-bills/0.73 Russell benchmark. Sound benchmarks are well defined, tradeable, replicable and risk adjusted. CalPERS: alpha -1.11% (t -1.16) against 32% bonds/68% stocks: fail to reject. Nonlinear (short-option) strategies produce illusory alpha; include tradeable nonlinear factors or test nonlinear terms. See also A18 for alpha, tracking error, IR, Sharpe and the fundamental law (IM-3 b, d).

Formulas

CAPM: $E(r_m) - r_f = \bar{\gamma}\,\sigma_m^2; \qquad E(r_i) - r_f = \beta_i\,(E(r_m) - r_f)\,, \quad \beta_i = \dfrac{\text{cov}(r_i, r_m)}{\text{var}(r_m)} = \rho_{i,m}\dfrac{\sigma_i}{\sigma_m}$

SDF: $P_i = E[m \times \text{payoff}_i]; \quad 1 = E[m(1 + r_i)]; \quad \dfrac{1}{1 + r_f} = E[m]; \quad E(r_i) - r_f = \dfrac{\text{cov}(r_i, m)}{\text{var}(m)}\left(-\dfrac{\text{var}(m)}{E(m)}\right)$

Multi-beta: $E(r_i) = r_f + \sum_k \beta_{i,k} E(f_k)$

Fama-French (1993): $E(r_i) = r_f + \beta_{i,MKT}E(r_m - r_f) + \beta_{i,SMB}E(SMB) + \beta_{i,HML}E(HML)$; add $\beta_{i,WML}E(WML)$ for momentum (Carhart, 1997)

Factor regression: $r_{it} - r_{ft} = \alpha + \beta(r_{mt} - r_{ft}) + s\,SMB_t + h\,HML_t + u\,UMD_t + \varepsilon_{it}$; benchmark: $(1 - \beta)$ in T-bills, β in the market, long-short positions of size s, h, u

Style analysis: $r_{t+1} = \alpha_t + \beta_{SPY,t}SPY_{t+1} + \beta_{SPYV,t}SPYV_{t+1} + \beta_{SPYG,t}SPYG_{t+1} + \varepsilon_{t+1}, \quad \beta_{SPY,t} + \beta_{SPYV,t} + \beta_{SPYG,t} = 1$

Low-risk factors: $BAB_{t+1} = \dfrac{r_{L,t+1} - r_f}{\beta_{L,t}} - \dfrac{r_{H,t+1} - r_f}{\beta_{H,t}}; \qquad VOL_{t+1} = \sigma_{target}\left(\dfrac{r_{L,t+1} - r_f}{\sigma_{L,t}} - \dfrac{r_{H,t+1} - r_f}{\sigma_{H,t}}\right)$

Exam traps

- **Sibling swap, SMB vs SML:** SMB is the Fama-French size factor (small minus big); the SML is the CAPM's beta pricing line. A "loading on SML" does not exist.
- **Scope error on the market's loadings:** the market (and the average stock) has *zero* SMB and HML loadings and a beta of one; only tilted stocks carry nonzero s and h.
- **Sibling swap, size vs value:** size vanished after the mid-1980s; value has been robust for fifty years; momentum earns the most but crashes. Swapping these robustness profiles is the standard distractor.
- **Polarity flip on feedback:** value is negative feedback (stabilizing); momentum is positive feedback (destabilizing). Momentum is not the opposite of value (correlation -16%).
- **Sibling swap, beta vs volatility anomaly:** the beta anomaly lives in Sharpe ratios (average returns are flat), so BAB must lever the low-beta leg; the volatility anomaly shows in average returns, so VOL trades directly. The source sentence on SMB signs runs opposite to its own table: rely on the conclusion, beta anomaly in small stocks, volatility anomaly in large stocks.
- **Absolute claim on explanations:** leverage constraints explain only the overpricing of high-beta stocks; no single explanation covers the whole anomaly (lottery preferences, the best group, explain close to half).

Question B4.1

IM-2 d; IM-1 b · S1 · [Map #18]

A consultant to a Singapore pension fund is reviewing a junior analyst's research note on the Fama-French (1993) model before it goes to the investment committee. The note's section heading reads "Portfolio composition (SML, HML)", and the committee wants to know whether to add a size tilt to the fund's SGD-denominated equity mandate. Which of the following statements should the consultant accept as correct?

A. The SML in the heading is the security market line, so a fund's loading on it measures beta-adjusted exposure to the market risk premium and can replace the market factor in the regression.

B. Because value stocks beat growth stocks on average, the market portfolio itself carries a positive HML loading, so the average stock in the economy is tilted toward value.

C. The size factor is SMB, a long-short factor-mimicking portfolio long small caps and short large caps, and since the mid-1980s it has shown no significant premium after adjusting for beta.

D. Like the value premium, the size premium has stayed robust out of sample since Banz (1981), so an unconstrained investor should tilt toward small caps for higher risk-adjusted returns.

Solution

Answer: C

Step 1 – Identify the object in the heading

Fama-French adds two factors to the CAPM market factor: SMB (small minus big, by market capitalization) and HML (high minus low book-to-market). The SML is the CAPM pricing relationship $E(r_i) - r_f = \beta_i(E(r_m) - r_f)$, not a factor portfolio.

⇒ *"SML" in the note is a transcription error for SMB; the committee should be reading about the size factor.*

Step 2 – How SMB is built

SMB = USD 1 in small caps minus USD 1 in large caps: a long-short, factor-mimicking portfolio that diversifies across many stocks to isolate the size effect.

Step 3 – What the size factor has delivered

After Banz (1981) and Reinganum (1981), the compound returns of market-adjusted SMB peaked in the early 1980s; since the mid-1980s there has been no significant size premium, in the US or internationally.

⇒ *The weak size effect means an asset owner should not tilt toward small stocks solely for higher risk-adjusted returns.*

Interpretation

C is correct. The consultant should correct the heading to SMB and advise the committee that a size tilt is hard to justify on risk-adjusted grounds alone; a case exists only for investors who want higher returns but cannot short the market or cannot lever.

Why the other options are wrong

A is incorrect. Sibling swap: substitutes the CAPM's security market line for the Fama-French size factor. The SML describes how beta is priced; it is not a factor with its own loading, and the market factor stays in the regression.

B is incorrect. Scope/condition error: extends a property of value stocks to the market. In the Fama-French model the SMB and HML betas are centered around zero; the market is size neutral and value neutral.

D is incorrect. Sibling swap: gives size the robustness profile of value. Unlike size, the value premium is robust; the size effect has not been significant since the mid-1980s (data mining or arbitraged away).

LO: IM-2 d: Explain how dynamic risk factors can be used in a multifactor model of asset returns, using the Fama-French model as an example. · IM-1 b: Discuss the capital asset pricing model (CAPM) including its assumptions and explain how factor risk is addressed in the CAPM. · Reference: Ang, Asset Management, Ch.7 (Section 2.4 Dynamic Factors, Fama-French (1993) Model, Size Factor) and Ch.6 (Section 1.4, CAPM Lesson 5)

Question B4.2

IM-2 e; IM-3 h, i · S2, S3, S4 · [Map #18]

The CIO of a Canadian university endowment is reviewing a consultant's proposal to add three investing-style tilts (value, momentum and low volatility) to the endowment's CAD 2.4 billion equity portfolio. Before the investment committee meets, the CIO checks four statements in the proposal. Which statement is correct?

A. Momentum is a negative feedback strategy that buys stocks after they have fallen, so like value it is inherently stabilizing and rarely suffers crashes.

B. The beta anomaly shows that high-beta stocks earn much lower average returns than low-beta stocks, so a long-short of beta quintiles captures it without leverage.

C. Value and momentum returns are only weakly negatively correlated, and momentum's largest crashes came when policymakers interrupted trends, as with bailouts of loser stocks in 2008.

D. Leverage constraints fully explain the low-risk anomaly, because constrained institutions overweight high-beta stocks, which both overprices high-beta and underprices low-beta stocks.

Solution **Answer: C**

Step 1 – Value vs momentum (IM-2 e)

Value buys stocks whose prices have fallen far enough to look cheap: a negative feedback, stabilizing strategy. Momentum buys past winners and shorts past losers: a positive feedback, destabilizing strategy. The correlation of HML with WML is only -16%.

$\Rightarrow$ *Momentum is not the opposite of value, so the two tilts are close to complementary, not offsetting.*

Step 2 – Momentum's risk profile

Momentum is negatively skewed (-1.43). Of the eleven largest crashes, seven came in the Great Depression, one in 2001 and three in 2008, when bailouts put a floor under loser financials (Citi, Bank of America, Goldman, Morgan Stanley) that momentum was short.

$\Rightarrow$ *The endowment must be able to tolerate large drawdowns concentrated in periods of policy intervention.*

Step 3 – Low-risk tilt (IM-3 h, i)

The beta anomaly shows up in Sharpe ratios (0.9 falling to 0.4), not in average returns, which are roughly flat across beta quintiles; BAB therefore levers the low-beta leg to unit beta. Leverage constraints explain why high-beta stocks are overpriced but not why low-beta stocks are underpriced.

Interpretation

C is correct. The CIO can accept the value/momentum diversification argument and the crash warning, but should ask the consultant to correct the description of the beta anomaly and to drop the claim that one explanation accounts for the whole risk anomaly.

Why the other options are wrong

A is incorrect. Polarity flip: momentum is a positive feedback strategy that buys stocks after they have risen; positive feedback is destabilizing and subject to periodic crashes.

B is incorrect. Sibling swap: gives the beta anomaly the volatility anomaly's profile. High-beta stocks do not have low average returns; they have high volatility and low Sharpe ratios, so the BAB factor must lever the low-beta portfolio.

D is incorrect. Absolute claim: the leverage-constraint story explains only the overpricing of high-beta stocks, not the underpricing of low-beta stocks, and institutions in fact underweight high-risk stocks.

LO: IM-2 e: Compare value and momentum investment strategies, including their return and risk profiles. · IM-3 h: Compare the volatility anomaly and the beta anomaly and analyze evidence of each anomaly. · IM-3 i: Describe potential explanations for the risk anomaly. · Reference: Ang, Asset Management, Ch.7 (Section 2.5, Momentum; Characterizing Momentum Risk) and Ch.10 (Section 3.5, Beta Anomaly; Betting against Beta; Explanations)

B5 Climate Risk (not in 2026 LOs: notes only)

Readings: Not in the 2026 LOs: evidence comes from an older Current Issues reading (map: R103, Global Financial Stability Report; the 2026 CI-4 is geopolitical risk, see D30). Related 2026 text: ORR-3 a · Risk Identification; ORR-5 g · Risk Mitigation; ORR-6 c · Risk Reporting; CI-2 c; CI-5 c; CI-8 a (CI sources: SchweserNotes Book 5, one rung below a GARP text)

MAP TIER B · #19 · 5/16 SOURCES TELEGRAM NOTES ONLY

Specifics reported

Map #19: 'Reported: 'climate risk drivers'; 'climate risk: 3 questions'; 'climate risk and inflation'; and one specific answer recall: 'climate risk question, I chose catastrophe bond as answer', indicating a risk-transfer-instrument framing. Keep physical risk drivers distinct from transition risk drivers; that split is the natural sibling swap.'
Shapes covered here: S1 climate risk drivers: physical vs transition; S2 climate risk and inflation; S3 catastrophe bonds and risk transfer; S4 where climate still appears in 2026 readings (operational risk identification, reporting, ICT outages, AI energy demand, fiscal pressure).

Revision notes

Status

- **Not examinable by LO in 2026.** The climate questions in past recalls came from an older GFSR reading. The 2026 GFSR reading (CI-4) covers geopolitical risk (see D30). Climate can still appear inside the 2026 readings listed in the srcline, so the drivers and the risk-transfer framing below are worth one careful read.

The specifics: drivers, inflation, catastrophe bonds (outside-source unless stated)

- **Physical risk drivers (S1).** Damage from the climate itself (outside-source): **acute** events (storms, floods, wildfires, heatwaves) and **chronic** shifts (rising temperatures, sea-level rise, changing rainfall). They destroy assets and collateral, interrupt business and raise insurance claims.
- **Transition risk drivers (S1).** Losses from the move to a low-carbon economy (outside-source): **policy and regulation** (carbon prices, emission limits), **technology** (cheaper clean substitutes), **market and consumer preferences**, and **litigation and reputation**. They reprice carbon-intensive firms and can strand assets such as fossil-fuel reserves.
- **The sibling swap.** A flood that damages mortgage collateral is *physical*; a carbon tax that strands coal reserves is *transition*. The same firm can face both, but the driver decides the label (outside-source).
- **Transmission to financial risks.** Both drivers reach banks and investors as existing risk types rather than a new one (outside-source): **credit** (higher PD, lower collateral values), **market** (repricing of exposed sectors), **liquidity** (deposit or funding outflows after a disaster), **operational** (damage to premises and systems) and, for insurers, **underwriting** losses.
- **Climate risk and inflation (S2).** Physical shocks to harvests and energy supply push up food and energy prices; transition policies such as carbon pricing raise energy costs (outside-source). In the 2026 text, CI-5 c lists "the global push for sustainability" as very costly for fiscal budgets, one of the spending pressures that, with inflation and high debt, sharpens the tension between fiscal and monetary policy (see D16).
- **Catastrophe bonds (S3).** An insurance-linked security (outside-source): investors buy a bond whose principal is lost, in part or in full, if a defined catastrophe (for example a hurricane or earthquake above a set severity) occurs during the term. The sponsor (an insurer, reinsurer, corporate or government) pays a coupon that works as an insurance premium; the proceeds sit in a collateral account, so the protection carries little counterparty risk. Triggers can be indemnity (sponsor's actual losses), industry-loss or parametric (a physical measure such as wind speed). A cat bond transfers **physical** (tail) risk to capital markets; it does not hedge transition risk.
- **The risk-transfer logic in GARP's text (ORR-5 g).** External insurance reduces profit and loss volatility. It makes sense for risks that are (1) quite predictable and known in their distribution of likelihood and impact, so they can be underwritten and priced, and (2) easily transferable. If correctly priced, the premium is roughly equal to or greater than expected losses, so large firms self-insure small losses (sometimes via a captive) and buy external cover for very large **tail** events (cyber, business discontinuity, major class actions).

- **Transfer is never perfect (ORR-5 g).** The insured remains exposed to the insurer's **capacity and willingness** to pay, and compensation may arrive months after the event, leaving a liquidity gap. Under the Basel standardized approach, insurance recoveries are deducted from gross losses to compute net losses in the loss multiplier. Typical policies include property insurance and business interruption, the two closest to physical climate damage.

Where climate appears in 2026 readings (S4)

- **Known emerging risk (ORR-3 a).** Horizon scanning separates **known** emerging risks (cyber, compliance, employee well-being, mental health and **climate change**), identified sometimes decades ago and where "emerging" means "increasing", from truly **nascent** emerging risks just appearing on the radar. Sources: the WEF Global Risks Report, risk.net's Top Operational Risks, ORX's Operational Risk Horizon, big-four top-10 lists.
- **Accelerator of operational losses (ORR-6 c).** Over the past decade, climate change is "the best example of conditions that may accelerate losses": volatility in environmental factors has caused unanticipated operational losses, such as weather damage causing significant system downtime and lost customer revenue, especially where business continuity plans are inappropriate. Robust loss analysis considers low-, medium- and high-risk scenarios that may alter the loss profile.
- **ICT outages (CI-8 a).** About 80% of cloud customers suffered at least one outage in the prior three years, and roughly 20% of those outages were significant enough to noticeably disrupt operations; climate change and rising energy costs may be causing longer and more frequent outages.
- **AI energy demand (CI-2 c).** AI-related energy use is about 1% of global energy but expected to rise; as finance depends more on AI it becomes more exposed to energy supply and cost pressures, and clean, reliable energy is needed to avoid exacerbating climate-related risks.
- **Scenario horizon.** Climate scenario analysis typically runs over decades, far beyond the one- to two-year horizons of capital stress tests, and mixes orderly, disorderly and "hot house" pathways (outside-source). For the 2026 exam, the safe link is ORR-6's scenario-based view of loss profiles and the ORR-5 transfer criteria.
- **Overlap with geopolitics (CI-4 b).** Commodity shocks after conflicts favor energy exporters and hurt importers (depreciation, wider spreads); the same importer/exporter asymmetry is the right instinct for energy-price shocks from climate events (see D30).

Likely distractors

- **Corrupted:** "A carbon tax that strands a utility's coal reserves is a physical risk driver." **True version:** it is a *transition* risk driver (policy and regulation); physical drivers are acute events and chronic climate shifts (outside-source).
- **Corrupted:** "Floods that destroy mortgage collateral are a transition risk because they change the value of the bank's portfolio." **True version:** damage from the climate itself is *physical* risk; any risk changes portfolio value, so that is not the test (outside-source).
- **Corrupted:** "A catastrophe bond lets a sponsor hedge rising carbon prices: investors lose principal if carbon prices exceed a trigger." **True version:** principal is lost if a defined *catastrophe* occurs; the bond transfers physical tail risk to capital markets (outside-source).
- **Corrupted:** "External insurance is a perfect transfer of operational risk, so the bank carries no residual exposure after a climate disaster." **True version:** the bank stays exposed to the insurer's capacity and willingness to pay, and payment can come months later, leaving a liquidity gap (ORR-5 g).
- **Corrupted:** "Climate change is a nascent emerging risk that has only just appeared on firms' risk radar." **True version:** it is a *known* emerging risk, identified long ago, where "emerging" means "increasing" (ORR-3 a).
- **Corrupted:** "External insurance is best used for small, frequent losses, while tail events should be self-insured through a captive." **True version:** the reverse: large firms self-insure small losses and buy external cover for extreme tail events (ORR-5 g).

Tier C: Moderate Corroboration

Map Tier C topics not already in Tier A.

C1 Regression Hedging, DV01 and PCA

Readings: MR-11 a, b, c, d, e, f, g, h · Regression Hedging and Principal Component Analysis (Tuckman and Serrat, Fixed Income Securities 4e, Ch.6)

MAP TIER C · #23 · 5/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #23: "Reported: 'hedge ratio given beta, alpha, DV01'; 'regression hedge: face amount if offsetting'; 'DV01 hedging'; 'regression-based hedge'. LO 11.a, the drawbacks of a plain DV01-neutral hedge, is the qualitative counterpart and the reason the regression hedge exists. PCA as a hedging-portfolio construction (LO 11.h) is the adjacent item."
Shapes covered here:
S1 hedge ratio from a regression output that lists alpha, beta and both DV01s, with alpha as the tempting extra input (Q1)
S2 face amount of the offsetting position in a regression hedge, paired with the hedged P&L volatility (Q1)
S3 DV01 hedging: the DV01-neutral amount as the planted distractor against the regression amount (Q1, Q2)
S4 drawbacks of a DV01-neutral hedge and why the regression hedge improves it (Q2)
S5 PCA as the construction of a hedging portfolio: how many PCs two hedging securities can neutralize, and the risk-weight sum (Q2).

Revision notes

- **DV01-neutral hedge and its drawback** (MR-11 a, S3, S4): the trader sells face = face hedged × DV01 ratio (book: USD 100 million × $0.2124/0.1910$ = USD 111.2 million of the 30-year Treasury against 40-year JNJ bonds). The hedge "assumes that the yields of the JNJ and Treasury bonds move up or down in parallel". Because the corporate bond trades at a changing spread and 40- and 30-year rates are not perfectly correlated, the parallel-shift assumption is questionable.
- **Regression model** (MR-11 b): $\Delta y_t^{\text{JNJ}} = \alpha + \beta\,\Delta y_t^{30} + \varepsilon_t$. Least squares picks $\hat{\alpha}$, $\hat{\beta}$ to minimise the sum of squared residuals; squaring penalises large errors heavily and stops positive and negative errors offsetting. Assumptions: the model is a true description, errors are identically distributed, independent of each other and uncorrelated with the independent variable; then estimates are linear, unbiased, consistent and efficient.
- **Reading the output** (MR-11 c): book $\hat{\beta} = 0.842$ (std error 0.051), $\hat{\alpha} = 0.060$ (std error 0.223), $R^2 = 77.5\%$, standard error 2.00 bp. The 95% interval for α contains zero, so $\alpha = 0$ cannot be rejected; the interval for β, (0.740, 0.944), excludes one, so parallel shifts are rejected. In a one-variable regression R^2 is the squared correlation (88.0%). The intercept is economically odd if nonzero and does not enter the hedge (S1).
- **Regression hedge face amount** (MR-11 d, S2): $F^{30} = -F^{\text{JNJ}}\,\dfrac{\text{DV01}^{\text{JNJ}}}{\text{DV01}^{30}}\,\hat{\beta}$. Book: $-100 \times (0.2124/0.1910) \times 0.842 = -$USD 93.6 million, so fewer Treasuries are sold than the DV01 hedge's USD 111.2 million.
- **Risk weight** (MR-11 c): the DV01 of the hedge as a fraction of the DV01 of the hedged bonds equals $\hat{\beta}$ (84.2%). The risk weight of a DV01-neutral hedge is always 100%.
- **Why the regression hedge is better** (MR-11 b, S4): it minimises the variance of the hedged P&L (Appendix A6.1), whether or not yields move exactly by the regression line. Hedged P&L volatility $= |F^{\text{JNJ}}\,\text{DV01}^{\text{JNJ}}/100| \times \sigma_\varepsilon$: book USD 212,400 × 2 bp = USD 424,800 per day. Whether that is acceptable depends on how much and how fast the trade makes money.
- **Estimation window and stability:** too short a window gives unreliable estimates, too long includes less relevant data; an earlier episode (for example a past Fed tightening) may be more relevant than the recent past. Results are in-sample; poor out-of-sample performance questions the stability of the coefficients (book out-of-sample standard error 1.5 bp versus 2.0 in-sample).
- **Two-variable regression hedge** (MR-11 e): 20-year yield on 10- and 30-year yields ($\hat{\beta}^{10} = 0.465$, $\hat{\beta}^{30} = 0.669$, $R^2 = 94.7\%$, standard error 1.15 bp). The butterfly buys 20s and sells 10s and 30s: both shorts defend against general rate rises, the 10-year short against flattening, the 30-year short against steepening. Face amounts $F^{10} = -F^{20}(\text{DV01}^{20}/\text{DV01}^{10})\hat{\beta}^{10}$ and similarly for 30s: USD 79.44 million and USD 50.69

million against USD 100 million 20s. Risk weights 46.5% and 66.9% sum to 113.4%: the hedge carries 13.4% more DV01 than the position.

- **Level versus change regressions** (MR-11 f): if the least-squares assumptions hold for one specification they hold for the other. If error independence fails, estimates are still unbiased and consistent but may not be efficient. Level errors tend to be persistent (serially correlated); change-on-change errors can also stay positive while a yield trends toward its model value. Alternative: a level model with error dynamics $\varepsilon_t = \rho\varepsilon_{t-1} + \nu_t$, $\rho < 1$ (book: -4% yesterday, $\rho = 75$%, expected error -3% today).
- **Reverse regression** (MR-11 g): regressing the Treasury on JNJ gives $\hat{\beta} = 0.921$ (regression standard error 2.09 bp), a risk weight of $1/0.921 = 108.6$% and USD 82.8 million JNJ against USD 93.6 million Treasuries. Neither hedge is "wrong": choosing to hedge USD 100 million JNJ fixes the trade and its risk (about USD 425,000 versus 374,000). The regression hedge minimises variance *given* the face amount of the bond being hedged; other objectives give other constructions.
- **Why PCA** (MR-11 h): regression hedging is ad hoc (different bonds, windows and variable counts across traders are different models of the term structure). PCA gives one empirical description usable across a portfolio.
- **PCA properties** (MR-11 h): (1) the sum of the PC variances equals the sum of the rate variances; (2) the PCs are *uncorrelated* with each other; (3) each PC has the maximum variance given the earlier ones. The first three PCs explain an overwhelming fraction (USD swaps: 99.84%; level over 90%, slope about 8%, short rate under 1%). Three PCs explain 74.54% of the one-year rate's variation but over 99% for terms beyond three years.
- **Level, slope, short rate** (MR-11 h): a one-standard-deviation level move is not particularly level from 1 to 7 years in the 2020 to 2021 sample; slope moves short rates down and long rates up (or the reverse); the short-rate PC (also called curvature) adds explanatory power at the short end. A trader in 8- to 10-year swaps can defend a one-factor approach; 3- to 6-year or long-end traders may need two factors; the very short end may need three.
- **Hedging with PCA** (MR-11 h, S5): shift the curve by each PC, reprice, compute a '01 per PC, and choose hedges that neutralise them. Book: receive 100 in 20-year swaps, pay 49.56 in 10s and 53.60 in 30s to neutralise level and slope; risk weights 27.1% and 74.1% sum to 101.2%. "Only under the assumption of parallel shifts do the risk weights always sum to 100%." With two hedging securities only two PCs can be hedged: the residual short-rate exposure was 0.007 per 100, so the trader may skip a third (15-year) swap given transaction costs and the relative value view.
- **See also** A7 and A25 for term structure models whose factor shapes (parallel versus mean-reverting) give the model-based counterpart of these empirical hedges.

Formulas

DV01-neutral hedge. $F^H = -F^P \dfrac{\text{DV01}^P}{\text{DV01}^H}$ (risk weight 100%)

Single-variable regression hedge (eq. 11.8). $F^H = -F^P \dfrac{\text{DV01}^P}{\text{DV01}^H}\hat{\beta}$ risk weight $\dfrac{-F^H\,\text{DV01}^H/100}{F^P\,\text{DV01}^P/100} = \hat{\beta}$

Hedged P&L volatility (eq. 11.12). $\sigma_{\text{P\&L}} = \left|\dfrac{F^P\,\text{DV01}^P}{100}\right|\sigma_\varepsilon$ (DV01 of the *hedged* position $\times$ regression standard error)

Two-variable regression hedge (eqs. 11.18 to 11.21). $F^{10} = -F^{20}\dfrac{\text{DV01}^{20}}{\text{DV01}^{10}}\hat{\beta}^{10}$, $F^{30} = -F^{20}\dfrac{\text{DV01}^{20}}{\text{DV01}^{30}}\hat{\beta}^{30}$

PCA hedge (eqs. 11.30, 11.31). For each hedged PC k: $-F^{10}\dfrac{\text{DV01}^{10}}{100}\text{PC}_k^{10} - F^{30}\dfrac{\text{DV01}^{30}}{100}\text{PC}_k^{30} - 100\dfrac{\text{DV01}^{20}}{100}\text{PC}_k^{20} = 0$

PC volatility of a rate (eq. 11.26). $\sqrt{\text{PC}_1^2 + \text{PC}_2^2 + \text{PC}_3^2}$ (PCs uncorrelated); variance share of PC $k = \text{PC}_k^2 / \sum_j \text{PC}_j^2$

Exam traps

- **Dropped term (beta):** the DV01 ratio alone gives the DV01-neutral amount (risk weight 100%), not the regression hedge. With $\hat{\beta} < 1$ the regression hedge sells *fewer* hedging bonds.
- **Red herring (alpha):** the intercept is reported but does not enter the face amount. Adding it to $\hat{\beta}$ is a named mistake.
- **Wrong DV01 in the P&L volatility:** $\sigma_{\text{P\&L}}$ uses the DV01 of the position being hedged, not of the hedge leg. Using the hedge leg gives $\hat{\beta}$ times the right answer.
- **Direction of the regression:** the dependent variable is the yield of the bond being hedged. A reverse regression gives a different risk weight (108.6% versus 84.2% in the book), not the reciprocal of the same hedge.
- **PCA polarity and absolutes:** PCs are *uncorrelated*; two hedging securities neutralise only two PCs; PCA risk weights need not sum to 100% (book 101.2%), only parallel-shift hedges do.

Question C1.1 MR-11 c, d; MR-11 a · S1, S2, S3 · [Map #23]

A relative value trader at a Canadian dealer buys CAD 50 million face amount of a 30-year provincial bond for a client and hedges the interest rate risk by selling the Government of Canada 30-year benchmark bond. The desk's risk analyst supplies the following:

- DV01 per 100 face: provincial bond 0.1835; Government of Canada bond 0.2010
- Regression of daily changes in the provincial yield on daily changes in the Government of Canada yield (bp), 90 observations: constant 0.08 (std. error 0.21); slope 0.91 (std. error 0.04); R^2 81%; standard error of the regression 1.6 bp

Which of the following is closest to the face amount of the Government of Canada bond the trader should sell under a regression hedge, and to the standard deviation of the daily P&L of the hedged position?

A. CAD 41.54 million; CAD 133,588

B. CAD 41.54 million; CAD 146,800

C. CAD 45.19 million; CAD 146,800

D. CAD 45.65 million; CAD 146,800

Solution **Answer: B**

Step 1 – DV01 ratio

$$\frac{\text{DV01}^P}{\text{DV01}^H} = \frac{0.1835}{0.2010} = 0.91294$$

$\Rightarrow$ *Per unit of face, the provincial bond moves 0.913 times as much as the benchmark for the same yield change.*

Step 2 – Regression hedge face amount

$$F^H = -F^P \frac{\text{DV01}^P}{\text{DV01}^H}\hat{\beta} = -50 \times 0.91294 \times 0.91 = -41.54$$

$\Rightarrow$ *Sell CAD 41.54 million. The hedge ratio per unit of face is* $0.91294 \times 0.91 = 0.831$*, and the risk weight (hedge DV01 over position DV01) equals* $\hat{\beta} = 91\%$.

Step 3 – DV01 of the hedged position

$$\frac{F^P\,\text{DV01}^P}{100} = \frac{50{,}000{,}000 \times 0.1835}{100} = 91{,}750$$

Step 4 – Hedged P&L volatility

$$\sigma_{\text{P\&L}} = 91{,}750 \times 1.6 = 146{,}800$$

$\Rightarrow$ *Each 1 bp residual (the provincial yield moving 1 bp more than 0.91 times the benchmark change) costs CAD 91,750, so a 1.6 bp residual volatility gives about CAD 146,800 per day.*

Interpretation

The slope's 95% interval, $0.91 \pm 2(0.04) = (0.83,\ 0.99)$, excludes one, so the data reject parallel shifts and the DV01-neutral hedge over-hedges. The constant's interval, 0.08 ± 0.42, includes zero and the constant plays no part in the hedge. Whether CAD 146,800 per day is acceptable depends on the spread the trader earns and the holding period.

Why the other options are wrong

A is incorrect. Correct face amount, but computed the P&L volatility with the DV01 of the hedge leg: $41.54\text{m} \times 0.2010/100 \times 1.6 = 133{,}588$, which is $\hat{\beta}$ times the right answer.

C is incorrect. Added the intercept to the slope: $50 \times 0.91294 \times (0.08 + 0.91) = 45.19$. The constant is a red herring for the hedge ratio.

D is incorrect. Dropped $\hat{\beta}$: $50 \times 0.91294 = 45.65$ is the DV01-neutral hedge, which assumes parallel yield shifts.

LO: MR-11 c: Calculate the regression hedge adjustment factor, beta. MR-11 d: Calculate the face value of an offsetting position needed to carry out a regression hedge. MR-11 a: Explain the drawbacks to using a DV01-neutral hedge for a bond position. · Reference: Tuckman and Serrat 4e, Ch.6, 11.1, eqs. 11.8 to 11.12, Tables 11.1 and 11.2

Question C1.2

MR-11 a, b, h · S3, S4, S5 · [Map #23]

A risk consultant is reviewing the hedging methodology of a bank's swap desk. Single-bond positions are currently hedged DV01-neutral, and the desk proposes to hedge its 10s/20s/30s butterflies using principal component analysis (PCA) of swap rates, with the 10-year and 30-year swaps as the two hedging instruments. Which of the following statements should the consultant include in the report?

A. A DV01-neutral hedge minimizes the variance of the hedged P&L, because by construction the DV01 of the hedge equals the DV01 of the position being hedged.

B. A PCA hedge is efficient because the principal components are constructed to be highly correlated, so neutralizing the level PC also neutralizes the others.

C. A PCA hedge using two hedging swaps can neutralize exposure to only two PCs, so the residual exposure to the third PC must be measured and either accepted or hedged.

D. A PCA hedge of the butterfly sets the risk weights of the two hedging swaps so that they always sum to 100% of the DV01 of the position being hedged.

Solution

Answer: C

Step 1 – What a DV01-neutral hedge assumes

Equal DV01s on both sides (risk weight 100%) is the right hedge only if the two yields move in parallel. The book rejects $\beta = 1$ for JNJ versus the 30-year Treasury (interval 0.740 to 0.944).

Step 2 – Which hedge minimises variance

The regression hedge, with risk weight $\hat{\beta}$, minimises the variance of the hedged P&L (Appendix A6.1). The DV01 hedge does so only in the special case $\hat{\beta} = 1$.

Step 3 – How many PCs two securities can hedge

Each hedged PC adds one equation (11.30, 11.31); two hedging swaps solve two equations. The book measures the residual short-rate exposure at 0.007 per 100 and notes the trader may accept it or add a 15-year swap and a third equation.

Interpretation

C is correct: it states the counting rule and the follow-up decision that the book models.

Why the other options are wrong

A is incorrect. Sibling swap. Variance minimisation is the property of the regression hedge; equal DV01s minimise variance only if yields shift in parallel.

B is incorrect. Polarity flip. PCs are constructed to be *uncorrelated*; that is why each needs its own neutralising equation.

D is incorrect. Absolute claim. Only under parallel shifts do risk weights always sum to 100%; the book's PCA butterfly needs $27.1\% + 74.1\% = 101.2\%$.

LO: MR-11 a: Explain the drawbacks to using a DV01-neutral hedge for a bond position. MR-11 b: Describe a regression hedge and explain how it can improve a standard DV01-neutral hedge. MR-11 h: Describe principal component analysis and explain how it is applied to constructing a hedging portfolio. · Reference: Tuckman and Serrat 4e, Ch.6, 11.1 and 11.5, eqs. 11.30 to 11.34

C2 Backtesting VaR: Exceedance vs PIT

Readings: MR-4 a-f · Backtesting VaR (Jorion, Ch.6); MR-7 a-d · Beyond Exceedance-Based Backtesting of Value-at-Risk Models (Iercosan, Shcherbakova, McArthur and Alper, Ch.4); MR-18 c · Fundamental Review of the Trading Book (Hull, RMFI Ch.18, 18.3.1); ORR-21 c · Capital Regulation Before the Global Financial Crisis (Carey)

MAP TIER C · #21 · 4/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #21: "Reported: 'backtesting VaR'; 'PIT vs exceedance': the point-in-time vs. exceedance-based distinction is the direct subject of Reading 7 and appears twice in the consolidated list. Kupiec's POF tests unconditional coverage only; Christoffersen adds independence, giving conditional coverage. Type I = rejecting a correct model; Type II = accepting a flawed one. Basel penalty zones (green/yellow/red) and their multiplier consequences are separately examinable. This pair remains uncleared in your Confusables Registry."
Shapes covered here:
S1 backtesting VaR: counting exceptions against the model (Q1)
S2 PIT-based versus exceedance-based backtesting (Q2)
S3 Kupiec POF (unconditional coverage) versus Christoffersen (independence, conditional coverage) (Q3)
S4 Type I versus Type II errors (Q3; trade-off in Q1)
S5 Basel penalty zones and the multiplier they set (Q1).

Revision notes

- **Backtesting** (MR-4 a): "a formal statistical framework that consists of verifying that actual losses are in line with projected losses". An exception (exceedance) is a day when the loss exceeds the previous day's VaR. Too many exceptions: the model underestimates risk (too little capital, regulatory penalties); too few: excess, inefficient capital allocation.
- **Which return?** (MR-4 b): VaR assumes a frozen portfolio. The actual return is contaminated by intraday trades, fees, commissions, spreads and net interest income; the hypothetical return applies fixed positions to close-to-close returns; the cleaned return is actual minus non-mark-to-market items. Use both actual and hypothetical: passing with hypothetical but failing with actual points to intraday trading; failing with hypothetical means the methodology must be reexamined.
- **Failure rate test** (MR-4 c): under the null, exceptions x are binomial with mean pT and variance $p(1-p)T$; $z = (x - pT)/\sqrt{p(1-p)T}$, reject if $|z| > 1.96$ (JPM 1998: 20 exceptions at 95% over 252 days, $z = 2.14$, reject). The test is fully nonparametric. Kupiec's LR_{uc} is $\chi^2(1)$: reject if $LR_{uc} > 3.841$ (JPM: 3.91). Nonrejection region at 99%, $T = 252$: $N < 7$; at 95%, $T = 252$: $6 < N < 20$.
- **The test confidence level is not the VaR confidence level** (MR-4 c): Basel backtests a 99% VaR with a decision rule at, for example, 95% test confidence.
- **Type 1 and Type 2 errors** (MR-4 d; S4): Type 1 is rejecting a correct model (bad luck penalised); Type 2 is not rejecting an incorrect model. At $p = 0.01$, $T = 250$: more than four exceptions occur 10.8% of the time under a correct model (Type 1); a model with true 97% coverage passes 12.8% of the time (Type 2). Power $= 1-$ Type 2 rate.
- **Raising power** (MR-4 d): exception tests have low power at high VaR confidence and short samples. Lowering the VaR confidence level to 95% (cutoff 17 exceptions, Type 1 12.5%) cuts Type 2 to 7.4%; raising T to 1,000 (cutoff 14) gives Type 1 13.4% and Type 2 0.03%. Detection of systematic bias is hardest for small p (rare events).
- **Conditional coverage** (MR-4 e; S3): exceptions should be evenly spread; bunching should invalidate the model. Christoffersen's LR_{ind} compares π_0 (exception after a non-exception day) with π_1 (exception after an exception day); reject independence if $LR_{ind} > 3.841$. $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2(2)$: reject if $LR_{cc} > 5.991$. JPM: $\pi_0 = 14/232 = 6.0\%$, $\pi_1 = 6/20 = 30.0\%$, $LR_{ind} = 9.53$: exceptions cluster.
- **Basel rules** (MR-4 f; S5): daily exceptions of the 99% one-day VaR over the last 250 days; 2.5 expected. Green 0 to 4 ($k = 3$); yellow 5 to 9 with plus factors 0.40, 0.50, 0.65, 0.75, 0.85; red 10 or more (plus 1.00, automatic). In the yellow zone the penalty depends on the cause: basic integrity of the model and model accuracy could be improved: penalty "should" apply; intraday trading: "should be considered"; bad luck:

expected some of the time, and exceptions from sudden abnormal changes in rates, major political events or natural disasters may be excluded.

- **The capital formula** (ORR-21 c): MR $= \max(\text{VaR}_{t-1},\ m \times \text{VaR}_{avg})$, with 10-day 99% VaR and the 60-day average; $m \geq 3$. Carey's summary ("five or less exceptions enabled the multiplier m to be 3", 10 or more required 4) is looser than Jorion's Table 4.3; for MR-4 f questions use Jorion's table.
- **Two properties of an accurate VaR model** (MR-7 a; S2): unconditional coverage (Kupiec) and independence (Christoffersen). In a PIT backtest these become *uniformity* of the PITs on $U(0,1)$ and independence: a perfectly accurate model gives i.i.d. $U(0,1)$ PITs (Rosenblatt).
- **Deriving PITs** (MR-7 b): $p_{t+1} = F_t(PL_{t+1})$, the model's (previous day's forecast) probability of observing a loss below the actual P&L. Exceedances test one percentile; PITs carry information "at any percentile of the forecast distribution". Hull (MR-18 c) calls the same quantity the p-value, uniformly distributed if the model works perfectly.
- **Shape of the PIT histogram** (MR-7 c): conservative misspecification (model distribution too wide) puts P&L in the middle, a hump-shaped PIT histogram; aggressive misspecification (too narrow) puts realised P&L in the tails, spikes at 0 and 1. The sample showed models conservative in the body but with tails thinner than realised P&L at subportfolio level.
- **Goodness-of-fit tests** (MR-7 d): Kolmogorov-Smirnov measures the maximum distance between empirical and uniform CDFs but lacks sensitivity in the tails; Anderson-Darling puts more weight on the tails (more power for a 99% VaR); Cramer-von Mises uses the mean squared deviation. Independence of PITs: transform to normals and regress on lags.
- **Results and limits** (MR-7 a, d): 67% of subportfolios passed Kupiec, 92% Christoffersen independence, 71% conditional coverage, but only 21% passed KS and 11% A-D. Christoffersen ignores dependence beyond one lag; Ljung-Box tests several lags; the Christoffersen and Pelletier duration test (exponential versus Weibull) adds power. Banks do better on exceedance tests than on distribution tests.
- **FRTB backtesting** (MR-18 c): stressed ES is not backtested; each desk backtests a one-day VaR at 99% and 97.5% over 12 months; more than 12 exceptions at 99% or more than 30 at 97.5% sends the desk to the standardized approach. See also B2 for ES.

Formulas

Normal approximation. $z = \dfrac{x - pT}{\sqrt{p(1-p)T}}$, reject at 95% (two-tailed) if $|z| > 1.96$.

Kupiec POF. $LR_{uc} = -2\ln\left[(1-p)^{T-N}p^N\right] + 2\ln\left[\left(1-\frac{N}{T}\right)^{T-N}\left(\frac{N}{T}\right)^N\right] \sim \chi^2(1)$, critical value 3.841.

Christoffersen. $\pi_0 = \dfrac{T_{01}}{T_{00}+T_{01}}$, $\pi_1 = \dfrac{T_{11}}{T_{10}+T_{11}}$; $LR_{ind} \sim \chi^2(1)$ (3.841); $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2(2)$ (5.991).

Basel plus factors (Jorion Table 4.3). Exceptions 0 to 4: 0.00; 5: 0.40; 6: 0.50; 7: 0.65; 8: 0.75; 9: 0.85; 10+: 1.00. $k = 3 +$ plus factor.

Market risk charge (Carey). MR $= \max\left(\text{VaR}_{t-1},\ k \times \text{VaR}_{avg,60}\right)$

PIT. $p_{t+1} = F_t(PL_{t+1}) \sim$ i.i.d. $U(0,1)$ under a correct model.

Duration test. $\Pr(D = d) = (1-p)^{d-1}p$; null: exponential durations (Weibull $b = 1$, $a = p$).

Exam traps

- **Type 1 versus Type 2 (polarity flip):** rejecting a *correct* model is Type 1 (the 10.8% for five or more exceptions); accepting a flawed one is Type 2 (the 12.8%).
- **Kupiec versus Christoffersen (sibling swap):** Kupiec cannot see clustering if the count is right; Christoffersen's independence test sees only one lag, not breaches a week apart.
- **Excludable exceptions (scope error):** only exceptions from sudden abnormal market moves, major political events or natural disasters may be excluded; ordinary bad luck ("markets were particularly volatile or correlations changed") still counts.
- **Plus-factor row off-by-one:** 6 exceptions is +0.50, 7 is +0.65, 8 is +0.75; 5 is already yellow (+0.40) in Jorion's table.

- **PIT histogram polarity:** PIT is the probability integral transform, not a point-in-time measure. A hump in the middle means a *conservative* (too wide) model; spikes at the ends mean an *aggressive* (too narrow) one.
- **KS versus A-D (sibling swap):** Anderson-Darling, not Kolmogorov-Smirnov, weights the tails.

Question C2.1

MR-4 f; ORR-21 c · S1, S5 · [Map #21]

A regulatory analyst at a German bank is computing the market risk capital charge for a trading book under the internal models approach. Over the last 250 trading days the one-day 99% VaR was exceeded on 8 days, with the following causes identified:

- 4 days: markets were particularly volatile and correlations changed
- 2 days: the model uses too few maturity buckets for the yield curve
- 1 day: positions changed materially during the day through intraday trading
- 1 day: a major earthquake caused large losses across local markets

The supervisor will exclude any exception that the Basel rules allow to be excluded. The 10-day 99% VaR for the previous day is EUR 52 million and the average 10-day 99% VaR over the last 60 days is EUR 40 million. Which of the following is closest to the multiplier k and the market risk capital charge?

A. $k = 3.00$; EUR 120 million

B. $k = 3.50$; EUR 140 million

C. $k = 3.65$; EUR 146 million

D. $k = 3.75$; EUR 150 million

Solution

Answer: C

Step 1 – Count the exceptions that remain

Exceptions "may be excluded if they are the result of such occurrences as sudden abnormal changes in interest rates or exchange rates, major political events, or natural disasters". Only the earthquake qualifies. Volatile markets and changed correlations are bad luck that should "be expected to occur at least some of the time"; model accuracy and intraday trading exceptions are penalised.

$$N = 8 - 1 = 7$$

Step 2 – Zone and plus factor

Seven exceptions fall in the yellow zone (5 to 9); Jorion's Table 4.3 gives an increase in k of 0.65:

$$k = 3 + 0.65 = 3.65$$

$\Rightarrow$ *With a correct 99% model the expected count is* $0.01 \times 250 = 2.5$*; under a correct model the probability of seven or more exceptions is only 1.4% (Table 4.4), so a penalty is appropriate.*

Step 3 – Capital charge

$$\text{MR} = \max\left(\text{VaR}_{t-1},\ k \times \text{VaR}_{avg}\right) = \max(52,\ 3.65 \times 40) = \max(52,\ 146) = 146$$

Interpretation

The bank holds EUR 146 million, EUR 26 million more than with $k = 3$. Fixing the maturity buckets ("model accuracy could be improved") is the actionable item: those exceptions carry a penalty that "should" apply.

Why the other options are wrong

A is incorrect. Excluded every "bad luck" exception as well as the earthquake: $8 - 1 - 4 = 3$ exceptions, green zone, $k = 3$, $3 \times 40 = 120$.

B is incorrect. Read the plus factor one row too high in the table (0.50 belongs to 6 exceptions): $3.50 \times 40 = 140$.

D is incorrect. Did not exclude the earthquake: 8 exceptions, $k = 3.75$, $3.75 \times 40 = 150$.

LO: MR-4 f: Describe the Basel rules for backtesting. ORR-21 c: Describe measures introduced in the 1995 and 1996 amendments, including guidelines for netting of credit exposures and methods for calculating market risk capital for assets in the trading book. · Reference: Jorion, Value at Risk 3e, Ch.6, The Basel Rules, Tables 4.3 and 4.4; Carey, Capital Regulation Before the Global Financial Crisis, 21.1

Question C2.2 MR-7 a, b, c, d · S2 · [Map #21]

A model validation analyst at a US bank holding company is presenting to newly hired analysts on how the bank's subportfolio VaR models are validated. The bank stores, for each day, the one-day 99% VaR, the clean P&L and the corresponding probability integral transform (PIT). Which of the following statements is correct for the analyst to make?

A. A hump-shaped PIT histogram, with realised P&L clustering in the middle of the forecast distribution, signals aggressive misspecification because the model distribution is too narrow.

B. Uniformity of the PITs on the interval from 0 to 1 is the counterpart of unconditional coverage, so a PIT test uses information from every percentile of the forecast distribution.

C. The Kolmogorov-Smirnov test is the preferred uniformity test for a 99% VaR model, because it places more weight on the tails of the PIT distribution than the Anderson-Darling test.

D. Because PITs use the whole forecast distribution, a subportfolio that passes Kupiec's unconditional coverage test will also pass a Kolmogorov-Smirnov test of PIT uniformity.

Solution **Answer: B**

Step 1 – What the PIT is

$p_{t+1} = F_t(PL_{t+1})$: the previous day's forecast distribution evaluated at the realised P&L. A correctly specified model produces i.i.d. $U(0,1)$ PITs.

Step 2 – Map the two exceedance properties

Unconditional coverage (exceedances at 1% VaR observed 1% of the time, at 3% VaR 3% of the time, and so on) "is analogous to the uniformity distributional property of a series of PITs"; independence of exceedances maps to independence of the PITs.

Interpretation

B is correct. An exceedance count tests the model at a single percentile; PITs, when well estimated, "provide information on the accuracy of the risk model at any percentile of the forecast distribution". That is why PIT tests can reveal misspecifications that pure exceedance analysis misses.

Why the other options are wrong

A is incorrect. Polarity flip. P&L clustering in the middle is *conservative* misspecification: the model distribution is too wide. Aggressive misspecification (too narrow) shows realised P&L in the tails.

C is incorrect. Sibling swap. The KS test lacks sensitivity at the extremes; Anderson-Darling is the modification that puts more weight on the tails and so better detects tail misspecification of a 99% VaR model.

D is incorrect. Absolute claim. In the study 67% of subportfolios passed Kupiec but only 21% passed KS: "banks perform better on exceedence tests than on tests of distribution".

LO: MR-7 a: Identify the properties of an exceedance-based backtest that indicate a VaR model is accurate, and describe how these properties are reflected in a PIT-based backtest. MR-7 b: Explain how to derive probability integral transforms (PITs) in the context of validating a VaR model. MR-7 c: Describe how the shape of the distribution of PITs can be used as an indicator of the quality of a VaR model. MR-7 d: Describe backtesting using PITs, and compare the various goodness-of-fit tests that can be used to evaluate the distribution of PITs: the Kolmogorov-Smirnov test, the Anderson-Darling test, and the Cramér-von Mises test. · Reference: Iercosan, Shcherbakova, McArthur and Alper, Ch.4, 7.1, 7.2, 7.6, 7.7

Question C2.3 MR-4 c, d, e; MR-7 a · S3, S4 · [Map #21]

An internal auditor at a Singapore bank is reviewing the backtesting report for a desk's 99% VaR model. Over 250 days the desk recorded 3 exceptions, on three consecutive trading days of a volatile month. The auditor drafts comments for the risk committee. Which of the following comments is correct?

A. Kupiec's proportion of failures test should reject the model, because exceptions bunched within two weeks raise the failure rate measured by the test.

B. If the model is in fact correct and is rejected because of bad luck, the error committed is a Type 2 error, the risk that the Basel zones are designed to keep low.

C. Christoffersen's independence test would also reject a model whose exceptions occur exactly one week apart, because it detects dependence between exceptions at any lag.

D. The conditional coverage statistic adds Kupiec's LR_{uc} to Christoffersen's LR_{ind} and can reject this model

even though its exception count is acceptable.

Solution **Answer: D**

Step 1 – Unconditional coverage

Expected exceptions: $0.01 \times 250 = 2.5$. Three exceptions lies inside the nonrejection region ($N < 7$ at $T = 252$), so Kupiec's test does not reject: $LR_{uc} = 0.10 < 3.841$. The count is fine.

Step 2 – Independence

Exceptions "could cluster or 'bunch' closely in time, which also should invalidate the model". Christoffersen's LR_{ind} compares π_1, the probability of an exception after an exception day, with π_0; bunching raises π_1. Here $T_{00} = 245$, $T_{01} = 1$, $T_{10} = 1$, $T_{11} = 2$: $\pi_0 = 1/246 = 0.4\%$ but $\pi_1 = 2/3 = 66.7\%$, giving $LR_{ind} = 15.65 > 3.841$.

Step 3 – Conditional coverage

$$LR_{cc} = LR_{uc} + LR_{ind} = 0.10 + 15.65 = 15.75 > 5.991 = \chi^2_{0.95}(2)$$

Interpretation

D is correct. The joint test checks "whether the proportion of violations following an exceedance is the same as the proportion of violations following a non-exceedance" while also checking the level, so it can catch a model whose exceptions are right in number but wrong in timing, the signal that the VaR is not responding to changing volatility.

Why the other options are wrong

A is incorrect. Sibling swap. Kupiec's test counts exceptions only and "ignores the time variation in the data"; three consecutive exceptions have the same failure rate as three spread-out ones. Detecting bunching is Christoffersen's job.

B is incorrect. Polarity flip. Rejecting a correct model is a Type 1 error (the 10.8% probability of five or more exceptions under a correct model); a Type 2 error is failing to reject an incorrect model.

C is incorrect. Scope/condition error. Christoffersen's test uses a first-order Markov chain and "does not allow for the possibility that independence may be violated at a distance greater than one lag"; breaches one week apart need a test such as Ljung-Box or the duration test.

LO: MR-4 c: Evaluate the accuracy of a VaR model based on exceptions or failure rates by using a model verification test. MR-4 d: Identify and describe Type I and Type II errors in the context of a backtesting process. MR-4 e: Explain the need to consider conditional coverage in the backtesting framework. MR-7 a: Identify the properties of an exceedance-based backtest that indicate a VaR model is accurate, and describe how these properties are reflected in a PIT-based backtest. · Reference: Jorion, Value at Risk 3e, Ch.6, 4.2 (Model Verification, Conditional Coverage Models), Tables 4.1 and 4.2; Iercosan et al., Ch.4, 7.5

C3 VaR Mapping

Readings: MR-5 a-g · VaR Mapping (Jorion, Value at Risk 3e, Ch.11)

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Specifics reported

Map #22: "Reported twice in identical wording: 'difference in VaR mapping: principal, duration, cash flow.' The sheet additionally singles out LO 5.f, 'especially interpretation of variance improvement': VaR relative to a performance benchmark. Principal mapping uses only the timing of principal; duration mapping uses portfolio duration; cash-flow mapping decomposes every payment onto the grid. Ranking their risk-capture fidelity is the classic comparison."
Shapes covered here:
S1 difference among principal, duration and cash-flow mapping (Q2)
S2 VaR relative to a benchmark: tracking error VaR and the interpretation of variance improvement (Q1)
S3 ranking the three methods by how faithfully they capture risk (Q2).

Revision notes

- **Why mapping** (MR-5 a): "Mapping is the process by which the current values of the portfolio positions are replaced by exposures on the risk factors." VaR is aggregation at the highest level (J.P. Morgan Chase: 2.1 million positions, 240,000 pricing series). Thousands of USD/EUR forwards collapse onto one risk factor, the spot rate: acceptable for risk measurement, not for pricing.
- **Mapping as a data solution** (MR-5 a): instruments whose profile changes as they age (bonds must be mapped on yields that represent their current profile; options on their primary risk factors); positions with no history (IPOs replaced by similar risk factors); stale prices across time zones (Tokyo closes 15 hours before New York), which make daily cross-market correlations too low.
- **The process** (MR-5 a): mark every position to market in the reference currency, allocate each market value to the risk factors (any unallocated part is cash, which has no risk), then sum exposures by factor into the vector x. Allocation is exact for derivatives priced off the factors (partial derivatives) and estimated for others (a stock's regression beta on the index).
- **General and specific risk** (MR-5 b): with each stock mapped on the index, $\beta_p = \sum w_i \beta_i$, the exposure is $x = W\beta_p$, and $V(R_p) = \beta_p^2 V(R_m) + \sum w_i^2 \sigma_{\epsilon,i}^2$: general market risk plus aggregate specific risk. "With more detail on the primitive or general-market risk factors, there will be less specific risk for a fixed amount of total risk." The choice of factors trades approximation quality against speed.
- **Three fixed-income methods** (MR-5 c; S1): principal mapping uses one risk factor at the average portfolio maturity (redemption timing only); duration mapping uses one risk factor at the portfolio duration; cash-flow mapping groups the portfolio cash flows into maturity buckets (vertices). "Mapping should preserve the market value of the position. Ideally, it also should preserve its market risk."
- **Jorion's two-bond example** (MR-5 c, d; S3): USD 100 million 5-year 6% plus USD 100 million 1-year 4%, both at par; average maturity 3 years, duration 2.733 years. Principal VaR $= 200 \times 1.484\% =$ USD 2.97 million ("overstates the true risk because it ignores intervening coupon payments"). Duration VaR: interpolated risk $0.987 + (1.484 - 0.987)(0.733) = 1.351\%$, USD 2.70 million. Cash flow: undiversified USD 2.63 million, diversified USD 2.57 million.
- **Decomposing the gap** (MR-5 c): of the USD 130,000 between duration (2.70) and diversified cash-flow (2.57) VaR, USD 70,000 comes from risk not being linear in maturity (2.70 versus 2.63) and USD 60,000 from correlations below one (2.63 versus 2.57). Ranking by fidelity: cash flow best, then duration, principal crudest.
- **Cash-flow mapping mechanics** (MR-5 d): each cash flow is represented by its present value at the zero-coupon rate and placed on the vertex; VaR $= \sqrt{(x \times V)' R (x \times V)}$ with $V = \alpha\sigma$ the vector of zero VaRs; undiversified VaR $= \sum |x_i| V_i$. Cash flows between vertices have their present values apportioned.
- **Stress test** (MR-5 e): with perfect correlation, shock every zero by its VaR (1-year zero $0.9615 \times (1 - 0.4696\%) = 0.9570$) and reprice: USD 197.37 million, exactly USD 2.63 million lower, the undiversified VaR. Matrix multiplication is more direct and allows imperfect correlations.

- **VaR relative to a benchmark** (MR-5 f; S2): TE-VaR $= \alpha\sqrt{(x-x_0)'\Sigma(x-x_0)}$. J.P. Morgan US bond index (duration 4.62): absolute monthly 95% VaR USD 1.99 million on USD 100 million. A duration-matched 4- and 5-year zero portfolio (USD 38 million and 62 million) has TE-VaR USD 0.43 million.
- **Variance improvement** (MR-5 f; S2): $1-(\text{TE-VaR}/\text{VaR}_{index})^2 = 1-(0.43/1.99)^2 = 95.4\%$, "similar to an R^2 in a regression" and "in line with the explanatory power of the first factor". The remaining tracking error "is due to nonparallel moves in the term structure".
- **Absolute is not relative** (MR-5 f): Jorion's portfolios 2 to 5 have TE-VaR 0.29, 0.16, 0.20, 0.81. Portfolio 3 (2- and 9-year) tracks best because it comes closest to the index cash flows; portfolio 5 (1-month bills and 30-year zeros, a barbell) has the *lowest absolute* VaR (1.71) but the *worst* tracking error. "Minimizing absolute market risk is not the same as minimizing relative market risk"; duration hedging is only a first approximation.
- **Linear derivatives** (MR-5 g): long currency forward = long foreign currency spot + long foreign currency bill + short USD bill (EUR 100 million example: USD 125.89 million each; spot VaR 5.713, total 5.735). Commodity forwards map on the forward price (convenience yield is not tied to another financial variable): 12-month oil, USD 43.743 million $\times$ 14.05% = USD 6.146 million. Long 6×12 FRA = long 6-month bill + short 12-month bill (VaR 0.327). A swap is a fixed bond against an FRN (FRN on cash just before reset) or a string of forwards; VaR USD 2.152 million either way, 1.763 after the first reset.
- **Options** (MR-5 g): delta-normal mapping: long call = long Δ of the asset + short $(\Delta S - c)$ in a bill (ATM: $\Delta = 0.536$, $c = 4.20$, asset 53.60, loan 49.40). Delta is not constant; a linear approximation is acceptable for long-dated options over short risk horizons. Deep in the money, the option behaves like a forward.

Formulas

General and specific risk. $\beta_p = \sum_i w_i\beta_i, \quad x = W\beta_p, \quad V(R_p) = \beta_p^2 V(R_m) + \sum_i w_i^2\sigma_{\epsilon,i}^2$
Principal mapping. VaR $= W \times V_{\bar{M}}$, $\bar{M}$ = average maturity of principal.
Duration mapping. VaR $= W \times V_D$, with $V_D = V_j + (V_{j+1} - V_j)(D - j)$ by linear interpolation.
Cash-flow mapping. VaR $= \sqrt{(x\times V)'R\,(x\times V)}$; undiversified $= \sum_i |x_i|V_i$.
Tracking error VaR. TE-VaR $= \alpha\sqrt{(x-x_0)'\,\Sigma\,(x-x_0)}$
Variance improvement. $1 - \left(\dfrac{\text{TE-VaR}}{\text{VaR}_{benchmark}}\right)^2$
Currency forward. $f_t = S_t e^{-r^*\tau} - Ke^{-r\tau}$: long spot + long foreign bill + short domestic bill. **Option.** Long call = long ΔS in the asset + short $(\Delta S - c)$ in a bill, $\Delta = e^{-r^*\tau}N(d_1)$.
Book checks. Principal 2.97, duration 2.70, cash flow 2.63 (undiversified) and 2.57 (diversified); index 1.99, TE-VaR 0.43, variance improvement 95.4%.

Exam traps

- **Principal mapping polarity:** it *overstates* risk (ignores intervening coupons), so it gives the highest VaR of the three.
- **Duration versus cash flow (sibling swap):** duration mapping uses *one* vertex at the portfolio duration; only cash-flow mapping spreads each payment over the grid and captures imperfect correlation.
- **Variance improvement, not VaR improvement:** square the ratio; $1 -$ TE-VaR/VaR understates the improvement.
- **Absolute versus relative risk:** the barbell (cash and a 30-year zero) has the lowest absolute VaR and the worst tracking error; matching absolute VaR to the index is not tracking.
- **Market value is preserved by all three methods (scope error):** what distinguishes them is how well they preserve market *risk*.

Question C3.1

MR-5 f · S2 · [Map #22]

A portfolio manager at a German pension fund must track a government bond index with EUR 250 million. The index's monthly 95% absolute VaR is EUR 4.80 million. The risk team has built five candidate portfolios, all with the same duration as the index, and computed their monthly 95% VaRs (EUR million):

Portfolio	1	2	3	4	5
Vertices used	4Y, 5Y	3Y, 7Y	2Y, 9Y	1Y, 10Y	1M, 30Y
Absolute VaR	5.40	5.10	4.95	4.75	4.10
Tracking error VaR	1.05	0.62	0.38	0.55	1.95

The manager's mandate is to track the index as closely as possible. Which portfolio should the manager choose, and what is its variance improvement relative to the index?

A. Portfolio 3; variance improvement 99.4%

B. Portfolio 3; variance improvement 92.1%

C. Portfolio 4; variance improvement 98.7%

D. Portfolio 5; variance improvement 83.5%

Solution — Answer: A

Step 1 – Choose the right risk measure

The mandate is relative, so the selection criterion is the tracking error VaR, $\alpha\sqrt{(x-x_0)'\Sigma(x-x_0)}$, not the absolute VaR. Portfolio 3 has the lowest TE-VaR, EUR 0.38 million.

Step 2 – Variance improvement for each candidate

$$\text{VI} = 1 - \left(\frac{\text{TE-VaR}}{\text{VaR}_{index}}\right)^2$$

Portfolio 1: $1-(1.05/4.80)^2 = 95.2\%$; portfolio 2: 98.3%; portfolio 3: $1-(0.38/4.80)^2 = 1-0.00627 = 99.4\%$; portfolio 4: 98.7%; portfolio 5: 83.5%.

$\Rightarrow$ *The ranking by variance improvement is the same as the ranking by TE-VaR, and portfolio 3 wins.*

Interpretation

Like an R^2, a 99.4% variance improvement means the portfolio reproduces all but 0.6% of the variance of the index's returns; the residual comes from non-parallel moves in the term structure. Spreading the two positions to 2 and 9 years comes closest to the index's cash-flow profile, which is why it beats the narrow 4- and 5-year bracket even though all five have the same duration.

Why the other options are wrong

B is incorrect. Right portfolio, but used the ratio of VaRs instead of the ratio of variances: $1-0.38/4.80 = 92.1\%$.
C is incorrect. Picked the portfolio whose absolute VaR is closest to the index's (4.75 against 4.80) instead of the lowest tracking error; matching absolute risk is not matching the index.
D is incorrect. Picked the lowest absolute VaR (the barbell of 1-month bills and 30-year zeros). "Minimizing absolute market risk is not the same as minimizing relative market risk": its TE-VaR is the worst, $1-(1.95/4.80)^2 = 83.5\%$.

LO: MR-5 f: Explain how VaR can be calculated and used relative to a performance benchmark. · Reference: Jorion, Value at Risk 3e, Ch.11, 5.2 Benchmarking, Table 5.5 and eq. 5.8

Question C3.2 — MR-5 c, d · S1, S3 · [Map #22]

A junior risk analyst at a Japanese life insurer is preparing a report comparing principal, duration and cash-flow mapping for a bond portfolio, using the textbook example of a 5-year 6% bond and a 1-year 4% bond held in equal amounts at par. Which of the following statements is correct for the analyst to include in the report?

A. Principal mapping understates the true risk of the portfolio, because it maps the market value to the average maturity of principal and ignores the intervening coupon payments.

B. Duration mapping places the present value of each cash flow on the vertex nearest its payment date, which allows it to capture imperfect correlation along the curve.

C. Cash-flow mapping is preferred because it preserves the market value of the portfolio, a property that principal mapping and duration mapping do not have.

D. The diversified cash-flow VaR is below the duration VaR, because risk is not linear in maturity and because correlations between the vertices are below one.

Solution **Answer: D**

Step 1 – The three numbers

Principal (3-year vertex): USD 2.97 million. Duration (2.733 years, interpolated risk 1.351%): USD 2.70 million. Cash flow: USD 2.63 million undiversified, USD 2.57 million diversified.

Step 2 – Decompose the duration to cash-flow gap

$$2.70 - 2.57 = 0.13 = \underbrace{(2.70 - 2.63)}_{0.07 \text{ nonlinear risk in maturity}} + \underbrace{(2.63 - 2.57)}_{0.06 \text{ correlations below one}}$$

Interpretation

D is correct. The ranking of fidelity runs from cash-flow mapping (finest, captures both effects), through duration mapping (one vertex, ignores the curvature of risk across maturities and assumes perfect correlation), to principal mapping (crudest, ignores coupons).

Why the other options are wrong

A is incorrect. Polarity flip. Principal mapping "overstates the true risk because it ignores intervening coupon payments": it gives the highest VaR, USD 2.97 million.

B is incorrect. Sibling swap. Grouping each cash flow on term-structure vertices is cash-flow mapping; duration mapping replaces the whole portfolio by one zero-coupon bond with maturity equal to the portfolio duration.

C is incorrect. Scope/condition error. All three methods preserve market value (USD 200 million is mapped in each case); "ideally, it also should preserve its market risk", and it is risk preservation that separates them.

LO: MR-5 c: Differentiate among the three methods for mapping portfolios of fixed-income securities. MR-5 d: Summarize how to map a fixed-income portfolio into positions of standard instruments. · Reference: Jorion, Value at Risk 3e, Ch.11, 5.2 Mapping Approaches, Tables 5.2 to 5.4

C4 Policy-Mix VaR and Risk Budgeting

Readings: IM-6 a, b, c, d, e, f, g, h · VaR and Risk Budgeting in Investment Management (Jorion, Value at Risk 3e, Ch.17; RMIM Ch.6, 6.1 to 6.5)

MAP TIER C · #35 · 4/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #35: "Reported verbatim: 'then VaR was asking to be calculated by policy mix method.' LO 88.e is the exact home: absolute, relative, policy-mix, active-management, funding and sponsor risk as a six-way sibling set. Also reported: '3-4 on portfolio return computation' and 'IR/2 active risk numerical'. Risk budgets across asset classes and active managers (LO 88.h) is the computational form."
Shapes covered here:
S1 VaR calculated by the policy-mix method: policy weights on benchmark returns, diversified (Q1)
S2 the six-way sibling set: absolute, relative, policy-mix, active-management, funding, sponsor risk (Q3; policy-mix versus asset and active VaR in Q1)
S3 portfolio return computation: expected excess return of a pool of managers (Q2)
S4 IR and active risk numerical: IR from expected excess return and TEV, portfolio IR (Q2)
S5 risk budgets across asset classes and across active managers (Q1, Q2)

Revision notes

- **Risk budgeting** (IM-6 a): "the process of allocating and managing risk using a top-down approach to different aspects of the investment process". It builds on VaR measures applied to asset classes, asset managers and even securities, and is "equivalent to a top-down allocation of economic risk capital". It gives a dynamic comparison of current risk profiles with prespecified risk budgets.
- **Sell side versus buy side** (IM-6 b): bank trading has a short horizon (1 day, intraday), rapid turnover and high leverage, so historical risk measures are "basically useless" and VaR, position limits, VaR limits and stop-loss rules dominate. Investors have long horizons (month, quarter, years), slow turnover and low leverage; risk is managed through asset allocation, tracking error, diversification, benchmarking and investment guidelines.
- **Investment process of large investors** (IM-6 c): step one, a consultant's strategic long-term asset-allocation study, usually mean-variance, sets the amounts per asset class using passive benchmarks; step two, money is delegated to active managers reviewed on tracking error and controlled by investment guidelines. Risk is usually measured *ex post*. VaR is needed because investing is global, instruments are complex (Wisconsin's MexSpread swap: thought USD 10 million at risk, lost USD 35 million on one contract) and positions are dynamic across many managers.
- **Hedge funds** (IM-6 d): heterogeneous, mostly leveraged, some with high turnover (LTCM), so closer to bank desks. Funds holding illiquid assets (e.g. convertibles) report a stale NAV, which artificially *lowers* correlations (fix: regressions with lagged market factors, summing coefficients, Dimson) and artificially *lowers* volatility (shows up as positive serial autocorrelation). Lack of transparency on positions is the third problem. See also D9 and A24.
- **Absolute versus relative risk** (IM-6 e): absolute (asset) risk is the risk of a dollar loss over the horizon, return R_{asset}. Relative risk is the dollar loss of the fund relative to its benchmark; the relevant return is Jorion's tracking error $e = R_{\text{asset}} - R_B$, and with normality VaR $= \alpha W \sigma_e$.
- **Policy-mix versus active-management risk** (IM-6 e, S1): policy-mix risk is the risk of a dollar loss owing to the policy mix selected by the fund; since it can be implemented with passive funds, it is the risk of a passive strategy. Active-management risk is the risk of a dollar loss owing to the total deviations from the policy mix, summed across all managers (with diversification across styles), and includes temporary deviations of the current asset mix from the policy mix.
- **The decomposition** (eq. 6.2): $R_{\text{asset}} = \sum_i w_i^p R_i^b + \sum_i (w_i R_i - w_i^p R_i^b)$. Total VaR comes from the policy-mix VaR, the active-management VaR and a cross-product term. OTPPB (99%, annual): policy-mix 19.6%, active 1.6%, asset 19.3%. Most risk is policy mix (Brinson et al. 1986); active VaR is small (diversified styles, near-indexed assets); the two do *not* add up, and a slightly negative correlation means managers could take larger deviations without raising total VaR.

- **Funding risk** (IM-6 e): the risk that assets will not be sufficient to cover the fund's liabilities, measured on the surplus $S = A - L$. Nominal liabilities behave like a short position in a long-term bond, so falling rates can raise liabilities more than equities rise. Minimum-risk position: immunisation (asset duration equal to liability duration), with an opportunity cost. Defined-benefit plans leave the shortfall to the fund's owner; defined-contribution plans put the risk on the employees. Also called surplus at risk (SAR).
- **PERF example:** assets 1,000, liabilities 900, surplus 100; expected surplus growth 5% of assets $= 50$; surplus volatility 9.4% gives a 99% VaR of 22% or 220. Expected surplus 150 minus 220 $= -70$: a 1% chance of a deficit of 70 or more. OTPPB: 99% over 1 year is "equivalent" to 90% over 4 years because $(1.28/2.33)\sqrt{4} \approx 1$.
- **Sponsor risk** (IM-6 e): the risk to the plan sponsor, split into *cash-flow risk* (year-to-year fluctuations in contributions) and *economic risk* (variation in the sponsor's total economic earnings). Pension management should be integrated with the sponsor's financial goals, in line with enterprisewide risk management.
- **VaR to monitor risk** (IM-6 f): VaR checks compliance (Common Fund: USD 138 million unauthorised loss, then an independent risk oversight officer and fewer managers) and gives a central position repository. Passive benchmarks do not keep risk constant (the late-1990s tech weight in the S&P 500). A VaR jump has three causes, each needing a different action: a manager taking more risk, different managers taking similar bets (visible only at portfolio level), or more volatile markets. Global custodians are the natural centralisation point; large plans may build their own systems.
- **VaR for guidelines and the investment process** (IM-6 g): limits on notionals ignore risk and correlations; limits on sensitivities have blind spots (hedged portfolios); structured notes evade both. VaR limits are comparable across assets and capture diversification, leverage and derivatives, e.g. "tracking-error volatility of at most 3%". Marginal VaR picks between equal-return stocks (utility with portfolio beta 0.5 versus Internet stock with 2.0); Chrysler's currency position had individual VaR of 44 but incremental VaR of only 3, so it was not hedged. See also A12.
- **Budgeting across asset classes** (IM-6 h): PERF's 10% total volatility on 1,000 gives $2.33 \times 0.10 \times 1{,}000 = 233$. Individual risk budgets are $\alpha\sigma_i w_i W$ (U.S. stocks $2.33 \times 0.151 \times 525 = 184$); they sum to 308, the undiversified VaR. Two equal managers in U.S. stocks (16% volatility, correlation 0.78) get 98 each, recombining to 184. Managers within budget trade without approval: no micromanaging.
- **Budgeting across active managers** (IM-6 h): IR $= \mu/\omega$ (Grinold and Kahn: 0.50 is "good", top quartile). Maximise portfolio IR subject to a TEV target with independent active returns: $x_i\omega_i = (\text{IR}_i/\text{IR}_P)\,\omega_P$, so relative risk budgets are proportional to IRs and any slack sits in the index (TEV 0). Jorion's example: 55%, 37%, 8% index, portfolio IR 0.72, above either manager. Equal IRs give IR$\sqrt{N}$ (law of active management). Footnote: TE-only budgeting is second-best because it fails to control total risk. See also A18.

Formulas

Return decomposition (eq. 6.2). $R_{\text{asset}} = \underbrace{\sum_i w_i^p R_i^b}_{\text{policy mix}} + \underbrace{\sum_i (w_i R_i - w_i^p R_i^b)}_{\text{active management}}$

Policy-mix VaR (normal). $\text{VaR}_{\text{policy}} = \alpha\, W \sqrt{\sum_i \sum_j w_i^p w_j^p \sigma_i \sigma_j \rho_{ij}}$ using *policy* weights and *benchmark* volatilities

Relative VaR. $\text{VaR} = \alpha W \sigma_e, \quad e = R_{\text{asset}} - R_B$

Surplus return (eq. 6.3). $R_S = \dfrac{\Delta S}{A} = R_{\text{asset}} - R_{\text{liabilities}} \dfrac{L}{A}$

Individual risk budget. $\alpha\, \sigma_i\, w_i W$; sum of budgets = undiversified VaR $\geq$ fund VaR

Two managers in one class. $\text{VaR} = \sqrt{V_1^2 + V_2^2 + 2\rho V_1 V_2}$

Information ratio (eq. 6.4). IR $= \mu/\omega$ (μ expected TE, ω TEV)

Portfolio TEV with independent managers (eq. 6.6). $\omega_P = \sqrt{\sum_i x_i^2 \omega_i^2}$

Optimal relative risk budgets (eq. 6.7). $x_i\omega_i = \dfrac{\text{IR}_i}{\text{IR}_P}\,\omega_P, \quad \text{IR}_P = \sqrt{\sum_i \text{IR}_i^2}$

Expected excess return of the pool. $\mu_P = \sum_i x_i \mu_i = \text{IR}_P\, \omega_P$

Law of active management. Equal IRs: $\text{IR}_P = \text{IR}\sqrt{N}$

Exam traps

- **Policy weights versus current weights (wrong-input twin):** policy-mix VaR uses the *policy* weights on *benchmark* returns. The current mix's deviation from policy belongs to active-management risk.
- **Non-additivity (absolute claim):** policy-mix VaR plus active-management VaR is not the asset VaR; OTPPB's 19.6% and 1.6% gave 19.3% because of a slightly negative correlation.
- **Undiversified versus diversified:** the sum of individual risk budgets (308 for PERF) is the undiversified VaR; the fund VaR (233) is lower. A distractor plants the sum.
- **Cash-flow versus economic risk (sibling swap):** cash-flow risk is fluctuation in *contributions*; economic risk is variation in the sponsor's *total economic earnings*.
- **Who bears funding risk (role):** the fund's owner in a defined-benefit plan; the employees in a defined-contribution plan.
- **Jorion's "tracking error":** Jorion's TE is the active return itself and TEV its volatility; IR $= \mu/\omega$. Portfolio IR with independent managers is $\sqrt{\sum \text{IR}_i^2}$, not the sum or the average.

Question C4.1

IM-6 e, h · S1, S2, S5 · [Map #35]

A risk analyst at an Australian superannuation fund with AUD 8,000 million of assets is asked by the investment committee for the fund's one-year VaR at the 99% confidence level, calculated by the policy-mix method. Returns are assumed normal and the analyst uses $\alpha = 2.33$.

Asset class	Policy weight	Current weight	Benchmark volatility	Managers' TEV
Domestic equities	50%	45%	15%	3.0%
Global equities	20%	20%	17%	4.0%
Fixed income	30%	35%	6%	1.5%

Benchmark correlations: domestic/global equities 0.60; domestic equities/fixed income 0.10; global equities/fixed income 0.05. Which of the following is closest to the policy-mix VaR?

A. AUD 1,571 million

B. AUD 1,794 million

C. AUD 1,910 million

D. AUD 2,367 million

Solution

Answer: C

Step 1 – Choose the inputs

Policy-mix risk is "the risk of a dollar loss owing to the policy mix selected by the fund": the first term of eq. 6.2, $\sum_i w_i^p R_i^b$. So the weights are the *policy* weights and the volatilities are the *benchmark* volatilities. The current weights and the managers' TEVs describe active-management risk and are not inputs here.

Step 2 – Variance of the policy-mix return

$$\begin{aligned}\sigma_p^2 &= (0.50\times 0.15)^2 + (0.20\times 0.17)^2 + (0.30\times 0.06)^2 \\ &\quad + 2(0.50)(0.20)(0.60)(0.15)(0.17) + 2(0.50)(0.30)(0.10)(0.15)(0.06) \\ &\quad + 2(0.20)(0.30)(0.05)(0.17)(0.06) \\ &= 0.005625 + 0.001156 + 0.000324 + 0.003060 + 0.000270 + 0.000061 = 0.010496\end{aligned}$$

$$\sigma_p = \sqrt{0.010496} = 10.25\%$$

$\Rightarrow$ *A passive implementation of the policy mix would have an annual volatility of about 10.25%.*

Step 3 – Policy-mix VaR

$$\text{VaR}_{\text{policy}} = \alpha\,\sigma_p\,W = 2.33\times 0.10245\times 8{,}000 = 1{,}909.7$$

⇒ *There is a 1% chance that the chosen policy mix alone loses more than about AUD 1,910 million over a year.*

Interpretation

The individual risk budgets are $2.33 \times 0.15 \times 0.50 \times 8{,}000 = 1{,}398$, $2.33 \times 0.17 \times 0.20 \times 8{,}000 = 634$ and $2.33 \times 0.06 \times 0.30 \times 8{,}000 = 336$; their sum, 2,367, is the undiversified VaR, and diversification removes about 458. The managers' TEVs would feed a separate active-management VaR; as at OTPPB, the fund's asset VaR need not equal the sum of the two, so the committee should not add them.

Why the other options are wrong

A is incorrect. Ignored the correlations (treated benchmark returns as uncorrelated): $\sqrt{0.005625 + 0.001156 + 0.000324} = 8.43\%$, $2.33 \times 0.0843 \times 8{,}000 = 1{,}571$.

B is incorrect. Used the current weights (45/20/35) instead of the policy weights: volatility 9.62%, VaR $2.33 \times 0.0962 \times 8{,}000 = 1{,}794$. The tilt away from policy is a deviation from the policy mix, which Jorion places in active-management risk.

D is incorrect. Summed the individual risk budgets, $1{,}398 + 634 + 336 = 2{,}367$: the undiversified VaR, as if all correlations were 1.

LO: IM-6 e: Describe and compare the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk, funding risk, and sponsor risk. IM-6 h: Describe the risk budgeting process and calculate risk budgets across asset classes and active managers. · Reference: Jorion, Value at Risk 3e, Ch.17, 6.2 eqs. 6.1, 6.2 and 6.5

Question C4.2

IM-6 h · S3, S4, S5 · [Map #35]

A senior analyst at a Dutch pension fund is setting risk budgets for the EUR 600 million domestic equity sleeve. The sleeve can be split between two active managers and a passive index fund (TEV 0%). The investment committee wants to maximise the sleeve's information ratio subject to an overall tracking-error volatility of 3%. The managers' active returns are assumed independent of each other.

- Manager X: expected excess return over the benchmark 2.5%; TEV 5%
- Manager Y: expected excess return over the benchmark 2.4%; TEV 8%

Using the optimal relative risk budgets, which of the following is closest to the weight allocated to Manager Y and the expected excess return of the sleeve?

A. Weight to Y 14.1%; expected excess return 1.28%

B. Weight to Y 19.3%; expected excess return 1.75%

C. Weight to Y 19.3%; expected excess return 1.20%

D. Weight to Y 37.5%; expected excess return 2.46%

Solution

Answer: B

Step 1 – Managers' information ratios

$$\mathrm{IR}_X = \frac{\mu_X}{\omega_X} = \frac{2.5\%}{5\%} = 0.50, \qquad \mathrm{IR}_Y = \frac{2.4\%}{8\%} = 0.30$$

⇒ *X earns 0.50 units of excess return per unit of active risk ("good", top quartile); Y earns 0.30.*

Step 2 – Portfolio IR with independent managers

$$\mathrm{IR}_P = \sqrt{0.50^2 + 0.30^2} = \sqrt{0.34} = 0.5831$$

Step 3 – Relative risk budgets and weights (eq. 6.7)

$$x_Y \omega_Y = \frac{\mathrm{IR}_Y}{\mathrm{IR}_P}\,\omega_P = \frac{0.30}{0.5831}(3\%) = 1.5435\% \;\Rightarrow\; x_Y = \frac{1.5435\%}{8\%} = 19.29\%$$

$$x_X \omega_X = \frac{0.50}{0.5831}(3\%) = 2.5725\% \;\Rightarrow\; x_X = \frac{2.5725\%}{5\%} = 51.45\%$$

$$\text{index} = 100\% - 51.45\% - 19.29\% = 29.26\%$$

⇒ *Check:* $\sqrt{2.5725^2 + 1.5435^2} = 3.00\%$*, the TEV target. The 29.3% left in the index is the slack needed to hit exactly 3%.*

Step 4 – Expected excess return of the sleeve

$$\mu_P = x_X\mu_X + x_Y\mu_Y = 0.5145(2.5\%) + 0.1929(2.4\%) = 1.286\% + 0.463\% = 1.749\%$$

$\Rightarrow$ *Equivalently* $IR_P \times \omega_P = 0.5831 \times 3\% = 1.749\%$*: the sleeve's IR of 0.58 exceeds both managers' because independent active bets diversify.*

Interpretation

In money terms the 99% relative risk budgets are $2.33 \times 2.5725\% \times 600 =$ EUR 36.0 million for X and $2.33 \times 1.5435\% \times 600 =$ EUR 21.6 million for Y; the sleeve's budget is $2.33 \times 3\% \times 600 =$ EUR 41.9 million, smaller than their sum because of diversification. Within those budgets the managers trade without seeking approval.

Why the other options are wrong

A is incorrect. Took IR_P as the sum $0.50 + 0.30 = 0.80$, as if active returns were perfectly correlated: $x_Y = (0.30/0.80)(3\%)/8\% = 14.1\%$, $x_X = 37.5\%$, $\mu_P = 0.375(2.5\%) + 0.141(2.4\%) = 1.28\%$.

C is incorrect. Correct weights, but computed the sleeve's excess return as the *average* manager IR times the TEV: $0.40 \times 3\% = 1.20\%$, ignoring the diversification of independent active bets.

D is incorrect. Split the whole sleeve in proportion to the IRs with nothing in the index: $x_Y = 0.30/0.80 = 37.5\%$, $\mu_P = 0.625(2.5\%) + 0.375(2.4\%) = 2.46\%$. This ignores the TEV constraint: the sleeve's TEV would be $\sqrt{3.125^2 + 3.0^2} = 4.33\%$.

LO: IM-6 h: Describe the risk budgeting process and calculate risk budgets across asset classes and active managers. · Reference: Jorion, Value at Risk 3e, Ch.17, 6.5 eqs. 6.4 to 6.7, Table 6.4

Question C4.3

IM-6 e · S2 · [Map #35]

The CRO of a Canadian defined-benefit teachers' pension plan is preparing a report for the board's risk committee, which has asked for a clear statement of how the plan's risk types differ. Which of the following statements is correct to include in the report?

A. Funding risk is measured on the surplus; because nominal pension liabilities behave like a short position in a long-term bond, a fall in interest rates can reduce the surplus even in a year when the plan's equities rise.

B. Sponsor risk has two parts: cash-flow risk, the variation in the total economic earnings of the plan sponsor, and economic risk, the year-to-year fluctuation in contributions to the fund.

C. Policy-mix risk and active-management risk are additive, so the plan's total asset VaR always equals the policy-mix VaR plus the active-management VaR reported for its managers.

D. Funding risk is borne by the plan sponsor in a defined-contribution plan, which must make up any shortfall of the fund's assets relative to the benefits promised to retirees.

Solution

Answer: A

Why A is correct

Funding risk is "the risk that the value of assets will not be sufficient to cover the liabilities of the fund"; the relevant variable is the surplus $S = A - L$, with $R_S = R_{\text{asset}} - R_{\text{liabilities}}(L/A)$. Jorion: when liabilities consist mainly of nominal payments, "their value in general will behave like a short position in a long-term bond. Thus decreases in interest rates, while beneficial for equities on the asset side, can increase even more the value of liabilities, thereby negatively affecting the surplus."

Interpretation

The board should judge the plan on the surplus, not on asset returns alone; the minimum-risk position is an immunised portfolio matching asset and liability durations, at an opportunity cost.

Why the other options are wrong

B is incorrect. Sibling swap. The labels are reversed: cash-flow risk is "the risk of year-to-year fluctuations in contributions", economic risk is "the risk of variation in total economic earnings of the plan sponsor".

C is incorrect. Absolute claim. Total VaR comes from the policy-mix VaR, the active-management VaR *and a cross-product term*. At OTPPB, 19.6% and 1.6% gave an asset VaR of 19.3% because of a slightly negative correlation.

D is incorrect. Role misassignment. A defined-benefit plan's shortfall "will have to be made up by the fund's owner"; "defined-contribution plans put the risk on the employees".

LO: IM-6 e: Describe and compare the following types of risk: absolute risk, relative risk, policy-mix risk, active management risk,

funding risk, and sponsor risk. · Reference: Jorion, Value at Risk 3e, Ch.17, 6.2 (Funding Risk, Sponsor Risk, Policy Mix and Active Management Risk)

C5 Basel III Post-Crisis Reforms and Stress-Test Regimes

Readings: ORR-21 a, b, c, d, g, h · Capital Regulation Before the Global Financial Crisis (Carey, GARP Risk Institute 2019); ORR-22 a, d, e, h · Solvency, Liquidity, and Other Regulation After the Global Financial Crisis (Carey, GARP Risk Institute 2019); ORR-23 a, b, c · High-Level Summary of Basel III Reforms (BCBS, December 2017); ORR-24 a, b, c · Basel III: Finalising Post-Crisis Reforms (BCBS, December 2017, pp. 128 to 136); ORR-17 a · Stress Testing Banks (Schuermann, 2014)

MAP TIER C · #30 · 4/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #30: "Examinable comparisons: pre-SCAP vs. post-SCAP; the standardised vs. internal-models split; output floors; and the Basel 1 / 2 / 2.5 / III progression. The sibling swap across SCAP, EBA and CCAR stress-test regimes, which banks failed which test, base vs. stress scenario requirements, is separately reported."
Shapes covered here:
S1 pre-SCAP versus post-SCAP features of stress testing (Q2)
S2 the standardised versus internal-models split across accords: who uses which, and where the December 2017 reforms removed models (Q1)
S3 output floors: the Basel II floor on Basel I and the Basel III floor on the revised standardised approaches (Q1)
S4 the Basel I / 1996 / II / 2.5 / III progression, each change attached to the right accord (Q1)
S5 the SCAP / CEBS / EBA / CCAR sibling swap: which banks failed, base versus stress disclosure, which risk weights (Q2).

Revision notes

- **Why Basel exists** (ORR-21 a): private support arrangements (the early Bank of England, clearinghouses) failed in panics large enough that only an entity able to print money could help; crises imposed large costs on the economy; customers complained of unfairness; and globalization interlinked banks. Herstatt's failure in 1974 (currency settlement across time zones) led to the BCBS in 1974; IOSCO followed in 1983 and the IAIS in 1994.
- **Basel I motivations and ratio** (ORR-21 a, b): a common G10 interest in banks holding enough equity, and a level playing field (avoid a race to the bottom on capital minimums). Published December 1987, agreed July 1988, fully implemented by end 1992; no legal standing of its own. A *risk-based* ratio was chosen over a leverage ratio, which would disadvantage low-risk portfolios. Tier 1/RWA $\geq 4\%$, total capital/RWA $\geq 8\%$ ("Cooke" ratios); Tier 2 may be no more than *half* of total capital.
- **Basel I capital and weights** (ORR-21 b): Tier 1 = common equity + disclosed reserves − goodwill; Tier 2 = general loan loss reserves (capped at 2%, later 1.25% of RWA), undisclosed reserves, hybrids. Risk weights 0% (cash, OECD governments), 20% (OECD banks and public sector entities), 50% (uninsured residential mortgages), 100% (all other). Off-balance-sheet credit conversion factors 100%, 50%, 20% (commitments of one year or more), 0% (under one year). Example 21.1: book assets CAD 500 million, RWA CAD 250 million.
- **1995 Amendment: netting** (ORR-21 c): original Basel I gave almost no credit for netting (master agreements untested in bankruptcy courts as of 1988), so offsetting swaps each drew an add-on and hedging was disincentivized. From 1995, enforceable bilateral netting reduces the credit equivalent amount: CEA $= \max(\sum V_i, 0) + \sum_j (0.4D_j + 0.6D_j \,\text{NRR})$. Example 21.3: NRR $= 5/10 = 0.5$, CEA $= 40.7$.
- **1996 Amendment: market risk** (ORR-21 c): standardized approach (five categories, assets assumed perfectly correlated, specific and general market risk) versus the internal models approach (generally smaller charges). 10-day 99% VaR from at least one year of data, $m \geq 3$ on the 60-day average; specific risk multiplier 4; Tier 3 capital (subordinated debt of at least two years) for about 70% of market risk capital; daily backtesting. Full working in D4.
- **Basel II and the three pillars** (ORR-21 d): reaction to Basel I being "not risk-based enough" (the 100% weight covered very safe and very risky loans) and to Nordic crises in well-capitalized systems. Four innovations: credit risk weights from banks' internal risk measures; operational risk capital; Pillar 2 (supervisory review: ICAAP, capital above the minimum, early intervention) and Pillar 3 (disclosure); repeated Quantitative Impact Studies. Final text 2004.

- **Basel II credit: the standardised versus internal-models split** (ORR-21 d, e): the standardised approach was intended for banks whose internal risk measures were *insufficient* to support IRB; Foundation IRB (bank supplies PD) and Advanced IRB (bank supplies PD, LGD, EAD). The United States did not implement the standardised approach: internationally active banks used IRB. IRB detail: see also A31.
- **Basel II operational risk** (ORR-21 g): Basic Indicator Approach $= 15\%$ of average annual gross income over three years, ignoring negative years; Standardized Approach uses business-line multipliers of 12%, 15% or 18% (negatives offset within a year, negative-total years ignored); Example 21.4: SA USD 9.21 billion versus BIA USD 9.83 billion. AMA: internal one-year 99.9th percentile less expected losses, four data elements (internal, external, scenario, business environment and controls), insurance offset capped at 20%. AMA "has lost favor and is no longer permitted".
- **Solvency II** (ORR-21 h): EIOPA; one-year VaR at the *99.5th* percentile; three pillars; breach of the MCR can stop new business or trigger resolution, breach of the SCR requires a restoration plan; internal models must be sound, calibrated to target criteria and used in actual decisions (use test); Tier 1, 2, 3 capital.
- **Basel 2.5 and Basel III** (ORR-22 a, d, e): Basel 2.5 (by end 2011) added stressed VaR, the incremental risk charge and the comprehensive risk measure (D4). Basel III eliminated Tier 3, split Tier 1 into Core (CET1) and Additional Tier 1: CET1 $\geq 4.5\%$, Tier 1 $\geq 6\%$, total $\geq 8\%$; buffers: capital conservation 2.5%, G-SIB 1 to 3.5%, countercyclical 0 to 2.5%; 3% leverage ratio (A32). CoCos: D25.
- **Why the December 2017 reforms** (ORR-23 a): restore credibility in RWA calculations and comparability of capital ratios by (i) more robust, risk-sensitive standardised approaches for credit, CVA and operational risk; (ii) constraining internal models: IRB input floors and *removal* of internal models for CVA and operational risk; (iii) a leverage ratio buffer for G-SIBs; (iv) a more robust output floor.
- **Revised standardised approach for credit risk** (ORR-23 b): mortgage weights vary with LTV (previously one flat weight); less mechanistic reliance on ratings (due diligence; SCRA grades A/B/C at 40%/75%/150% for banks); corporates 20% to 150% by rating, or 65% investment grade and 100% other where ratings are not permitted, SMEs 85%; subordinated debt 150%, equity 250% (400% speculative unlisted); positive CCF of 10% for unconditionally cancellable commitments.
- **CVA risk and leverage ratio** (ORR-23 b): CVA losses exceeded default losses in some instances during the crisis; the revised framework adds the *exposure* component, removes the internal model and leaves SA-CVA (fair value sensitivities, consistent with the market risk framework) and BA-CVA (benchmarked to SA-CVA); banks with non-centrally cleared notional of EUR 100 billion or less may use a multiplier of the CCR charge. G-SIB leverage ratio buffer $= 50\%$ of the risk-weighted higher-loss absorbency requirement (2% gives 1%), met with Tier 1, with distribution constraints.
- **Output floor** (ORR-23 c): the Basel II floor was 80% of Basel I requirements and was implemented inconsistently. The Basel III floor: RWA $= \max(\text{RWA}_{\text{approved approaches}},\ 72.5\% \times \text{RWA}_{\text{standardised only}})$, phased in 50% (2022), 55%, 60%, 65%, 70%, 72.5% (1 January 2027); national discretion to cap the floor-driven increase at 25% of pre-floor RWA during phase-in; the IRB 1.06 scaling factor is removed. Floor mechanics: see also A31.
- **SMA: the new standardised approach for operational risk** (ORR-24 a): BI $=$ ILDC $+$ SC $+$ FC, each a three-year average with absolute values taken year by year first. BIC applies marginal coefficients 12% (up to EUR 1 billion), 15% (1 to 30 billion), 18% (above 30 billion); BI of EUR 35 billion gives BIC EUR 5.37 billion. LC $= 15\times$ average annual losses over 10 years; ILM $= \ln(e - 1 + (\text{LC}/\text{BIC})^{0.8})$, equal to 1 when LC $=$ BIC, above 1 when LC $>$ BIC. Bucket 1 banks: ILM $= 1$. ORC $=$ BIC $\times$ ILM; RWA $= 12.5\times$ ORC.
- **SMA versus AMA** (ORR-24 b): the AMA (internal models) and the three Basel II standardised approaches are replaced by one risk-sensitive standardised approach for all banks, because capital proved insufficient for some banks' losses and misconduct and control failures were hard to model. The SMA assumes op risk rises at an increasing rate with income and that past losses predict future losses. Supervisors may set ILM $= 1$ for all banks, but loss disclosure still applies.
- **Loss data criteria** (ORR-24 c): 10-year window (five years on transition; under five years, BIC only); EUR 20,000 inclusion threshold (EUR 100,000 at discretion for buckets 2 and 3); record dates of occurrence, discovery and accounting, and build the data set on the accounting date; net losses of recoveries only once payment is received (receivables do not count, tax effects are not recoveries); include direct charges, external costs, provisions, pending and timing losses; exclude maintenance contracts, post-loss upgrades and insurance premiums; credit-related losses already in credit RWA are excluded, market-related losses are included; exclusions need supervisory approval, materiality (for example over 5% of average losses) and a minimum retention period.

- **Pre-SCAP versus post-SCAP** (ORR-17 a, Table 17.2): pre-SCAP stress testing was mostly single-shock, product or business-unit level, static, not usually tied to capital adequacy, and losses only. Post-SCAP: broad macro scenario plus market stress, comprehensive and firm-wide, dynamic and path dependent, an explicit post-stress common equity threshold, and losses, revenues and costs.
- **Regimes in one line each** (ORR-17 a): SCAP 2009, 19 BHCs, 4% T1C, ten required to raise USD 75 billion; CEBS 2010, 91 banks, seven raised EUR 3.5 billion, Irish banks passed then needed EUR 24 billion; EBA 2011, 90 banks, 5% T1C, eight raised EUR 2.5 billion, Basel II weights; CCAR 2011 published only the macro scenario. Full comparison, disclosure and design: see also C9.

Formulas

Basel I. $\dfrac{\text{Tier 1}}{\text{RWA}} \geq 4\%, \quad \dfrac{\text{Tier 1} + \text{Tier 2}}{\text{RWA}} \geq 8\%, \quad \text{Tier 2} \leq \frac{1}{2}\,\text{Total capital}, \quad \text{RWA} = \sum_i w_i A_i$

1995 netting. $\text{CEA} = \max\left(\sum_i V_i, 0\right) + \sum_j \left(0.4 D_j + 0.6 D_j\,\text{NRR}\right), \qquad \text{NRR} = \dfrac{\max(\sum_i V_i, 0)}{\sum_i \max(V_i, 0)}$

1996 and Basel 2.5 market risk. $MR = \max(VaR_{t-1},\, m\,VaR_{avg})$; $MR_{2.5} = \max(VaR_{t-1}, m_c VaR_{avg}) + \max(SVaR_{t-1}, m_s SVaR_{avg}),\ m \geq 3$

Basel II operational risk. BIA $= 15\% \times$ average positive annual gross income over three years; SA: business-line multipliers 12%, 15%, 18%.

SMA (ORR-24). $\text{BI} = \text{ILDC} + \text{SC} + \text{FC}$; $\text{BIC} = 12\%\,\min(\text{BI}, 1) + 15\%\,[\min(\text{BI}, 30) - 1]^+ + 18\%\,(\text{BI} - 30)^+$ (EUR billion)

$$\text{ILM} = \ln\left(e - 1 + \left(\frac{\text{LC}}{\text{BIC}}\right)^{0.8}\right), \qquad \text{LC} = 15 \times \overline{\text{annual loss}}_{10y}, \qquad \text{ORC} = \text{BIC} \times \text{ILM}, \qquad \text{RWA}_{op} = 12.5 \times \text{ORC}$$

Output floor (ORR-23). $\text{RWA} = \max\left(\text{RWA}_{\text{models}},\ 72.5\% \times \text{RWA}_{\text{SA}}\right)$; Basel II floor: 80% of Basel I.

Exam traps

- **Wrong accord (sibling swap):** three pillars and op risk capital are Basel II; stressed VaR, IRC and CRM are Basel 2.5; CET1, buffers, leverage ratio, LCR and NSFR are Basel III; SMA, SA-CVA and the 72.5% floor are December 2017.
- **Two floors:** the Basel II floor was 80% of *Basel I* requirements; the Basel III floor is 72.5% of RWA under the *revised standardised* approaches, phased in from 50% in 2022.
- **Direction of the 2017 reforms (polarity flip):** internal models were *removed* for CVA and operational risk and *constrained* for IRB; a statement that they were extended is false.
- **Who uses the standardised approach (role swap):** it was meant for banks whose internal measures were insufficient; US internationally active banks were required to use IRB.
- **ILM direction and LC scale:** LC $>$ BIC gives ILM $>$ 1; LC is 15 times the *average* annual loss, not the 10-year sum; bucket 1 banks have ILM $= 1$.
- **Regime facts:** the Irish banks *passed* CEBS 2010; EBA 2011 used Basel II (rating-sensitive) weights while SCAP and CCAR used Basel I; CCAR 2011 published no bank-level results.

Question C5.1

ORR-21 a, d, e; ORR-23 a, b, c · S2, S3, S4 · [Map #30]

A regulatory policy analyst at a Canadian bank is presenting a timeline of the Basel accords, from Basel I to the December 2017 reforms, to a group of newly hired analysts. Which of the following statements is correct for the analyst to include?

A. Basel I required Tier 1 capital of at least 4% and total capital of at least 8% of risk-weighted assets, with Tier 2 capital allowed to make up as much as two-thirds of total capital.

B. The December 2017 reforms replaced the Basel II output floor, set at 80% of Basel I capital requirements, with a floor on standardised-approach RWA phased in from 50% in 2022 to 72.5% in 2027.

C. Basel II required internationally active banks to use the standardised approach for credit risk and reserved the IRB approaches for banks whose internal risk measures could not support more granular weights.

D. The December 2017 reforms widened the use of internal models by introducing an internally modelled ap-

proach for CVA risk and by retaining the AMA for banks with ten years of loss data.

Solution **Answer: B**

Step 1 – Place each claim in its accord

Basel I (1988): risk-based ratios and capital definitions. Basel II (2004): credit risk options and the three pillars. December 2017 Basel III reforms: revised standardised approaches, constrained internal models, output floor.

Step 2 – Test the output floor claim against ORR-23 c

The Basel II floor was "calibrated at 80% of the relevant Basel I capital requirements" and was applied inconsistently. It is replaced by a floor based on the revised Basel III standardised approaches: RWA is the higher of RWA under approved approaches and 72.5% of RWA under the standardised approaches only, with the transitional percentage rising 50%, 55%, 60%, 65%, 70% and 72.5% from 1 January 2022 to 1 January 2027 (Table 23.5).
⇒ *The floor limits the capital benefit an internal-models bank can draw relative to a standardised bank, keeping a level playing field and supporting comparability.*

Interpretation

B is correct. It joins S3 (the two floors) to S4 (Basel II to Basel III progression); the other three statements each corrupt one attribute of the standardised versus internal-models split or of Basel I.

Why the other options are wrong

A is incorrect. Scope/condition error: Basel I capped Tier 2 at no more than *half* of total capital; Tier 1 above 4% could be counted toward the 8% requirement, not the other way round.
C is incorrect. Role misassignment: the roles are reversed. The standardised approach was "intended for banks with internal risk measures and risk management practices that were insufficient to support the IRB approaches"; the United States did not implement it and required internationally active banks to use IRB.
D is incorrect. Polarity flip: the reforms *removed* the internal model for CVA risk (leaving SA-CVA and BA-CVA) and replaced the AMA with a single standardised approach for all banks.

LO: ORR-21 a: Explain the motivations for introducing the Basel regulations, including key risk exposures addressed, and explain the reasons for revisions to Basel regulations over time. ORR-23 c: Describe the revised output floor introduced as part of the Basel III reforms and approaches to be used when calculating the output floor. · Reference: Carey, Capital Regulation Before the Global Financial Crisis (Basel I; Basel II, Capital for Credit Risk); BCBS, High-Level Summary of Basel III Reforms (Output Floor; Table 23.5; CVA Risk Framework; Operational Risk Framework)

Question C5.2 ORR-17 a · S1, S5 · [Map #30]

An internal auditor at a European bank is reviewing a training module on the supervisory stress tests run in the US and Europe between 2009 and 2012. Which of the following statements in the module is correct?

A. Post-SCAP stress testing replaced single-factor shocks with a broad macro scenario applied firm-wide, but like pre-SCAP testing it remained static and confined to projected losses.

B. The 2011 CCAR asked banks to submit results under their own baseline and stress scenarios in addition to the common supervisory scenario, but published no bank-level results.

C. The 2011 EBA stress test applied Basel I risk weights, so banks needed only stressed default rates rather than full stressed rating migration matrices to project RWA.

D. All four Irish banks failed the 2010 CEBS stress test, which led directly to the March 2011 Irish exercise that revealed a capital need of EUR 24 billion.

Solution **Answer: B**

Step 1 – Pre- versus post-SCAP (Table 17.2)

Post-SCAP testing is a broad macro scenario with market stress, comprehensive and firm-wide, *dynamic and path dependent*, tied to an explicit post-stress common equity threshold, and covers *losses, revenues and costs*.

Step 2 – Regime facts (Tables 17.1 and 17.3, Sections 17.3 and 17.5)

CCAR, in operation since 2011, "asks banks to submit results using their own scenarios (baseline and stress) in addition to results under the common supervisory stress scenario". The 2011 CCAR disclosure: "only the macro-scenario was published, with no bank-level results"; banks allowed to raise dividends were read as having passed.
⇒ *The dual approach lets supervisors compare banks on the common scenario while learning what each bank thinks its own high-risk scenarios are.*

Interpretation

B is correct. Contrast the 2009 SCAP (bank-level stress losses by asset class, supervisory estimates, stress results only) and the 2012 CCAR (near-SCAP detail; four of 19 did not pass in the sense of receiving an objection to their capital plans).

Why the other options are wrong

A is incorrect. Scope/condition error: the first clause is true, but post-SCAP testing is dynamic and path dependent and covers revenues and costs as well as losses; "static, losses only" describes pre-SCAP testing.

C is incorrect. Sibling swap: Basel I weights belong to the US exercises (SCAP, CCAR). The EBA test used Basel II weights, which are ratings sensitive, so banks had to use stress migration matrices to project the whole future ratings distribution, not just defaults.

D is incorrect. Polarity flip (case cross-mapping): "all of these banks had previously passed the CEBS stress test". The EUR 24 billion need for the four banks came from the 2011 exercise run largely by outside advisors after Ireland requested EU and IMF assistance.

LO: ORR-17 a: Describe the evolution of the stress testing process and compare the methodologies of historical European Banking Association (EBA), Comprehensive Capital Analysis and Review (CCAR), and Supervisory Capital Assessment Program (SCAP) stress tests. · Reference: Schuermann, Stress Testing Banks, International Journal of Forecasting 30:3 (2014), Sections 17.1, 17.3, 17.5; Tables 17.1, 17.2, 17.3

C6 Model Risk Management

Readings: ORR-15 a, b, c, d · Supervisory Guidance on Model Risk Management (FDIC FIL-22-2017, adopting the Federal Reserve's SR 11-7; GARP ORR Ch.15: 15.3 Overview, 15.4 Development, Implementation and Use, 15.5 Model Validation, 15.6 Governance, Policies and Controls); ORR-16 a, b, c · Case Study: Model Risk and Model Validation (GARP ORR Ch.16); ORR-3 c · Risk Identification (ORX reference taxonomy: model risk at Level 1)

MAP TIER C · #32 · 3/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #32: "Reported: 'model risk'; and one specific item: 'friction between investor and rating agency… I chose model error as the answer', which maps model risk onto the securitisation-frictions taxonomy. The three components of model risk management (development/implementation/use, validation, governance) and the independence requirement for validation are the examinable core."
Shapes covered here:
S1 the friction between investor and rating agency labelled model error, tied to the GARP Gaussian copula case (Q1)
S2 the three components of model risk management: development, implementation and use; validation; governance, policies and controls (Q2)
S3 the independence requirement for validation, with effective challenge (incentives, competence, influence) (Q2)

Revision notes

The specific: the investor and rating agency friction is model error (S1)

- **Frictions taxonomy.** The subprime securitisation chain is often described as a set of frictions between pairs of parties (Ashcraft and Schuermann, 2008): originator and borrower (predatory lending), arranger and third parties (adverse selection), servicer and mortgagor or third parties (moral hazard), asset manager and investor (principal-agent), and investor and credit rating agency, labelled **model error**, because the investor relies on the agency's rating model (outside-source). The GARP text that carries this is the ORR-16 Gaussian copula case.
- **Gaussian copula and CDO pricing (ORR-16 c).** Li's approach used **CDS prices**, not observed historical correlations, to infer correlation. CDSs had about a decade of history: a short, benign sample (rising house prices, record-low defaults), so implied correlations were very low and extremely sensitive to house prices. The method was adopted "by everybody from bond investors and Wall Street banks to ratings agencies and regulators"; warnings were ignored. The copula was valid and accurate in *normal* conditions; when house prices reversed, implied correlations shot up. The problem was not the formula itself but that banks did not update it with the new correlations and kept pricing on old assumptions. Blame falls on quants who should have warned and users who relied on a black box.
- **Two other cases (ORR-16 c).** *Barclays and Lehman*: an Excel file of assets with 179 trading contracts hidden rather than deleted; converted to PDF by a junior associate, the rows reappeared; spotted on 1 October after the deal was approved. A simple **implementation error** in a tool that is not even a model, yet it nearly cost millions. *NASA Mars orbiter* (USD 125 million): English versus metric units, an error in the model's "assumptions"; in finance, the wrong currency or discount factor. It is impossible to know in advance which benign-looking mistakes will be catastrophic.

What a model is, and how model risk arises (ORR-15 a; ORR-16 a)

- **Model.** "A quantitative method, system, or approach that applies statistical, economic, financial, or mathematical theories, techniques, and assumptions to process input data into quantitative estimates"; also covers approaches with partly or wholly qualitative or expert-judgment *inputs*, provided the *output* is quantitative. Three components: information input, processing, reporting. Industry consensus now stresses the *uncertain* nature of the estimate rather than the technique.
- **Model risk.** "The potential for adverse consequences from decisions based on incorrect or misused model outputs and reports": financial loss, poor decisions, reputational damage. Two reasons: (1) **fundamental errors** anywhere from design to implementation, including shortcuts, approximations and poor inputs; (2) **incorrect or inappropriate use**, even of a sound model, especially outside the environment it was designed for (new products or markets, or changed conditions). ORR-16 names the same pair **execution risk** (the

model does not do what it was designed to do: coding, implementation, wrong data) and **conceptual errors** (wrong assumptions or techniques, often matters of judgment, where MRM's role is transparency).

- **Size and aggregation.** Model risk increases with complexity, input and assumption uncertainty, broader use and larger potential impact. Consider it per model and **in aggregate**: common assumptions, data or methods can hit several models at once. It is a tail risk, so it is often underestimated; most errors are benign, which breeds misplaced confidence.
- **Effective challenge.** The guiding principle: critical analysis by objective, informed parties. It depends on **incentives** (stronger with *greater separation* from development, and with compensation and culture), **competence** and **influence** (explicit authority, stature, support from senior management). Model risk **cannot be eliminated**; other tools are limits on use, monitoring, revision and supplementary analysis.

The three components (S2; ORR-15 b, c)

- **Component 1: development, implementation and use.** Clear statement of purpose; documented design, theory and logic; comparison with alternative theories; rigorous data assessment (proxies identified and justified, especially for external or vendor data); testing of accuracy, robustness, stability, extreme inputs and the impact on downstream models. Any single test is rarely sufficient. User challenge is useful but tends to be weak, partial and **asymmetric** (users rarely challenge favourable outcomes). Report ranges, not spurious point estimates. Conservatism must be defined and measured; an extreme point on a misestimated distribution is not conservative.
- **Component 2: validation (ORR-15 d).** Three core elements: **conceptual soundness** (including developmental evidence, sensitivity analysis, stress testing of inputs); **ongoing monitoring** (process verification of data, code and systems, which starts when the model enters production; benchmarking; analysis of overrides, where a high override rate signals redevelopment); **outcomes analysis** (including back-testing on data not used in development, at a frequency matching the forecast horizon; parallel outcomes analysis for adjusted models; early-warning metrics for long-horizon models). All components, in-house or vendor, are validated. Periodic review at least annually; material changes validated.
- **Component 3: governance, policies and controls.** The board is ultimately responsible and delegates execution to senior management, who report on significant model risk, individually and in aggregate. Policies approved by the board or its delegates and reviewed annually. Roles: **ownership** (business units; model owner accountable for use and performance), **controls** (risk-control staff with authority to **restrict model use**), **compliance**. A firm-wide **model inventory** and documentation sufficient for someone unfamiliar with the model.

Validation independence (S3) and the MRM function (ORR-15 d; ORR-16 b)

- **Independence.** Validation is generally done by people not responsible for development or use, with no stake in the result. Independence is not an end in itself; it may be supported by separate reporting lines but is **judged by actions and outcomes**, for example models actually changed after validation. Some validation work may be done best by developers and users, provided it is **critically reviewed by an independent party** who adds further work. The developer is an important source of information but not an objective or sole one.
- **Before and after first use.** Rigor matches the risk of use. Significant deficiencies: do not use, or use only under tight constraints; too severe: reject. If validation cannot be completed before use (data paucity), document it, tell users and senior management, and apply compensating controls. Validation is an important check in *benign* periods, when risk estimates turn optimistic.
- **Vendor models.** Same principles. Require developmental evidence and testing from the vendor; the bank validates its *own use*, relying more on sensitivity analysis and benchmarking where code is proprietary; document customisation; keep knowledge in-house and a contingency plan if the vendor disappears. See also D2 for the outsourcing of model risk management services.
- **Internal audit.** Assesses the overall effectiveness of the framework (both types of model risk, individual and aggregate); does not duplicate MRM. Staff who perform validation work must not assess the overall framework. Verifies inventory, policy compliance, timeliness of validations, data reliability, and the objectivity, competence and standing of validators. External resources: a designated internal party verifies scope, tracks issues and integrates results.
- **MRM function in practice (ORR-16 b).** Sits in the **second line**, independent of development; sets documentation, data and versioning standards. **Tiering** by materiality, complexity, client-facing use and regulatory use: top tier full-scope validation every 2 to 3 years, mid tier every 4 or 5, low tier more rarely; every model has an **annual review**. Monitoring reports from model owners (quarterly or annual for capital models,

monthly or weekly for fraud). Move from point-in-time to **continuous** MRM. Developers and owners are the first line and own the risk; mature firms add a first-line QA/QC team. The ORX taxonomy makes model risk a Level 1 risk (design, implementation and application errors); Basel lists "model errors" as a Level 3 example (see A22).

Exam traps

- **Scope/condition on the copula:** it was accurate in normal markets; it failed when correlations spiked and users did not update or understand it. "Always wrong" and "the formula itself was the problem" both drop the condition.
- **Sibling swap, execution versus conceptual:** Barclays (hidden rows) is an implementation error in a non-model tool; the copula is a conceptual, assumption-driven failure; NASA is a unit assumption. Case cross-mapping uses one case's lesson for another.
- **Polarity flip on incentives:** effective challenge is *stronger* with *greater* separation from development, not with proximity.
- **Role misassignment:** the model owner (first line) is accountable for use and performance; risk-control staff hold the authority to restrict use; internal audit evaluates the framework and must not both validate and assess it; the board is ultimately responsible.
- **Absolute claims:** validation does not eliminate model risk; independence does not mean developers may do no validation work; a separate reporting line alone does not prove independence.

Question C6.1 ORR-16 a, c; ORR-15 a · S1 · [Map #32]

A credit risk consultant is presenting to newly hired analysts at a Japanese trust bank on lessons from the 2007 to 2009 crisis. One slide maps the subprime securitisation chain as a series of frictions between pairs of parties, and labels the friction between CDO investors and the credit rating agencies as *model error*. An analyst asks how that label connects to the GARP case study on Gaussian copula pricing of CDOs. Which of the following statements should the consultant give as correct?

A. Investors relied on ratings and prices from a Gaussian copula calibrated to a short, benign CDS history; when correlations spiked the assumption failed, yet users kept relying on outputs whose limitations they did not understand.

B. The friction is an execution error: like the hidden rows in the Barclays spreadsheet, the copula failed to do what it was designed to do, so replicating its code would have exposed the problem.

C. The friction would have been removed by a sufficiently rigorous validation of the copula before use, because robust validation of a sound model eliminates its model risk.

D. The copula produced inaccurate CDO prices in all market conditions, so the losses came from a fundamental flaw in Li's formula rather than from how users relied on it.

Solution **Answer: A**

Step 1 – Name the friction

In the securitisation-frictions taxonomy, the investor depends on the rating agency's model to judge the credit quality of the tranche it buys; the friction is that the model can be wrong or wrongly used, hence "model error" (outside-source framing).

⇒ *The investor's risk sits inside someone else's model, and it is model risk in the ORR-15 sense: adverse consequences from decisions based on incorrect or misused model outputs.*

Step 2 – Map it to the GARP case

Li's Gaussian copula inferred correlation from CDS prices. CDSs had only about a decade of history in a benign period, so implied correlations were very low and sensitive to house prices. The method was adopted by investors, banks, rating agencies and regulators. It worked in normal conditions; when house prices reversed, implied correlations shot up, and banks kept pricing with the old assumptions.

⇒ *This is a* conceptual *error (an assumption that held only in one context) combined with* misuse *(blind reliance on a black box).*

Step 3 – The MRM lesson

MRM must replicate the model to rule out coding errors, but its more critical job is to challenge assumptions and explain limitations to non-quantitative users in plain terms.

Interpretation

A is correct. The consultant's answer: the investor and rating agency friction is model risk of the conceptual-and-misuse type, exactly the failure in the Gaussian copula case, and the remedy is transparency about assumptions, not just better code.

Why the other options are wrong

B is incorrect. Sibling swap (case cross-mapping): execution error and the Barclays hidden-rows lesson describe an implementation failure in a tool that is not even a model; the copula did what it was designed to do, on an assumption that stopped holding.

C is incorrect. Absolute claim: even with skilled modelling and robust validation, model risk cannot be eliminated; other tools (limits on use, monitoring, revision, supplementary analysis) are still needed.

D is incorrect. Scope/condition error: the copula assumption was valid and accurate in normal market conditions; dropping that condition also reverses the case's verdict that the problem was not the formula itself but its blind use without updating correlations.

LO: ORR-16 a: Define a model and describe different ways that financial institutions can become exposed to model risk. · ORR-16 c: Describe lessons learned from the three case studies involving model risk. · ORR-15 a: Describe model risk and explain how it can arise in the implementation of a model. · Reference: GARP, Operational Risk and Resilience, Ch.16, Model Risks and Gaussian Copula and CDO Pricing; Ch.15, 15.3

Question C6.2

ORR-15 b, d; ORR-16 b · S2, S3 · [Map #32]

The head of model risk at a Singapore bank is drafting the charter of a newly created model risk management (MRM) function. The charter is organised around the three elements named in the supervisory guidance: model development, implementation and use; model validation; and governance, policies and controls. The validation team has four staff and the bank runs over 300 models, so some testing of lower-tier models is currently performed by the model developers. Which of the following statements is appropriate to include in the charter?

A. Model owners in the business units, who carry ultimate accountability for model use and performance, should hold the authority to restrict a model's use once validation finds significant deficiencies.

B. To preserve independence, no validation work may be performed by developers or users, and any testing they have already done must be discarded and repeated in full by the MRM team.

C. Incentives for effective challenge are strongest when validators sit close to the development process, because that proximity is what gives them the competence to critique a model.

D. Developers may perform some validation work if an independent party critically reviews it and adds further testing; independence may be supported by reporting lines but is judged by actions and outcomes.

Solution

Answer: D

Step 1 – Place the problem in the three components

Development, implementation and use (first line: developers and model owners); validation (independent effective challenge); governance, policies and controls (board, senior management, policies, roles, inventory, documentation, internal audit). The staffing gap sits in the second component, but the cure is written into the third.

Step 2 – What independence requires

Validation is generally done by people not responsible for development or use. As a practical matter some validation work may be done most effectively by developers and users, provided it is subject to critical review by an independent party who conducts additional activities. Independence may be supported by separate reporting lines but should be judged by actions and outcomes.

⇒ *The charter can keep developer testing on lower-tier models, but only with independent review on top, and must evidence independence through results (models changed after validation).*

Step 3 – Effective challenge

It needs incentives (stronger with greater separation from development, plus compensation and culture), competence, and influence (authority, stature, senior support).

Interpretation

D is correct. It is the guidance's own answer to scarce validation resources, combined with tiering so that the full-scope effort goes to the highest-tier models.

Why the other options are wrong

A is incorrect. Role misassignment: the model owner is accountable for use and performance, but the authority to restrict the use of models belongs to risk-control staff, whose temporary exceptions are themselves subject to other controls.

B is incorrect. Absolute claim: the guidance expressly allows some validation work by developers and users if an independent party critically reviews it; a blanket ban and full repetition is not required.

C is incorrect. Polarity flip: incentives to provide effective challenge are stronger when there is *greater separation* of that challenge from the model development process; competence comes from technical skill and business familiarity, not proximity.

LO: ORR-15 b: Describe elements of an effective model risk management process. · ORR-15 d: Describe elements of a strong model validation process and challenges to an effective validation process. · ORR-16 b: Describe the role of the model risk management function and explain best practices in the model risk management and validation processes. · Reference: GARP, Operational Risk and Resilience, Ch.15, 15.3 and 15.5 (Model Validation), 15.6 (Roles and Responsibilities); Ch.16, Model Risk Management

C7 CMT Swap Valuation and OAS on a Rate Tree

Readings: MR-12 a, b, c, d, e, f, g, h, i, j · Arbitrage Pricing with Term Structure Models (Tuckman and Serrat, Fixed Income Securities 4e, Ch.7)

MAP TIER C · #33 · 3/16 SOURCES VARDEEZ

Specifics reported

Map #33: "Reported: 'decision tree question with OAS of CMT swap'. LO 12.h is the exact objective: value a constant-maturity Treasury swap given an interest-rate tree and risk-neutral probabilities. LO 12.d, the true-vs-risk-neutral probability distinction applied to interest-rate drift, is the conceptual partner and a standing polarity trap."
Shapes covered here:
S1 value a CMT swap backward through a two-period rate tree with risk-neutral probabilities, inside an OAS wrapper (Q1)
S2 OAS: shift the discount rates only, then read cheap versus rich from the sign (Q1)
S3 true versus risk-neutral probabilities and the drift they imply, with the real-world probabilities supplied as a temptation (Q1).

Revision notes

- **Expected discounted value is not price** (MR-12 a): book tree, six-month rate 2% today, 2.50% or 1.50% in six months with true probability 0.5 each; one-year spot 2.15%. The one-year zero's expected discounted value is 980.302, but its market price is 978.842: investors dislike the price uncertainty and penalise the risky bond.
- **Arbitrage pricing by replication** (MR-12 b): a six-month call on a then six-month zero struck at 990 pays 0 (up) or 2.556 (down). Solve $F_{.5} + .987654F_1 = 0$, $F_{.5} + .992556F_1 = 2.556$: long 521.4375 one-year zeros, short 515.0000 six-month zeros; price 0.504. Expected discounted value with true probabilities, 1.2653, overstates it. The probabilities never enter the replication; they matter only indirectly through current bond prices.
- **Risk-neutral pricing** (MR-12 c): find the probabilities that make expected discounted values equal market prices ($p = 0.8009$ recovers 978.842), then price any claim by expected discounted value under them: $(0.1991 \times 2.556)/1.01 = 0.504$. It works because an arbitrage-priced security's price does not depend on risk preferences; an imaginary risk-neutral economy with the same bond prices gives the same price.
- **True versus risk-neutral probabilities and drift** (MR-12 d, S3): with true probabilities 0.5 the drift of the six-month rate is zero; under $p = 0.8009$ it is $0.8009(50) - 0.1991(50) = 30.09$ bp. From date 1 to 2, $q = 0.6520$ gives a risk-neutral drift of 15.20 bp. Risk-neutral probabilities put more weight on the high-rate (low-price) states; MR-13 explains that a drift of a given number of basis points has the same pricing effect as a risk premium of that size per year of duration.
- **Multi-period extension** (MR-12 e): add the 1.5-year zero (spot 2.25%, price 966.995) and solve for q at date 1 (assumed equal from both states). The tree then implies the one-year rate in six months (2.5754% or 1.5754%): in a one-factor model the short-rate process determines every other rate. Replicating portfolios change by date and state: *dynamic* replication.
- **Recombining trees** (MR-12 g): up-down lands on down-up. A nonrecombining tree has 2^N end states (over 500,000 for 10 years of semiannual steps, over 500 billion for 20), so it is computationally unwieldy; effects such as state-dependent volatility must be implemented more efficiently.
- **CMT swap** (MR-12 h, S1): pays $N(y_{\text{CMT}} - K)/2$ every six months, where y_{CMT} is the semiannually compounded yield of a fixed maturity on the payment date (in practice 2-, 5- or 10-year Treasury yields; the book uses the six-month yield). Value at each date-1 node = expected discounted value of date-2 payoffs *plus* that node's own date-1 payoff; then discount back to date 0.
- **Book CMT result:** USD 1,000,000 struck at 2%: date-1 values 5,719.52 and $-4,227.29$, date-0 value 3,702.11. With true probabilities the value is close to zero (-6.07): the risk-neutral probabilities, not the true ones, determine the arbitrage price even though 2% is the average rate under the true probabilities.
- **Option-adjusted spread** (MR-12 f, S2): the spread that, added to the model's discount rates, recovers the market price. Rates are shifted *only for discounting*, never for computing cash flows. Book: an OAS of 10 bp

takes the CMT value from 3,702.11 to 3,699.18. Positive OAS: market price below model price, the security trades *cheap*; negative OAS: *rich*.

- **OAS and expected return:** under the risk-neutral process a security's expected return is the short rate plus OAS per period. Book: 1.00% over six months for the fairly priced swap, 1.05% (half of 2.10%) with a 10 bp OAS.
- **P&L attribution with an OAS** (MR-12 f): $dP = (r_0 + \text{OAS})P\,dt + \frac{\partial P}{\partial r}(dr - E[dr]) + \frac{\partial P}{\partial \text{OAS}}d\text{OAS}$: carry-roll-down, rate-change gains or losses, spread-change gains or losses. A hedged ($\partial P/\partial r = 0$) and financed cheap security earns the OAS plus any convergence; risk premium terms cancel when factor risk is hedged.
- **Reducing the time step** (MR-12 i): smaller steps place cash flows close to tree dates and give a less coarse, more realistic rate distribution, but demand more numerical care and computation. Weekly or monthly steps may suffice for a 30-year callable; a one-month bond option needs much smaller steps. Step size can vary across dates.
- **Why not BSM for bonds** (MR-12 j): a bond's price converges to par at maturity; its price volatility must decline toward maturity; and assuming a constant short rate while the bond price is random is inconsistent. Hence models of the short rate (or the whole term structure). Practitioners still use versions of BSM for some fixed income derivatives for simplicity.
- **See also** D12 and D18 (MR-13: how expectations, risk premium and convexity shape the tree's term structure) and A7 (MR-14: the drift in Ho-Lee trees).

Formulas

Expected discounted value on a semiannual tree. $V_{\text{node}} = \dfrac{p\,V_{\text{up}} + (1-p)\,V_{\text{down}}}{1 + r_{\text{node}}/2} + \text{CF}_{\text{node}}$

Risk-neutral probability from a zero price (eq. 12.7). $\dfrac{p\,P_{\text{up}} + (1-p)\,P_{\text{down}}}{1 + r_0/2} = P_{\text{market}}$

Risk-neutral drift over a step of $\pm\Delta$. $\text{drift} = p\,\Delta - (1-p)\,\Delta = (2p-1)\,\Delta$

CMT swap payment (eq. 12.18). $N\,\dfrac{y_{\text{CMT}} - K}{2}$ each half-year

Valuation at an OAS. discount at $r_{\text{node}} + \text{OAS}$; cash flows still from r_{node}. OAS $> 0 \Rightarrow$ cheap; OAS $< 0 \Rightarrow$ rich

Expected return with an OAS (eq. 12.35). $E[dP/P] = (r_0 + \text{OAS})\,dt$

Book check (Figure 12.8). $p = 0.8009$, $q = 0.6520$, USD 1 million struck at 2%: 3,702.11 (model); 3,699.18 at OAS 10 bp; -6.07 with true probabilities.

Exam traps

- **True versus risk-neutral (wrong-input twin):** the real-world probabilities (often 0.5) give the wrong price. Only the risk-neutral probabilities recover market prices.
- **OAS shifts discounting only:** adding the OAS to the rates that set the CMT payments changes the cash flows and inflates the value.
- **Cheap versus rich (polarity flip):** a positive OAS means the market price is *below* the model value: cheap. A negative OAS means rich.
- **Dropped term:** each date-1 node value includes that node's own payment as well as the discounted date-2 expectation.
- **Same probability at every date:** q at date 1 is solved separately (book 0.6520 versus 0.8009 at date 0); reusing p is a named mistake.
- **Intermediate result:** the model value at zero OAS is a legitimate step, not the market price when an OAS is quoted.

Question C7.1

MR-12 h, f, d · S1, S2, S3 · [Map #33]

A risk analyst at a US insurer is checking the desk's mark on a one-year stylized constant-maturity Treasury (CMT) swap on the six-month yield. The insurer receives, every six months for one year, USD 5,000,000 $\times\,(y_{\text{CMT}} - 2.80\%)/2$, where y_{CMT} is the six-month rate on the payment date (a negative amount is paid). The desk's recombining tree of six-month rates (semiannually compounded) and its probabilities are:

- Date 0: 3.00%. Date 1: 3.60% (up) or 2.40% (down). Date 2: 4.20%, 3.00% or 1.80%
- Risk-neutral probability of an up move: 0.70 from date 0; 0.60 from either date-1 node
- Estimated true probability of an up move at every node: 0.50

The pricing system reports that the swap trades in the market at an option-adjusted spread (OAS) of $+25$ bp. Which of the following is closest to the swap's market price, and how does it trade relative to the model?

A. USD 24,328; the swap trades cheap

B. USD 24,328; the swap trades rich

C. USD 9,719; the swap trades cheap

D. USD 36,523; the swap trades cheap

Solution **Answer: A**

Step 1 – CMT payments (from the unshifted tree rates)

$$\text{Date 1: } 5{,}000{,}000 \times \frac{3.60\% - 2.80\%}{2} = 20{,}000; \qquad 5{,}000{,}000 \times \frac{2.40\% - 2.80\%}{2} = -10{,}000$$

$$\text{Date 2: } 35{,}000\ (4.20\%); \qquad 5{,}000\ (3.00\%); \qquad -25{,}000\ (1.80\%)$$

$\Rightarrow$ *The OAS does not touch these: rates are shifted only for discounting.*

Step 2 – Date-1 values, discounting at the tree rate plus 25 bp

$$V_{1,u} = \frac{0.60(35{,}000) + 0.40(5{,}000)}{1 + 0.0385/2} + 20{,}000 = \frac{23{,}000}{1.01925} + 20{,}000 = 22{,}565.61 + 20{,}000 = 42{,}565.61$$

$$V_{1,d} = \frac{0.60(5{,}000) + 0.40(-25{,}000)}{1 + 0.0265/2} - 10{,}000 = \frac{-7{,}000}{1.01325} - 10{,}000 = -6{,}908.46 - 10{,}000 = -16{,}908.46$$

$\Rightarrow$ *Each node value is the discounted risk-neutral expectation of the next payment plus the payment made at that node.*

Step 3 – Date-0 value at the OAS

$$V_0 = \frac{0.70(42{,}565.61) + 0.30(-16{,}908.46)}{1 + 0.0325/2} = \frac{29{,}795.93 - 5{,}072.54}{1.01625} = \frac{24{,}723.39}{1.01625} = 24{,}328.06$$

Step 4 – Compare with the model value (zero OAS)

Repeating Steps 2 and 3 at the unshifted rates gives date-1 values 42,593.32 and $-16{,}916.99$ and a model value of USD 24,374.61. The market price, 24,328, is USD 46.55 below the model value, consistent with a positive OAS: the swap trades *cheap*.

Interpretation

The swap is worth far more than zero although its strike, 2.80%, sits below a 3.00% starting rate and the true probabilities are symmetric. The reason is the risk-neutral drift: $(2 \times 0.70 - 1) \times 60 = 24$ bp from date 0 and $(2 \times 0.60 - 1) \times 60 = 12$ bp from date 1, against zero under the true probabilities. If the model is right, a hedged, financed long position earns the 25 bp OAS plus any convergence of the OAS to zero; under the risk-neutral process the expected six-month return is $(3.00\% + 0.25\%)/2 = 1.625\%$.

Why the other options are wrong

B is incorrect. Polarity flip on the OAS sign. A positive OAS means the market price is below the model price, so the swap is cheap, not rich.

C is incorrect. Used the true probabilities of 0.50 at every node (with the OAS): date-1 values 39,622.27 and $-19{,}869.23$, date-0 value 9,718.59. Real-world probabilities do not recover market prices.

D is incorrect. Added the OAS to the CMT yields that set the payments as well as to the discount rates: payments become 26,250, $-3{,}750$, 41,250, 11,250 and $-18{,}750$, giving 36,522.75.

LO: MR-12 h: Calculate the value of a constant-maturity Treasury swap, given an interest rate tree and the risk-neutral probabilities. MR-12 f: Define option-adjusted spread (OAS) and apply it to security pricing. MR-12 d: Explain the difference between true and risk-neutral probabilities and apply this difference to interest rate drift. · Reference: Tuckman and Serrat 4e, Ch.7, 12.3, 12.5 and 12.6, eqs. 12.18 to 12.31, Figure 12.8

C8 Counterparty Exposure Profiles: EE, PFE, EPE

Readings: CR-19 a, b, c, d, e, f, g, h, i · Future Value and Exposure (Gregory, The xVA Challenge 4e, Ch.11: 19.1 Credit Exposure; 19.2 Drivers of Exposure; 19.3 Aggregation, Portfolio Effects, and the Impact of Collateralisation; 19.4 Funding, Rehypothecation, and Segregation); CR-12 a · Credit Risk (Hull, OFOD 11e, Ch.24, 12.1)

MAP TIER C · #37 · 3/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #37: "Your formula card lists 'potential credit risk exposure' and 'netting' as separate rows. The examinable set is expected MtM, expected exposure, PFE, EPE, effective EE/EPE and maximum PFE, a six-way sibling family, plus the netting factor. Red-herring risk: a sold option carries zero counterparty exposure regardless of how much Black-Scholes data the stem supplies."
Shapes covered here:
S1 the six-way metric family: expected MtM (EFV), expected exposure / EPE, PFE, ENE, effective EE/EPE and average EPE, maximum PFE (Q2; EPE and EFV computed in Q1)
S2 the netting row: aggregation, EFV additive but EPE not, netting factor (Q1, Q2)
S3 sold option red herring: zero exposure whatever Black-Scholes data the stem supplies (Q1).

Revision notes

- **Positive and negative exposure** (CR-19 a): on a counterparty default the survivor closes out, nets, and applies margin. If the net value is *negative* it still owes the amount and neither gains nor loses; if *positive* it has an unsecured claim and recovers only a fraction. Positive exposure $= \max(\text{value}, 0)$ (19.1); negative exposure $= \min(\text{value}, 0)$ (19.2), the exposure the counterparty has to the party's own default (DVA). Recovery is, by convention, *not* included in exposure.
- **Sold option (S3)**: a sold option with the premium paid upfront has a value to the seller that can never be positive, so $\max(\text{value}, 0) = 0$ on every path. Hull (CR-12 a): if the transactions have a negative value to the bank there is no exposure. Spot, strike, volatility and a Black-Scholes price in the stem price the option; they do not create exposure for the seller. The *buyer* has the exposure (Hull: a bought European option is the special case where exposure equals the no-default value).
- **Exposure as a short option** (CR-19 a, b): the payoff is asymmetric, like a short option for the survivor (the defaulter has the option not to pay). Two consequences: volatility of the contract value is a key driver, and even simple instruments have option-like, complex exposure. A netting set behaves like a basket option; margin shifts the strike.
- **The metrics (S1)**: *expected future value* (EFV, also called expected mark-to-market or, confusingly, EE) is the average of *all* values and equals current value at time zero; *PFE* is the exposure exceeded with probability no more than 100% minus the confidence level, "an equivalent metric to VAR"; *EPE* is the average of the positive exposures with zero and negative values counted as zero (not dropped), sometimes called expected exposure (EE); *ENE* is the same for negative values, sometimes called NEE. Book example: five scenarios $70, 50, 30, -10, -30$: EFV 22, PFE 70, EPE 30, ENE -8.
- **Sensitivity to volatility**: normal future value, mean 2.0: at $\sigma = 2.0$, PFE(99%) 6.65, EPE 2.17, ENE 0.17; at $\sigma = 4.0$, PFE 11.31, EPE 2.79, ENE 0.79, while EFV stays 2.0. ENE rises almost fivefold because it is out of the money.
- **Across time**: *average EPE* is the time-weighted average of the EPE profile (a simple average if points are equally spaced), often called the *loan equivalent*: a crude summary that averages out market randomness and time, but with a strong theoretical basis for capital and xVA. *Maximum PFE* is the peak of the PFE profile (Hull: *maximum peak exposure*, the maximum of the peak exposures at all midpoints). Gregory warns that Basel definitions differ: Basel's *effective EE* is EE made non-decreasing over time, and *effective EPE* is the average of effective EE over the first year (outside-source).
- **Exposure versus VaR** (CR-19 b): same quantile problem, but exposure needs (i) many horizons, so ageing, cash flows, exercise and margin path dependency matter, and the drift cannot be ignored as it can over 10 days; (ii) risk mitigants (netting, margin, termination features); (iii) both real-world (risk management) and risk-neutral (pricing) calculations.

- **Why EFV drifts from today's value** (CR-19 c): cash flow differentials (fixed exceeds floating early in a swap on an upward-sloping curve), forward rates differing from spot (implied drift), and asymmetric (one-way) margin agreements.
- **Typical profiles** (CR-19 d, e): forward or FX forward, and vanilla options with upfront premium: $\propto \sqrt{t}$, monotonic, maturity irrelevant except that exposure ends at T. Interest rate swap: $\propto (T-t)\sqrt{t}$, peaked, maximum at $T/3$. Receive more often than you pay: less risk (and vice versa). Receiver swap on an upward curve: lower EPE, more negative ENE, negative EFV; payer swap the reverse. Cross-currency swap: IRS plus FX forward with a final notional exchange, large exposure for the payer of the high-rate currency; low correlation raises it. ITM swap: EPE close to EFV, small ENE. Physically settled swaption: higher exposure than the forward swap before exercise, lower after. Long-protection CDS: the 99% PFE jumps (60% at 40% recovery) once the reference entity PD exceeds 1%; EPE, like expected shortfall, does not.
- **Aggregation and netting (S2)** (CR-19 f): EFV is additive across transactions; EPE and ENE are not: aggregated EPE or ENE can never exceed the sum of the parts, since $\max(\sum v_i, 0) \le \sum \max(v_i, 0)$. Benefit arises only in scenarios where values have opposite signs: small with positive correlation (Table 19.1: EPEs 18 and 4, aggregated 21), large with negative correlation (Table 19.2: aggregated 9). Off-market portfolios also produce benefit: an OTM portfolio damps a new trade's positive exposure. Hull's netting example: $+10, +30, -25$ gives 40 without netting, 15 with it.
- **Netting factor (S2)**: for n transactions with equal exposure and common correlation ρ, netted exposure is approximately $\sqrt{n + n(n-1)\rho}\,/\,n$ times the gross exposure (outside-source: Gregory's Appendix on aggregation is not in the GARP excerpt). Lower correlation and more transactions mean a smaller factor (greater benefit), which matches the reading's statement that exposure reduction increases with portfolio size and falls as correlation rises.
- **Margin** (CR-19 c, h): positive exposure $= \max(\text{value} - \text{margin}, 0)$; received margin reduces positive exposure, posted margin can create it. Imperfections: granularity (threshold, MTA; a threshold roughly caps the exposure), delay (the MPoR), and variability of non-cash margin; path dependency matters. Profiles: partially collateralised (threshold), strongly collateralised (zero threshold, MPoR leaves residual exposure), overcollateralised (initial margin, possibly to zero). See also A21 for threshold, MTA and MPoR mechanics.
- **Funding versus credit exposure** (CR-19 g): funding $=$ value $-$ margin; a positive value must be funded, a negative value is a funding benefit. Differences: value definition (close-out versus objective), MPoR (applies only to default; funding uses normal posting frequency, so a collateralised trade's FVA may be zero while its CVA is not), aggregation (netting set versus whole portfolio), WWR (mainly credit), segregation.
- **Segregation and rehypothecation** (CR-19 i): positive exposure $= \max(\text{value} - \text{margin}_R + \text{margin}_P^{NS}, 0)$; funding $=$ value $-$ $\text{margin}_R^{RH} + \text{margin}_P$. Specifically, positive exposure $= \max(\text{value} - \text{VM} - \text{IM}_R, 0)$, funding $=$ value $-$ VM $+$ IM_P. Example: value 20, VM 18, segregated IM received 3: exposure 0, funding 2. Segregated margin reduces counterparty risk but gives no funding benefit; a counterparty's own bonds give a funding benefit but questionable risk reduction (Table 19.4).

Formulas

Exposure (19.1, 19.2). $\text{PE} = \max(V, 0)$, $\text{NE} = \min(V, 0)$
Metrics over n equally weighted scenarios. $\text{EFV} = \frac{1}{n}\sum V_k$, $\text{EPE} = \frac{1}{n}\sum \max(V_k, 0)$, $\text{ENE} = \frac{1}{n}\sum \min(V_k, 0)$ (divide by n, not by the number of positive scenarios)
PFE at confidence α. the α-quantile of the exposure distribution (the highest value when α exceeds $1 - 1/n$)
Average EPE. $\overline{\text{EPE}} = \dfrac{\sum_j \text{EPE}(t_j)\,\Delta t_j}{\sum_j \Delta t_j}$; maximum PFE $= \max_j \text{PFE}(t_j)$
Profiles. Forward or option: Exposure $\propto \sqrt{t}$; swap: Exposure $\propto (T-t)\sqrt{t}$, peak at $t = T/3$
Aggregation. $\text{EFV}_{\text{agg}} = \sum \text{EFV}_i$, $\text{EPE}_{\text{agg}} \le \sum \text{EPE}_i$
Netting factor (outside-source). $\text{NF} = \dfrac{\sqrt{n + n(n-1)\rho}}{n}$
With margin (19.5, 19.10, 19.11). $\text{PE} = \max(V - \text{VM} - \text{IM}_R, 0)$, $\text{Funding} = V - \text{VM} + \text{IM}_P$

Exam traps

- **Sold option red herring (S3):** the seller of an option with upfront premium has zero exposure. A Black-Scholes value in the stem is a value, not an exposure; never add it to EPE and never use it to offset another

trade's exposure.

- **Netting needs a legal agreement:** trades aggregate only within an enforceable netting set. Without one, total exposure is the *sum* of standalone exposures; netting scenario by scenario understates it.
- **EPE denominator:** EPE averages over *all* scenarios with negatives set to zero. Averaging only the positive scenarios overstates it (in the five-scenario example, 50 instead of 30).
- **Additivity (sibling swap):** EFV adds across trades; EPE and ENE do not. A claim that aggregated EPE equals the sum of standalone EPEs borrows EFV's property.
- **Volatility polarity:** higher volatility raises EPE, ENE (in magnitude) and PFE; it leaves EFV unchanged.
- **Name collisions:** Gregory's EFV is "sometimes referred to as expected mark-to-market or expected exposure", and his EPE is "sometimes known as expected exposure"; average EPE is "sometimes known simply as EPE". Read the definition in the stem, not the acronym.

Question C8.1

CR-19 a, f; CR-12 a · S1, S2, S3 · [Map #37]

A counterparty risk analyst at a Canadian bank is estimating exposure metrics to a pension fund counterparty at the six-month horizon. The bank has two trades with the fund:

- A receive-fixed interest rate swap.
- A one-year European put on an equity index that the bank *sold* to the fund on 20,000 units, premium received in full at inception. Inputs: index level 4,000; strike 3,800; implied volatility 20%; risk-free rate 3%. Black-Scholes value today: CAD 3.5 million.

The two trades were booked under different master agreements, and legal counsel has confirmed that close-out netting between them is not enforceable. The bank's simulation produces five equally likely scenarios for the value of each trade to the bank in six months (CAD million):

Scenario	1	2	3	4	5
Swap	18.0	9.0	3.0	-6.0	-14.0
Sold put	-0.5	-1.0	-2.0	-4.5	-9.0

Which of the following is closest to the bank's expected positive exposure (EPE) to the fund and the expected future value (EFV) of the two trades at the six-month horizon?

A. EPE CAD 2.5 million; EFV CAD -1.4 million

B. EPE CAD 5.3 million; EFV CAD -1.4 million

C. EPE CAD 6.0 million; EFV CAD 2.0 million

D. EPE CAD 6.0 million; EFV CAD -1.4 million

Solution

Answer: D

Step 1 – Decide the aggregation level

Netting is not enforceable, so each trade is its own netting set. Exposure is computed trade by trade with $\max(\text{value}, 0)$ and then summed.

$\Rightarrow$ *The two trades cannot offset one another in a default; the bank would have to pay the put's value in full while claiming the swap's value as an unsecured creditor.*

Step 2 – Swap EPE

$$\text{EPE}_{\text{swap}} = \tfrac{1}{5}\big(18 + 9 + 3 + 0 + 0\big) = \tfrac{30}{5} = 6.0$$

$\Rightarrow$ *Negative scenarios count as zero but stay in the denominator: on average CAD 6.0 million is at risk from the swap.*

Step 3 – Sold put EPE

$$\text{EPE}_{\text{put}} = \tfrac{1}{5}\big(0 + 0 + 0 + 0 + 0\big) = 0$$

⇒ *Every value to the bank is negative or zero: having received the premium, the bank is owed nothing by the fund, so the put adds no exposure. The index level, strike, volatility and the CAD 3.5 million Black-Scholes value price the option but are irrelevant to the seller's exposure.*

Step 4 – Total EPE

$$\text{EPE} = 6.0 + 0 = 6.0$$

Step 5 – EFV (additive, includes every value)

$$\text{EFV}_{\text{swap}} = \tfrac{18+9+3-6-14}{5} = 2.0, \qquad \text{EFV}_{\text{put}} = \tfrac{-0.5-1.0-2.0-4.5-9.0}{5} = -3.4, \qquad \text{EFV} = 2.0 - 3.4 = -1.4$$

⇒ *The two trades are expected to be worth CAD 1.4 million* against *the bank in six months, yet the bank still has CAD 6.0 million of expected positive exposure.*

Interpretation

EFV and EPE answer different questions. EFV is the average of all values and is additive, so the sold put pulls it negative. EPE keeps only what the counterparty would owe the bank, so the sold put contributes nothing to it and, without enforceable netting, cannot reduce the swap's exposure either. The CVA desk should price CAD 6.0 million of EPE, and the legal team's view on netting is worth CAD 0.7 million of EPE here.

Why the other options are wrong

A is incorrect. Red herring used as an offset: subtracted the sold put's Black-Scholes value from the swap EPE, $6.0 - 3.5 = 2.5$. Today's option value is neither an exposure nor a netting offset.

B is incorrect. Assumed netting where none is enforceable: summed values scenario by scenario, $17.5, 8.0, 1.0, -10.5, -23.0$, giving EPE $(17.5 + 8.0 + 1.0)/5 = 5.3$. This is the netted EPE (a netting factor below 1), not the exposure the bank actually has.

C is incorrect. Correct EPE, but dropped the sold put from EFV because it has zero exposure, reporting only the swap's EFV of 2.0. Zero exposure does not mean zero value; EFV averages all values.

LO: CR-19 a: Describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive exposure, and maximum exposure. CR-19 f: Explain the general impact of aggregation on exposure, and the impact of aggregation on exposure when there is correlation between transaction values. CR-12 a: Assess the credit risks of derivatives. · Reference: Gregory, The xVA Challenge 4e, Ch.11, 19.1.1, 19.1.5, 19.3.1, Tables 19.1 and 19.2; Hull, OFOD 11e, Ch.24, 12.1

Question C8.2

CR-19 a, b, f · S1, S2 · [Map #37]

A senior counterparty risk analyst at a Singapore bank is reviewing a glossary of exposure metrics drafted by a junior colleague for the bank's xVA desk. Which of the following statements is correct to keep in the glossary?

A. Expected future value and expected positive exposure are both additive across transactions, so the EPE of a netting set equals the sum of the EPEs of its trades.

B. Average EPE is the time-weighted average of the EPE profile, often called the loan equivalent, while maximum PFE is the peak of the PFE profile over the life of the portfolio.

C. Expected negative exposure is the exposure seen from the counterparty's point of view, so it drives the credit value adjustment the bank charges that counterparty.

D. Doubling the volatility of future value leaves the expected future value unchanged but lowers expected positive exposure, because more scenarios end below zero.

Solution

Answer: B

Step 1 – Check each statement against the metric definitions

Gregory (19.1.5): average EPE is "the average exposure across all time horizons", "the weighted average (the weights being the time intervals) of the EPE profile", and "this single average EPE number is often called a 'loan equivalent'". Maximum PFE is the highest point of the PFE profile (Figure 19.4; Hull's maximum peak exposure).
⇒ *B is correct. The trap inside it is naming: average EPE is "sometimes known simply as EPE", which differs from the single-horizon EPE.*

Interpretation

The six-way family splits on two axes: what is averaged (all values: EFV; positive values with negatives set to zero: EPE; negative values: ENE; a quantile: PFE) and how time is handled (one horizon, a time-weighted average, or a maximum). A glossary should state both axes for every metric and flag that Basel's effective EE and effective EPE are non-decreasing versions defined for capital (outside-source).

Why the other options are wrong

A is incorrect. Sibling swap: additivity belongs to EFV only. EPE and ENE are not additive; aggregated EPE can never exceed the sum, since $\max(\sum v_i, 0) \leq \sum \max(v_i, 0)$ (Table 19.1: 18 and 4 versus 21). This sub-additivity is the whole netting benefit.

C is incorrect. Role misassignment: ENE "represents the positive exposure from the counterparty's point of view", which is the bank's *own* default risk and drives DVA. CVA is driven by EPE, the bank's exposure to the counterparty.

D is incorrect. Polarity flip: higher volatility *raises* EPE. With mean 2.0, EPE rises from 2.17 to 2.79 when the standard deviation doubles from 2.0 to 4.0, because exposure is option-like; only EFV is unchanged.

LO: CR-19 a: Describe and calculate the following metrics for credit exposure: expected mark-to-market, expected exposure, potential future exposure, expected positive exposure and negative exposure, effective expected positive exposure, and maximum exposure. CR-19 b: Compare the characterization of credit exposure to VaR methods and describe additional considerations used in the determination of credit exposure. CR-19 f: Explain the general impact of aggregation on exposure, and the impact of aggregation on exposure when there is correlation between transaction values. · Reference: Gregory, The xVA Challenge 4e, Ch.11, 19.1.4, 19.1.5, 19.3.1

C9 Stress-Testing Practices and Scenario Design

Readings: CR-21 a to h · The Evolution of Stress Testing Counterparty Exposures (Lynch, in Siddique and Hasan (eds.), Stress Testing, Ch.4); ORR-17 a, b, c · Stress Testing Banks (Schuermann, International Journal of Forecasting, 2014); IM-15 a, b, c, d · Market-Driven Scenarios: An Approach for Plausible Scenario Construction (Golub (ed.), BlackRock's Guide to Fixed-Income Risk Management, Ch.5) (source: SchweserNotes Book 5, one rung below a GARP text)

MAP TIER C · #38 · 3/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #38: "Reported: '2 to 3 questions on stress testing related: appropriate practices', i.e. best-practice audits rather than computation. Market-driven scenarios (Reading 97) supply the plausible-scenario-construction angle. The SCAP / EBA / CCAR sibling swap on which regime required what, and which banks failed, is the historical-fact layer."
Shapes covered here:
S1 appropriate stress-testing practices as a best-practice audit: what to stress, how to aggregate, what to avoid (Q1)
S2 market-driven scenarios: plausible scenario construction, z-scores and covariance choice (Q2)
S3 the SCAP / EBA / CCAR sibling swap on which regime did what (Q2; regime comparison also in C5.2).

Revision notes

- **Exposure measures** (CR-21 a, BCBS 2005 definitions): *current exposure* (replacement cost) = max(0, market value of the netting set), lost on default with no recovery; *peak exposure* = a high percentile (95% or 99%) of the exposure distribution at a future date; *expected exposure* = the mean of that distribution at a future date; *EPE* = the time-weighted average of expected exposures (over the first year, or the longest maturity if shorter, for minimum capital). See also C8.
- **CCR as credit risk and as market risk** (CR-21 b): statistical measures progressed faster than stress testing. Credit view (2000 to 2006, the basis of Basel II): PFE limits, collateral, netting, set at inception; at default the bank replaces all trades at once. Market view: CVA (FAS 157), hedged dynamically by replacing trades in proportion to the counterparty's PD before default, so default becomes a non-event. Either view alone leaves the bank blind to the other risk; a dealer must consider both. The number of stress results is at least twice the number of counterparties plus one.
- **Stressing current exposure** (CR-21 c): reprice each counterparty's trades under a scenario (for example equity down 25%) and report the top counterparties. Three problems: (i) aggregation: summing stressed exposures is the loss if *every* counterparty defaulted, and running current exposure through a portfolio credit model is wrong (expected exposures belong there); (ii) no credit quality (USD 200 million to a start-up hedge fund versus to a AAA corporate); (iii) no insight into wrong-way risk.
- **Loan portfolio stress** (CR-21 c, d): EL $= \sum p_i \, ead_i \, lgd_i$; hold EAD and LGD fixed and stress the PD (directly, or through the macro and balance-sheet variables it depends on) to get EL^S. The *stress loss* is $\text{EL}^S - \text{EL}$, not the stressed EL itself.
- **Derivative portfolio in the credit framework** (CR-21 c, d): replace EAD by $\alpha \times$EPE, which lets derivatives sit in a portfolio credit model with loans; stress the PD (Table 21.2, dotcom: aggregate EL 1.36, stressed EL 2.63, stress loss 1.27, USD million) or stress EPE through market variables (Table 21.3). PD stresses move all counterparties the same way; EPE stresses may raise or lower the aggregate (directional bias, margining). Stress losses on loans and derivatives can be *added*. Joint credit and market stresses are hard to make consistent; the best wrong-way check is to stress current exposure, find the most exposed counterparties, then examine them for wrong-way risk.
- **Stressing CVA** (CR-21 e, f): CVA $= \sum_n \text{LGD}^*_n \sum_j EE^*_n(t_j) \, q^*_n(t_{j-1}, t_j)$ with risk-neutral inputs; shock market variables instantaneously (exposures EE^* re-simulated, marginal PDs from credit spread shocks) to get CVA^S; the stress loss is $\text{CVA}^S - \text{CVA}$. Differences from the EL view: risk-neutral versus physical values, life of the trades versus a fixed horizon, market-implied PD. Market-implied PD allows a stress of the exposure-PD correlation (Hull and White, 2012).
- **DVA and BCVA in aggregation** (CR-21 g): DVA uses negative expected exposure (exposure from the counterparty's side) and the counterparty's survival probability; CVA terms are weighted by the bank's own survival. Losses can arise because the bank's *own* credit quality improves. Stress loss = stressed BCVA − BCVA; with the same market shocks it is added to trading-position stress results by simple addition.

- **Common pitfalls** (CR-21 h): using stressed *current* exposure where EE or EPE belongs (for at-the-money trades the gap is almost half the shock); delta approximations for highly nonlinear CCR, worst for at-the-money portfolios under large moves; too many stresses can hide the risk; combining default stresses and fair value stresses into one test remains the true difficulty. CCR is rarely aggregated consistently with loan or trading stresses.
- **Why supervisory stress tests** (ORR-17 a): three kinds of capital (what you have, what you need, what regulators think you need); Bear Stearns, Lehman, WaMu, Wachovia, Fannie and Freddie were adequately capitalized by regulatory rules before failing. A credible program needs a credible estimate of the capital hole and a credible way to fill it (the Treasury's Capital Assistance Program backed SCAP; none drew on it).
- **The regimes** (ORR-17 a, Table 17.3): **SCAP** (Mar 2009): 19 BHCs with at least USD 100 billion assets, about two-thirds of banking assets; 4% T1C and 6% T1 post-stress; two-year scenario; ten required to raise USD 75 billion, raised USD 77 billion within six months; disclosed bank-level stress losses for eight asset classes, supervisory estimates. **CEBS** (Jul 2010): 91 banks, 6% T1, seven raised EUR 3.5 billion; loss rates only for retail and corporate; the Irish banks all passed, then needed EUR 24 billion. **EBA** (Jul 2011): 90 banks in 21 countries, 5% T1C, eight raised EUR 2.5 billion; downloadable bank-level spreadsheets; five of 25 Spanish banks did not pass but needed EUR 0 after provisions and conversions. **CCAR** (Mar 2011): SCAP-19, 5% T1C, macro scenario only published. **CCAR** (Mar 2012): SCAP-19 plus 11 BHCs over USD 50 billion; four of 19 did not pass (objection to capital plans); near-SCAP disclosure.
- **Scenario design and coherence** (ORR-17 b): the authorities' risk appetite sets severity, length and the post-stress hurdle (SCAP: two years, 4% T1C; Spain bottom-up: three years, 6% core Tier 1, a lower risk appetite). Severity is hard to compare (no credible "1 in 100" statements). Coherence: not all exchange rates can depreciate at once, high inflation implies a policy rate response, a flight to quality needs safe havens; acute for the trading book (tens of thousands of risk factors). SCAP used three macro dimensions plus a historical market shock (June to December 2008): loose coherence, but it tests nothing new. EBA 2011 specified over 70 trading-book factors and eight macro factors for each of 21 countries. Every supervisory test imposed one scenario on all banks; CCAR added banks' own baseline and stress scenarios for risk discovery.
- **Modelling losses, revenues, balance sheet** (ORR-17 c): map few macro factors to many intermediate drivers (geography: 2011 state unemployment 3.3% North Dakota to 12.9% Nevada; used-car Manheim index up 4% while new auto sales fell 37%, so auto LGD need not rise; industry distress drives corporate LGD). EBA's Basel II weights required full stress migration matrices; the US tests used Basel I. Trading book: reprice; CCR needs two simulations (PD and exposure). Revenues are "a black box" (non-interest income share 25% in 1985 to 43% in 2001); median loss coverage by profits 58% (SCAP), 63% (CCAR 2012), 66% (EBA). Balance sheet: static or dynamic, never below required ratios in any quarter, T+1 years for end-horizon reserves (CCAR horizon nine quarters).
- **Disclosure** (ORR-17 a): SCAP "opened Pandora's box" with projected bank-level losses; CCAR 2011 disclosed nothing bank-level ("symmetric ignorance"); Goldstein and Sapra: in normal times aggregated results limit window dressing and preserve private information production; in a crisis detailed bank-level disclosure is valuable. Supervisory tests work as a "pop quiz".
- **Market-driven scenarios** (IM-15 a): pre-GFC stress tests were add-ons to VaR or tracking error built on historical relationships that broke down in the crisis. MDS meets Surowiecki's three criteria (specialized private knowledge, independent opinions, an objective aggregation mechanism): diverse experts propose views, risk management aggregates them, outcomes become instantaneous shocks on policy variables, Mahalanobis distance checks plausibility, the covariance matrix implies the rest, iterate, then revalue. Subjectivity remains but becomes transparent and open to challenge.
- **Conditional framework and z-scores** (IM-15 b): shocks to a small set of K policy variables imply shocks to the other $M - K$ factors by multivariate regression (equivalent to estimating the covariance matrix); a small K reduces arbitrariness; nonlinear positions are fully revalued. Mahalanobis distance: a multivariate z-score that identifies outliers. Scenario z-score: severity in standard deviations, normalized so it does not grow with the number of policy variables; lower means more plausible; positioned on a risk ruler of historical events. Volatility z-score: magnitude, computed with zero correlations. Correlation z-score: consistency with the covariance matrix, independent of absolute shock size; near 0 consistent, above 1.0 a warning.
- **Covariance choice and the five steps** (IM-15 b, c): slow data (five years weekly), fast data (one year weekly), recent data; each gives a different risk ruler. For a crash, use data from similar crash periods, since correlations among risky assets spike in crises. Steps: (1) define a low-probability but plausible, precise, relevant event with a catalyst; (2) select few policy variables (do not add one with adjusted R^2 of 90% or more on

existing ones; "asset blocks" if more are needed); (3) calibrate shocks to a historical analogue or a target frequency; (4) imply nonpolicy shocks through the covariance matrix; (5) estimate P&L, fully revaluing optionality.

- **Brexit example and challenges** (IM-15 c, d): Remain, Soft Brexit, Adverse Brexit; Soft Brexit meant risk-off and a dovish Bank of England; policy variables FTSE All Share, 10-year gilt yield, GBP; gilts a safe haven in Soft Brexit but yields up in Adverse; recent covariance matrix (no comparable history); 60/40 FTSE 100 and iBoxx GBP portfolio. Challenges: subjectivity and groupthink, translating events into shocks, simplicity versus accuracy (multicollinearity), covariance calibration, judging plausibility, organisational process.
- **See also** C5 (Basel progression and C5.2 on SCAP versus CCAR), C8 (EE, PFE, EPE profiles), A6 (CVA, DVA, BCVA), A20 (wrong-way risk), D10 (BHC scenario design under the Capital Plan Rule).

Formulas

Loans (CR-21). $\text{EL} = \sum_i p_i\, ead_i\, lgd_i, \qquad \text{EL}^S = \sum_i p_i^S\, ead_i\, lgd_i, \qquad \text{Stress loss} = \text{EL}^S - \text{EL}$
Derivatives, credit view. $\text{EL} = \sum_i p_i\, \alpha\, epe_i\, lgd_i, \qquad \text{EL}^S = \sum_i p_i^S\, \alpha\, epe_i^S\, lgd_i$
CVA, market view. $\text{CVA} = \sum_n \text{LGD}_n^* \sum_j EE_n^*(t_j)\, q_n^*(t_{j-1}, t_j), \qquad \text{Stress loss} = \text{CVA}^S - \text{CVA}$
Bilateral. CVA terms weighted by own survival $S_I(t_{j-1})$; DVA terms use NEE, own marginal PD and the counterparty's survival; stress loss $= \text{BCVA}^S - \text{BCVA}$.
Mahalanobis distance (outside-source). $\text{MD} = \sqrt{(r-\mu)^\top \Sigma^{-1} (r-\mu)}$, with r the shock vector, μ the mean vector, Σ^{-1} the inverse covariance matrix of policy variables.

Exam traps

- **Stressed EL versus stress loss (intermediate result):** the stress loss is the *difference* $\text{EL}^S - \text{EL}$ (or $\text{CVA}^S - \text{CVA}$).
- **Summing stressed current exposures (absolute claim):** the total assumes every counterparty defaults; it is not a portfolio stress loss.
- **Current exposure versus EE/EPE (sibling swap):** portfolio credit models and BCVA aggregation need expected exposure; delta shortcuts fail most for at-the-money books under large moves.
- **Z-score siblings:** scenario z-score = severity (lower = more plausible); volatility z-score = magnitude with zero correlations; correlation z-score = consistency with the covariance matrix.
- **Few policy variables (polarity flip):** MDS keeps the policy set *small* to limit multicollinearity and arbitrariness.
- **Regime swap:** SCAP had three macro dimensions plus a historical market shock; the 70-plus factor scenario is EBA 2011; Basel I weights are SCAP and CCAR, Basel II weights are EBA.

Question C9.1

CR-21 c, d, g, h · S1 · [Map #38]

The head of counterparty credit risk at a Japanese bank is reviewing the bank's CCR stress-testing program after supervisors asked for derivative stress results that can be combined with the stress results of the loan book. Which of the following changes should the head recommend?

A. Run the PD and market stresses through expected positive exposure scaled by alpha in the expected-loss framework, so that derivative stress losses can be added to the loan stress losses.

B. Sum the stressed current exposures across all counterparties in each scenario and report that total as the stress loss of the derivatives book for comparison with the loan book.

C. Reprice the derivatives under the large scenario shocks using delta sensitivities, since the linear approximation is most reliable for portfolios that are close to at-the-money.

D. Leave DVA out of the CVA stress test, since a bilateral view can only reduce reported stress losses and never creates a loss from changes in the bank's own credit quality.

Solution

Answer: A

Step 1 – What can be combined with a loan stress

Loan stress loss $= \text{EL}^S - \text{EL}$ with $\text{EL} = \sum p_i\, ead_i\, lgd_i$. For derivatives, $\alpha \times$ EPE "is the value that allows CCR

exposures to be placed in a portfolio credit model along with loans"; then EL $= \sum p_i \, \alpha \, epe_i \, lgd_i$.
$\Rightarrow$ *The derivative stress loss is measured in the same units and on the same physical-PD basis as the loan stress loss, so the bank "can also combine the stress losses on the loan portfolio and the stress losses on its derivatives portfolio by adding these stress losses together".*

Step 2 – Which stresses to run

Stress PDs (or the macro and balance-sheet variables behind them) and stress EPE through market variables; note that EPE stresses can raise or lower the aggregate depending on directional bias and margining.

Interpretation

A is correct. It moves the program beyond simple current-exposure stresses: aggregation with loans becomes possible and counterparty credit quality enters the result.

Why the other options are wrong

B is incorrect. Absolute claim / scope error: the sum of stressed current exposures "would represent the loss that would occur if every counterparty defaulted in the stress scenario"; and current exposure ignores credit quality, so it cannot be compared with an expected-loss stress on loans.
C is incorrect. Polarity flip: CCR is highly nonlinear, and "at-the-money portfolios with large price moves applied to the portfolio are especially prone to errors from using delta approximations".
D is incorrect. Absolute claim: the bilateral formula "may lead to counterintuitive results such as losses occurring because the firm's own credit quality improves"; to capture the full impact of a scenario on CVA P&L the liability side (DVA) should be included.

LO: CR-21 d: Differentiate between stressed expected loss and stress loss of a credit portfolio, and calculate the stress loss on a loan portfolio and the stress loss on a derivative portfolio. CR-21 h: Describe the common pitfalls in stress testing CCR. · Reference: Lynch, The Evolution of Stress Testing Counterparty Exposures, Sections 21.3, 21.4, 21.5, 21.6

Question C9.2

IM-15 a, b, c; ORR-17 b · S2, S3 · [Map #38]

A portfolio risk manager at a London asset manager is drafting internal guidance on building plausible stress scenarios, drawing on the market-driven scenario (MDS) approach and on lessons from the supervisory bank stress tests. Which of the following statements should the manager keep in the guidance?

A. The scenario z-score measures how consistent a set of policy shocks is with the chosen covariance matrix, so a value close to zero signals a coherent scenario.

B. The set of policy variables should be kept large, so that the regression that implies the nonpolicy factor shocks leaves less room for arbitrariness.

C. The 2009 SCAP achieved coherence with a rich scenario of more than 70 trading-book risk factors and eight macro factors for each participating country.

D. For a market crash scenario, the covariance matrix should be estimated from periods with similar crashes, since correlations among risky assets spike in crises.

Solution

Answer: D

Step 1 – Covariance choice (IM-15 b, c)

Z-scores and implied nonpolicy shocks both depend on the covariance matrix. "If you are modeling a market crash, you should use data or a covariance matrix from periods that experienced similar crashes, because correlations among risky assets typically spike during market crises. Relying only on 'normal' data would likely understate the severity of the stress."

Step 2 – Map the other statements to their true owners

Consistency with the covariance structure belongs to the correlation z-score; a small policy set is the MDS rule; the 70-plus factor scenario belongs to EBA 2011, while SCAP used GDP, unemployment and house prices plus a historical market shock.

Interpretation

D is correct. Its supervisory counterpart in ORR-17 is the same lesson from the other side: a historical shock gives loose coherence but "does not test for anything new", so the data window must match the stress being modelled.

Why the other options are wrong

A is incorrect. Sibling swap: the scenario z-score measures severity (lower means more plausible). The *correlation* z-score normalizes for magnitude and shows consistency with the covariance matrix: near zero consistent, above

1.0 a warning.

B is incorrect. Polarity flip: "keeping the number of policy variables small helps minimize arbitrariness in the resulting perturbation vector"; a large set invites multicollinearity (do not add a variable with adjusted R^2 of 90% or more).

C is incorrect. Sibling swap (case cross-mapping): over 70 trading-book factors and eight macro factors for each of 21 countries describe the 2011 EBA test. The 2009 SCAP state space had only three dimensions with a historical market shock from June to December 2008.

LO: IM-15 b: Describe the scenario z-score, the volatility z-score, the correlation z-score, and the Mahalanobis distance, and explain how these are used in estimating the likelihood of occurrence and magnitude of scenarios. IM-15 c: Explain steps in the process of defining and developing market-driven scenarios, and describe the application of the MDS approach in the example provided. ORR-17 b: Explain challenges in designing stress test scenarios, including the problem of coherence in modeling risk factors. · Reference: Golub (ed.), BlackRock's Guide to Fixed-Income Risk Management, Ch.5 (via SchweserNotes Book 5, Reading 97, LO 97.b and 97.c); Schuermann, Stress Testing Banks, Section 17.3

Tier D: Thin Evidence

Map Tier D topics, including notes-only sections for entries with no reported question.

D1 CIR and Lognormal Short-Rate Models

Readings: MR-15 d, e, f (and a, b, c for context) · The Art of Term Structure Models: Volatility and Distribution (Tuckman and Serrat, Fixed Income Securities 3e, Ch.10); MR-14 c, d, f · The Art of Term Structure Models: Drift (Ch.9) for the Vasicek and Ho-Lee siblings

MAP TIER D · #39 · 4/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #39: "Reported: 'CIR model solving'. CIR: mean-reverting, basis-point volatility proportional to $\sqrt{r}$, rates cannot go negative. Lognormal with deterministic drift: volatility proportional to r. This three-way set against Vasicek (mean-reverting, can go negative) and Ho-Lee (no mean reversion, normal) is the highest-yield sibling swap in Market Risk."
Shapes covered here:
S1 CIR model solving: one step of the short-rate change with drift, volatility and a random draw (Q1)
S2 CIR basis-point volatility $\sigma\sqrt{r}$ versus the constant parameter σ (Q1, Q2)
S3 non-negativity: CIR and lognormal versus normal models (Q2)
S4 lognormal models: basis-point volatility proportional to r, with deterministic drift or with mean reversion (Q2)
S5 the sibling set against Vasicek and Ho-Lee (Q2).

Revision notes

- **Why volatility should depend on the rate** (MR-15 d): earlier models assume basis-point volatility independent of the rate level, which is "almost certainly not true at extreme levels". Periods of high inflation and high short rates are inherently unstable, so basis-point volatility tends to be high; at very low rates it is limited because rates cannot decline much below zero.
- **CIR process** (MR-15 d, S1): risk-neutral dynamics $dr = k(\theta - r)dt + \sigma\sqrt{r}\,dw$. The drift term is not random and the standard deviation of dw is $\sqrt{dt}$, so the annualised standard deviation of dr, the basis-point volatility, is $\sigma\sqrt{r}$. The parameter σ is constant; basis-point volatility is not, and increases with the level of the short rate (S2).
- **Proportional volatility** (MR-15 d, S4): Courtadon $dr = k(\theta - r)dt + \sigma r\,dw$ and the simplest lognormal model, Model 4, $dr = ar\,dt + \sigma r\,dw$. Here σ is called *yield volatility*: constant as a percentage of the rate, while basis-point volatility σr rises in proportion to the rate.
- **Units and calibration** (MR-15 e): basis-point volatility is in rate units (100 bp); yield volatility is a percentage of the rate (12.5%). Setting all three specifications to 100 bp at an 8% rate: $\sigma_{bp} = 0.01$; $\sigma_{\text{CIR}}\sqrt{8\%} = 1\% \Rightarrow \sigma_{\text{CIR}} = 0.0354$; $\sigma_Y \times 8\% = 1\% \Rightarrow \sigma_Y = 12.5\%$. CIR and proportional volatility both rise with the rate but at different speeds, and both equal zero at a zero rate (Figure 15.2).
- **Why CIR rates cannot go negative** (MR-15 d, S3): basis-point volatility equals zero when the rate is zero, and the drift is positive when the rate is zero. This is "in some respects" an improvement over constant-volatility models that allow negative rates, but model choice depends on purpose: a trader who believes constant volatility fits the current environment, that negative rates barely affect the securities, and who values computational simplicity may well accept some probability of negative rates.
- **Distribution shape matters** (MR-15 d): with the same mean (5%) and standard deviation (2.32%) after 10 years, the CIR, normal and lognormal terminal distributions differ enough away from the mean to give significantly different out-of-the-money option prices. The shape of the distribution is "an important determinant of that model's performance".
- **Salomon Brothers model: lognormal with deterministic drift** (MR-15 f, S4): $dr = a(t)r\,dt + \sigma r\,dw$; by Ito's lemma $d[\ln r] = [a(t) - \frac{1}{2}\sigma^2]dt + \sigma\,dw$, so $\ln r$ is normal and r is lognormal. It is "the Ho-Lee model based on the natural logarithm of the short rate". Tree nodes: $r_0 e^{a_1 dt \pm \sigma\sqrt{dt}}$.
- **Multiplicative perturbations** (MR-15 f): because $e^x > 0$, every node is positive whenever $r_0 > 0$. Since $e^x \approx 1 + x$, the up node is about $r_0(1 + \sigma)$ for $a_1 = 0$, $dt = 1$: with $\sigma = 12.5\%$ the up-state rate is 12.5% above the initial rate, which is why lognormal volatility is quoted as a percentage of the rate. The drift constants a_1, a_2 are fitted to bond prices, as in Ho-Lee.

- **Black-Karasinski: lognormal with mean reversion** (MR-15 f, S4): $d[\ln r] = k(t)(\ln\theta(t) - \ln r)dt + \sigma(t)dw$: the log of the rate follows a time-dependent Vasicek process, reverting to $\ln\theta(t)$ at speed $k(t)$; it is in the arbitrage-free class. The tree recombines only if $k(2) = (\sigma(1) - \sigma(2))/(\sigma(1)dt)$, which would tie mean reversion to the volatility function; instead the model varies the time step: $k(2) = \frac{1}{dt_2}\left[1 - \frac{\sigma(2)\sqrt{dt_2}}{\sigma(1)\sqrt{dt_1}}\right]$. Mean reversion controls the term structure of volatility; time-dependent volatility controls future short-rate volatility (option prices by expiry).
- **Time-dependent volatility, Model 3** (MR-15 a, b, c): $dr = \lambda(t)dt + \sigma e^{-\alpha t}dw$ fits caplet prices across expiries; its horizon standard deviation resembles Vasicek's, but it remains a parallel-shift model with a flat (time-varying) term structure of volatility, and relies on the market forecasting volatility in the distant future. See also A7.
- **Siblings from MR-14** (S5): Ho-Lee $dr = \lambda(t)dt + \sigma\, dw$: normal, time-dependent drift fitted to bond prices, no mean reversion, parallel shifts. Vasicek $dr = k(\theta - r)dt + \sigma\, dw$: normal with mean reversion toward θ, constant basis-point volatility, so rates can become negative; mean reversion gives a downward-sloping term structure of volatility and a non-parallel factor. Methods for negative rates (MR-14 c): change the distribution (CIR, lognormal) or set negative tree rates to zero. See also A7 and A25.
- **Inflation link:** the economic argument for rate-dependent volatility (high inflation, high rates, high basis-point volatility) is also the stress channel in D18.

Formulas

CIR (eq. 15.3). $dr = k(\theta - r)dt + \sigma\sqrt{r}\, dw, \qquad dw = z\sqrt{dt}, \qquad \text{bp vol} = \sigma\sqrt{r}$
One step. $\Delta r = k(\theta - r)\, dt + \sigma\sqrt{r}\, z\sqrt{dt}$
Courtadon (15.4) and Model 4 (15.5). $dr = k(\theta - r)dt + \sigma r\, dw; \qquad dr = ar\, dt + \sigma r\, dw; \qquad \text{bp vol} = \sigma r$
Calibration to a common bp vol at rate r^* (15.6 to 15.8). $\sigma_{\text{CIR}} = \sigma_{bp}/\sqrt{r^*}, \qquad \sigma_Y = \sigma_{bp}/r^*$ (book: 0.0354 and 12.5% at 8%)
Scaling of bp vol with the rate. normal: constant; CIR: $\propto \sqrt{r}$; lognormal: $\propto r$
Salomon Brothers (15.9, 15.11). $dr = a(t)r\, dt + \sigma r\, dw, \qquad d[\ln r] = \left[a(t) - \frac{1}{2}\sigma^2\right]dt + \sigma\, dw$
Black-Karasinski (15.15). $d[\ln r] = k(t)\big(\ln\theta(t) - \ln r\big)dt + \sigma(t)\, dw$

Exam traps

- **σ is not the bp vol in CIR:** the constant parameter (for example 0.05) must be multiplied by $\sqrt{r}$ (95 bp at 3.6%), not read as 500 bp.
- **Unit/time conversion:** drift scales with dt, the random term with $\sqrt{dt}$. Forgetting dt on the drift or $\sqrt{dt}$ on the shock are separate named mistakes.
- **Sibling swap $\sqrt{r}$ versus r:** CIR basis-point volatility rises with the square root of the rate; lognormal (Courtadon, Model 4, Salomon Brothers, Black-Karasinski) rises in proportion to the rate.
- **Mean reversion does not prevent negative rates:** Vasicek mean-reverts but has constant basis-point volatility and a normal distribution, so rates can go negative. Non-negativity comes from volatility falling to zero at zero (CIR) or from multiplicative shocks (lognormal).
- **Salomon Brothers versus Black-Karasinski:** deterministic drift only versus mean reversion in the log rate. Ho-Lee has no mean reversion and no long-run level.
- **Absolute claim:** "CIR is always the better model" is false; the book's trader may rationally choose a constant-volatility model.

Question D1.1

MR-15 d, e · S1, S2 · [Map #39]

A risk analyst at an Australian bank is simulating monthly paths of the short rate under the risk-neutral Cox-Ingersoll-Ross (CIR) process $dr = k(\theta - r)dt + \sigma\sqrt{r}\, dw$, where dw is normal with mean zero and standard deviation $\sqrt{dt}$. For the first step of one path the inputs are:

- Current short rate: 3.60%; long-run value θ: 5.00%; speed of mean reversion k: 0.40
- Volatility parameter σ: 0.05

- Time step: one month ($dt = 1/12$); the standard normal draw for this step is $+1.2$

Which of the following is closest to the change in the short rate over the month and to the annualized basis-point volatility of the short rate at the start of the month?

A. $+37.5$ bp; 94.9 bp

B. $+88.9$ bp; 94.9 bp

C. $+118.5$ bp; 94.9 bp

D. $+177.9$ bp; 500.0 bp

Solution **Answer: A**

Step 1 – Basis-point volatility

$$\sigma\sqrt{r} = 0.05 \times \sqrt{0.036} = 0.05 \times 0.18974 = 0.0094868 = 94.9 \text{ bp}$$

⇒ At a 3.60% short rate, one standard deviation of annual rate change is about 95 bp. The parameter 0.05 is not itself a basis-point volatility.

Step 2 – Drift over the month

$$k(\theta - r)\,dt = 0.40 \times (0.0500 - 0.0360) \times \tfrac{1}{12} = 0.000467 = 4.67 \text{ bp}$$

⇒ The rate is below θ, so mean reversion pulls it up.

Step 3 – Random term

$$dw = 1.2\sqrt{1/12} = 0.34641, \qquad \sigma\sqrt{r}\,dw = 0.0094868 \times 0.34641 = 0.003286 = 32.86 \text{ bp}$$

Step 4 – Rate change

$$\Delta r = 4.67 + 32.86 = 37.53 \text{ bp}, \qquad r_1 = 3.975\%$$

Interpretation

At the new rate the basis-point volatility is $0.05\sqrt{0.03975} = 99.7$ bp: CIR volatility rises with the square root of the rate. Had the draw driven the rate toward zero, volatility would have shrunk toward zero while the drift $k\theta > 0$ stayed positive, so the rate cannot become negative.

Why the other options are wrong

B is incorrect. Did not scale the drift by dt: $0.40 \times 0.014 = 0.0056$, plus 0.003286, gives 88.9 bp.
C is incorrect. Did not scale the shock by $\sqrt{dt}$: $0.000467 + 0.0094868 \times 1.2 = 0.011851$, or 118.5 bp.
D is incorrect. Treated the constant parameter $\sigma = 0.05$ as the basis-point volatility (500 bp) and omitted $\sqrt{r}$: $0.000467 + 0.05 \times 0.34641 = 0.017787$, or 177.9 bp.

LO: MR-15 d: Describe the short-term rate process under the Cox-Ingersoll-Ross (CIR) and lognormal models. MR-15 e: Calculate the short-term rate change and describe the basis point volatility using the CIR and lognormal models. · Reference: Tuckman and Serrat 3e, Ch.10, 15.2, eqs. 15.3 and 15.6 to 15.8

Question D1.2 MR-15 d, f; MR-14 c, f · S2, S3, S4, S5 · [Map #39]

A model validation analyst at a Japanese bank is asked to recommend a one-factor short-rate model for the bank's interest rate desk. The desk sets three requirements: (i) the short rate cannot become negative; (ii) the basis-point volatility of the short rate rises with the level of the rate, but less than proportionally; and (iii) the short rate is pulled toward a long-run level. Which recommendation, with its justification, should the analyst make?

A. The Vasicek model, because mean reversion pulls the short rate toward its long-run level and therefore prevents the short rate from becoming negative.

B. The CIR model, because its basis-point volatility rises with the square root of the rate and falls to zero at a zero rate, where the drift is positive.

C. The Black-Karasinski model, because the logarithm of the rate mean-reverts and its basis-point volatility rises with the square root of the rate.

D. The Ho-Lee model, because fitting its time-dependent drift to market bond prices guarantees that no node of

the rate tree becomes negative.

Solution **Answer: B**

Step 1 – Requirement (ii) sorts the volatility specifications

Normal models (Ho-Lee, Vasicek) have constant basis-point volatility; lognormal models (Courtadon, Model 4, Salomon Brothers, Black-Karasinski) have basis-point volatility σr, proportional to the rate; CIR has $\sigma\sqrt{r}$, rising less than proportionally.

Step 2 – Requirement (i)

CIR: basis-point volatility is zero at a zero rate and the drift $k(\theta - 0)$ is positive there, so the rate cannot go negative.

Step 3 – Requirement (iii)

CIR's drift $k(\theta - r)$ pulls the rate toward θ. Only CIR meets all three.

Why the other options are wrong

A is incorrect. Scope/condition error. Vasicek does mean-revert, but its basis-point volatility is constant and rates are normally distributed, so negative rates remain possible; mean reversion only damps the spread of future rates.

C is incorrect. Sibling swap. Black-Karasinski does mean-revert in the log rate and stays positive, but as a lognormal model its basis-point volatility is proportional to the rate, the property CIR does not have.

D is incorrect. Absolute claim. Ho-Lee rates are normal with constant volatility; fitting the drift matches bond prices but does not stop low nodes from going negative, and Ho-Lee has no long-run level at all.

LO: MR-15 d: Describe the short-term rate process under the Cox-Ingersoll-Ross (CIR) and lognormal models. MR-15 f: Describe lognormal models with deterministic drift and mean reversion. MR-14 c: Describe methods for addressing the possibility of negative short-term rates in term structure models. MR-14 f: Describe the process of constructing a simple and recombining tree for a short-term rate under the Vasicek Model with mean reversion. · Reference: Tuckman and Serrat 3e, Ch.10, 15.2 to 15.4; Ch.9, 14.3 and 14.4

D2 Outsourcing and Third-Party Risk

Readings: ORR-12 a, b, c · Guidance on Managing Outsourcing Risk (Federal Reserve SR 13-19/CA 13-21, December 2013; GARP ORR Ch.12); ORR-13 a, b · Case Study: Third-Party Risk Management (GARP ORR Ch.13); ORR-5 g · Risk Mitigation (Ch.5, 5.6 Risk Transfer); ORR-4 g · Risk Measurement and Assessment (Ch.4, KRIs for Resilience); ORR-3 d · Risk Identification (Basel EDPM, Vendors and Suppliers)

MAP TIER D · #51 · 3/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #51: "Reported: 'outsourcing'. Core principle and standing exam trap: a firm may outsource the activity but never the accountability. Concentration risk in critical service providers, and the resilience implications, are the modern angle."
Shapes covered here:
S1 outsource the activity, never the accountability, applied to a vendor and subcontractor failure (Q1)
S2 concentration risk in a critical service provider and its resilience implications (Q2)

Revision notes

The specifics: accountability and concentration (S1, S2)

- **Accountability is not transferable (S1).** SR 13-19: the use of service providers "does not relieve a financial institution's board of directors and senior management of their responsibility to ensure that outsourced activities are conducted in a safe-and-sound manner and in compliance with applicable laws and regulations." ORR-5: outsourcing transfers the execution of a process and some of its operational risks, but "accountability, however, is not transferable"; the bank remains accountable to its customers. Reputational damage cannot really be outsourced. Increasingly, outsourcing is seen as **risk sharing** rather than risk transfer.
- **Outsourcing can add or cut risk.** Outsourcing to cut cost (offshore call centres) is closer to **risk acquisition** in exchange for margin; outsourcing a non-core activity to a better-controlled specialist (payments, cloud data centres) can reduce net operational risk at a higher cost. Either way the firm gains **third-party risk**: the risk of failed controls at the provider. Contracting with providers "does not fully mitigate operational risk".
- **Concentration risk (S2).** SR 13-19 lists six risks from service providers: **compliance**, **concentration** ("outsourced services or products are provided by a limited number of service providers or are concentrated in limited geographic locations"), **reputational**, **country** (foreign-based provider), **operational**, **legal**. In the cyber-resilience reading, concentration risk in this context means a single firm's lack of control over one provider to which many activities are outsourced, not systemic risk to the industry (see A23; AI provider concentration in A14).
- **Resilience link (S2; ORR-4 g).** Single points of failure are resilience KRIs: key supplier dependency, low substitutability and high switching costs (IT and cloud providers, data custodians, payment platforms, utilities). Remove them through back-ups and redundancy, or reflect them in impact tolerances and stress tests. Resilience is organised by **important business service** and impact tolerance; business continuity by process and RTO. Contingency planning (SR 13-19): ensure a DR/BCP exists for the contracted service, assess the provider's plan and its **alignment** with the firm's own, test it periodically, and keep an **exit strategy** with a pool of comparable providers.

The SR 13-19 programme (ORR-12 a)

- **Board and senior management.** Policies approved by the board (or an executive committee of the board) establish the programme: risk assessments, due diligence, contract standards, ongoing monitoring, business continuity. Senior management executes the policies and reports adherence to the board.
- **Six core elements.** (A) risk assessments; (B) due diligence and selection; (C) contract provisions; (D) incentive compensation review; (E) oversight and monitoring; (F) business continuity and contingency plans. The programme is **risk-focused**: it concentrates on activities with a substantial impact on financial condition, critical to operations, involving sensitive customer information or new products, or posing material compliance risk. A community bank with a few critical activities outsourced to reputable providers may use fewer elements than a bank with thousands of providers.

- **Risk assessment and incentives.** Decide first whether outsourcing fits strategy, then weigh benefits, risks, provider risk, cost, availability of qualified providers and the firm's own ability to oversee the relationship; update periodically. Incentive compensation in provider contracts must be reviewed: variable fees that steer customers to higher-margin products without regard to suitability are inappropriate (see D21).
- **Oversight and escalation.** Metrics set by the business line or relationship management; overseers with expertise and stature; higher-risk providers get more frequent and stringent monitoring and a designated point of contact; lower-risk providers less. Monitor financial condition (and subcontractor controls), and for significant relationships the control environment, including SOC 2 reports. Triggers escalate oversight, on-site reviews and the right to audit, with criteria for alternatives and termination.
- **Additional considerations.** SAR functions (Bank Secrecy Act confidentiality); foreign providers (US law, on-site review rights, home-supervisor access to customer data); internal audit outsourcing (Sarbanes-Oxley bars the external auditor from certain non-audit services); outsourced risk management, including model validation, must follow SR 11-7 (see C6).

Due diligence and contracts (ORR-12 b, c)

- **Due diligence before engagement.** Depth depends on scope, complexity, importance, familiarity and reputation. Three areas: (1) business background, reputation and strategy (history, principals, background checks on employees, references, licences, pending litigation, resiliency of the business model); (2) financial performance and condition (statements, sustainability, effect of the relationship on the provider, commitment for the contract term, insurance, review of subcontractors); (3) operations and internal controls (facilities, training, systems security, privacy, records, business resumption, background checks, legal compliance).
- **Contract provisions.** Reviewed by legal counsel before execution. Scope (rights and responsibilities, including the ability to subcontract); cost and compensation (no incentives to take imprudent risks); right to audit; measurable performance standards; confidentiality and security (NPPI, breach notification, GLBA 501(b)); ownership and licence (escrow for source code); indemnification; default and termination (with time to transfer services and return of data); dispute resolution; limits on liability (the board judges reasonableness); insurance; customer complaints; provider's business resumption plan; foreign providers (choice of law and jurisdiction); **subcontracting**: the same provisions apply to the subcontractor and the primary provider has overall accountability for all services it and its subcontractors provide.

TPRM and the cases (ORR-13 a, b)

- **TPRM scope and life cycle.** Third parties include vendors, suppliers, partners, contractors; fourth parties (subcontractors) and fifth parties extend the chain. "Third-party failure" is a Level 2 category of Basel's seventh event type (EDPM); ORX makes third-party risk Level 1. Shared Assessments life cycle: business model decision; evaluation, risk rating and due diligence (proportionate; SIG questionnaires); contracts, SLAs and contract management (sign only after open items are remediated; impose standards on vendor outsourcing to address fourth parties; audit rights); continuous monitoring with **trigger events** for reassessment; remediation or termination with an exit strategy. Risk-based vendor rating criteria: process criticality, data access, system access, regulated sector, **substitutability**.
- **Capital One and AWS (2019).** About 100 million people affected; a former AWS employee exploited a misconfigured web application firewall (SSRF); data were not encrypted because of Capital One's own configuration. OCC fine USD 80 million for "failure to establish effective risk assessment processes prior to migrating significant information technology operations to the public cloud". Lesson: cloud providers offer the controls, but the customer must use them; risk management stays with the business.
- **Morgan Stanley data-centre decommissioning (2016, 2019).** OCC fine USD 60 million: the bank failed to assess the risks of decommissioning hardware, to assess the risk of vendors *including subcontractors*, and to keep an inventory of customer data on its devices; inadequate vendor due diligence and monitoring. Lesson: outsourcing may transfer some operational risk, but the accountability is not transferable.

Exam traps

- **Role misassignment (the standing trap):** contracts, indemnities and subcontracting clauses allocate duties between firm and provider; they never move the board's and senior management's accountability to the provider, the subcontractor or the supervisor.
- **Absolute claim:** outsourcing does not eliminate or fully mitigate operational risk; it trades it for third-party risk and is better seen as risk sharing.

- **Polarity flip on oversight:** higher-risk, critical or data-sensitive providers get *more* frequent and stringent monitoring, whatever the provider's reputation.
- **Scope/condition error on concentration:** SR 13-19 concentration means few providers or few geographic locations for the firm's services, a firm-level risk; it is not only an industry-wide systemic concern left to supervisors.
- **Sibling swap, BCM versus resilience:** business continuity works by process and RTO; operational resilience works by important business service and impact tolerance. Case cross-mapping: Capital One (cloud misconfiguration, no pre-migration risk assessment) versus Morgan Stanley (vendor and subcontractor oversight in decommissioning).

Question D2.1 ORR-12 a, c; ORR-13 a, b · S1 · [Map #51]

The chief compliance officer of an Australian wealth manager is briefing the board after a vendor engaged to decommission two client data servers passed the work to a subcontractor, and the hardware was later resold with client data still on it. The firm had not assessed the subcontractor and kept no inventory of the client data held on the servers. The CEO argues that because the master services agreement transferred the decommissioning task, and states that the vendor has overall accountability for its subcontractors, the regulatory exposure now sits with the vendor. Which of the following statements should the CCO include in the briefing?

A. Because the agreement gives the vendor overall accountability for its subcontractors, the firm's own accountability to its clients and its regulator for the breach passes to the vendor.

B. Outsourcing the decommissioning to a specialist eliminated the operational risk of the activity for the firm, leaving only the vendor's performance against the contract to be managed.

C. As decommissioning is infrequent and done by a specialist, lighter oversight than for routine processes was appropriate, so not reviewing the subcontractor was a proportionate choice.

D. Outsourcing can transfer the task and some of its operational risk, but the board and senior management remain responsible for its safe and lawful conduct; the accountability is not transferable.

Solution **Answer: D**

Step 1 – What the contract can and cannot do

The subcontracting provision makes the primary provider accountable *to the firm* for services delivered by its subcontractors. SR 13-19 is explicit that the use of service providers does not relieve the board and senior management of responsibility for safe-and-sound, lawful conduct of the outsourced activity.

⇒ *The clause allocates duties between the two companies; it does not move the firm's accountability to its clients or its regulator.*

Step 2 – Compare with the GARP case

The facts mirror the Morgan Stanley case: the OCC fined the bank USD 60 million for failing to assess the risks of decommissioning, to assess vendors including subcontractors, and to keep an inventory of customer data stored on the devices. The case's stated lesson is that outsourcing may transfer some operational risk, but accountability is not transferable.

Step 3 – What the programme required

The activity involved sensitive customer information, which puts it squarely inside the risk-focused part of the programme: due diligence that covers subcontractors, contract terms on data security, return and destruction, and monitoring scaled to the risk.

Interpretation

D is correct. The CCO should tell the board that the firm owns the breach, must remediate (customer notification, data inventory, subcontractor due diligence standards in vendor contracts), and may pursue the vendor contractually, for example under indemnification, without that changing who answers to the regulator.

Why the other options are wrong

A is incorrect. Role misassignment: the subcontracting clause makes the vendor accountable to the firm for its subcontractors; it does not transfer the firm's accountability to clients and regulators, which remains with its board and senior management.

B is incorrect. Absolute claim: contracting with a third party does not fully mitigate operational risk; it adds third-party risk, and outsourcing is better seen as risk sharing than risk transfer.

C is incorrect. Polarity flip: the programme is risk-focused on activities that involve sensitive customer information

or are critical; such activities call for *more* rigorous due diligence and monitoring, including of subcontractors, not less.

LO: ORR-12 a: Explain how risks can arise through outsourcing activities to third-party service providers and describe elements of an effective program to manage outsourcing risk. · ORR-12 c: Describe topics and provisions that should be addressed in a contract with a third-party service provider. · ORR-13 a: Explain how risks related to the use of third parties can arise and describe characteristics of an effective third-party risk management framework. · ORR-13 b: Describe the lessons learned from the presented case studies. · Reference: Federal Reserve SR 13-19 (GARP ORR Ch.12, 12.3 and 12.4 C, Subcontracting); GARP ORR Ch.13, OCC Fines Morgan Stanley; Ch.5, 5.1 Risk Response

Question D2.2

ORR-12 a; ORR-13 a; ORR-4 g · S2 · [Map #51]

An operational risk manager at a Canadian bank is reviewing the bank's third-party register. Three important business services (retail payments, client data hosting and the mortgage servicing platform) all run on one cloud provider from a single geographic region. The bank's risk-based vendor rating scores the provider "critical business operations would be halted without the vendor" on process criticality and "none or very limited" on substitutability. Which of the following recommendations should the manager make?

A. Concentration risk here refers only to systemic risk for the banking industry from many banks using the same provider, so it is a matter for supervisors rather than for the bank to manage.

B. Because the provider is a large, reputable firm with strong built-in security, the bank can reduce the frequency of its monitoring and place reliance on the provider's own controls.

C. The bank should set a recovery time objective for each internal process the provider supports, since under resilience rules the process-level RTO is the unit of tolerance to be tested.

D. The bank should treat the provider as a single point of failure: map each important business service end to end, track key-supplier dependency as a KRI, and keep an exit strategy with comparable providers.

Solution

Answer: D

Step 1 – Identify the risk

SR 13-19: concentration risk arises when outsourced services are provided by a limited number of providers or are concentrated in limited geographic locations. Here one provider in one region supports three important business services.

⇒ *This is a firm-level concentration the bank must manage, not only an industry issue.*

Step 2 – Read the vendor rating

Process criticality at the top level and substitutability "none or very limited" put the provider in the highest risk class, which calls for the most frequent and stringent monitoring, a designated point of contact and a tailored mitigation plan.

Step 3 – The resilience response

Single points of failure, including key supplier dependency with low substitutability, are KRIs for resilience; they should be removed through back-ups and redundancy where possible, otherwise reflected in impact tolerances and stress tests. Resilience is organised around important business services, mapped end to end, with impact tolerances. Contingency planning requires an exit strategy with a pool of comparable providers and alignment of the provider's DR/BCP with the bank's own.

Interpretation

D is correct. The recommendation reduces dependency where it can (redundancy, multi-region or second provider), measures what remains, and prepares the exit.

Why the other options are wrong

A is incorrect. Scope/condition error: SR 13-19 defines concentration as few providers or few locations for the firm's own services, and the cyber-resilience reading stresses the firm's lack of control over a single provider, not systemic risk; the bank's board remains responsible for managing it.

B is incorrect. Polarity flip: the risk-based rating makes this a higher-risk provider, which requires *more* frequent and stringent monitoring; Capital One shows that built-in cloud controls still have to be used and checked by the customer.

C is incorrect. Sibling swap: process-level recovery time objectives belong to business continuity management; operational resilience is articulated around important business services and their impact tolerances.

LO: ORR-12 a: Explain how risks can arise through outsourcing activities to third-party service providers and describe elements

of an effective program to manage outsourcing risk. · ORR-13 a: Explain how risks related to the use of third parties can arise and describe characteristics of an effective third-party risk management framework. · ORR-4 g: Describe and explain the steps to ensure a strong level of operational resilience, and to test the operational resilience of important business services. · Reference: Federal Reserve SR 13-19 (GARP ORR Ch.12, 12.2, 12.4 E and F); GARP ORR Ch.13, Risk-Based Vendor Rating and Capital One case; Ch.4, 4.5 Quantification of Operational Resilience (KRIs for Resilience)

D3 Regulatory vs Economic Capital and Capital Adequacy

Readings: CR-4 a, b, c, d, g, h, i · Capital Structure in Banks (Schroeck Ch.5, Box 4.1); CR-5 a, b, c · Introduction to Credit Risk Modeling and Assessment (Doumpos et al., Ch.1, 5.4 and 5.5); ORR-18 a · Risk Capital Attribution and Risk-Adjusted Performance Measurement (Crouhy, Galai and Mark, Ch.17, 18.1)

MAP TIER D · #61 · 3/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #61: "Reported: 'capital structure in banks'; 'unexpected loss, regulatory capital, economic capital.' The distinction, regulatory capital is a formula-driven minimum, economic capital is the institution's own UL-based estimate, is the discrimination. Risk-weighted assets and the capital adequacy ratio (LO 23.c) are the computational partner."
Shapes covered here:
S1 regulatory capital as a formula-driven minimum versus economic capital as the bank's own estimate (Q2)
S2 unexpected loss into economic capital: EC $=$ UL$_p$ $\times$ CM, loan level ULC$_i$ $\times$ CM (Q2; red-herring EC input in Q1)
S3 risk-weighted assets (standardized weights plus IRB $K \times 12.5 \times$ EAD) and the capital adequacy ratio against a minimum (Q1).

Revision notes

- **Regulatory versus economic capital** (ORR-18 a, S1): regulatory capital applies only to a few regulated industries (banking, insurance) where regulators protect small depositors or policyholders; it is "calculated according to a set of industrywide rules and formulas and sets only a minimum required level of capital adequacy". It rarely captures the true level of risk, and the gap to risk capital "can be quite wide". Even if the two are similar at firm level, they may differ by business line; some practitioners allocate the *higher* of the two. Under Basel III, for activities such as securitization regulatory capital may end up much higher than economic capital, and the activity is then likely to migrate to the shadow banking sector.
- **Economic capital defined** (CR-4 a, Box 4.1): "an estimate of the overall capital reserve needed to guarantee the solvency of a bank for a given confidence level", with the confidence level set consistent with the bank's target credit rating. Crouhy: risk capital is often called economic capital, and the two are generally treated as identical.
- **The two drivers of EC** (CR-4 a): the confidence level and the riskiness of the assets. A bank targeting a very high rating chooses a very high confidence level (e.g. 99.97%), which means a *higher* capital multiplier (CM); a riskier asset book has a higher UL and needs more EC for the same confidence level. Holding less capital frees it to earn returns elsewhere, so EC balances risk and return.
- **Expected loss** (CR-4 c): EL $=$ PD $\times$ EA $\times$ LR. It is the foreseeable cost of doing business, covered by loan loss reserves (standard risk costs) and adequate loan pricing; it is the long-run average loss across typical economic conditions, not next year's cyclical forecast. "By definition, EL does not itself constitute risk."
- **The three factors** (CR-4 b): PD is borrower-specific, linked to the rating, usually annual (migration matrices give multiperiod PDs; second-year PD is typically higher for initially low PDs); EA is the expected exposure at default including all outstanding payments and interest; LR (severity, LGD, LIED) is largely a function of collateral and equals $1-$ recovery rate.
- **Unexpected loss** (CR-4 d, e): the standard deviation of credit losses around EL, UL $= \text{EA}\sqrt{\text{PD}\,\sigma_{LR}^2 + \text{LR}^2\sigma_{PD}^2}$, with $\sigma_{PD}^2 = \text{PD}(1-\text{PD})$ from the binomial default indicator. For the loss rate, the binomial overstates the variance, the uniform assumes any loss from 0 to 100% is equally likely, and the normal is the most reasonable assumption. ULs "cannot be adequately priced for in a loan's interest rate": they need a capital cushion.
- **Portfolio UL and contributions** (CR-4 f): for n similar loans UL$_p$ $=$ UL$_i\sqrt{n+\rho(n^2-n)}$, and ULC$_i$ $\approx$ UL$_i\sqrt{\rho}$ for large n; the ULCs sum to UL$_p$. See also A1 for the full numerical treatment.
- **Deriving economic capital** (CR-4 g, S2): EC is the distance between the expected outcome and the unexpected outcome at the chosen confidence level, EC$_p$ $=$ UL$_p$ $\times$ CM and EC$_i$ $=$ ULC$_i$ $\times$ CM: a loan's capital is proportional to its contribution to portfolio risk. Schroeck's footnote: "contrary to the regulatory approach

that assigns roughly 8% equity capital to credits on a standalone basis", this reflects both borrower quality and diversification.

- **Modeling the loss distribution** (CR-4 h): credit losses are highly skewed (upside capped at the promised payments), so the crucial task is the choice of distribution for the tail. The beta distribution (a continuous approximation of the binomial) is fully characterised by EL_p and UL_p; tail fitting combines it with Monte Carlo simulation. At 99.97%, CM is between 7.0 and 7.5, far above the multiples for normally distributed market risk. The beta overestimates EC for high-quality portfolios ($\text{EL}_p > \text{UL}_p$) and has too thin a tail for lower-quality ones.
- **Challenges** (CR-4 i): the approach treats credits as illiquid (a value approach would suit traded credit better); multi-period modeling is cumbersome, so almost all practical models use a one-year horizon; other risk types are assumed to be measured separately. Country risk is "lumpier", with fewer diversification benefits.
- **Capital adequacy ratio** (CR-5 c, S3): $\text{CAR} = \text{Capital}/\text{RWA} \geq \alpha$, with $\alpha = 8\%$ under Basel II; capital is Tier 1 plus Tier 2. Tier 1 is core capital (equity and reserves), the primary source for unexpected losses; Tier 2 is supplementary and "not as secure". Doumpos: Basel III imposes higher standards (CAR $\geq 10.5\%$), which matches ORR-22's 8% total capital plus the 2.5% capital conservation buffer. See also D25 for the Basel III tiers.
- **Risk-weighted assets** (CR-5 c): under the standardized approach, prescribed risk weights set by the supervisor apply to all cases and risk assessment relies on external models; it is *costlier* because capital requirements are higher to account for its ambiguity. Under IRB, PD comes from internal rating/scoring systems; LGD is set by supervisors (foundation) or modeled internally (advanced); $\text{RWA} = K \times 12.5 \times \text{EAD}$, with K from the ASRF model, which assumes a completely diversified portfolio so only systematic risk matters. Asset correlation falls with PD and rises with firm size; the maturity adjustment rises with maturity and is larger for low-PD borrowers; granularity adjustments (positive or negative) address residual concentration. See also A31.
- **CAMEL** (CR-5 a): Capital, Assets, Management, Earnings, Liquidity; the most widely used empirical evaluation system in early credit risk management. It was soon realised that such empirical systems cannot provide a solid and objective basis, which led to analytical models, encouraged by regulators through relaxed capital requirements for sophisticated models.
- **Elements of credit risk modeling** (CR-5 b): $\text{EL} = \text{PD} \times \text{EAD} \times \text{LGD}$ over (usually) one year; the standard default definition is a payment delay of at least 90 days; EAD depends mostly on the loan rather than the obligor (fixed for a bond, variable for a credit card); $\text{LGD} = 100\% -$ recovery rate, increasingly from statistical models since Basel II.

Formulas

Expected and unexpected loss (Schroeck 4.1, 4.2, 4.3). $\text{EL} = \text{PD} \cdot \text{EA} \cdot \text{LR}$, $\text{UL} = \text{EA}\sqrt{\text{PD}\,\sigma_{LR}^2 + \text{LR}^2\sigma_{PD}^2}$, $\sigma_{PD}^2 = \text{PD}(1-\text{PD})$

Portfolio and contribution (4.13, 4.15). $\text{UL}_p = \text{UL}_i\sqrt{n + \rho(n^2-n)}$, $\text{ULC}_i \approx \text{UL}_i\sqrt{\rho}$

Economic capital (4.17, 4.18). $\text{EC}_p = \text{UL}_p \times \text{CM}$, $\text{EC}_i = \text{ULC}_i \times \text{CM}$ (CM about 7.0 to 7.5 at 99.97%)

Capital adequacy ratio (Doumpos 5.5). $\text{CAR} = \dfrac{\text{Tier 1} + \text{Tier 2}}{\text{RWA}} \geq \alpha$ (8% Basel II; 10.5% Basel III)

Standardized RWA. $\text{RWA} = \sum_j w_j \times \text{Exposure}_j$ with prescribed weights w_j

IRB RWA. $\text{RWA} = K \times 12.5 \times \text{EAD}$ ($12.5 = 1/8\%$), where

$$K = \left[\text{LGD} \cdot N\left(\frac{N^{-1}(\text{PD}) + \sqrt{R}\,N^{-1}(0.999)}{\sqrt{1-R}}\right) - \text{PD} \cdot \text{LGD}\right]\frac{1 + (M-2.5)\beta}{1 - 1.5\beta}$$

with maturity adjustment $\beta = \left[0.11852 - 0.05478\ln(\text{PD})\right]^2$.

Exam traps

- **Sibling swap, regulatory versus economic:** "industrywide rules and formulas" and "only a minimum" describe *regulatory* capital; "own estimate at a confidence level tied to the target rating" describes *economic* capital.
- **Polarity of the target rating:** a higher target rating means a higher confidence level, a higher CM and *more* economic capital, not less.

- **EL is not in EC (scope error):** EL is covered by pricing and loan loss reserves; EC is the distance from EL to the confidence level, $\text{UL}_p \times \text{CM}$.
- **Dropped 12.5 (dropped term):** K is a capital percentage; $K \times \text{EAD}$ is capital, not RWA. Forgetting 12.5 shrinks RWA and inflates the CAR.
- **Wrong numerator or denominator:** the CAR uses Tier 1 *plus* Tier 2 over RWA. Tier 1 alone gives the Tier 1 ratio; dividing by unweighted total assets gives a leverage-style ratio (ORR-22 f sibling).
- **IRB exposure at face value (wrong input):** an IRB book enters RWA through $K \times 12.5$, not at a 100% weight on EAD.

Question D3.1 CR-5 c; CR-4 g · S2, S3 · [Map #61]

A regulatory reporting analyst at a Singapore bank is preparing the year-end capital adequacy return. The bank's banking book and capital are summarised below (SGD million):

- Corporate loans under foundation IRB: EAD 2,000; IRB capital requirement K = 7.5% of EAD
- Residential mortgages under the standardized approach: exposure 3,000; prescribed risk weight 35%
- Claims on the domestic sovereign: exposure 1,500; prescribed risk weight 0%
- Claims on other banks: exposure 800; prescribed risk weight 20%
- Tier 1 capital: 262; Tier 2 capital: 70
- Internal economic capital for credit risk ($\text{UL}_p \times \text{CM}$ at 99.97%): 300

The supervisor applies the Basel III minimum capital adequacy ratio of 10.5%. Which of the following is closest to the bank's risk-weighted assets and capital adequacy ratio, together with the correct conclusion?

A. RWA 7,300; CAR 4.5%; the bank falls short of the 10.5% minimum

B. RWA 3,085; CAR 8.5%; the bank falls short of the 10.5% minimum

C. RWA 3,210; CAR 10.3%; the bank falls short of the 10.5% minimum

D. RWA 3,085; CAR 10.8%; the bank meets the 10.5% minimum

Solution **Answer: D**

Step 1 – RWA of the IRB book

$$\text{RWA}_{\text{IRB}} = K \times 12.5 \times \text{EAD} = 0.075 \times 12.5 \times 2{,}000 = 1{,}875$$

$\Rightarrow$ The corporate book carries a 93.75% effective risk weight: K is a capital charge, and 12.5 ($= 1/8\%$) converts it back into risk-weighted assets.

Step 2 – RWA of the standardized books

$$\text{RWA}_{\text{SA}} = 0.35(3{,}000) + 0(1{,}500) + 0.20(800) = 1{,}050 + 0 + 160 = 1{,}210$$

$\Rightarrow$ Prescribed weights set by the supervisor: the sovereign book adds nothing, the interbank book only one fifth of its face value.

Step 3 – Total RWA

$$\text{RWA} = 1{,}875 + 1{,}210 = 3{,}085$$

Step 4 – Capital adequacy ratio

$$\text{CAR} = \frac{\text{Tier 1} + \text{Tier 2}}{\text{RWA}} = \frac{262 + 70}{3{,}085} = \frac{332}{3{,}085} = 10.76\%$$

$\Rightarrow$ The bank holds SGD 10.76 of regulatory capital for every SGD 100 of risk-weighted assets, about 0.26 percentage points above the 10.5% minimum.

Interpretation

The bank meets the Basel III level (8% total capital plus the 2.5% conservation buffer), but with a thin cushion of about SGD 8 million ($332 - 0.105 \times 3{,}085 = 8.1$). The SGD 300 million of economic capital is a red herring: it

is the bank's own UL-based estimate at its chosen confidence level and plays no part in the regulatory ratio, which is a formula-driven minimum.

Why the other options are wrong

A is incorrect. Divided capital by unweighted total exposures, $2{,}000 + 3{,}000 + 1{,}500 + 800 = 7{,}300$, giving $332/7{,}300 = 4.5\%$: a leverage-style ratio, not the risk-based CAR.

B is incorrect. Correct RWA, but used Tier 1 only in the numerator: $262/3{,}085 = 8.5\%$. That is the Tier 1 ratio; the CAR counts Tier 1 plus Tier 2.

C is incorrect. Entered the IRB loans at their EAD (an implicit 100% weight) instead of $K \times 12.5 \times \text{EAD}$: RWA $= 2{,}000 + 1{,}210 = 3{,}210$, CAR $= 332/3{,}210 = 10.3\%$, which wrongly flips the conclusion.

LO: CR-5 c: Estimate risk-weighted assets and capital adequacy ratio of a financial institution. CR-4 g: Describe how economic capital is derived. · Reference: Doumpos et al., Ch.1, 5.5; Schroeck, Ch.5, Box 4.1 and eq. 4.17

Question D3.2

CR-4 a, g; ORR-18 a · S1, S2 · [Map #61]

An internal auditor at an Australian bank is reviewing a draft board paper that compares the bank's regulatory capital for credit risk with the economic capital produced by its internal credit portfolio model. The bank targets an AA rating and calibrates its model at the 99.97% confidence level. Which of the following statements is correct for the paper to include?

A. A bank targeting a higher credit rating can choose a lower confidence level, which lowers the capital multiplier applied to unexpected loss and so reduces the economic capital it must hold.

B. Economic capital is calculated with industrywide rules and formulas and sets only a minimum level of capital adequacy, whereas regulatory capital reflects the bank's own estimate of its risk.

C. Economic capital for credit risk is the capital needed to absorb the expected loss plus the unexpected loss up to the confidence level, so it is measured from zero loss rather than from EL.

D. Economic capital at the loan level equals the loan's unexpected loss contribution times the capital multiplier, which reflects diversification, unlike a regulatory approach assigning roughly 8% capital to each credit standalone.

Solution

Answer: D

Why D is correct

Schroeck derives $\text{EC}_i = \text{ULC}_i \times \text{CM}$ (eq. 4.18): "the required economic capital at the single credit transaction level is directly proportional to its contribution to the overall portfolio credit risk". Footnote 62 contrasts this with "the regulatory approach that assigns roughly 8% equity capital to credits on a standalone basis", whereas the economic approach differentiates capital by borrower quality and reflects the benefits of diversification.

$\Rightarrow$ *This is the map's discrimination: economic capital is the bank's own UL-based estimate; regulatory capital is a formula-driven minimum.*

Why the other options are wrong

A is incorrect. Polarity flip. Box 4.1: a bank targeting a very high rating "must choose a very high confidence level (e.g., 99.97%), which corresponds to a higher capital multiplier", so economic capital *rises*. A lower CM belongs to a more aggressive bank targeting a lower rating.

B is incorrect. Sibling swap. Crouhy assigns "industrywide rules and formulas" and "only a minimum required level of capital adequacy" to *regulatory* capital; risk (economic) capital is intended to capture the economic realities of the risks the firm runs.

C is incorrect. Scope error. EL is "the foreseeable cost of doing business", covered by pricing and loan loss reserves, and "does not itself constitute risk". Economic capital is "the distance between the expected outcome and the unexpected (negative) outcome at a certain confidence level", i.e. $\text{UL}_p \times \text{CM}$ measured from EL.

LO: CR-4 a: Evaluate a bank's economic capital relative to its level of credit risk. CR-4 g: Describe how economic capital is derived. ORR-18 a: Define, compare, and contrast risk capital, economic capital, and regulatory capital, and explain methods and motivations for using economic capital approaches to allocate risk capital. · Reference: Schroeck, Ch.5, Box 4.1, eqs. 4.17 and 4.18, footnote 62; Crouhy, Galai and Mark, Ch.17, 18.1

D4 Stressed VaR and Market Risk Capital

Readings: ORR-22 a, b, c · Solvency, Liquidity, and Other Regulation After the Global Financial Crisis (Carey, GARP Risk Institute 2019: Basel 2.5); ORR-21 c, f · Capital Regulation Before the Global Financial Crisis (Carey, GARP Risk Institute 2019: the 1995 and 1996 Amendments; the IRB Approach)

MAP TIER D · #73 · 3/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #73: "Reported: 'stressed market VaR calc'; 'market risk capital charge calculations'; and a cross-domain item, 'mixed question: market risk stressed VaR, credit risk VaR given default rate.' That last one is a genuine cross-section integration vignette of the kind that resolves on one line of reasoning per option."
Shapes covered here:
S1 stressed market VaR calculation: the SVaR term with its own max and multiplier (Q1, Q2)
S2 market risk capital charge calculation: $\max(VaR_{t-1}, m\,VaR_{avg})$, $\sqrt{10}$ scaling, multiplier from backtesting (Q1, Q2)
S3 mixed vignette: market stressed VaR charge plus credit capital given the 99.9th percentile default rate, one error per option (Q2).

Revision notes

- **1996 Amendment** (ORR-21 c): two methods, standardized (five categories; specific and general market risk; positions assumed perfectly correlated) and internal models (preferred by banks with material trading books because charges were generally smaller). Internal models: 10-day VaR at the 99th percentile, at least one year of daily data, usually one-day VaR $\times\sqrt{10}$; $MR = \max(VaR_{t-1},\, m \times VaR_{avg})$ with VaR_{avg} the 60-day average and $m \geq 3$. With $m = 3$ the second term is usually the larger.
- **Specific risk and Tier 3** (ORR-21 c): specific risk by internal model uses a multiplier of 4 and may not be below half the standardized charge (so internal-model banks also compute the standardized charge). Tier 3 capital (unsecured subordinated debt, original maturity of at least two years) could meet about 70% of market risk capital. Qualitative criteria: independent risk management, limits, board involvement.
- **Backtesting and the multiplier** (ORR-21 c): daily, one-day 99% VaR over the most recent 250 days; a day with a loss above VaR is an exception. Five or fewer exceptions: $m = 3$; more than five: larger at supervisory discretion; 10 or more: $m = 4$ required.
- **From capital to RWA** (ORR-21 c): charges for specific and general market risk across the five categories are summed and multiplied by 12.5 (the inverse of 8%), so trading capital $= 0.08 \times 12.5 \times \sum_j (MR_j + SR_j)$ and the usual RWA multipliers apply.
- **Why stressed VaR** (ORR-22 a): most banks used historical simulation; in calm periods measured VaR drifts down because nearly all observations are small, and when volatility returned in 2007 VaR was slow to follow. Basel 2.5 (implemented by end 2011) added stressed VaR, the incremental risk charge and the comprehensive risk charge.
- **Stressed VaR definition** (ORR-22 a): draw daily observations from the one-year (250-day) period in the most recent seven years that was most stressful for the *current* portfolio. Result: a relatively large one-day VaR that changes little over time (unless calm persists for seven years); scaled to 10 days like VaR.
- **Basel 2.5 charge** (ORR-22 a): $MR_{2.5} = \max(VaR_{t-1}, m_c VaR_{avg}) + \max(SVaR_{t-1}, m_s SVaR_{avg})$, both multipliers at least 3. Because the recent period cannot be more stressed than the stressed period and the two terms are summed, $MR_{2.5}$ is at least twice the 1996 charge when the multipliers are equal. Each max is a separate decision: after a sharp position build-up the previous day's VaR can exceed m times the average.
- **Incremental risk charge** (ORR-22 b): banks had placed illiquid default-risky instruments in the trading book to exploit lower capital than the banking book. First an incremental default risk charge (internal model at the IRB 99.9th percentile, one-year horizon, or a standardized or current exposure approach; trading capital the greater of market risk and banking book capital). Then extended to rating changes, because most crisis credit losses came from downgrades, spreads and liquidity rather than defaults: positions downgraded or defaulted are replaced at the pre-downgrade rating (constant credit quality), with a liquidity horizon of at least three months.

- **Comprehensive risk charge** (ORR-22 c): Basel II assumes a constant Gordy correlation, unreasonable for the correlation book (securitizations, re-securitizations, derivatives on them), whose tranche values move with correlation. The CR charge replaces the IRC and specific risk charge for the correlation book. Standardized capital as a % of exposure (not risk weights): securitizations 1.6%, 4%, 8%, 28%, 100% for AAA/AA, A, BBB, BB, below BB or unrated; re-securitizations 3.2%, 8%, 18%, 52%, 100%. Internal models need supervisory approval, a floor at a fraction of the standardized charge, and must capture multiple defaults and rating changes, correlation and spread volatility, basis risk, hedge dynamics and recovery volatility.
- **Credit capital given a default rate** (ORR-21 f): capital $= \sum_i \text{EAD}_i \times \text{LGD}_i \times \text{DR}_{99.9,i} - \text{EL}$, with $\text{EL} = \sum_i \text{EAD}_i \times \text{LGD}_i \times \text{PD}_i$, one-year horizon, 99.9th percentile; capital covers unexpected loss only because reserves are taken to cover EL. Hull's credit VaR (CR-10) is the loss percentile itself, EAD × LGD × WCDR; Malz (CR-11) defines portfolio credit VaR as the quantile minus EL. IRB detail: see also A31; credit VaR: B1.
- **Mixed vignettes** (ORR-22 a with ORR-21 f): the trading book charge and the banking book charge are computed separately on their own conventions (10-day 99% with multipliers versus one-year 99.9% net of EL) and only then added. One error per option is the usual design: drop SVaR, drop a multiplier, forget $\sqrt{10}$, forget to subtract EL.
- **See also** A4 (FRTB replaces VaR plus stressed VaR with stressed ES at 97.5% and liquidity horizons), C2 (backtesting tests), C5 (the Basel progression), A31 (IRB capital).

Formulas

1996 Amendment. $MR = \max\left(VaR_{t-1},\ m \times VaR_{avg}\right), \quad m \geq 3, \quad VaR_{\text{10-day}} = VaR_{\text{1-day}} \times \sqrt{10}$
Basel 2.5. $MR_{2.5} = \max\left(VaR_{t-1},\ m_c\, VaR_{avg}\right) + \max\left(SVaR_{t-1},\ m_s\, SVaR_{avg}\right), \quad m_c, m_s \geq 3$
Multiplier from backtesting (250 days, one-day 99%). ≤ 5 exceptions: $m = 3$; 6 to 9: supervisory discretion; ≥ 10: $m = 4$.
Capital and RWA. Trading capital $= 0.08 \times 12.5 \times \sum_j (MR_j + SR_j)$, RWA $= 12.5 \times$ capital
Credit capital given DR$_{99.9}$ (ORR-21 f). Capital $= \text{EAD} \times \text{LGD} \times \text{DR}_{99.9} - \text{EAD} \times \text{LGD} \times \text{PD}$

Exam traps

- **The max is a decision (threshold):** use VaR_{t-1} when it exceeds $m \times VaR_{avg}$; always multiplying the average understates the charge after a spike.
- **Intermediate result:** the VaR term alone is the 1996 charge; the Basel 2.5 charge adds the SVaR term.
- **Unit conversion:** one-day figures need $\sqrt{10}$; 10-day figures must not be scaled again.
- **Multiplier choice:** four exceptions permit $m = 3$; 10 or more require 4. Do not apply 4 to a clean backtest.
- **Credit side, dropped term:** subtract EL from the 99.9th percentile loss; the undeducted figure is Hull's credit VaR, a planted intermediate.
- **CR table:** the percentages are capital as a fraction of exposure, not risk weights; re-securitizations carry roughly double the securitization charge.

Question D4.1

ORR-22 a; ORR-21 c · S1, S2 · [Map #73]

A market risk analyst at a Canadian bank is calculating the bank's Basel 2.5 market risk capital charge for the trading book. The bank computes one-day 99% figures by historical simulation and scales them to 10 days with the square-root-of-time rule. A large position was added to the book two days ago. Data (CAD million, one-day 99%):

- VaR on the previous day: 6.5; average VaR over the last 60 trading days: 1.6
- Stressed VaR on the previous day: 7.4; average stressed VaR over the last 60 trading days: 6.9
- Backtesting exceptions over the last 250 trading days: 4

The supervisor applies, to both terms, the multiplier that this backtesting result permits under the 1996 Amendment. Which of the following is closest to the market risk capital charge?

A. CAD 20.6 million

B. CAD 27.2 million

C. CAD 80.6 million

D. CAD 86.0 million

Solution **Answer: D**

Step 1 – Multiplier

Four exceptions is five or fewer, so $m_c = m_s = 3$.

$\Rightarrow$ *The backtest is clean; a multiplier of 4 is required only at 10 or more exceptions.*

Step 2 – VaR term (one-day)

$$\max(VaR_{t-1},\ m_c VaR_{avg}) = \max(6.5,\ 3 \times 1.6) = \max(6.5,\ 4.8) = 6.5$$

$\Rightarrow$ *After the position build-up, yesterday's VaR exceeds three times the 60-day average, so the previous-day term binds.*

Step 3 – Stressed VaR term (one-day)

$$\max(SVaR_{t-1},\ m_s SVaR_{avg}) = \max(7.4,\ 3 \times 6.9) = \max(7.4,\ 20.7) = 20.7$$

$\Rightarrow$ *Stressed VaR is stable, so the multiplied average binds as usual.*

Step 4 – Scale to 10 days and add

$$MR_{2.5} = (6.5 + 20.7) \times \sqrt{10} = 27.2 \times 3.1623 = 86.01$$

Interpretation

The charge is CAD 86.0 million, an RWA equivalent of 12.5×86.01 = CAD 1,075 million. The 1996 charge would have been only $6.5 \times \sqrt{10} = 20.55$; the SVaR term makes the Basel 2.5 charge about 4.2 times larger, well above the "at least twice" minimum.

Why the other options are wrong

A is incorrect. Omitted the stressed VaR term (the 1996 charge, an intermediate result): $6.5 \times \sqrt{10} = 20.55$.
B is incorrect. Forgot the $\sqrt{10}$ scaling to a 10-day horizon: $6.5 + 20.7 = 27.2$.
C is incorrect. Skipped the max decision on the VaR term and always used $m \times VaR_{avg}$: $(4.8 + 20.7) \times \sqrt{10} = 80.64$.

LO: ORR-22 a: Describe and calculate the stressed VaR introduced in Basel 2.5 and calculate the market risk capital charge. ORR-21 c: Describe measures introduced in the 1995 and 1996 amendments, including guidelines for netting of credit exposures and methods for calculating market risk capital for assets in the trading book. · Reference: Carey, Solvency, Liquidity, and Other Regulation After the Global Financial Crisis, Section 22.2 (Stressed VaR); Carey, Capital Regulation Before the Global Financial Crisis, Section 21.1 (Capital for Market Risks Associated with Trading Activities)

Question D4.2 ORR-22 a; ORR-21 f · S1, S2, S3 · [Map #73]

The CRO of a Singapore bank asks a senior risk analyst for two numbers for the risk committee pack: the Basel 2.5 market risk capital charge for the trading book and the Basel II IRB credit risk capital for a large, homogeneous corporate loan portfolio in the banking book. Data (SGD million):

- Trading book, 10-day 99%: VaR on the previous day 12; 60-day average VaR 10; stressed VaR on the previous day 26; 60-day average stressed VaR 24; multipliers $m_c = m_s = 3$
- Loan portfolio: EAD 400; LGD 45%; PD 2%; one-year default rate at the 99.9th percentile 12%

Which of the following pairs is closest to the market risk capital charge and the credit risk capital?

A. Market risk SGD 38.0 million; credit risk SGD 18.0 million

B. Market risk SGD 30.0 million; credit risk SGD 18.0 million

C. Market risk SGD 102.0 million; credit risk SGD 21.6 million

D. Market risk SGD 102.0 million; credit risk SGD 18.0 million

Solution **Answer: D**

Step 1 – Market risk: VaR term (already 10-day)

$$\max(VaR_{t-1},\, m_c VaR_{avg}) = \max(12,\; 3 \times 10) = 30$$

Step 2 – Market risk: stressed VaR term

$$\max(SVaR_{t-1},\, m_s SVaR_{avg}) = \max(26,\; 3 \times 24) = 72$$

Step 3 – Market risk capital charge

$$MR_{2.5} = 30 + 72 = 102.0$$

$\Rightarrow$ *The figures are already 10-day, so no* $\sqrt{10}$ *is applied. The stressed term supplies about 70% of the charge.*

Step 4 – Credit: loss at the 99.9th percentile default rate

$$\text{EAD} \times \text{LGD} \times \text{DR}_{99.9} = 400 \times 0.45 \times 0.12 = 21.6$$

$\Rightarrow$ *SGD 21.6 million is the one-year 99.9% loss, which is Hull's credit VaR; it is not yet IRB capital.*

Step 5 – Credit: subtract expected loss

$$\text{EL} = 400 \times 0.45 \times 0.02 = 3.6, \qquad \text{Capital} = 21.6 - 3.6 = 18.0$$

$\Rightarrow$ *IRB capital covers unexpected loss only, because loan loss reserves are taken to cover EL.*

Interpretation

Market risk SGD 102.0 million and credit risk SGD 18.0 million: SGD 120.0 million in total, an RWA equivalent of SGD 1,500 million. Each wrong option contains exactly one error, so the pair can be checked line by line.

Why the other options are wrong

A is incorrect. Market side: used the previous-day values without the multiplied averages, $12 + 26 = 38$; the credit figure is correct.

B is incorrect. Market side: omitted the stressed VaR term, leaving the 1996-style charge $\max(12, 30) = 30$ (intermediate result); the credit figure is correct.

C is incorrect. Credit side: did not subtract EL, reporting the 99.9th percentile loss (credit VaR) of 21.6 as capital; the market figure is correct.

LO: ORR-22 a: Describe and calculate the stressed VaR introduced in Basel 2.5 and calculate the market risk capital charge. ORR-21 f: Calculate credit risk capital under Basel II utilizing the IRB approach. · Reference: Carey, Solvency, Liquidity, and Other Regulation After the Global Financial Crisis, Section 22.2; Carey, Capital Regulation Before the Global Financial Crisis, Section 21.2 (The IRB Approach); Hull, Credit VaR (CR-10)

D5 Credit Spread and PD from Spreads

Readings: CR-10 f · Credit Value at Risk (Hull, RMFI 6e, Ch.19, 10.5 Credit Spread Risk); CR-12 c · Credit Risk (Hull, OFOD 11e, Ch.24, 12.1 CVA and DVA, Special Cases); CR-9 h, j, l · Estimating Default Probabilities (Hull, RMFI 6e, Ch.17, 9.5, 9.6, 9.7, Problem 9.17)

MAP TIER D · #42 · 2/16 SOURCES VARDEEZ

Specifics reported

Map #42: "Verify your card before the exam. It writes CS $= (1/T)\cdot\ln(D/F) - R_f$. The standard form is CS $= -(1/T)\cdot\ln(D_t/F) - r$, where D_t is the current market value of the debt. As written, the sign is inverted: precisely the polarity flip GARP builds distractors from. Also examinable: deriving PD from credit spreads (LO 30.c) and PD $= (1/\text{LGD})\big(1-(1+r_f)/(1+y)\big)$."
Shapes covered here:
S1 credit spread from the market value of debt, with the card's inverted sign as the distractor (Q1)
S2 deriving PD from credit spreads (Q2)
S3 one-period PD $= (1/\text{LGD})\big(1-(1+r_f)/(1+y)\big)$, where recovery decides the ranking (Q2).

Revision notes

- **The specific: spread from debt value** (S1). For zero-coupon debt with face F, market value D_t and continuous compounding, $D_t = Fe^{-yT}$, so $y = -\frac{1}{T}\ln(D_t/F)$ and the credit spread is CS $= -\frac{1}{T}\ln(D_t/F) - r$. Since $D_t < F$, $\ln(D_t/F) < 0$ and the minus sign makes the yield positive. The card's $(1/T)\ln(D/F) - R_f$ is always negative: a polarity flip. Hull (OFOD, Special Cases) writes the same relation as $B = e^{-yT}$, $B_{nd} = e^{-y_{nd}T}$, so a derivative's value is $f = f_{nd}\,e^{-(y-y_{nd})T}$: discount at the risk-free rate plus the T-year credit spread (Example 12.2: $3e^{-0.015\times 2} = 2.91$).
- **Merton spread** (CR-9 l): under Merton the debt value is $V_0 - E_0$, and Hull's Problem 9.17 gives the zero-coupon credit spread as $-\ln\big[N(d_2) + N(-d_1)/L\big]/T$ with $L = De^{-rT}/V_0$. See also A5.
- **Credit spread defined** (CR-9 h): "the extra rate of interest per annum required by investors for bearing a particular credit risk". Measures: CDS spread, bond yield spread (over a similar risk-free bond), asset swap spread. Buying a 7% bond plus 200 bps protection approximates a 5% risk-free bond, so the n-year CDS spread $\approx$ the n-year par yield spread.
- **Which risk-free rate**: traders quote over Treasuries; rates implied from CDS and bonds are higher than Treasuries, near AA short-term borrowing rates (bond 4.7%, CDS 80 bps implies 3.9%). Asset swap spread: its PV equals the risk-free bond price minus the corporate bond price.
- **CDS-bond basis** = CDS spread − bond yield spread. Drivers: bond price away from par (above par negative, below par positive); CDS counterparty risk (negative); cheapest-to-deliver option (positive); no accrued interest in the payoff (negative); restructuring clause (positive); a different risk-free reference. Positive pre-2007 (about 16 bps), very negative in the crisis, small and negative since.
- **PD from spreads, approximate** (CR-9 h, CR-12 c; S2): the spread is roughly an average loss rate, so $\bar{\lambda}(T) = s(T)/(1-R)$ (240 bps, $R = 40\%$: 4%). Bootstrap (Example 9.3: 50, 60, 100 bps at 3, 5, 10 years, $R = 60\%$): 1.25%, 1.5%, 2.5%; years 3 to 5: $(5\times 0.015 - 3\times 0.0125)/2 = 1.875\%$; years 5 to 10: 3.5%. Works well for CDS spreads and for bonds near par. See also A15.
- **PD from spreads, exact** (S2): Hull 9.6.2: a 6% bond yielding 7% against a 5% risk-free bond: prices 95.34 and 104.09, expected loss 8.75; with defaults at mid-years and 40% recovery, expected loss $= 288.48Q$, so $Q = 3.03\%$ per year (unconditional).
- **PD per interval for CVA** (CR-12 c): with interpolated spreads $s(t_i)$, $q_i = \exp\big(-s(t_{i-1})t_{i-1}/(1-R)\big) - \exp\big(-s(t_i)t_i/(1-R)\big)$. These are *risk-neutral*, because CVA values future cash flows.
- **One-period formula (S3)**: price a one-year zero two ways, $\dfrac{(1-\text{PD})+\text{PD}\cdot R}{1+r_f} = \dfrac{1}{1+y}$, so PD · LGD $= 1 - \dfrac{1+r_f}{1+y}$ and PD $= \dfrac{1}{\text{LGD}}\left(1-\dfrac{1+r_f}{1+y}\right)$ (outside-source formula; derivation from risk-neutral valuation as above). $1-(1+r_f)/(1+y) = (y-r_f)/(1+y)$ is the expected loss rate; dividing by LGD turns it into a PD. Recovery matters: at the same spread, higher recovery means a higher implied PD.

- **Real-world versus spread-implied** (CR-9 j, k): seven-year hazard rates for A-rated bonds: 0.109% from default history versus 1.145% from a 68.68 bps spread; the gap widens as quality falls (CCC/C: 10.118% versus 18.395%). Causes: illiquidity (about 25 bps), traders' depression scenarios, and above all systematic risk from correlated defaults, plus undiversifiable skewed idiosyncratic risk. Use spread-implied (risk-neutral) PDs for valuation, historical (real-world) PDs for scenario analysis.
- **Credit spread risk** (CR-10 f): trading book credit products depend on spreads, so VaR or ES must capture spread changes. Historical simulation of 500 days of spread changes for a one-day 99% VaR scaled by $\sqrt{10}$ has two problems: a surviving company shows no past default (so no future default probability), and spreads are not updated daily for all names. Alternative: CreditMetrics-style 10-day rating transitions with a distribution of spreads per rating; correlation via a Gaussian copula on rating changes and (nearly) perfectly correlated spread moves across rating categories. Regulators handle jump-to-default separately.
- **Example 10.1**: two-year zero, USD 1,000, risk-free 3%, spread 200 bps, so price $1{,}000/1.05^2 = 907.03$. Default (0.1%) leaves 400, a loss of 507.03; B at 500 bps prices $1{,}000/1.08^{1.917} = 862.85$, a loss of 44.17. VaR is 507.03 above 99.9% confidence, 44.17 between 99.633% and 99.9%, and only 2.92 at 99%.
- **Constant level of risk**: rebalancing monthly to keep BBB lowers the default probability ($12 \times 0.0001 \times 0.9999^{11} = 0.0012$ versus 0.0017 buy-and-hold) but raises the chance of a small downgrade (0.04404 versus 0.0378); VaR and ES are generally smaller.

Formulas

Credit spread from debt value (continuous). $\text{CS} = -\dfrac{1}{T}\ln\left(\dfrac{D_t}{F}\right) - r$ (not $+\frac{1}{T}\ln(D_t/F)$)

Credit-risky discounting (Hull 12.5, 12.6). $f = f_{nd}\, e^{-(y-y_{nd})T}$

Average hazard from a spread (Hull 9.3). $\bar{\lambda}(T) = \dfrac{s(T)}{1-R}$; forward hazard t_1 to t_2: $\dfrac{t_2\bar{\lambda}_2 - t_1\bar{\lambda}_1}{t_2 - t_1}$

Interval PD (CR-12). $q_i = e^{-s(t_{i-1})t_{i-1}/(1-R)} - e^{-s(t_i)t_i/(1-R)}$

One-period PD (annual compounding; outside-source). $\text{PD} = \dfrac{1}{\text{LGD}}\left(1 - \dfrac{1+r_f}{1+y}\right) = \dfrac{y - r_f}{(1+y)\,\text{LGD}}$

Merton spread (Problem 9.17). $-\dfrac{1}{T}\ln\left[N(d_2) + \dfrac{N(-d_1)}{L}\right], \quad L = \dfrac{De^{-rT}}{V_0}$

Exam traps

- **Polarity flip (S1):** $\ln(D_t/F)$ is negative for risky debt, so the yield needs the minus sign. A negative "spread" on risky debt is the card's inverted formula at work.
- **Intermediate result:** $-\frac{1}{T}\ln(D_t/F)$ is the *yield*; the spread needs $-r$.
- **Annualising:** forgetting $1/T$ reports a T-year cumulative log loss as an annual spread.
- **Wrong-input twin:** PD divides by LGD $= 1 - R$, not by R.
- **Spread is not PD:** the spread (or $1 - (1+r_f)/(1+y)$) is an expected loss rate; ranking issuers by spread ignores recovery and can reverse the PD ranking.
- **Risk-neutral versus real-world:** spread-implied PDs are risk-neutral and much higher than historical ones; use them for valuation and CVA, not for scenario analysis.

Question D5.1

CR-12 c; CR-9 l · S1 · [Map #42]

A credit analyst at a Japanese life insurer is checking a colleague's formula card, which gives the credit spread on zero-coupon debt as $\text{CS} = (1/T)\ln(D/F) - R_f$. The analyst applies the standard definition to an issuer whose only debt is a zero-coupon bond. Data (JPY billion, rates continuously compounded):

- Market value of the firm's assets from a calibrated Merton model: 68.0
- Market value of equity: 24.5; equity volatility: 55%
- Face value of the zero-coupon bond: 50.0, maturing in 4 years
- Risk-free rate: 1.5% per annum

Which of the following is closest to the credit spread on the issuer's debt?

A. -498 bps

B. 198 bps

C. 348 bps

D. 1,243 bps

Solution **Answer: B**

Step 1 – Market value of debt

$$D_t = V_0 - E_0 = 68.0 - 24.5 = 43.5$$

$\Rightarrow$ *Debt with a JPY 50.0 billion face value is worth JPY 43.5 billion today.*

Step 2 – Yield on the debt

$$D_t = Fe^{-yT} \;\Rightarrow\; y = -\frac{1}{T}\ln\frac{D_t}{F} = -\frac{1}{4}\ln\frac{43.5}{50.0} = -\frac{1}{4}(-0.13926) = 0.03482$$

$\Rightarrow$ *The debt yields 3.48% per annum. The logarithm is negative because $D_t < F$; the leading minus sign turns it into a positive yield.*

Step 3 – Credit spread

$$\text{CS} = y - r = 0.03482 - 0.015 = 0.01982 \approx 198 \text{ bps}$$

$\Rightarrow$ *Investors demand 1.98% a year over the risk-free rate for this issuer's default risk.*

Interpretation

The promised payment is worth $50e^{-0.06} = 47.09$ risk-free, so the market charges $(47.09 - 43.5)/47.09 = 7.6\%$ of its value for expected default losses. The 198 bps spread is the rate that, added to r, discounts the promised JPY 50.0 billion back to its market value; it is the input a risk-neutral PD or CVA would use. Equity volatility is a red herring here: it calibrates V_0 and σ_V but plays no role once the debt value is known.

Why the other options are wrong

A is incorrect. Used the card's inverted sign: $\frac{1}{4}\ln(0.87) - 0.015 = -0.0348 - 0.015 = -0.0498$. A negative spread on risky debt is the polarity flip.

C is incorrect. Stopped at the intermediate result: 348 bps is the yield y, not the spread; the risk-free rate was not subtracted.

D is incorrect. Did not annualise: $-\ln(0.87) - 0.015 = 0.1393 - 0.015 = 0.1243$. The 13.9% is a four-year cumulative log loss, not a per-annum yield.

LO: CR-12 c: Calculate the probability of default using credit spreads. CR-9 l: Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model. · Reference: Hull, OFOD 11e, Ch.24, 12.1 Special Cases (eqs. 12.5, 12.6, Example 12.2); Hull, RMFI 6e, Ch.17, 9.5, 9.8, Problem 9.17

Question D5.2

CR-12 c; CR-9 h, j · S2, S3 · [Map #42]

A risk analyst at an Australian bank is comparing three issuers using their one-year zero-coupon bonds. The analyst assumes risk-neutral pricing with a single default date at maturity. The one-year risk-free rate is 3.0%; all rates use annual compounding.

Issuer	One-year yield	Expected recovery rate	Bond seniority
P	6.0%	60%	Senior secured
Q	7.0%	20%	Subordinated
R	6.5%	40%	Senior unsecured

Which of the following correctly identifies the issuer with the highest implied one-year probability of default and that probability?

A. Issuer P; 7.08%

B. Issuer P; 7.50%

C. Issuer Q; 3.74%

D. Issuer Q; 18.69%

Solution **Answer: A**

Step 1 – Expected loss rate implied by each yield

$$\text{PD} \cdot \text{LGD} = 1 - \frac{1 + r_f}{1 + y}$$

$$P:\ 1 - \frac{1.03}{1.060} = 0.02830, \qquad Q:\ 1 - \frac{1.03}{1.070} = 0.03738, \qquad R:\ 1 - \frac{1.03}{1.065} = 0.03286$$

$\Rightarrow$ *Q has the widest spread and the largest expected loss rate: the market expects to lose 3.74% of Q's face value.*

Step 2 – Divide by LGD $= 1 - R$

$$P:\ \frac{0.02830}{0.40} = 7.08\%, \qquad Q:\ \frac{0.03738}{0.80} = 4.67\%, \qquad R:\ \frac{0.03286}{0.60} = 5.48\%$$

$\Rightarrow$ *P loses only 40% when it defaults, so its smaller expected loss rate needs a* larger *default probability to explain it.*

Step 3 – Rank

P (7.08%) > R (5.48%) > Q (4.67%).

Interpretation

The ranking by spread (Q, R, P) reverses once recovery is taken into account: the spread prices expected loss, which is PD times LGD. These are risk-neutral PDs, suited to valuation and CVA; Hull's evidence (Table 9.5) is that they sit well above real-world PDs from default history, so a stress scenario would use historical rates instead. The formula is the one-period version of the CR-12 idea that the hazard rate is roughly $s/(1 - R)$ (outside-source in this exact discrete form).

Why the other options are wrong

B is incorrect. Dropped the $1/(1 + y)$ term and treated the raw spread as the expected loss rate: $(0.060 - 0.030)/0.40 = 7.50\%$. That is the approximation $s/(1 - R)$, not the exact one-period PD the stem asks for.

C is incorrect. Dropped the division by LGD and ranked on the expected loss rate: Q's $1 - 1.03/1.07 = 3.74\%$ is PD times LGD, not PD.

D is incorrect. Wrong-input twin: divided by the recovery rate instead of LGD: $0.03738/0.20 = 18.69\%$ for Q (with P at 7.08% becoming 4.72%).

LO: CR-12 c: Calculate the probability of default using credit spreads. CR-9 h: Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates. CR-9 j: Compare default probabilities calculated from historical data with those calculated from credit yield spreads. · Reference: Hull, OFOD 11e, Ch.24, 12.1; Hull, RMFI 6e, Ch.17, 9.6.1 eq. 9.3, 9.7, Table 9.5

D6 Repurchase Agreements

Readings: LTR-13 a-g · Repurchase Agreements and Financing (Tuckman and Serrat, Fixed Income Securities 3e, Ch.12)

MAP TIER D · #43 · 2/16 SOURCES VARDEEZ

Specifics reported

Map #43: "Reported: 'repo calculation: which party is more vulnerable if one bank is not allowing the repo to roll over?' The candidate's own answer was the hedge fund. Closed-book settlement amount: [(price + accrued) × (1 − haircut)] × (1 + repo·t). This template is confirmed to recur across GARP practice papers."
Shapes covered here: S1 repo settlement amount from the closed-book template, with accrued interest and haircut, and the repurchase price at the unwind (Q1); S2 which party is more vulnerable when a bank will not roll the repo over (Q2).

Revision notes

- **Mechanics** (LTR-13 a): a repo is "a contract in which a security is traded at some initial price with the understanding that the trade will be reversed at some future date at some fixed price." Book example: counterparty A sells EUR 100 million face of the DBR 4s of 2037 to B at an invoice price of EUR 111.772 million and agrees to repurchase it 92 days later at the invoice price plus interest at a repo rate of 0.23%, actual/360: $111{,}772{,}000(1 + 0.23\% \times 92/360) =$ EUR 111,837,697.10 (eq. 14.1).
- **Haircut and margin calls** (LTR-13 a, c): because the lender of cash faces a borrower default at the same time as a fall in the collateral, it lends less than the collateral value (EUR 106 million against EUR 111.772 million). Repos are also subject to margin calls: if the bund falls to EUR 110 million the borrower posts the EUR 1.772 million difference, so the investor still holds EUR 111.772 million of collateral against a EUR 106 million loan. A dealer financing inventory must fund the haircut from its own capital.
- **Motivation 1, cash management** (LTR-13 b): money market funds, municipalities (tax receipts arrive before spending), mutual funds, insurers, pension funds and some corporates lend cash in repo for a short-term rate "without sacrificing much liquidity or incurring significant default risk." They prefer overnight or open repo (a one-day repo that renews daily until cancelled) and take *general collateral*: government, GSE and government-guaranteed MBS, any issue within the class. Many crisis-era lenders found they lacked the expertise or operations to take possession of and liquidate collateral.
- **Motivation 2, long financing** (LTR-13 b): financial institutions are the typical cash borrowers. A trading desk that buys bonds from a client "repos out" (sells the repo) the bonds to finance them until a buyer is found. If no buyer appears it must roll the repo or unwind it and repo the bonds with a new counterparty C; because the new cash depends on the bond price at the roll, the roll leaves a cash surplus or deficit (offset by any hedge P&L). Customer financing (for example a hedge fund) is done back-to-back, with each leg's haircut set by that counterparty's creditworthiness and negotiating power.
- **Motivation 3, shorting through a reverse repo** (LTR-13 b, e): a hedge fund that sells a bond short borrows it by doing a reverse repo ("reverses in" the security, "buys the repo"): it lends cash and takes the specific bond. It profits on the outright short if the bond's return is below the repo rate. The same trade is a repo for the lender of the security, so always say whether cash or securities are being lent.
- **Counterparty risk** (LTR-13 c): for the lender of cash, the risk "that the borrower defaults and the value of the collateral turns out to be insufficient to cover the loan amount." With high-grade government collateral this is a *right-way* risk: a dealer default would likely trigger a flight to quality that raises the collateral's value.
- **Liquidity risk and the funding spectrum** (LTR-13 c): equity is the most stable funding, then long-term debt; short-term unsecured funding (commercial paper) is the least stable. Repo is cheap and sits on the less stable side, though more stable than unsecured short-term debt because collateral should stop lenders "bolting too quickly". If repo investors lose confidence, the financing "can disappear as fast as the repos mature, which is mostly overnight", forcing fire sales. "The loss of financing is often the killing blow", and the same applies to all leveraged investors, "like part of the hedge fund world". Borrowers want longer terms, lenders shorter; the system cannot do both.
- **Pre-crisis collateral drift** (LTR-13 c, d): borrowers financed lower-quality corporate bonds and MBS at low repo rates and haircuts; in the crisis the worst-hit could not meet margin calls, post collateral against higher

haircuts, or replace lost financing, and suffered collateral liquidations, capital depletion and failure.

- **Bear Stearns** (LTR-13 d): in late 2006 to 2007 it cut short-term unsecured funding from USD 25.8 billion to USD 11.6 billion (commercial paper from USD 20.7 billion to USD 3.9 billion) and replaced it with secured repo, lengthening terms to six months or more for whole loans and non-agency MBS. In the week of 10 March 2008 a loss of confidence brought three consequences: prime brokerage clients withdrew cash and unencumbered securities; repo lenders declined to roll even on agency collateral; FX counterparties refused to pay until Bear paid first.
- **Lehman and JPMorgan** (LTR-13 d): JPM was Lehman's tri-party clearing agent. Investors' tri-party loans are overnight, but during the day the agent itself lends to the dealer; JPM's intraday lending to Lehman typically exceeded USD 100 billion. JPM historically took no haircut on intraday advances, then phased haircuts in to match overnight investors by mid-August 2008, demanded USD 5 billion of cash collateral in September and seized USD 8.6 billion of cash in the last four business days. Lehman's estate: JPM used its "life and death power" to become over-collateralised. JPM's CRO Zubrow: JPM bore the dealer's *entire* tri-party book each day with no assurance investors would return each evening, and much collateral was illiquid and overvalued.
- **General versus special collateral** (LTR-13 e): in GC trades the cash lender accepts any security in a class; in specials the cash lender initiates the trade to obtain a particular security, so every special rate is typically below the GC rate. Special spread = GC rate − special rate. Funding trades are predominantly GC, but a holder of a bond trading special lends it in the specials market and borrows cash below GC. US GC trades close to but below fed funds (on average about 16 bp below between 23 October and 1 July 2010 in the book's sample); the gap widens when Treasuries are scarce and in flights to quality.
- **Special spreads and the auction cycle** (LTR-13 f): the on-the-run (OTR) issue trades most special, then the old, then the double-old; the OTR 10-year trades more special than the OTR 30-year. Spreads are volatile and often hundreds of basis points; they are small just after an auction (shorts can stay in the old OTR; reopenings add supply and get less special), then rise as shorts migrate and spike into the next auction as dealers short the OTR to hedge purchases of the new issue. Before May 2009 the special rate could not fall below zero (failing to deliver cost the same as borrowing at 0%), so the spread was capped by the GC rate; the TMPG fails penalty of $\max(3\% - \text{fed funds target}, 0)$ made the penalty rate the new cap.
- **Financing advantage of a special bond** (LTR-13 g): value of lending 100 of cash at the term special spread until the bond is expected to trade as GC. Book: 0.22% for 125 days gives $100 \times 125 \times 0.22\%/360 = 0.076$ per 100 market value, about 7.7 cents per 100 face at a price of 101.90, against a total premium of over 2 points; divided by DV01 0.085 it is only about 0.9 bp. Most of the OTR's richness is liquidity, not financing.
- **See also** D27 for Duffie's dealer-bank runs (repo creditors, tri-party clearing banks) and A26 for how a repo moves the TSECF and TSCLGC.

Formulas

Settlement at initiation (cash lent). $\text{Cash}_0 = \text{Face} \times \dfrac{\text{clean price} + \text{accrued}}{100} \times (1 - h)$

Repurchase price at the unwind (actual/360). $\text{Cash}_T = \text{Cash}_0 \left(1 + r_{\text{repo}} \times \dfrac{\text{days}}{360}\right)$

Closed-book template. $[(\text{price} + \text{accrued}) \times (1 - \text{haircut})] \times (1 + \text{repo} \cdot t)$

Borrower's own capital. invoice value $\times\ h$ (the haircut is funded by the borrower of cash)

Special spread. GC rate − special repo rate

Forward special spread (eq. 14.2). $33 \times 0.32\% + (61 - 33)\, s_{fwd} = 61 \times 0.28\% \ \Rightarrow\ s_{fwd} = 0.23\%$

Financing advantage (eq. 14.3). $100 \times \dfrac{\text{days} \times \text{term special spread}}{360}$ per 100 market value; $\times$ price$/100$ for per 100 face; $\div$ DV01 for its yield value

Fails penalty (TMPG, May 2009). $\max(3\% - \text{fed funds target},\ 0)$; e.g. USD 100 million $\times 3\%/360 =$ USD 8,333 per day

Exam traps

- **Dropped accrued (dropped term):** the collateral is valued at the invoice (full) price; using the clean price understates both the cash lent and the repurchase price.
- **Haircut direction (role):** the haircut protects the *lender* of cash; the *borrower* of cash funds it from its own

capital. Forgetting it gives zero own capital and a repurchase price on the full invoice value.

- **GC versus special rate (wrong-input twin):** a funding trade on a bond that is not trading special pays the GC rate; the special rate belongs to the bond in demand and is always below GC.
- **Who is exposed to the roll (role misassignment):** non-renewal hurts the *borrower* of cash (refinance or fire-sell); the lender gets its cash back and holds over-collateralisation in the meantime.
- **"Collateral makes repo funding stable" (absolute claim):** repo is more stable than unsecured short-term funding, but it "can disappear as fast as the repos mature".
- **Special spread cap (scope):** before the 2009 fails penalty the cap was the GC rate (special rate floored at 0%); after it, the cap is the penalty rate.

Question D6.1

LTR-13 a, e · S1 · [Map #43]

A treasury analyst at a Canadian hedge fund is arranging the financing of a new long position. The fund buys CAD 50 million face of a 4% semi-annual Government of Canada bond and finances it with a 91-day term repo from a dealer bank, delivering the bond as collateral:

- Clean price: 97.50 per 100 face; 76 days have elapsed in the current 182-day coupon period
- Haircut: 4% of the invoice (full) price
- The bond is an off-the-run issue that is not trading special; the dealer's 91-day general collateral repo rate is 3.20%
- The on-the-run issue of similar maturity trades special at a 91-day repo rate of 1.40%
- Repo interest accrues on an actual/360 basis

Which of the following is closest to the cash the fund must supply from its own capital at initiation and to the repurchase price it must pay at the end of the repo?

A. Own capital CAD 0.000 million; repurchase price CAD 49.565 million

B. Own capital CAD 1.950 million; repurchase price CAD 47.179 million

C. Own capital CAD 1.967 million; repurchase price CAD 47.368 million

D. Own capital CAD 1.967 million; repurchase price CAD 47.583 million

Solution

Answer: D

Step 1 – Accrued interest and full price

$$\text{accrued} = \frac{4}{2} \times \frac{76}{182} = 0.8352, \qquad \text{full price} = 97.50 + 0.8352 = 98.3352$$

⇒ *The repo lender values the collateral at the invoice price the buyer actually pays, coupon accrual included.*

Step 2 – Invoice value of the collateral

$$50{,}000{,}000 \times 0.983352 = \text{CAD } 49{,}167{,}582$$

Step 3 – Cash lent and the fund's own capital

$$\text{Cash}_0 = 49{,}167{,}582 \times (1 - 0.04) = \text{CAD } 47{,}200{,}879$$
$$\text{own capital} = 49{,}167{,}582 - 47{,}200{,}879 = \text{CAD } 1{,}966{,}703$$

⇒ *The dealer lends 96% of the full value; the fund pays the 4% haircut, CAD 1.967 million, from its own capital.*

Step 4 – Choose the repo rate

This is a funding trade on a bond that is not trading special, so it is done at the GC rate of 3.20%. The 1.40% special rate applies only to the on-the-run bond that cash lenders want to borrow.

Step 5 – Repurchase price

$$\text{Cash}_T = 47{,}200{,}879 \times \left(1 + 0.032 \times \frac{91}{360}\right) = 47{,}200{,}879 \times 1.0080889 = \text{CAD } 47{,}582{,}682$$

⇒ *The fund pays CAD 381,803 of repo interest over 91 days to get its bonds back.*

Interpretation

The fund controls CAD 49.17 million of bonds with CAD 1.97 million of its own money (leverage of 25 times). That leverage is only as durable as the dealer's willingness to roll the repo at maturity, which is the vulnerability examined in D6.2.

Why the other options are wrong

A is incorrect. Dropped the haircut: the dealer is assumed to lend the full invoice value, so own capital is zero and the repurchase price is $49{,}167{,}582 \times 1.0080889 = 49{,}565{,}294$.

B is incorrect. Dropped the accrued interest: invoice $= 50\text{m} \times 0.975 = 48{,}750{,}000$, own capital $= 0.04 \times 48{,}750{,}000 = 1{,}950{,}000$, repurchase $= 46{,}800{,}000 \times 1.0080889 = 47{,}178{,}560$.

C is incorrect. Wrong-input twin: used the on-the-run special rate instead of GC: $47{,}200{,}879 \times (1 + 0.014 \times 91/360) = 47{,}367{,}918$. Own capital is unaffected, so the error shows only in the second number.

LO: LTR-13 a: Describe the mechanics of repurchase agreements (repos) and calculate the settlement for a repo transaction. LTR-13 e: Compare the use of general and special collateral in repo transactions. · Reference: Tuckman and Serrat, Fixed Income Securities 3e, Ch.12 (reading sections 14.1 and 14.3, eq. 14.1)

Question D6.2

LTR-13 c · S2 · [Map #43]

A hedge fund has financed a leveraged long position in BBB-rated corporate bonds for several months through overnight repo with Bank K, rolling the repo each day at a 5% haircut. After market rumours about the fund's trading losses, Bank K tells the fund that it will not roll the repo when it matures tomorrow morning. A risk consultant reviewing the episode is asked which party is more vulnerable to the non-renewal. Which assessment is correct?

A. The hedge fund is more vulnerable: as the borrower of cash it must replace the financing or sell the bonds at short notice, possibly at fire-sale prices, while Bank K's loan is secured by collateral worth more than the cash it lent.

B. Bank K is more vulnerable: as the lender of cash it must find a buyer for the bonds at short notice, possibly at fire-sale prices, while the fund's cash is protected by the haircut it received at the start of the repo.

C. Neither party is materially vulnerable: because the repo is fully collateralised with a haircut, the non-renewal simply unwinds the trade, and the fund can always refinance the same bonds with another repo lender.

D. The hedge fund is more vulnerable, but only if the bonds have fallen in value since the repo began, because otherwise returning the cash and taking back the bonds leaves the fund with no loss and no funding need.

Solution

Answer: A

Step 1 – Name the roles

The fund sold the bonds to Bank K with an agreement to buy them back: it is the *borrower of cash* (it did the repo). Bank K is the *lender of cash* and holds the bonds as collateral, worth 105% of its loan.

Step 2 – What non-renewal does to each side

Bank K simply receives its cash back plus interest; until then its loan is over-collateralised and it can sell the collateral "to recover any amounts owed" if the fund defaults. The fund must hand over cash it does not have: Tuckman and Serrat note that a borrower's repo financing "can disappear as fast as the repos mature, which is mostly overnight", after which it "would have to sell inventory" in sales that "would likely turn into fire sales", and that the argument "applies to all leveraged investors, like part of the hedge fund world."

⇒ *The exposure created by a refusal to roll is funding liquidity risk, and it sits with the borrower of cash.*

Interpretation

This is the Bear Stearns pattern in miniature: repo lenders "declined to roll over or renew repo loans, even when the loans were supported by high-quality collateral". "The loss of financing is often the killing blow."

Why the other options are wrong

B is incorrect. Role misassignment. The roles are reversed: the lender of cash gets its cash back and is protected by the haircut it *charges*; it is the borrower of cash who must raise funds or sell into a fire sale.

C is incorrect. Absolute claim. Collateral makes repo more stable than unsecured funding but does not guarantee refinancing; in the crisis borrowers "were unable to replace lost financing arrangements when lower-quality collateral found fewer and fewer takers".

D is incorrect. Scope/condition error. The fund's problem exists even at unchanged prices: it must still find the

full cash to repay the loan, and only then does it get the bonds back. A price fall adds a loss but is not the condition for vulnerability.

LO: LTR-13 c: Explain how counterparty risk and liquidity risk can arise through the use of repo transactions. · Reference: Tuckman and Serrat, Fixed Income Securities 3e, Ch.12 (reading sections 14.1, Repos and Cash Management, and 14.2, Repo, Liquidity Management, and the Financial Crisis)

D7 Early Warning Indicators and PIT vs TTC

Readings: LTR-3 a, b, c · Early Warning Indicators (Venkat and Baird, Ch.6); CR-6 a, b, d · Credit Scoring and Rating (Doumpos et al., Ch.2, 6.1, 6.2.1 and the criticisms of CRAs); ORR-18 d · Risk Capital Attribution and Risk-Adjusted Performance Measurement (Crouhy, Galai and Mark, Ch.17)

MAP TIER D · #45 · 2/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #45: "Reported: 'E.W.I.S.: point in time & through the cycle', appearing twice in the same list. Point-in-time ratings respond to current conditions and are volatile; through-the-cycle ratings are stressed-scenario-anchored and stable. The polarity of which is more procyclical is the trap."
Shapes covered here:
S1 characteristics of sound EWIs: leading, sharp, calibrated thresholds (Q1)
S2 applications of EWIs in the liquidity risk process: escalation to the right authority (Q1)
S3 PIT versus TTC definitions and users (Q2)
S4 the procyclicality polarity: PIT is the volatile, procyclical one (Q2).

Revision notes

- **EWIs as a dashboard** (LTR-3 a): a turn signal is an intentional risky activity (running a deposit special to raise rate-sensitive liabilities); a low oil light means action is needed soon (the LCR has dropped below a threshold); a check engine light calls for investigation (a dramatic rise in call center volumes). "EWI triggers are used to initiate management discussions and actions that should be formally documented."
- **Beyond compliance** (LTR-3 c): holding liquid assets is a cost-benefit decision; a "set and forget" buffer cannot manage risk against a dynamic market. Increasing buffers *before* stress extends the survival horizon cost-effectively, and a strong balance sheet in stress opens market share and acquisition opportunities.
- **BCBS 2008 Sound Principles** (LTR-3 b, Table 3.1): EWIs may include rapid asset growth funded with potentially volatile liabilities; growing concentrations; increases in currency mismatches; a decrease in the weighted average maturity of liabilities; repeated incidents of approaching or breaching limits; negative trends in a product line; significant deterioration in financial condition; negative publicity; a rating downgrade; stock price declines; rising debt costs; widening debt or CDS spreads; rising wholesale or retail funding costs; counterparties requesting collateral or resisting new transactions; a drop in credit lines; increasing retail deposit outflows; early CD redemptions; difficulty accessing longer-term funding or placing short-term liabilities.
- **OCC 2012 and SR 10-6** (LTR-3 b, Table 3.2): the OCC wants EWIs that signal whether embedded triggers in products (callable public debt, OTC derivatives) are about to be breached or contingent risks are likely to materialise, e.g. reluctance of traditional fund providers, pending regulatory action, CAMELS downgrades, rising funding costs in an otherwise stable market, elimination of committed credit lines. The 2010 Interagency Statement (SR 10-6): tailor indicators to the institution's specific liquidity risk profile and move into "progressive states of readiness" as an event evolves. BCBS 2012 adds intraday monitoring indicators (see D27).
- **Where EWIs sit** (LTR-3 c, FRB SR 12-7): the framework must be end-to-end. Risks inherent in business activities are identified, recorded in a liquidity risk inventory, assessed by the businesses, treasury, independent risk management and others, then measured and monitored through stress tests, EWIs, and a limit and response framework, linked to the funding playbook.
- **M.E.R.I.T.** (LTR-3 a): Measures, Escalation, Reporting, Integrated systems, Thresholds. Measures without escalation are "a mere academic exercise".
- **Measures** (LTR-3 a): BCBS Principle 5 asks for metrics that assess the structure of the balance sheet, project cash flows and future liquidity positions, and take into account off-balance sheet risks, spanning business-as-usual and stressed conditions over various horizons.
- **Leading and sharp** (LTR-3 a, S1): a leading indicator signals stress before the event; lagging indicators such as government-reported GDP report what has already happened, so build proxies from internal loan portfolio metrics. Sharpness is granularity: total deposit balances is acceptable, but deposits of volatile

segments (high-net-worth, rate-sensitive products) are sharper. Sharp, leading EWIs give management more time to act.

- **Internal and external** (LTR-3 a): balance internal measures (customised to the balance sheet; they catch internally driven stress before it becomes public, e.g. loan performance, planned accounting adjustments, operational losses) with external ones that signal systemic change (sovereign default, banking crisis).
- **Environments and horizons** (LTR-3 a): track EWIs in business as usual and add stressed measures and limits; typical cycles are daily, weekly or monthly. "An indicator that is not updated for a period of more than a month is unlikely to satisfy the 'early' aspect"; some are intraday.
- **Escalation** (LTR-3 c, S2): breaches go to the level of management with authority to act. Modest responses (increasing the liquid asset buffer) usually sit within the corporate treasurer's delegation; a signal of a potentially large cash outflow may need ALCO to authorise a check on asset growth; the most extreme cases may require the contingency funding plan. Escalation governance must be formalised, documented and included in the CFP (see D26).
- **Reporting and integrated systems** (LTR-3 a): EWI dashboards are commonly daily, with leaders moving to intraday for some measures (especially trading-focused firms); reporting must be broad enough for coverage yet specific enough to carry only key messages. Stressed and non-stressed metrics should come from the same source system.
- **Thresholds** (LTR-3 a, S1): green, amber, red. Bands should not be so wide that movements go undetected, nor so narrow that constant investigation desensitises management ("boy who cried wolf"). Calibrate on historical volatility over a series long enough to be significant, recent enough to be relevant, ideally including a stress period such as 2007 to 2008 (at least one year when available); back-test and allow a judgment-based management override (e.g. the FOMC target range).
- **Scoring versus rating** (CR-6 a): a credit score is a numerical indicator from automated models, solely for internal use; a credit rating is an ordinal grade, each grade tied to an empirically estimated PD, often public, and may combine analytical and judgmental assessment. Types: internal versus external; issuer versus issue-specific; behavioral and profit scoring.
- **PIT versus TTC** (CR-6 b, S3): Basel distinguishes two philosophies. TTC assessments are "primarily of interest for major credit rating agencies", have a long-term orientation covering at least one business cycle, are robust to short-term changes and updated less often. PIT assessments are "usually employed by internal systems developed by credit institutions and banks", focus on the current and short-term condition (usually no more than one year), are more volatile and updated promptly, and suit short-term transactions and loans. Research on their relative predictive power is mixed; default is easier to define PIT, and extending to TTC is hard because the time until default must be modeled.
- **Procyclicality** (CR-6 d, ORR-18 d, S4): Crouhy: a PIT PD (KMV and structural approaches) suits near-term EL and pricing; a TTC PD (largely the agencies' approach) suits economic capital, profitability and strategic decisions. A firm is *less* likely to stay in the same rating under PIT, so "the TTC approach therefore reduces the volatility of economic capital"; compare both in RAROC at a normal and at the worst point of the cycle. Calibrating parameters on a shorter period makes capital "more procyclical (capital goes down when the credit environment improves and goes up when it deteriorates)". Doumpos: agencies claim TTC ratings, yet critics found them too optimistic in growth and too pessimistic in recessions, i.e. pro-cyclical. ORR-7: capital, regulatory or economic, is a TTC concept; stress testing is a PIT exercise.

Exam traps

- **Polarity flip, PIT versus TTC:** PIT is the volatile, promptly updated, more procyclical philosophy; TTC is stable and reduces economic capital volatility.
- **Role misassignment, who uses which:** PIT belongs to banks' internal systems and short-term loans; TTC to the major rating agencies (and to economic capital).
- **Absolute claim on accuracy:** research on PIT versus TTC predictive power is *mixed*; the choice depends on the scope of the analysis.
- **Sibling swap, leading versus lagging:** GDP releases are the book's example of a lagging indicator; EWIs should be leading and sharp.
- **Absolute claim on thresholds:** neither "as narrow as possible" (boy who cried wolf) nor so wide that move-

ments go undetected.

- **Role misassignment in escalation:** treasurer for modest buffer moves, ALCO for asset-growth checks after a large-outflow signal, CFP for the extreme.

Question D7.1 LTR-3 a, c · S1, S2 · [Map #45]

A treasury risk manager at a mid-sized Canadian bank has been asked by the CFO to review the bank's liquidity early warning indicator dashboard and recommend improvements. The current dashboard tracks total deposit balances as a single line, uses quarterly GDP releases as its main macroeconomic signal, and has no documented escalation rules. Which of the following recommendations is consistent with sound EWI practice?

A. Set each amber threshold as narrowly as possible, so that every movement in an indicator triggers an investigation and no deterioration can go undetected.

B. Retain the quarterly GDP releases as the main macroeconomic signal, because lagging indicators confirm that stress has occurred before management commits to action.

C. Track deposit balances of high-net-worth and rate-sensitive segments separately from total deposits, so that the indicator becomes sharper and gives management more time to act.

D. Route every EWI breach to the corporate treasurer, including a signal of a potentially large cash outflow, since all EWI-triggered actions sit within the treasurer's delegated authority.

Solution **Answer: C**

Why C is correct

Venkat and Baird: "detecting a drop in overall deposit balances is an acceptable EWI; however, detecting drops in deposit balances of more volatile segments, such as high-net worth customers or rate-sensitive products balances, brings into focus that certain important classes of customers are leaving the bank." Sharp, forward-looking EWIs allow "more time for management to take corrective or mitigating actions".

⇒ *Sharpness (granularity) and a leading bias are the two hallmarks of a sound measure; the bank's single deposit line has neither.*

Why the other options are wrong

A is incorrect. Absolute claim. Thresholds should not be so wide that movements go unnoticed, but institutions "should also be careful not to set boundaries that are so narrow as to force constant investigation", which desensitises management and creates a "boy who cried wolf" circumstance.

B is incorrect. Sibling swap. It credits a lagging indicator with the job of a leading one. The book names government-reported GDP as the lagging example and asks banks to build leading proxies from internal loan portfolio metrics instead.

D is incorrect. Role misassignment. Modest responses such as increasing the liquid asset buffer usually sit within the treasurer's delegation, but an EWI that signals a potentially large cash outflow "may require the asset-liability committee (ALCO) to authorize a check on future asset growth", and the most extreme cases the contingency funding plan.

LO: LTR-3 a: Evaluate the characteristics of sound Early Warning Indicators (EWI) measures. LTR-3 c: Discuss the applications of EWIs in the context of the liquidity risk management process. · Reference: Venkat and Baird, Liquidity Risk Management: A Practitioner's Perspective, Ch.6, Forward Looking Bias, Escalation and Thresholds

Question D7.2 CR-6 b; ORR-18 d · S3, S4 · [Map #45]

A credit risk manager at a German bank is presenting to a group of newly hired analysts on the rating philosophies used across the bank. Its internal rating system for the short-term trade finance book is point-in-time (PIT), its economic capital model currently uses through-the-cycle (TTC) PDs, and its bond portfolio relies on agency ratings. Which of the following statements is correct for the manager to make?

A. Because TTC ratings are updated promptly to reflect any new information about a borrower, they are the more volatile of the two philosophies and the more procyclical.

B. TTC assessments are mainly employed in banks' internal systems for short-term loans, while PIT assessments are primarily of interest for the major credit rating agencies.

C. Research shows that PIT assessments have unambiguously higher predictive power than TTC assessments, so

the bank should use PIT PDs for every purpose, including economic capital.

D. Switching the economic capital model from TTC to PIT PDs would make capital more volatile and procyclical, because a borrower is less likely to keep the same rating under a PIT approach.

Solution **Answer: D**

Why D is correct

Crouhy, Galai and Mark: "The probability of a firm's staying in the same rating when it is assessed using a PIT approach is smaller than when it is assessed using a TTC approach. The TTC approach therefore reduces the volatility of economic capital, compared to PIT approaches." Capital calibrated to shorter, current conditions is "more procyclical (capital goes down when the credit environment improves and goes up when it deteriorates)".
⇒ *The bank's current choice (TTC PDs for economic capital, PIT for short-term trade finance) matches Crouhy's split: TTC for economic capital and strategic decisions, PIT for near-term EL and pricing.*

Why the other options are wrong

A is incorrect. Polarity flip. Prompt updating and volatility describe PIT: "These ratings are more volatile, and they are updated in a timely manner." TTC assessments "are robust to short-term changes and they are subject to less frequent updates".

B is incorrect. Role misassignment. Doumpos assigns the users the other way round: TTC is "primarily of interest for major credit rating agencies"; PIT is "usually employed by internal systems developed by credit institutions and banks" and used for short-term transactions and loans.

C is incorrect. Absolute claim. "In terms of predictive power and accuracy of these two approaches, the findings from existing research are mixed", so the choice depends on the scope and requirements of the analysis; Crouhy reserves TTC for economic capital.

LO: CR-6 b: Differentiate between through-the-cycle credit rating system and point-in-time credit rating system. ORR-18 d: Explain challenges that arise when using RAROC for performance measurement, including choosing a time horizon, measuring default probability, and choosing a confidence level. · Reference: Doumpos et al., Ch.2, 6.2.1; Crouhy, Galai and Mark, The Essentials of Risk Management 2e, Ch.17, Default Probabilities: Point-in-Time vs. Through-the-Cycle

D8 US Dollar Shortage and the Cross-Currency Basis

Readings: LTR-15 a-c · The US Dollar Shortage in Global Banking and the International Policy Response (McGuire and von Peter, BIS Working Paper 291, 2009); LTR-16 a-c · Covered Interest Parity Lost: Understanding the Cross-Currency Basis (Borio, McCauley, McGuire and Sushko, BIS Quarterly Review, Q3 2016)

MAP TIER D · #46 · 2/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #46: "Reported verbatim: 'why non-US banks faced dollar shortage during GFC.' The mechanism, dollar-denominated assets funded by short-term wholesale dollar liabilities without a dollar deposit base or Fed access, is the answer. Covered interest parity and the cross-currency basis (Reading 82) is the adjacent reading."
Shapes covered here: S1 why non-US (chiefly European) banks faced a US dollar shortage in the GFC (Q1); S2 covered interest parity violations and the cross-currency basis, the adjacent reading (Q2).

Revision notes

- **Not a domestic bank run** (LTR-15 a): the open-economy run story (depositors convert domestic deposits into foreign currency and exhaust the country's FX reserves) still fits emerging markets, but the GFC dollar shortage came from the *international operations* of the major banking systems and the global interbank and FX swap markets they relied on.
- **Balance sheet expansion** (LTR-15 a): banks' foreign claims grew from USD 10 trillion (2000) to USD 34 trillion (end-2007), year-on-year growth near 30% by mid-2007, driven by structured finance, universal banking and prime brokerage to hedge funds. Swiss banks' foreign claims rose from about five to more than seven times Swiss GDP. US dollar and other non-euro positions were more than half of European banks' foreign asset growth.
- **Three ways to fund a foreign asset** (LTR-15 a): (1) borrow domestic currency and buy the asset spot (no later FX need, but unhedged currency risk); (2) borrow domestic currency and convert it with an FX swap; (3) borrow the foreign currency (interbank, non-banks, central banks). The second leg of (2) "is not that different" from (3): in both, the bank must *deliver* foreign currency when the contract matures. Funding risk is "the risk that funding liabilities cannot be rolled over"; a *foreign currency funding gap* exists when the investment horizon of foreign currency assets exceeds the maturity of the foreign currency funding or swaps.
- **Why foreign currency is special** (LTR-15 a, c): at home a central bank can be lender of last resort, but "central banks cannot create foreign currencies"; their help is limited by the exchange rate regime or by FX reserves.
- **The long dollar book** (LTR-15 a): Japanese and major European banking systems ran large net long positions in foreign currencies, mostly US dollars, mirrored by net borrowing in their own currency from home residents and bridged by FX swaps (78% of FX swap turnover is under seven days). European banks' long dollar positions reached about USD 700 billion by mid-2007 (USD 1.3 trillion with Japanese banks). Spanish banks were the exception: short dollars, funding euro lending at home.
- **Measuring the gap** (LTR-15 a, b): counterparty type proxies maturity (interbank short, non-bank claims the desired investment portfolio). Lower bound: net dollar claims on non-banks, USD 1.0 to 1.2 trillion for European banks by mid-2007, met by interbank borrowing (USD 432 billion), central bank deposits (USD 386 billion) and FX swaps (USD 315 billion). With money market fund paper (about USD 1 trillion) treated as short-term: USD 2.0 to 2.2 trillion; all non-bank liabilities short-term: USD 6.5 trillion.
- **How it turned into a shortage** (LTR-15 a): from August 2007 counterparty and liquidity concerns compromised interbank funding; FX swap dislocations made swapped dollars dearer; after Lehman, money market funds facing redemptions withdrew from bank paper and central banks withdrew reserve placements (deposits fell USD 257 billion). Assets could not be shed: structured products were hard to sell, off-balance-sheet vehicles came back on balance sheet and credit lines were drawn, so dollar claims even rose USD 248 billion (3%) to Q1 2008. "The effective holding period of assets lengthened just as the maturity of funding shortened." This endogenous rise in maturity mismatch, "difficult to hedge ex ante", generated the shortage.
- **Crowded trade** (LTR-15 a): "What pushed the system to the brink was not cross-currency funding per se, but rather too many large banks employing funding strategies in the same direction, the funding equivalent of a crowded trade."

- **Consolidated, not residency, view** (LTR-15 b): "bank nationality" and "bank residency" differ. Positions booked in one country are a small part of a banking system's global balance sheet, yet can be large relative to that host country's external position (foreign banks' offices held nearly 40% of UK external assets; only 18% of Swiss banks' foreign claims are booked in Switzerland). A mismatch booked in one office may be hedged in another, so vulnerabilities "can only be measured by taking into account the entire balance sheet of the consolidated global entity".
- **Policy response** (LTR-15 c): Fed swap lines with the ECB and SNB from December 2007; doubled after Lehman with new lines to the Bank of Canada, Bank of England and Bank of Japan (USD 247 billion), then extended across continents; unlimited for the BoE, ECB and SNB from 13 October 2008. The Fed acted as international lender of last resort, lending to central banks against foreign currency collateral; they auctioned the dollars locally, reaching banks with no US subsidiary or eligible collateral. The Fed "assumes no credit risk vis-a-vis the ultimate borrowers" and delegates monitoring to national authorities, so the design overcomes the resource and moral hazard problems of international lending of last resort. Euro area, Swiss and UK FX reserves were only USD 294 billion, an order of magnitude below the gap.
- **Effectiveness** (LTR-15 c): usage surged to a USD 583 billion peak in December 2008 and fell to USD 50 billion by early October 2009; Libor levels and spreads receded, auctions lowered the level and volatility of swap spreads and averted wider distress selling of dollar assets. Caveat: public dollar funding "may have delayed the necessary restructuring" of balance sheets.
- **CIP and the basis** (LTR-16 c): CIP "is the closest thing to a physical law in international finance". The cross-currency basis is the amount by which the interest paid to borrow a currency by swapping it against another differs from borrowing it directly in the cash market; a non-zero basis is a CIP violation. Since 2007 the basis for lending US dollars against the euro and yen has been negative: borrowing dollars through the FX swap market became *more expensive* than in the dollar cash market. For the Australian dollar it has been positive.
- **FX swap versus cross-currency swap** (LTR-16 a): FX swap: one party borrows one currency and lends another; amounts exchanged at spot S, repaid at the pre-agreed forward F; the implicit return is set by $F - S$ and quoted in forward points; mostly short-dated. Cross-currency swap: longer-term (typically above one year); periodic interest exchanges during the life (Libor plus the basis b in a basis swap); principal re-exchanged at the *initial spot rate* S.
- **What moves the basis** (LTR-16 b): market illiquidity (bid-ask between forward and spot), credit risk in the underlying (Libor-OIS spreads, sovereign CDS), a risk premium rp, and FX hedging demand scaled by per-dollar balance sheet costs. Three hedgers, all insensitive to the basis: banks (structural; Japanese banks fund dollar lending with yen deposits via swaps), institutional investors (Japanese life insurers hedge 60 to 70%), and non-financial issuers ("reverse yankee" euro bonds by US firms swapped into dollars). Where banks compete with insurers for swapped dollars (Japan, euro area) the basis is negative; where banks lend dollars in swaps and offset investors (Australia; Sweden against the euro) it is positive.
- **Why it stopped closing after 2014** (LTR-16 c): first-generation explanation: counterparty risk and dollar funding strains (GFC, euro debt crisis). The puzzle is persistence after 2014 with healthy banks: hedging demand plus tighter *limits to arbitrage*. "Balance sheet space is rented, not free": CVA charges, PFE add-ons in Basel III and leverage ratios, quarter-end spikes, the shift to repo funding (BoJ purchases made yen collateral scarce), internal limits on Japanese official sector exposure, and a slowdown in supranational issuance arbitrage after the USD swap spread inverted in Q3 2015. ECB and BoJ easing announcements widened it. Japanese banks' own reliance on swaps reduced their capacity to arbitrage. "CIP deviations look to be here to stay."

Formulas

Covered interest parity (Box 17.1; S, F in USD per unit of foreign currency). $\dfrac{F}{S} = \dfrac{1+r}{1+r^*}$

Basis as a CIP deviation (with $1 + r^* \approx 1$). $b = \dfrac{F-S}{S} - (r - r^*)$

Cross-currency basis swap, one period. $F - S = S\,\dfrac{1+r+b}{1+r^*} - S$; swap-implied dollar rate $\dfrac{F}{S}(1+r^*)$ exceeds $1 + r$ when the dollar borrower pays b

Illiquidity. $b = \dfrac{F_b - S_a}{S_a} - (r - r^*)$; credit/risk premium: $b = rp$; hedging pressure: $b \propto rp \times$ FX hedging demand

Foreign currency funding gap (eq. 16.2). $\underbrace{(A_i - L_i^{LT})}_{\text{short-term FX liabilities}} - \underbrace{(A_i - A_i^{LT})}_{\text{liquid FX assets}} = A_i^{LT} - L_i^{LT}$; net position in currency i: $A_i - L_i$ (long if positive)

Regression read (Table 17.1). 1% more hedging demand ⇒ about 1 bp wider two-year basis; Libor-OIS × demand interaction ⇒ about 45 bp on the three-month basis

Exam traps

- **Sign of the basis (polarity flip):** a negative EUR or JPY basis means dollars are *dearer* through the swap market than in the cash market. (Box 17.1 writes the same premium as a positive b paid on the dollar leg; market quotes put it on the non-dollar leg, outside-source.)
- **FX swap versus cross-currency swap (sibling swap):** re-exchange at the forward F, no interim interest, short-dated, versus re-exchange at the initial spot S, periodic interest, typically above one year.
- **Long versus short dollars (polarity flip):** the shortage hit banks that were *long* dollar assets funded short-term; Spanish banks' short dollar book is the exception, not the mechanism.
- **Who can create dollars (role misassignment):** home central banks cannot; the Fed supplied them through swap lines to central banks, which lent on and bore the credit risk of the ultimate borrowers.
- **Post-2014 cause (scope/condition error):** credit risk explains the GFC episodes, but the basis persisted when Libor-OIS normalised; hedging demand meeting balance sheet constraints is why it does not close.
- **Residency data (scope):** a country's national balance sheet can show a "mismatch" that is hedged elsewhere in the bank, and hide one that is booked abroad.

Question D8.1

LTR-15 a, b · S1 · [Map #46]

A risk consultant is presenting to the board risk committee of a German universal bank on lessons from the US dollar shortage of 2007 to 2009, drawing on the BIS international banking statistics. Which of the following statements best explains why European banks faced a US dollar shortage during the crisis?

A. They had built large net long US dollar positions in non-bank assets funded by short-term interbank borrowing, FX swaps and money market fund paper, so when those sources withdrew, their funding shortened just as the assets became harder to sell.

B. Their domestic retail depositors ran on them and converted home currency deposits into US dollars, so the resulting demand for foreign currency exhausted their home central banks' foreign exchange reserves within a few weeks.

C. They had built large net short US dollar positions, borrowing dollars short-term to fund long-term lending in their home currencies, so when dollar lenders withdrew they had no dollar assets left to sell to repay them.

D. The shortage arose only at banks with large losses on US structured finance products, since banks whose dollar assets were mainly US Treasury and agency securities could roll over their dollar funding throughout the crisis.

Solution

Answer: A

Step 1 – The position

By mid-2007 European banks' combined long US dollar position was about USD 700 billion, mirrored by net borrowing in their own currencies from home residents. Their lower-bound dollar funding gap (net dollar claims on non-banks) was USD 1.0 to 1.2 trillion, covered by interbank borrowing, central bank deposits and FX swaps, plus about USD 1 trillion from US money market funds.

Step 2 – The trigger

From August 2007 interbank funding was compromised; FX swap dislocations made swapped dollars dearer; after Lehman, money market funds and central bank reserve managers withdrew. Structured assets could not be sold without large losses, vehicles came back on balance sheet and credit lines were drawn.

⇒ *"The effective holding period of assets lengthened just as the maturity of funding shortened": an endogenous rise in maturity mismatch that generated the shortage, and that home central banks could not fix because they cannot create dollars.*

Interpretation

The consultant's point for the board: the vulnerability was visible only on the *consolidated* global balance sheet (LTR-15 b), and became systemic because many large banks ran the same funding strategy, "the funding equivalent of a crowded trade".

Why the other options are wrong

B is incorrect. Sibling swap. This is the open-economy domestic bank run story (depositors converting to foreign currency and draining FX reserves) that the paper says the dollar shortage requires "a departure from"; it remains relevant to emerging markets.

C is incorrect. Polarity flip. The major European systems were net *long* dollars (dollar assets funded in home currency and swapped). Borrowing dollars to fund home-currency lending describes Spanish banks until mid-2006, the exception.

D is incorrect. Scope/condition error. Writedowns worsened the picture, but the shortage was a funding mismatch across the system: banks with liquid US government securities still had to roll the short-term dollar funding behind all their dollar assets as suppliers withdrew.

LO: LTR-15 a: Identify the causes of the U.S. dollar shortage during the Great Financial Crisis. LTR-15 b: Evaluate the importance of assessing maturity/currency mismatches across the balance sheets of consolidated entities. · Reference: McGuire and von Peter, BIS Working Paper 291 (2009), sections 16.2 to 16.4 and 16.6

Question D8.2

LTR-16 a, b, c · S2 · [Map #46]

The treasurer of a Japanese life insurer that hedges its US dollar bond portfolio with three-month FX swaps is presenting to the investment committee. The committee asks why the cost of hedging has stayed high since 2014, even though concerns about bank credit quality have faded and wholesale dollar funding has recovered. Which statement is correct for the treasurer to make?

A. A persistently negative USD/JPY basis means that borrowing dollars through the FX swap market is cheaper than borrowing them directly in the dollar cash market, which has lowered the insurer's hedging cost.

B. The basis has persisted because heavy demand for dollar hedges from Japanese banks and insurers has met tighter limits to arbitrage, as balance sheet capacity has become costly for the banks that would close it.

C. Unlike the insurer's FX swaps, a cross-currency swap exchanges no interest during its life and re-exchanges the principal at a pre-agreed forward rate rather than at the initial spot rate.

D. Japanese banks, which fund their dollar assets through FX swaps, act mainly as arbitrageurs in the yen/dollar market, so their growing reliance on swaps has narrowed the basis since 2014.

Solution

Answer: B

Step 1 – Read the sign

Since 2007 the basis for lending dollars against the yen has been negative: dollars obtained by swapping yen cost *more* than dollars borrowed in the cash market. The insurer pays that premium in its forward points.

Step 2 – Why it opens

Hedging demand that is "largely insensitive to the size of the basis" presses on it: Japanese banks fund dollar lending with yen deposits via swaps, and life insurers hedge 60 to 70% of their US dollar bonds. Aggregate Japanese hedging need rose from USD 0.9 trillion (2009) to over USD 1.2 trillion (2015), and the basis tracked it.

Step 3 – Why it does not close

Arbitrage now "incurs a cost per unit of balance sheet": CVA and PFE capital charges, leverage ratios, quarter-end reporting, costlier dollar repo relative to yen repo, and internal limits on Japanese official sector exposures. "Balance sheet space is rented, not free."

⇒ *With bank credit concerns gone, the persistence is explained by hedging demand meeting limits to arbitrage, not by counterparty risk.*

Why the other options are wrong

A is incorrect. Polarity flip. A negative basis makes swapped dollars *more* expensive than cash dollars, which raises the insurer's hedging cost.

C is incorrect. Sibling swap. These are FX swap features reversed: a cross-currency swap *does* exchange interest periodically (Libor plus the basis) and re-exchanges principal at the *initial spot* rate; it is the FX swap that settles at the forward rate with no interim interest.

D is incorrect. Role misassignment. Japanese banks are hedgers who add to dollar demand; their reliance on

swaps “has reduced their capacity to serve as counterparties to non-bank hedgers” and to arbitrage, widening, not narrowing, the basis.

LO: LTR-16 a: Differentiate between the mechanics of foreign exchange (FX) swaps and cross-currency swaps. LTR-16 b: Identify key factors that affect the cross-currency swap basis. LTR-16 c: Assess the causes of covered interest rate parity violations after the financial crisis of 2008. · Reference: Borio, McCauley, McGuire and Sushko, “Covered Interest Parity Lost”, BIS Quarterly Review (2016), Box 17.1, sections 17.1 and 17.3

D9 Illiquid Assets

Readings: IM-17 a, b, c, d, e, f · Illiquid Assets (Ang, Asset Management, Ch.13; LTR source text, Chapter 19, 19.1 to 19.7)

MAP TIER D · #48 · 2/16 SOURCES VARDEEZ

Specifics reported

Map #48: "Reported: 'illiquid asset question.' Examinable: the biases in reported illiquid-asset returns (survivorship, sample selection, infrequent trading and the resulting understated volatility and correlation), illiquidity risk premiums, and portfolio allocation with illiquid assets."
Shapes covered here:
S1 biases in reported illiquid asset returns: survivorship, selection, infrequent trading, unsmoothing (Q1)
S2 illiquidity risk premiums across versus within asset classes, and the four ways to harvest them (Q2)
S3 portfolio allocation with illiquid assets: optimal holdings, hurdle rates, rebalancing point (Q2)

Revision notes

- **Chapter verdict:** after taking into account biases induced by infrequent trading and selection, it is unlikely that illiquid asset classes have higher risk-adjusted returns, on average, than liquid stock and bond markets; but there are significant illiquidity premiums *within* asset classes. Portfolio choice models recommend only modest holdings and high required premiums.
- **Liquidating Harvard:** HMC held 55% in hedge funds, private equity and real assets; the endowment fell 22% from July to October 2008, and buyers of its USD 1.5 billion private equity stake asked discounts of about 50%. Its "worst failing" was asset-liability management: the duration of its liabilities was shorter than that of its assets. It borrowed (USD 2.5 billion in bonds).
- **Sources of illiquidity** (IM-17 b, Vayanos and Wang): (1) clientele effects and participation costs, (2) transaction costs, (3) search frictions, (4) asymmetric information (lemons, Akerlof; also a search friction), (5) price impact, (6) funding constraints.
- **Characteristics of illiquid markets** (IM-17 a): most asset classes are illiquid (infrequent trading, small amounts, low turnover; munis trade about twice a year, institutional real estate every 8 to 11 years, art every 40 to 70 years); illiquid markets are large (bigger in total than public stocks and bonds); investors hold lots of illiquid assets (about 90% of individuals' wealth); liquidity dries up regularly (commercial paper, repo, auction rate securities in 2008), with major crises at least once every ten years.
- **Reported returns are not returns** (IM-17 c): three biases overstate expected returns and understate risk: survivorship bias, infrequent sampling, selection bias. Survivorship: poorly performing funds stop reporting (the long-lived smokers analogy); for mutual funds it knocks at least 1% to 2% off expected returns, above 4% when comparing defunct and live funds; reporting bias adds to it. See also A24.
- **Infrequent trading**: volatilities, correlations and betas computed from reported returns are too low (Figure 19.1: quarterly vol 0.23 versus 0.28 from daily data). This is a "population" problem, not a small-sample problem. Geltner: real estate betas understated by a factor of 0.5.
- **Unsmoothing** (IM-17 d): Geltner, Ross and Zisler. Reported returns follow an AR(1); the filter inverts out true returns, which should then be close to IID. Unsmoothing *adds* noise (unlike ordinary filtering, which removes it), so the variance of true returns exceeds that of reported returns since $|\phi| < 1$.
- **Three properties of unsmoothing:** (1) it affects only risk estimates, not expected returns (the first and last prices are unchanged); (2) it has no effect if observed returns are uncorrelated ($\phi = 0$); (3) it is an art: real estate has a fourth-order lag from annual reappraisals, so a good procedure fits an ARMA model and needs economic knowledge of the market.
- **NCREIF evidence:** first-order autocorrelation 0.78; worst reported quarter -5.3% (Dec 1991) becomes -22.6% unsmoothed, and -8.3% (Dec 2008) becomes -36.3%; quarterly volatility rises from 2.25% to 6.26% (stocks about 7.5%); correlation with the S&P 500 rises from 9.2% to 15.8%. Art volatility: 6.5% to 11.5%.
- **Selection bias** (IM-17 c): returns are observed only when values are high (buildings sold when high, buyouts taken public when high, "zombie" VC companies). Fitting the SML to observed returns gives a positive alpha

when the true alpha is zero, a flatter slope (beta underestimated) and lower volatility. Correction: Heckman (1979). Cochrane: VC alpha above 90% falls to −7% after correction; real estate means 1.7% to 0.3%, standard deviations up by a factor of 1.5.

- **Premiums across asset classes** (IM-17 e): the conventional view (Ilmanen's chart) is flawed because of illiquidity biases, because it ignores other risks (NCREIF is beaten by 60/40; average hedge fund and PE fund give zero expected excess returns), because there is no investable market index, and because factor risk cannot be separated from manager skill.
- **Premiums within asset classes:** on-the-run versus off-the-run Treasuries (bonds more than 5% cheaper than notes of the same maturity in the crisis); illiquidity explains 7% of investment-grade and 22% of junk yield variation; equity premiums 1% to 8% (now diminished), pink sheets almost 20%; PE illiquidity premiums typically 3%. Explanation offered: limited integration across asset classes (silos), or investors overpaying for illiquid asset classes.
- **Four ways to harvest**: (i) passive allocation to illiquid asset classes, (ii) liquidity security selection within classes, (iii) market making (DFA in small caps; buying PE secondaries at 30% to 50% discounts in the 1990s, hedge fund discounts 6% to 8%), (iv) dynamic rebalancing at the aggregate level, which is simplest to implement and has the greatest impact. Large owners give up premiums by tracking standard indexes (inclusion effects of 3% to 5%).
- **Portfolio choice** (IM-17 f): Constantinides: with transaction costs there is a no-trade band (10% costs, 35% volatility: trade only outside 0% to 50% around a 25% optimum); premium above 5% for costs of 15% or more. Ang, Papanikolaou and Westerfield (liquidity events with Poisson intensity λ): optimal holding 59% with continuous trading, 44% (half year), 37% (1 year), 24% (2 years), 11% (5 years), 5% (10 years); required premiums 0.7%, 0.9%, 2.0%, 4.3%, 6.0%.
- **Behaviour with illiquid assets:** rebalance to a position *below* the long-run average holding (right skew); consume less; no illiquidity "arbitrage" (no plus or minus infinity positions); illiquidity induces time-varying, endogenous risk aversion and makes the investor more risk-averse toward both liquid and illiquid assets. Size and patience are necessary but not sufficient; the Swensen case rests on one word, "skilled".

Formulas

Reported return as AR(1) (eq. 19.1). $r_t^* = c + \phi\, r_{t-1}^* + \varepsilon_t, \qquad |\phi| < 1$

Unsmoothing filter (eq. 19.2). $r_t = \dfrac{1}{1-\phi}\, r_t^* - \dfrac{\phi}{1-\phi}\, r_{t-1}^*$

Equivalent smoothing view (eq. 19.4). $r_t^* = (1-\phi)\, r_t + \phi\, r_{t-1}^*$: reported returns are a weighted average of the true return and the lagged reported return.

Variance. $\text{var}(r_t) > \text{var}(r_t^*)$ for $|\phi| < 1$; with $\phi = 0$ the two series coincide.

Liquidity events. Poisson intensity λ; average interval between events $= 1/\lambda$ (ten years: $\lambda = 1/10$).

Exam traps

- **Direction of every bias (polarity):** all three biases overstate expected returns and understate risk. Survivorship drops *poor* performers; selection records *high* values.
- **Unsmoothing changes risk only (scope):** volatility, correlation and beta rise; the mean is unchanged.
- **Which correction for which bias (sibling swap):** unsmoothing (Geltner, Ross, Zisler) for infrequent trading; Heckman selection models for selection bias; only observing the full population removes survivorship bias.
- **Across versus within (polarity):** premiums are large *within* asset classes, not across them.
- **Which of the four harvesting routes (sibling swap):** dynamic rebalancing, not passive allocation, is simplest with the greatest impact.
- **Rebalancing point (polarity):** rebalance illiquid holdings to a level *below* the long-run average holding.

Question D9.1

IM-17 c, d · S1 · [Map #48]

A member of the investment committee of a Canadian university endowment is reviewing a consultant's report that relies on two data sets: an appraisal-based institutional real estate index whose quarterly returns show a first-

order autocorrelation of 0.75, and a commercial database of venture capital returns to which reporting is voluntary. Which of the following statements is correct to include in the committee's response to the consultant?

A. Unsmoothing the real estate index with an AR(1) filter will raise its estimated volatility, correlation and beta with equities, but will leave its estimated expected return essentially unchanged.

B. Survivorship bias in the venture capital database makes reported returns understate true returns, because funds that drop out of the database tend to be those with above-average returns.

C. Selection bias in the venture capital database, which arises because values are recorded mainly when they are high, is corrected by applying the same Geltner-Ross-Zisler unsmoothing filter.

D. The downward bias in the real estate index's volatility is a small-sample problem that will disappear once the consultant uses a sufficiently long history of reported quarterly returns.

Solution **Answer: A**

Why A is correct

An autocorrelation of 0.75 signals appraisal smoothing. Infrequent trading makes "estimates of risk, volatilities, correlations, and betas" too low; unsmoothing adds noise back to recover the true returns. Ang's first property: "unsmoothing affects only risk estimates and not expected returns", because smoothing spreads shocks over several periods but still counts all of them, so the first and last prices are unchanged. For NCREIF, volatility rose from 2.25% to 6.26% per quarter and the correlation with the S&P 500 from 9.2% to 15.8%.

Interpretation

The committee should ask for unsmoothed risk estimates before the real estate allocation enters a mean-variance optimiser; raw appraisal data make the asset look like a near-riskless diversifier.

Why the other options are wrong

B is incorrect. Polarity flip. Survivorship bias "results from the tendency of poorly performing funds to stop reporting", so it makes true returns *worse* than reported data; reported returns overstate true returns.

C is incorrect. Sibling swap. Unsmoothing corrects infrequent trading. Selection bias is addressed with Heckman's (1979) selection models, which let the threshold above which returns are observed vary over time.

D is incorrect. Scope/condition error. Ang states that the infrequent-observation bias "is not a 'small sample' problem, which goes away when our sample becomes very large; it is a 'population' problem".

LO: IM-17 c: Assess the impact of biases on reported returns for illiquid assets. IM-17 d: Explain the process of unsmoothing returns and the effects of unsmoothing. · Reference: Ang, Asset Management, Ch.13 (LTR source text Ch.19), 19.4

Question D9.2 IM-17 e, f · S2, S3 · [Map #48]

The CIO of a sovereign wealth fund proposes raising private equity from 10% to 35% of the portfolio, arguing that the fund's size and long horizon let it harvest a large illiquidity premium. A risk consultant is asked to reply using Ang's analysis of illiquid assets. Which of the following statements should the consultant include?

A. After correcting for data biases, illiquidity risk premiums are large across asset classes, while within asset classes the evidence for higher returns on less liquid securities is weak.

B. Of the four ways to harvest illiquidity premiums, a passive allocation to illiquid asset classes is the simplest to implement and has the greatest impact on portfolio returns.

C. In the Ang, Papanikolaou and Westerfield model, if liquidity events arrive on average once every ten years, the optimal holding falls to about 5% and the required illiquidity premium is about 6%.

D. Because illiquid wealth is right-skewed between liquidity events, the fund should rebalance its private equity holding to a level above its long-run average holding when trading is possible.

Solution **Answer: C**

Why C is correct

Table 19.2: with continuous trading the optimal risky holding is 59%; when the average interval between liquidity events is ten years it falls to 5% (Panel A), and the investor should demand a 6.0% illiquidity premium, defined as a certainty equivalent hurdle rate (Panel B). Ang adds that, given agency conflicts, capital calls and asset-liability mismatches, true hurdle rates are even higher, and most illiquid assets do not beat them.

Interpretation

A 35% target is far above what portfolio choice models with illiquidity risk support. "Size and patience are necessary but not sufficient": the premium Harvard earned came from skill in selecting managers, not from an asset class premium.

Why the other options are wrong

A is incorrect. Polarity flip (across versus within). Ang: "there do not seem to be significant illiquidity risk premiums across classes", but "there are large illiquidity risk premiums within asset classes" (Treasuries, corporate bonds, equities, hedge funds, PE, real estate).

B is incorrect. Sibling swap. Of the four routes (passive allocation, liquidity security selection, market making, dynamic rebalancing), it is dynamic rebalancing at the aggregate level that "is simplest to implement and has the greatest impact on portfolio returns".

D is incorrect. Polarity flip. Because illiquid holdings are right-skewed, the average holding exceeds the optimal rebalanced holding: "the optimal trading point of illiquid assets is lower than the long-run average holding".

LO: IM-17 e: Compare illiquidity risk premiums across and within asset categories. IM-17 f: Evaluate the impact of allocating illiquid assets to a portfolio, including the impact on rebalancing and trading and on optimizing the proportion of illiquid assets. · Reference: Ang, Asset Management, Ch.13 (LTR source text Ch.19), 19.5 and 19.6, Table 19.2

D10 BHC Capital Planning and VaR Model Validation

Readings: ORR-20 a, b · Capital Planning at Large Bank Holding Companies: Supervisory Expectations and Range of Current Practice (Board of Governors of the Federal Reserve System, August 2013); MR-6 a, b, c, d · Validating Bank Holding Companies' Value-at-Risk Models for Market Risk (Lynch, in Lynch, Hasan and Siddique (eds.), Validation of Risk Management Models, Ch.2)

MAP TIER D · #53 · 2/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #53: "Reported: 'BHC'. Both readings are new or newly emphasised, which historically correlates with heavier sampling. Capital planning governance, scenario design, and the validation standards applied to BHC VaR models are the examinable content."
Shapes covered here:
S1 capital planning governance: board versus senior management, reporting of overlays (Q1)
S2 BHC scenario design and estimation practices that the Federal Reserve calls stronger or weaker (Q1)
S3 validation standards for BHC VaR models: conceptual soundness, sensitivity analysis, confidence intervals, benchmarking (Q2).

Revision notes

- **Capital Plan Rule** (ORR-20 a): all US-domiciled top-tier BHCs with total consolidated assets of USD 50 billion or more must maintain a capital plan supported by a robust capital adequacy process (CAP). CCAR is the supervisory program that assesses the plans (2013: the 18 BHCs from the 2009 SCAP; others via a separate Capital Plan Review until 2014). Capital is "the first line of defense against losses". Supervisory scenarios are not all-encompassing: an internal process that simply mirrors the Federal Reserve's stress testing is a weak practice, and leading practices are not a safe harbor.
- **The seven CAP principles** (ORR-20 a): (1) sound foundational risk management; (2) effective loss-estimation methodologies; (3) solid resource-estimation methodologies (available capital and projected revenues over the same scenarios); (4) sufficient capital adequacy impact assessment; (5) comprehensive capital policy and capital planning; (6) robust internal controls (policies, change control, model validation and independent review, documentation, internal audit); (7) effective governance (board and senior management oversight, approval of capital decisions).
- **Risk identification** (ORR-20 b): systematic and repeatable; covers on- and off-balance-sheet positions including noncontractual support, and tests the enforceability of netting, guarantees and collateral under stress; a refreshed firm-wide risk inventory. "Other risks" (reputational, strategic, compliance) belong in net income projections or in a cushion above the capital target; a cushion set by a simple rule or unsupported judgment is lagging.
- **Internal controls and validation** (ORR-20 b): internal audit reviews the end-to-end process, not just components. Independent validation of all models under SR 11-7: conceptual soundness, ongoing monitoring (process verification and benchmarking) and outcomes analysis; keep an inventory including feeder models; since backtesting stress models is hard, compensate with sensitivity analysis and benchmark or "challenger" models; validation covers management overlays. Policies reviewed at least annually; robust MIS; full documentation.
- **Governance** (ORR-20 b): the board has ultimate oversight, *approves the capital plan* (and the capital policy, or a designated board committee does), receives capital information at least quarterly including post-stress pro forma measures, and should challenge results knowing they are estimates within a range. Senior management implements, ensures scenarios are severe enough, recommends capital goals and distributions, identifies weaknesses and remediation plans, and reports both to the board. Board minutes should show how decisions were made.
- **Capital policy, goals and targets** (ORR-20 b): internal goals for level and composition of capital, distribution guidelines (payout ratios), strategies for shortfalls, governance and escalation protocols; reevaluated at least annually. Targets sit *above* goals so capital stays above goals under stress. Weak: goals based solely on regulatory minimums. Contingency plan: triggers for baseline and stress, on *projected* as well as actual results and on market indicators (CDS spreads, ratings, stock price); actions ranked by ease of execution and

impact, realistic under stress.

- **BHC scenario design** (ORR-20 b): the BHC-designed scenario must reflect its *unique* vulnerabilities (business model, footprint, portfolio, revenue drivers); a generic recession similar to the supervisory severely adverse scenario does not qualify. Stronger: models combined with expert judgment, tailored vendor scenarios, a suite of scenarios (only one is submitted), sufficient variables, clear narratives. Weak: assuming the BHC gains market share in stress. Post-stress ratios below the supervisory scenario's are "not, in and of itself, a safe harbor".
- **Estimation: general expectations** (ORR-20 b): models or quantitative methods as the basis; internal data, external where internal is short; segment by common risk characteristics; sensitivity analysis and challenger models; vendor models understood, customized and validated. Overlays: quantify their impact against unadjusted estimates; "management should always provide both the initial results and the results after any judgmental adjustments" to the board; extensive overlays should trigger a review of the models. Conservatism: no selective exclusion of losses, no foresight of the scenario path, no favourable management actions that cannot be assured.
- **Credit loss methods** (ORR-20 b): EL approach (PD, LGD, EAD conditioned on the scenario; long-run average parameters "are not appropriate for projecting losses under stress"; aggregate weighted-average LGD without segmentation is lagging; static EAD common, loan-equivalent and CCF methods stronger); rating transition models (four steps; judgmentally picking past stress matrices is weak); roll-rate models (weak beyond the near future; mixing two models is weaker); vintage models; net charge-off models (avoid as primary for a material portfolio); scalars (immaterial portfolios only). AFS and HTM: test all credit-sensitive securities for OTTI.
- **Operational, market and counterparty risk** (ORR-20 b): AMA models not required; use complete internal loss data (do not exclude legal reserves, acquired-firm or discontinued-line losses); regression on frequency, modified LDA at lower percentiles (70th to 98th) with sensitivity of the percentile choice, scenario-analysis overlays, historical averages (backward-looking alone). Trading: deterministic scenarios dominate (design, translation to risk factors, revaluation); instantaneous shocks are suitably conservative; check plausibility (negative rates); generally full revaluation; explicit default of the largest counterparties; mitigating actions only if executable with high confidence.
- **PPNR, balance sheet, RWA, aggregation** (ORR-20 b): PPNR over the nine-quarter horizon consistent with the scenario, balances and RWA (one coherent story); weak: drastic expense cuts, same strategic assumptions in baseline and stress. Balance sheet: build from drivers (paydowns, prepayments, new business); avoid favourable asset-mix shifts, flight-to-quality deposit inflows, costless shrinkage, margin improvement; no RWA reductions from planned model enhancements. Keep ALLL adequate along and at the end of the path. Aggregate through a central group; spreadsheets are weak; rising PPNR amid falling balances needs investigation.
- **Three validation components and VaR models** (MR-6 a): conceptual soundness (data, assumptions, techniques, sensitivity analysis), outcomes analysis (backtesting) and benchmarking. Historical simulation is the most widely used VaR at commercial banks but is not very responsive to recent volatility; GARCH VaR (RiskMetrics a special case) and filtered historical simulation (GARCH volatility times the empirical quantile of standardized shocks) address that.
- **Conceptual soundness: fit for purpose** (MR-6 a): the key purpose of VaR is to show how risk changes when positions change. Berkowitz and O'Brien found a GARCH model on actual P&L outperformed banks' VaR, but a P&L-based VaR does not fall when a trader cuts positions (Lo's Capital Decimation Partners versus Jorion's positional VaR), so it is not fit for purpose and fails conceptual soundness. Supervisors spend effort checking the pseudo history of P&L.
- **Sensitivity analysis** (MR-6 b): VaR is linear homogeneous in positions (Euler): component VaR $=$ VaR $\times$ $w_i\beta_i$, marginal VaR $= \partial \text{VaR}/\partial V_i$. Challenge: the regression needs a full time series of the position's value changes, exactly what is missing; proxies may be less volatile or less correlated and understate. Tasche and Hallerbach: component VaR $\approx$ the position's change on the day that sets the HS VaR (or an average of ordered observations near it). If data were sufficient the risk would already be in VaR. Supervisors ask for risks not in VaR on both component and stand-alone bases (stand-alone is independent of portfolio composition).
- **Confidence intervals** (MR-6 c): few BHCs compute them. Jorion's asymptotic standard error needs the density at the VaR, and normality is a mistake. Nonparametric: order statistics (Dowd) and bootstrap (Christoffersen and Goncalves; assumes independence for HS; GARCH bootstraps re-estimate the model on each

sample). Table 6.1: intervals are asymmetric, more data tightens them, HS intervals are particularly wide in stress periods, FHS intervals are the narrowest (more efficient).

- **Backtesting** (MR-6, outcomes): Kupiec unconditional coverage, Christoffersen independence and conditional coverage ($LR_{cc} = LR_{uc} + LR_{ind}$), dynamic quantile, logistic DQ, VaR quantile regression (failed 19 of 20 BHCs). P&L should exclude fees and commissions; type II errors are expensive; small samples call for Monte Carlo p-values. 2013 to 2016: average exceedance 0.4%, maximum 2.1%, five firms with none. Detail: see also C2.
- **Benchmarking** (MR-6 d): the most neglected component. Two hindrances: changing portfolios make errors non-i.i.d., and banks rarely have two models. Fix: a GARCH(1,1) VaR on the bank's own P&L as a ready benchmark. Diebold and Mariano: the loss function matters. The regulatory loss function (Lopez) penalizes only exceedances, so it favours conservatism; the check (quantile) loss rewards accuracy; compare with the sign test. Results: positional VaR beats P&L VaR under the regulatory loss at every bank; P&L VaR beats positional VaR on accuracy at 16 of 19 banks.
- **See also** C2 (backtesting tests), C6 (model risk management and SR 11-7), A19 (FHS and bootstrap), A12 (component and marginal VaR), C9 (supervisory stress tests and scenario design), D19 (LDA).

Formulas

Euler decomposition (MR-6). $\text{VaR}(V_P) = \sum_{i \in P} \dfrac{\partial \text{VaR}}{\partial V_i} V_i, \qquad \text{Component VaR}_i = \text{VaR}(V_P)\, w_i\, \beta_i$

Asymptotic standard error of a VaR quantile (Jorion). $\text{SE}(\text{VaR}) = \dfrac{1}{f(\text{VaR})} \sqrt{\dfrac{c(1-c)}{T}}$

Conditional coverage. $LR_{cc} = LR_{uc} + LR_{ind}$

Sign test (Diebold and Mariano). $z_t = l_t^i - l_t^j, \qquad \dfrac{\sum_t I(z_t > 0) - 0.5N}{\sqrt{0.25N}} \sim N(0,1)$

Capital planning anchors (ORR-20). USD 50 billion threshold; nine-quarter horizon; board information at least quarterly; policies and capital policy reviewed at least annually.

Exam traps

- **Board versus senior management (role misassignment):** the board approves the capital plan and capital decisions; senior management implements, recommends and reports weaknesses.
- **Safe harbors (absolute claim):** neither leading practices nor a BHC scenario harsher than the supervisory severely adverse one is a safe harbor.
- **Parameters for stress (scope error):** long-run average PD, LGD, EAD are for a mix of conditions, not for stress; NCO models and scalars are not primary tools for material portfolios.
- **Accurate versus fit for purpose (sibling swap):** a GARCH VaR on actual P&L can be more accurate yet fail conceptual soundness because it ignores position changes.
- **Confidence interval width (polarity flip):** FHS intervals are the narrowest; HS intervals are widest, especially in stress periods.
- **Loss function (sibling swap):** the regulatory loss function rewards conservatism (positional VaR wins); the check loss rewards accuracy (P&L VaR wins at 16 of 19).

Question D10.1

ORR-20 a, b · S1, S2 · [Map #53]

An internal auditor at a large US bank holding company subject to the Capital Plan Rule is assessing the BHC's capital planning process against the Federal Reserve's description of the range of current practice. Which of the following observations should the auditor record as consistent with stronger practice?

A. The BHC's own stress scenario produces lower post-stress capital ratios than the supervisory severely adverse scenario, so its severity is accepted without further support.

B. Senior management, rather than the board, approves the capital plan, while the board is updated on capital adequacy and post-stress capital measures at least quarterly.

C. Where management overlays adjust modeled loss estimates, the board receives both the unadjusted model

results and the results after the judgmental adjustments.

D. Wholesale losses under stress are projected with long-run average PD, LGD, and EAD by rating grade, since these averages span both expansions and downturns.

Solution **Answer: C**

Step 1 – Overlays and reporting (Section 20.7)

If a BHC makes extensive adjustments to modeled estimates, "the impact of such adjustments should be quantified relative to unadjusted estimates" and documented for management and the board; "management should always provide both the initial results and the results after any judgmental adjustments".
⇒ *The board can only challenge the result if it sees how far judgment moved the model; extensive overlays should also prompt a review of whether better models are needed.*

Step 2 – Check the other observations against governance and scenario design

Board approval of the capital plan is required by the Rule (Section 20.4). Scenario severity must come from the BHC's unique vulnerabilities, and a harsher-than-supervisory outcome is not a safe harbor (Section 20.6). Long-run averages are excluded for stress projection (Section 20.7).

Interpretation

C is correct: it is a governance practice (S1) that keeps the board informed of model limitations before capital decisions are made.

Why the other options are wrong

A is incorrect. Absolute claim: "a BHC stress scenario that produces post-stress capital ratios lower than those under the supervisory severely adverse scenario is not, in and of itself, a safe harbor"; the scenario must still target the BHC's own vulnerabilities and be supported by a clear narrative.
B is incorrect. Role misassignment: "the board of directors is required to approve a BHC's capital plan under the Capital Plan Rule"; senior management implements board-authorized activities and makes recommendations to the board. The quarterly information clause is correct.
D is incorrect. Scope/condition error: "estimates based on long-run average behavior over a mix of conditions, including periods of economic expansion and downturn, are not appropriate for projecting losses under stress and should not be used for these purposes".

LO: ORR-20 b: Describe practices that can result in a strong and effective capital adequacy process for a BHC in the following areas: risk identification; internal controls, including model review and validation; corporate governance; capital policy, including setting of goals and targets and contingency planning; stress testing and stress scenario design; estimating losses, revenues, and expenses, including quantitative and qualitative methodologies; assessing the impact of capital adequacy, including risk-weighted asset (RWA) and balance sheet projections. · Reference: Board of Governors of the Federal Reserve System, Capital Planning at Large Bank Holding Companies (August 2013), Sections 20.4, 20.6, 20.7

Question D10.2

MR-6 a, b, c, d · S3 · [Map #53]

A model validation analyst at a US bank holding company is writing the validation report for the trading desk's historical-simulation VaR model, covering conceptual soundness, sensitivity analysis, confidence intervals and benchmarking. Which of the following statements is correct for the analyst to include in the report?

A. A VaR model fitted by applying a GARCH model to the desk's actual P&L history would fail the conceptual soundness test, because its VaR would not fall when the trader cuts positions.

B. Confidence intervals for historical-simulation VaR are typically narrower than those for filtered historical simulation, and they become narrower still when estimated over stress periods.

C. Under the regulatory loss function, which penalizes only exceedances, a P&L-based GARCH VaR outperformed the banks' positional VaR models at most of the banks studied.

D. The component VaR of a position with scarce valuation data is best estimated by regressing changes in portfolio value on the full time series of that position's value changes.

Solution **Answer: A**

Step 1 – Purpose test (Section 6.3)

"A crucial aspect of VaR modeling; its key purpose is to show how risk changes when positions change." A VaR

built on actual P&L history is unaffected when positions are cut, because the history does not change; a VaR on the pseudo history of the current portfolio does fall.
⇒ *Such a model "is not fit for purpose and thus fails a crucial conceptual soundness test", even though Berkowitz and O'Brien found the P&L GARCH model more accurate as a forecast.*

Step 2 – Check the remaining validation claims

Confidence intervals (Section 6.5, Table 6.1): FHS narrowest; HS "particularly wide" in stress periods. Benchmarking (Section 6.8): positional VaR wins under the regulatory loss function at every bank; P&L VaR wins under the check loss at 16 of 19. Sensitivity (Section 6.4): with scarce data the regression approach "is difficult to implement"; Tasche and Hallerbach use the position's change on the day that determines the HS VaR.

Interpretation

A is correct. It is the reason a validator may use the P&L GARCH VaR as a *benchmark* but not as the risk management model.

Why the other options are wrong

B is incorrect. Polarity flip: HS intervals are wider than FHS intervals (FHS "provides more efficient estimates of VaR"), and HS intervals widen, not narrow, in stress periods.

C is incorrect. Sibling swap: the P&L VaR outperformed on accuracy under the *check* loss function; under the regulatory loss function the positional VaR "significantly outperforms the P&L VaR in every case", reflecting conservatism.

D is incorrect. Scope/condition error: the regression of Equation 6.5 "requires that you have a full time series of the change in value of the positions", which is what is missing; the component is instead estimated from the position's loss on the VaR-determining day, or an average of ordered observations near it.

LO: MR-6 a: Describe some important considerations for a bank in assessing the conceptual soundness of a VaR model during the validation process. MR-6 b: Explain how to conduct sensitivity analysis for a VaR model, and describe the potential benefits and challenges of performing such an analysis. MR-6 c: Describe the challenges a financial institution could face when calculating confidence intervals for VaR. MR-6 d: Discuss the challenges in benchmarking VaR models and various approaches proposed to overcome them. · Reference: Lynch, Validating Bank Holding Companies' Value-at-Risk Models for Market Risk, Sections 6.3, 6.4, 6.5, 6.8

D11 Empirical Properties of Correlation

Readings: MR-9 a, b, c · Empirical Properties of Correlation: How Do Correlations Behave in the Real World? (Meissner, Correlation Risk Modeling and Management 2e, Ch.2)

MAP TIER D · #54 · 2/16 SOURCES VARDEEZ

Specifics reported

Map #54: "Named: 'identifying best-fit distribution (equity, bond and default correlation)' and 'know calculation of mean reversion and autocorrelation.' Which distribution best fits which correlation type is a three-way sibling swap; the mean-reversion rate is obtained by regressing the correlation change on the lagged level."
Shapes covered here:
S1 best-fit distribution for equity, bond and default probability correlations (Q2)
S2 mean reversion rate from regressing the correlation change on the lagged level (Q1)
S3 the corresponding autocorrelation (Q1).

Revision notes

- **The study** (MR-9 a): daily closes of the 30 Dow stocks, January 1972 to July 2017; monthly 30×30 Pearson correlation bins, diagonal removed and averaged: 534 months, 480,600 correlation values. States: expansion (GDP growth 3.5% or higher), normal (0% to 3.49%), recession (two consecutive quarters of negative growth).
- **Correlation levels by state** (MR-9 a): expansion 27.46%, normal 33.06%, recession 36.96%. "The worse the state of the economy the higher are equity correlations"; in strong growth, prices react to idiosyncratic factors, in recessions macroeconomic factors dominate. Dow correlations reached 96.97% in February 2009.
- **Correlation volatility by state** (MR-9 a): expansion 71.17%, normal 83.06%, recession 80.48%. Volatility is lowest in expansions; it is higher in normal periods than in recessions, against expectation, because "high correlation levels in a recession remain high without much additional volatility". Correlation level and correlation volatility are positively related.
- **Correlation volatility and recessions** (MR-9 a): before every recession except 1990-91 a downturn in correlation volatility occurred, but the relationship between the decline and the severity of the recession is statistically non-significant (R^2 close to zero).
- **Mean reversion** (MR-9 b; S2): "the tendency of a variable to be pulled back to its long-term mean"; present if the change $S_t - S_{t-1}$ is negatively related to S_{t-1}. Discrete Vasicek: $S_t - S_{t-1} = a(\mu_S - S_{t-1})\Delta t + \sigma_S \varepsilon \sqrt{\Delta t}$, with $0 \leq a \leq 1$: $a = 1$ pulls fully back to the mean each step, $a = 0.5$ halfway, $a = 0$ not at all.
- **Regression for the rate** (MR-9 b; S2): with $\Delta t = 1$ and no stochastic term, $S_t - S_{t-1} = a\mu_S - aS_{t-1}$. Regress $Y = S_t - S_{t-1}$ on $X = S_{t-1}$: the slope $\beta = -a$ and the intercept $= a\mu_S$. Dow 1972 to 2017: $y = -0.7903x + 0.256$, $R^2 = 0.3948$: mean reversion 79.03%, long-term mean $0.256/0.7903 \approx 32.38\%$.
- **Forecasting with it** (MR-9 b; Example 9.1): $S_t = a(\mu_S - S_{t-1}) + S_{t-1}$. February 2017 correlation 26.15%, mean 32.38%: March forecast $0.7903 \times (0.3238 - 0.2615) + 0.2615 = 31.07\%$.
- **Autocorrelation** (MR-9 b; S3): "the degree to which a variable is correlated to its past values", $AC(\rho_t, \rho_{t-1}) = \text{cov}(\rho_t, \rho_{t-1})/(\sigma(\rho_t)\sigma(\rho_{t-1}))$, algebraically the Pearson coefficient. It is "the opposite property" of mean reversion: one-lag autocorrelation 20.97% and mean reversion 79.03% "add up to 1". Positive autocorrelation is also called persistence. Autocorrelation decays with longer lags (the 2-month lag was the highest).
- **Equity correlation distribution** (MR-9 c; S1): 464,580 daily correlation values, 77.23% positive. Sixty-one distributions tested with Kolmogorov-Smirnov, Anderson-Darling and Chi-squared: the four-parameter Johnson SB (γ, δ shape; μ location; σ scale) fits best; normal, lognormal and beta fit poorly. Splitting by state of the economy changes the fit "slightly but not significantly".
- **Bond correlations** (MR-9 a, c; S1): level 41.67% (higher than equity's 34.83%), volatility 63.74% (lower than equity's 79.73%), mean reversion 25.79%. The distribution "shows a more normal shape" and is best fitted by the *generalised extreme value* distribution, "and quite well with the normal distribution".

- **Default probability correlations** (MR-9 a, c; S1): level 30.43% (slightly lower than equity), volatility 87.74% (slightly higher), mean reversion 29.97%. The distribution is similar to equity correlations and "can be replicated best by the Johnson SB distribution".
- **Mean reversion comparison** (MR-9 b): equity correlation mean reversion (77.51% in the bond and default comparison sample; 79.03% monthly in the main study) is far stronger than bond (25.79%) or default probability (29.97%) correlation mean reversion. Bond and default results are preliminary and being verified on a larger sample.
- **Implication**: higher correlation levels and correlation volatility in economic distress mean higher correlation risk in bad times; correlation models should include mean reversion.

Formulas

Discrete Vasicek (Meissner eq. 9.2). $S_t - S_{t-1} = a(\mu_S - S_{t-1})\Delta t + \sigma_S \varepsilon \sqrt{\Delta t}$

Regression form ($\Delta t = 1$, eq. 9.4 and 9.5). $\underbrace{S_t - S_{t-1}}_{Y} = \underbrace{a\mu_S}_{\text{intercept}} \underbrace{-a}_{\beta} \underbrace{S_{t-1}}_{X} \quad \Rightarrow \quad a = -\beta, \qquad \mu_S = \dfrac{\text{intercept}}{a}$

One-step forecast. $E[S_t] = S_{t-1} + a(\mu_S - S_{t-1})$

Autocorrelation (eq. 9.6). $AC(\rho_t, \rho_{t-1}) = \dfrac{\text{cov}(\rho_t, \rho_{t-1})}{\sigma(\rho_t)\,\sigma(\rho_{t-1})}$; in Meissner's study $AC \approx 1 - a$.

Book checks. $y = -0.7903x + 0.256$: $a = 79.03\%$, $\mu = 32.38\%$, $AC = 20.97\%$; forecast $26.15\% \to 31.07\%$.

Best fits. Equity: Johnson SB. Default probability: Johnson SB. Bond: generalised extreme value (normal also quite good).

Exam traps

- **Sign of the slope:** the regression slope is $-a$; the mean reversion rate is its absolute value. Plugging β with its sign into the forecast pushes the correlation *away* from the mean.
- **Intercept is not the mean:** the intercept equals $a\mu_S$; divide by a to get the long-term mean.
- **Mean reversion versus autocorrelation (sibling swap):** strong mean reversion means *low* autocorrelation (79.03% against 20.97%); using one in place of the other in the forecast is the classic slip.
- **Three-way best-fit swap:** equity and default probability correlations: Johnson SB; bond correlations: generalised extreme value. Swapping bond and default is the planted distractor.
- **Volatility by state (polarity flip):** correlation volatility is lowest in expansions, but it is higher in normal periods (83.06%) than in recessions (80.48%).
- **Absolute claims on distributions:** "normal, lognormal or beta" do not fit equity correlations; the economic state does not significantly change the equity fit.

Question D11.1

MR-9 b · S2, S3 · [Map #54]

A quantitative analyst at a Hong Kong asset manager models the monthly average pairwise correlation of a regional equity basket. Regressing the monthly change in the average correlation, $\rho_t - \rho_{t-1}$, on the previous month's level, ρ_{t-1}, gives:

- Intercept: 0.217; slope coefficient: -0.62; R^2: 0.31
- Current (last month's) average correlation: 24%

Ignoring the stochastic term, which of the following is closest to the one-lag autocorrelation of the correlation series implied by the regression and the expected average correlation for next month?

A. Autocorrelation 38%; expected correlation 30.8%

B. Autocorrelation 38%; expected correlation 28.2%

C. Autocorrelation 38%; expected correlation 22.6%

D. Autocorrelation 62%; expected correlation 30.8%

Solution **Answer: A**

Step 1 – Mean reversion rate

In $\rho_t - \rho_{t-1} = a\mu_\rho - a\rho_{t-1}$, the slope is $-a$:

$$a = -\beta = -(-0.62) = 0.62$$

⇒ *Each month, 62% of the gap between the current level and the long-term mean is closed.*

Step 2 – Long-term mean from the intercept

$$\mu_\rho = \frac{\text{intercept}}{a} = \frac{0.217}{0.62} = 0.35$$

Step 3 – Autocorrelation

$$AC = 1 - a = 1 - 0.62 = 0.38$$

⇒ *Mean reversion and one-lag autocorrelation are reverse properties; in Meissner's study they sum to 1 (79.03% and 20.97%).*

Step 4 – Expected correlation next month

$$E[\rho_t] = \rho_{t-1} + a(\mu_\rho - \rho_{t-1}) = 0.24 + 0.62 \times (0.35 - 0.24) = 0.24 + 0.0682 = 0.3082$$

Interpretation

The basket's correlation, currently well below its long-term mean of 35%, is expected to jump to about 30.8% next month. For a book that is short correlation (for example, sold basket options), that strong pull back towards the mean is the risk to hedge. The R^2 of 0.31 describes the regression's fit and is not an input.

Why the other options are wrong

B is incorrect. Used the autocorrelation (0.38) as the speed of mean reversion: $0.24 + 0.38 \times 0.11 = 0.2818$.
C is incorrect. Took the intercept (0.217) as the long-term mean instead of $a\mu_\rho$: $0.24 + 0.62 \times (0.217 - 0.24) = 0.2257$.
D is incorrect. Reported the absolute slope (0.62), which is the mean reversion rate, as the autocorrelation; the forecast is right but the autocorrelation is the sibling quantity.

LO: MR-9 b: Calculate a mean reversion rate using standard regression and calculate the corresponding autocorrelation. · Reference: Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.2 and 9.3, eqs. 9.2 to 9.6, Example 9.1

Question D11.2 MR-9 a, c · S1 · [Map #54]

A risk analyst at a UK bank is calibrating a correlation simulation engine that will draw correlation scenarios for three books: an equity basket options book, a corporate bond portfolio, and a credit book whose losses depend on default probability correlations. Following Meissner's empirical study, which of the following recommendations should the analyst make?

A. Use the Johnson SB distribution for all three correlation types, since it gave the best fit for equity, bond and default probability correlations alike.

B. Use the Johnson SB distribution for equity and default probability correlations, and the generalised extreme value distribution for bond correlations.

C. Use the Johnson SB distribution for equity and bond correlations, and the generalised extreme value distribution for default probability correlations.

D. Use the Johnson SB distribution for equity correlations in expansions, but switch to the normal distribution in recessions when correlations are high.

Solution **Answer: B**

Step 1 – Equity correlations

Of 61 distributions tested (Kolmogorov-Smirnov, Anderson-Darling, Chi-squared), the four-parameter Johnson SB gave the best fit; normal, lognormal and beta fit poorly.

Step 2 – Default probability correlations

"The default probability correlation distribution is similar to equity correlation distribution … and can be replicated best by the Johnson SB distribution."

Step 3 – Bond correlations

"The bond correlation distribution shows a more normal shape and can be best fitted with the generalised extreme value distribution and quite well with the normal distribution."

Interpretation

B is correct. The engine should use Johnson SB for the equity and credit books and the generalised extreme value distribution for the bond portfolio. The bond and default results are preliminary and "currently being verified with a larger sample", so the analyst should flag them for periodic review.

Why the other options are wrong

A is incorrect. Absolute claim. Johnson SB is not the best fit for every type: bond correlations have a more normal shape and are best fitted by the generalised extreme value distribution.

C is incorrect. Sibling swap. The bond and default probability fits are exchanged; it is default probability correlations that resemble equity correlations.

D is incorrect. Scope/condition error. Fitting equity correlations by state of the economy gave results that were "slightly but not significantly different", and the normal distribution fits equity correlations poorly in any state.

LO: MR-9 c: Identify the best-fit distribution for equity, bond, and default correlations. MR-9 a: Describe how equity correlations and correlation volatilities behave throughout various economic states. · Reference: Meissner, Correlation Risk Modeling and Management 2e, Ch.2, 9.4 and 9.6, Summary

D12 Convexity and Jensen's Inequality

Readings: MR-13 a, b, c, d · Expectations, Risk Premium, Convexity, and the Shape of the Term Structure (Tuckman and Serrat, Fixed Income Securities 4e, Ch.8); MR-14 (Tables 14.1, 14.2) for convexity effects by term

MAP TIER D · #55 · 2/16 SOURCES VARDEEZ

Specifics reported

Map #55: "Named: 'Jensen's inequality' and the memorised direction, 'the value of convexity increases with bond volatility and time to maturity.' One sheet adds that calculations on this are highly unlikely. This is a pure polarity item: get the direction right and the question is free."
Shapes covered here:
S1 Jensen's inequality applied to bond prices: the expected price exceeds the price at the expected rate (Q1)
S2 the value of convexity increases with volatility (Q1)
S3 the value of convexity increases with time to maturity, so forward rates bend down more at longer terms (Q1).

Revision notes

- **Expectations alone** (MR-13 a): with certainty that the one-year rate will be 8%, 7% and 6%, $P(1) = 1/1.08$, $P(2) = 1/(1.08 \times 1.07)$, $P(3) = 1/(1.08 \times 1.07 \times 1.06)$, so forward rates equal the expected rates. Without volatility, expectations completely determine the term structure, which can take any shape.
- **Granularity of expectations** (MR-13 a): views can be specific over short horizons (central bank meeting dates, tax dates, issuance calendars, quarter-end balance sheet management) but not over long ones: the expected one-year rate in 29 years is unlikely to differ from that in 30 years. Macroeconomics may anchor a long-run expectation (book: 4% = 1% real rate + 3% inflation).
- **Volatility in a risk-neutral tree** (MR-13 b): book tree with annual steps and 50% probabilities: 9% today; 13% or 5%; 17%, 9% or 1%. Expected rate is 9% at every horizon and each transition has a volatility of 400 bp. Prices of one-, two- and three-year zeros: 0.917431, 0.842815, 0.776366; forward rates 9.0000%, 8.8532%, 8.5590%. The forward curve slopes *down* although expectations are flat: interest rate volatility interacting with the convexity of bond prices.
- **Jensen's inequality** (MR-13 c, S1): $E\left[\frac{1}{1+r}\right] > \frac{1}{1+E[r]}$: "the expected price of a bond is greater than the price of a bond at the expected interest rate". Graphically, the chord between the two price points lies above the convex price-rate curve (Figure 13.2). Check: $0.5/1.13 + 0.5/1.05 = 0.918668 > 1/1.09 = 0.917431$.
- **From Jensen to forward rates** (MR-13 c): the two-year zero satisfies $\frac{1}{1.09(1+f)} = \frac{1}{1.09}\left[0.5\frac{1}{1.13} + 0.5\frac{1}{1.05}\right]$, and Jensen gives $\frac{1}{1+f} > \frac{1}{1.09}$, so $f < 9\%$: the one-year rate one year forward is *less* than the expected one-year rate in one year.
- **Size of the effect** (S2, S3): in the continuous decomposition the convexity term lowers the forward rate by $\frac{1}{2}C\sigma^2$: larger with rate volatility σ and with the convexity C of the zero of that term, which grows with maturity. Model 1 (113 bp volatility) convexity effects on par rates: -0.8, -5.1, -18.8, -135.3 bp at 2, 5, 10 and 30 years (Table 14.1).
- **Mean reversion damps it** (MR-14, Table 14.2): Vasicek at 126 bp volatility gives -1.0, -5.8, -19.1, -74.7 bp: slightly larger to 10 years (higher volatility), but much smaller at 30 years because mean reversion lowers the volatility of long rates. Same message: more volatility of the relevant rate, more convexity value. See also A25 (Vasicek effectiveness: mean reversion cuts the 30-year convexity effect to about 75 bp from 135 bp).
- **Return decomposition** (MR-13 d): $\frac{dP}{P} = f(T)dt - D\,dr + \frac{1}{2}C\sigma^2 dt$: return from the passage of time (the forward rate), from rate changes (minus duration), and from volatility (plus convexity). Across equal-duration portfolios, the more convex one gains more whether rates rise or fall.
- **Expected return for a risk-averse investor** (MR-13 d): risk-neutral investors require $E[dP/P] = r\,dt$; risk-averse investors require $E[dP/P] = r\,dt + \lambda D\,dt$. Book: $r = 1\%$, $D = 5$, $\lambda = 10$ bp per year of duration gives 1.5%. Sharpe ratio $= \lambda/\sigma$: 10 bp over 100 bp gives 10%.
- **Forward rate decomposition** (MR-13 b, c, d): $f(T) = r + D\,E[dr]/dt + \lambda D - \frac{1}{2}C\sigma^2$: expectations, risk premium, convexity. The forward rate is reduced by $\frac{1}{2}C\sigma^2$, which exactly offsets the direct convexity

gain in expected return: there is "no net advantage of convexity". See also D18 for the expectations and risk premium terms.

- **Drift equals risk premium in pricing** (MR-13): a drift of a number of basis points has the same pricing effect as a risk premium of that number per year of duration; bond prices alone cannot separate them, so only the risk-neutral process matters for pricing (link to C7).

Formulas

Jensen's inequality (eq. 13.5). $E\left[\frac{1}{1+r}\right] > \frac{1}{1+E[r]}$

Forward from zero prices. $1 + f(t) = P(t-1)/P(t)$: book $0.917431/0.842815 = 1.088532$, $0.842815/0.776366 = 1.085590$

Instantaneous return (eq. 13.15). $\frac{dP}{P} = f(T)\,dt - D\,dr + \frac{1}{2}C\sigma^2 dt$

Risk-averse expected return (eq. 13.18). $E\left[\frac{dP}{P}\right] = r_0\,dt + \lambda D\,dt \qquad SR = \frac{\lambda}{\sigma}$

Forward rate decomposition (eq. 13.20). $f(T) = r_0 + D\,\frac{E[dr]}{dt} + \lambda D - \frac{1}{2}C\sigma^2$

Exam traps

- **Polarity of Jensen:** the *expected price* exceeds the *price at the expected rate*; hence forward rates lie *below* expected rates once volatility is added.
- **Polarity of the convexity value:** it *increases* with volatility and with maturity. A statement that it falls with maturity, or is independent of volatility, is a flip.
- **Scope:** the convexity effect is not a parallel shift of forwards: equal per-step volatility still gives a larger effect at longer terms because C grows with term.
- **Absolute claim:** being long convexity does not earn a riskless excess return; the lower forward rate (lower yield) is the price paid for convexity.
- **Sign in the decomposition:** expectations and risk premium enter with a plus sign, convexity with a minus sign.

Question D12.1

MR-13 b, c · S1, S2, S3 · [Map #55]

A portfolio manager at a German insurer is presenting to a group of newly hired analysts. She uses a risk-neutral binomial tree with annual steps in which the expected one-year rate is the same at every future date, the one-year rate is volatile, and investors require no risk premium. She then compares the forward rates implied by the tree's zero-coupon bond prices with the flat expected rate. Which of the following statements is correct for her to make?

A. Because bond prices are convex in rates, the price of a zero at the expected rate exceeds its expected price, so forward rates lie above the expected short rate.

B. Convexity lowers every forward rate by the same number of basis points, because the volatility of each one-year rate transition in the tree is the same at every node.

C. The value of convexity increases with rate volatility and with maturity, so forward rates lie below the expected short rate and fall further at longer terms.

D. A position that is long convexity earns an expected return above the short rate in every scenario, because convexity adds to return whether rates rise or fall.

Solution

Answer: C

Step 1 – Jensen's inequality

$$E\left[\frac{1}{1+r}\right] > \frac{1}{1+E[r]}: \qquad 0.5 \times \frac{1}{1.13} + 0.5 \times \frac{1}{1.05} = 0.918668 > \frac{1}{1.09} = 0.917431$$

⇒ Averaging over volatile rates raises bond prices above the price at the average rate, which lowers the yields and forward rates they imply.

Step 2 – Effect grows with maturity

In the book's tree (expected rate 9% at every horizon, 400 bp per transition) the forward rates are 9.0000%, 8.8532% and 8.5590%: about 15 bp below 9% for year two and 44 bp for year three. In the decomposition the reduction is $\frac{1}{2}C\sigma^2$, and C rises with term.

Step 3 – Effect grows with volatility

The same $\frac{1}{2}C\sigma^2$ term rises with σ. The MR-14 tables show it: lowering long-rate volatility through mean reversion cuts the 30-year par-rate convexity effect from -135.3 bp (Model 1) to -74.7 bp (Vasicek).

Interpretation

With flat expectations and no risk premium, the forward curve slopes down purely because of convexity, more at longer terms and more in more volatile markets.

Why the other options are wrong

A is incorrect. Polarity flip of Jensen's inequality: the expected price exceeds the price at the expected rate, so forwards lie *below* the expected rate.

B is incorrect. Scope/condition error: equal per-transition volatility does not make the effect equal across terms; $\frac{1}{2}C\sigma^2$ grows with the convexity of longer zeros (15 bp versus 44 bp in the book).

D is incorrect. Absolute claim: the forward rate is reduced by exactly the convexity gain, so the expected return condition leaves no net advantage of convexity; nothing is earned "in every scenario".

LO: MR-13 b: Apply a risk-neutral interest rate tree to assess the effect of volatility on the shape of the term structure. MR-13 c: Estimate the convexity effect using Jensen's inequality. · Reference: Tuckman and Serrat 4e, Ch.8, 13.2 and 13.3, eqs. 13.5 to 13.8 and 13.20, Table 13.1; Tuckman and Serrat 3e, Ch.9, Tables 14.1 and 14.2

D13 Retail Credit Risk

Readings: CR-7 a-h · Credit Scoring and Retail Credit Risk Management (Crouhy, Galai and Mark, Ch.9); cross-link CR-6 a, b · Credit Scoring and Rating (Doumpos et al., Ch.2)

MAP TIER D · #57 · 2/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #57: "Reported: 'retail credit'. The examinable retail-specific material is the risk-based-pricing / adverse-selection dynamic and the dark-side-of-retail-credit argument. Rating agency methodologies (Reading 24) sit alongside as a read-only item."
Shapes covered here:
S1 risk-based pricing and adverse selection under a "one price fits all" policy; tiered pricing direction (Q1)
S2 the dark side of retail credit: a systematic factor that drives many credits at once (Q1)
S3 retail versus corporate credit risk: predictable EL, small but non-zero capital (Q1).

Revision notes

- **What counts as retail** (CR-7 a, Box 7.1): Basel defines retail exposures as homogeneous portfolios of a large number of small, low-value loans, with a consumer or business focus, where the incremental risk of any single exposure is small; small business loans qualify if total exposure is under EUR 1 million. Products: home mortgages (LTV is the key risk variable), home equity loans (HELOC), installment loans, unsecured credit card revolving loans, small business loans.
- **Retail versus corporate** (CR-7 b, S3): retail exposures "arrive in bite-sized pieces, so that default by a single customer is never expensive enough to threaten a bank". Portfolios behave like well-diversified portfolios in normal markets, so EL is predictable, treated as a cost of doing business and built into the price. Corporate portfolios contain large single-name exposures and concentrations, and their risk is dominated by credit losses rising to some *unexpected* level.
- **Diversification as a "fickle friend"** (CR-7 b): in 2007 to 2009 house prices fell at about the same time across the US; a systematic change in lending behaviour (advancing money without checking incomes) can plant a hidden systematic risk in whole credit industries.
- **The dark side** (CR-7 c, S2): "the danger that losses will suddenly rise to unexpected levels because of some unforeseen but systematic risk factor that influences the behavior of many of the credits". Four prime causes: new products without enough loss history; well-understood products behaving unexpectedly when risk factors worsen together (the "perfect storm", e.g. recession plus higher rates plus falling house prices); changing social and legal attitudes to default (personal bankruptcy in the 1990s); operational issues in the semiautomated credit process that hit the whole portfolio.
- **Measures against the dark side** (CR-7 c): keep only a limited number of retail portfolios exposed to new kinds of risk ("a little exposure … a lot makes the bank a hostage to fortune"); stress test large conventional portfolios. Post-crisis, the CFPB (from Dodd-Frank) requires an "ability to repay" check with eight underwriting standards; a "qualified mortgage" has no excess upfront points and fees, no toxic features, a DTI cap (e.g. below 43%) and no balloon payments.
- **Retail early warnings and the supertanker** (CR-7 c): rising defaults are often signalled by customer behaviour (missed minimum card payments), letting the bank tighten lending to existing customers, change marketing and acceptance rules, or price in the risk. A commercial portfolio is a "supertanker": by the time trouble is obvious it is often too late. Banks still ignore the signs to chase volume (Box 7.4: 2/28 and 3/27 teaser-rate ARMs designed to be refinanced).
- **Regulatory capital for retail** (CR-7 b): regulators accept that retail risk is relatively predictable and mortgages safer, so retail banks set aside a *relatively small* amount of capital under Basel II and III, but must provide PD, LGD and EAD by segment, segmented by credit score and by vintage. Basel uses three formulas for three subsectors: residential mortgages, revolving credit, other retail.
- **Other retail risks** (CR-7 a, Box 7.5): interest rate risk (transferred to treasury for ALM); asset valuation risk (mortgage prepayment, auto lease residual values); operational risk (customer fraud); business risk (volumes, strategy, M&A); reputation risk with customers and regulators.

- **Scoring model types** (CR-7 d): credit bureau (FICO) scores, low cost and quick, typically 300 to 850 with subprime below 660; pooled models built by vendors from several lenders' data, tailored to an industry but not company-specific; custom models built in-house for a specific product and population. Higher score means lower risk; characteristics (questions) take attributes (answers); about 50 items per application, up to 150; logistic regression is most common; population odds of 15:1 imply a 1 in 16 (6.25%) chance of a bad. Bureau files hold identifying information (not used in scoring), public records, collection information, trade lines and inquiries.
- **Mortgage variables, cutoffs and loss rates** (CR-7 e): risk score $= f$(doc type, transaction type, FICO, LTV, DTI, occupancy type, property type, payment type, economic cycle); pre-crisis originators leaned too heavily on bureau scores. Moving the cutoff score up cuts off more bads but forgoes more goods. By score band, implied PD = historical loss rate / LGD (2% with 50% LGD gives 4%).
- **CAP and accuracy ratio** (CR-7 f): the CAP plots cumulative actual defaults against the population sorted from highest to lowest risk score; the perfect model captures all defaults in the first (default-rate) slice, the random model is the 45-degree line. AR $= A_R/A_P$, closer to 1 is better. Monitor quarterly; performance deteriorates as population characteristics or behaviour change, and a product switch (auto loans to credit cards) justifies a new custom scorecard.
- **Customer relationship cycle** (CR-7 g): marketing initiatives, screening applications, managing accounts, cross-selling. Scorecards now include application, behavior, revenue, response, attrition, insurance and tax authority scores. Trade-off: marginally less creditworthy customers may be more profitable (they borrow often and pay higher rates); the question is whether that survives a downturn.
- **Risk-based pricing** (CR-7 h, S1): "explicitly incorporating risk-driven account economics into the annualized interest rate" at account level, using operating expenses, probability of take-up, PD, LGD, EAD, capital allocated and the cost of equity. "A 'one price fits all' policy in a competitive market leads to adverse selection": the bank attracts mainly high-risk customers and discourages low-risk ones, and the degree of adverse selection "may only become apparent when the economic climate deteriorates". Tiered pricing: the higher the score, the lower the price. RBP is still in its infancy; better pricing can improve key metrics by 10 to 20% or more, trading market share against the bad rate (strategy curve).
- **Securitization and reforms** (CR-7): about 50% of US mortgages were securitized before 2007; some structures aimed at regulatory capital arbitrage rather than transferring economic risk. Reforms: better disclosure, originator retention (e.g. 5%), public rating methodologies, capital that better reflects securitization risk.
- **Scoring versus rating, PIT versus TTC** (CR-6 a, b): see D7.

Formulas

Odds to probability. Population odds $G:B = k:1 \Rightarrow P(\text{bad}) = \dfrac{1}{k+1}$ (15:1 gives 6.25%)

Implied PD by score band. $\text{PD} = \dfrac{\text{loss rate}}{\text{LGD}}$ (2% / 50% = 4%)

Accuracy ratio. $\text{AR} = \dfrac{A_R}{A_P}$ (area under the actual model over area under the perfect model, both measured against the random line)

Debt-to-income. $\text{DTI} = \dfrac{\text{total monthly debt}}{\text{total monthly gross income}}$ (qualified mortgage cap e.g. 43%)

Exam traps

- **Sibling swap, dark side versus concentration:** the dark side is a *systematic* factor hitting many small credits at once; a large single-name loss is the corporate risk profile.
- **Polarity of adverse selection:** one price for all attracts the *high*-risk customers and repels the low-risk ones, and the damage shows up in a downturn.
- **Polarity of tiered pricing:** higher score band, *lower* price.
- **Absolute claim on capital:** predictable EL does not mean zero capital; retail carries "a relatively small amount" of regulatory capital, and the dark side is precisely the unexpected loss.
- **CAP axis:** the population is sorted from *highest* to lowest risk; the random model is the 45-degree line; AR closer to 1 is better.

- **Bureau data:** identifying information (gender, race, religion in the US) is not used in scoring models.

Question D13.1 CR-7 b, c, h · S1, S2, S3 · [Map #57]

The head of retail credit risk at a UK bank is preparing a report for the board risk committee. The credit card unit proposes to replace its single advertised interest rate with pricing by score band, and the mortgage unit plans to launch a near-prime product for which the bank has little loss history. Which of the following statements is correct for the report to include?

A. Keeping a single interest rate for all card applicants is likely to cause adverse selection, attracting mainly high-risk customers, and the extent of this may only become apparent when the economic climate deteriorates.

B. Because retail credit losses are relatively predictable, the dark side of retail credit is mainly the risk that a loss on a single large customer exceeds its expected level in a downturn.

C. Because a retail portfolio behaves like a well-diversified portfolio, its expected loss can be priced in and the bank need not hold any capital against retail unexpected losses.

D. Under tiered pricing above the cutoff score, the higher a customer's score band, the higher the interest rate charged, since high-scoring customers tend to use their credit lines more.

Solution **Answer: A**

Why A is correct

Crouhy, Galai and Mark: "a 'one price fits all' policy in a competitive market leads to adverse selection, i.e., the bank will primarily attract high-risk customers, to whom the product is attractive, and discourage low-risk customers (for the opposite reason). The degree of adverse selection suffered by a bank may only become apparent when the economic climate deteriorates." This is the case for the card unit's move to risk-based pricing.

⇒ *The near-prime mortgage product is the dark-side flag: little loss history means the bank should keep that exposure limited and stress test it.*

Why the other options are wrong

B is incorrect. Sibling swap. Large single-name exposures are the defining feature of corporate portfolios; retail default "by a single customer is never expensive enough to threaten a bank". The dark side is "some unforeseen but systematic risk factor that influences the behavior of many of the credits".

C is incorrect. Absolute claim. Predictable EL is priced in, but regulators ask retail banks to set aside "a relatively small amount of risk capital", not none, and the dark side is exactly the risk of losses rising to unexpected levels.

D is incorrect. Polarity flip. Tiered pricing means "the higher the score, the lower the price"; pricing as an increasing function of *riskier* score bands is what makes risk-based pricing effective.

LO: CR-7 h: Discuss the benefits of risk-based pricing of financial services. CR-7 c: Discuss the "dark side" of retail credit risk and the measures that attempt to address the problem. CR-7 b: Explain the differences between retail credit risk and corporate credit risk. · Reference: Crouhy, Galai and Mark, The Essentials of Risk Management 2e, Ch.9, 7.1, Box 7.2, 7.5 and 7.8

D14 Credit Derivatives and Credit-Linked Notes

Readings: CR-13 a, b, c, d, e, f, g, h · Credit Derivatives (Hull, OFOD 11e, Ch.25: 13.1 to 13.11); CR-9 g · Estimating Default Probabilities (Hull, RMFI 6e, Ch.17); CR-23 · An Introduction to Securitisation (Choudhry, SPV applications: credit-linked notes)

MAP TIER D · #58 · 2/16 SOURCES VARDEEZ

Specifics reported

Map #58: "Your formula card lists CLN as its final row. Know the CLN structure (funded, note investor bears credit risk), total return swaps, and the CDS mechanics from LO 27.g. LOs 31.f and 31.g are separately flagged as worth knowing."
Shapes covered here:
S1 CLN structure: funded, the note investor bears the credit risk (Q1, option D)
S2 total return swaps: who owns the bond, who earns the spread (Q1, option A)
S3 CDS mechanics: obligations of buyer and seller, settlement (Q1, option B)
S4 synthetic CDO valuation and the two implied correlations, compound and base (Q1, option C).

Revision notes

- **The specific: credit-linked notes** (S1). A CLN is a *funded* credit derivative: the investor pays the note principal upfront, earns a coupon that includes the credit premium, and is repaid less any credit loss, so the note investor is the protection seller and bears the credit risk, while the issuer (protection buyer) already holds the cash (outside-source: the CLN mechanics are not set out in the GARP excerpts). Choudhry (CR-23): CLNs issued through an SPV "do not have any credit-linkage to the sponsoring entity"; the note is linked to the assets sold to the SPV and its performance depends on those assets. A CLN issued directly by an originator also carries the issuer's credit risk.
- **CDS mechanics** (CR-9 g, CR-13 a; S3): the buyer pays the CDS spread (quarterly in arrears is standard) until maturity or a credit event (failure to pay, restructuring, bankruptcy); the seller pays $L(1-R)$. Physical settlement: deliver bonds for face value; now usually cash settlement, with an ISDA auction setting the cheapest-to-deliver bond price. Hull: 90 bps on USD 100 million is USD 225,000 per quarter; auction price 35 gives a USD 65 million payoff plus a final accrual (about USD 150,000). Unlike insurance, the buyer need not own the underlying (Lehman: about USD 400 billion of CDS against USD 155 billion of debt). Restructuring is sometimes excluded in North America.
- **CDS and bond yields**: buying a 7% bond and 200 bps protection approximates a 5% risk-free bond, so the CDS spread $\approx$ bond yield spread. CDS-bond basis = CDS spread $-$ bond yield spread; positive pre-2007, negative in the crisis, mixed since. See also A15 for hazard rates from CDS spreads.
- **Valuing a CDS** (CR-13 b, c): equate PV of expected spread payments plus accrual to PV of expected payoff. Hull: hazard 2%, $r = 5\%$, $R = 40\%$: $4.0728s + 0.0422s = 4.1150s = 0.0506$, so $s = 123$ bps. An old deal at 150 bps is worth $4.1150 \times 0.015 - 0.0506 = 0.0111$ to the *seller*. Default probabilities must be *risk-neutral*, implied from bond prices or CDS quotes (100 bps implies a 1.63% hazard). The value is insensitive to R when the same R is used to imply PDs (PDs $\propto 1/(1-R)$, payoff $\propto 1-R$); a binary CDS (fixed payoff) is not, priced at 205 bps.
- **Credit indices and fixed coupons** (CR-13 d): CDX NA IG and iTraxx Europe each hold 125 investment grade names, rolled on 20 March and 20 September. Index 66 bps on USD 800,000 per name costs USD 660,000 a year; a default cuts it by USD 5,280. The index is slightly below the average spread. Fixed coupons: price $P = 100 - 100 \times D \times (s - c)$ with D the CDS duration; the buyer pays $100 - P$ upfront, then the coupon. iTraxx example: 34 bps quote, 40 bps coupon, $D = 4.447$, $P = 100.27$, so the *seller* pays the buyer 0.27%.
- **CDS forwards and options** (CR-13 e): a forward CDS is an obligation, a CDS option a right (premium upfront), to buy or sell protection at a set spread at time T; both *cease to exist* if the reference entity defaults before T. A call (right to buy protection at 280 bps) is exercised if the spread is above 280 bps.
- **Basket CDS**: add-up pays on any default; first-, second-, kth-to-default pay only on the kth default, then terminate.

- **Total return swaps** (CR-13 a; S2): the *payer* passes the total return on a bond (coupons plus gain or loss) to the *receiver* against floating plus a spread. Used for financing: the payer buys and *retains ownership* of the bond; the receiver is in the position of having borrowed at floating plus spread. On default the swap usually ends with the receiver paying the excess of notional over the bond's market value. The spread compensates the *payer* for the risk that the *receiver* defaults when the bond has fallen, so it depends on both credit qualities and their correlation. Similar to a repo.
- **Cash and synthetic CDOs** (CR-13 a): a cash CDO funds a bond portfolio and tranche holders invest upfront; a synthetic CDO's originator *sells* CDS protection on each name, and tranches absorb payouts in order (Hull: equity 0 to 5% at 1,000 bps, mezzanine 5 to 20% at 100 bps, senior above 20% at 10 bps; USD 2 million of payouts cut equity principal to USD 3 million). Tranche holders post the tranche principal as collateral. Single-tranche trading uses standard tranches: iTraxx 0 to 3, 3 to 6, 6 to 9, 9 to 12, 12 to 22 and 22 to 100%; CDX 0 to 3, 3 to 7, 7 to 10, 10 to 15, 15 to 30 and 30 to 100%.
- **Correlation and tranche risk**: higher default correlation makes the equity tranche *less* risky and senior tranches *more* risky; with perfect correlation and zero recovery all tranches are equally risky. For a basket, higher correlation lowers the chance of one or more defaults and raises the chance of ten or more.
- **Valuing a synthetic CDO tranche** (CR-13 f; S4): with expected tranche principal E_j, $A = \sum(\tau_j - \tau_{j-1})E_j v(\tau_j)$, $B = \sum 0.5(\tau_j - \tau_{j-1})(E_{j-1} - E_j)v(\cdot)$, $C = \sum(E_{j-1} - E_j)v(\cdot)$; breakeven spread $s = C/(A + B)$, or upfront $C - s^*(A + B)$ for a fixed running spread. In the one-factor Gaussian copula, $Q(t \mid F) = N\big((N^{-1}[Q(t)] - \sqrt{\rho}F)/\sqrt{1-\rho}\big)$ with $Q(t) = 1 - e^{-\lambda t}$ from the index spread, binomial default counts conditional on F, then integration over F by Gaussian quadrature. Example 13.2: 3 to 6% tranche, $\rho = 0.15$, $A = 4.2846$, $B = 0.0187$, $C = 0.1496$, $s = 348$ bps.
- **Implied correlation** (CR-13 g; S4): with $R = 40\%$, ρ is the only unknown. *Compound (tranche) correlation* is the ρ that prices an individual tranche $\{a_{q-1}, a_q\}$; *base correlation* is the ρ that prices the $\{0, a_q\}$ tranche consistently with the market. Compound correlations show a *smile* (iTraxx January 2007: 17.7, 7.8, 14.0, 18.2, 23.3%); base correlations a *skew* rising with the detachment point (17.7 to 60.5%). Both would be flat if the model were right. For nonstandard tranches, interpolating expected losses beats interpolating base correlations, which can violate no-arbitrage.
- **Alternatives to the standard model** (CR-13 h): heterogeneous model; other copulas (Student t, Clayton, Archimedean, Marshall-Olkin; Hull and White's double t with four degrees of freedom fits well); random recovery and random factor loadings (Andersen and Sidenius: correlation rises as F falls); implied copula; dynamic models (structural, reduced form with jumps in hazard rates, top down). See also B3 and D15.

Formulas

CDS payoff. $L(1 - R)$, with R set by the ISDA auction price of the cheapest-to-deliver bond

CDS spread. $s = \dfrac{\text{PV(expected payoff)}}{\text{PV(payments per unit spread)} + \text{PV(accrual per unit spread)}}$ (Hull: $0.0506/4.1150 = 123$ bps)

Mark to market (to the seller). $(s_{\text{contract}} - s_{\text{market}}) \times$ PV of payments per unit spread

Fixed-coupon price. $P = 100 - 100 \times D \times (s - c)$; buyer pays $100 - P$ upfront

Tranche breakeven spread. $s = \dfrac{C}{A + B}$; upfront $= C - s^*(A + B)$

Gaussian copula conditional PD. $Q(t \mid F) = N\left(\dfrac{N^{-1}[Q(t)] - \sqrt{\rho}\,F}{\sqrt{1-\rho}}\right), \qquad Q(t) = 1 - e^{-\lambda t}$

Exam traps

- **Funded versus unfunded:** a CLN investor pays cash upfront and bears the credit risk (protection seller); a CDS seller pays nothing upfront. Calling the CLN investor the protection buyer is a role flip.
- **TRS roles:** the *payer* owns the bond and receives floating plus spread; the spread compensates the payer for the *receiver's* default risk.
- **CDS is not insurance:** the buyer need not own the reference obligation, and protection pays only on a credit event, not on a price fall.
- **Compound versus base (sibling swap):** compound prices a single tranche; base prices the 0 to a_q tranche. Compound shows a smile, base a skew.

- **Correlation polarity:** higher correlation helps the equity tranche and hurts the senior tranche.
- **Risk-neutral, not real-world:** CDS valuation uses risk-neutral default probabilities implied from bonds or CDS quotes.

Question D14.1 CR-13 a, f, g; CR-9 g; CR-23 · S1, S2, S3, S4 · [Map #58]

A structured credit specialist at a German bank is training newly hired analysts on four instruments the bank uses to transfer or take credit risk. Which of the following statements should the specialist present as correct?

A. In a total return swap used to finance a bond purchase, the total return receiver retains ownership of the bond, and the spread over the floating rate compensates the receiver for the risk that the payer defaults.

B. In a credit default swap, the protection buyer must own bonds of the reference entity, as with an insurance contract, and receives the notional times one minus the auction price after a credit event.

C. In a synthetic CDO priced with the one-factor Gaussian copula, base correlation is the correlation that prices each individual tranche, while compound correlation prices the tranche from zero to its detachment point.

D. In a credit-linked note issued through an SPV, the investor pays the principal upfront and bears the credit risk of the assets sold to the SPV, with no credit linkage to the sponsoring entity.

Solution **Answer: D**

Step 1 – The CLN statement

Choudhry (CR-23): "Unlike CLNs issued by originators direct, CLNs issued by SPVs do not have any credit-linkage to the sponsoring entity. The note is linked instead to assets that have been sold to the SPV, and its performance is dependent on the performance of these assets." The note is funded: the investor's principal is paid upfront and is repaid less credit losses, so the investor is the protection seller (outside-source for the funded mechanics).
$\Rightarrow$ *D is correct: funded, investor bears the reference credit risk, and the SPV removes the sponsor's own credit from the note.*

Interpretation

The four instruments split cleanly by who funds and who holds the asset: a CLN investor funds and bears credit risk; a CDS seller bears credit risk unfunded; a TRS payer holds the bond and lays off its total return; a synthetic CDO tranche holder sells protection on a slice of a CDS portfolio. Each wrong option misplaces one of those attributions.

Why the other options are wrong

A is incorrect. Role misassignment: in Hull's financing structure the *payer* buys and "retains ownership of the bond for the life of the swap", and "the spread over the floating rate received by the payer is compensation for bearing the risk that the receiver will default".
B is incorrect. Scope/condition error: the settlement clause is right, but ownership is not a condition. Hull: unlike an insurance contract, "in the case of a CDS, the underlying asset does not have to be owned"; CDS volume on a company can exceed its debt (Lehman).
C is incorrect. Sibling swap: compound (tranche) correlation is the ρ that prices an individual tranche $\{a_{q-1}, a_q\}$; base correlation is the ρ that prices the $\{0, a_q\}$ tranche. The definitions are reversed.

LO: CR-13 a: Describe a credit derivative, credit default swap (CDS), total return swap, and collateralized debt obligation (CDO). CR-13 f: Describe the process of valuing a synthetic CDO using the spread payments approach and the Gaussian copula model of time to default approach. CR-13 g: Define the two measures of implied correlation: compound (tranche) correlation and base correlation. CR-9 g: Define a credit default swap (CDS) and explain its mechanics including the obligations of both the default protection buyer and the default protection seller. · Reference: Hull, OFOD 11e, Ch.25, 13.1, 13.7, 13.10; Choudhry, Ch.12 (CR-23), SPV applications

D15 Structured Credit and Implied Correlation

Readings: CR-22 h, i, j, k, l · Structured Credit Risk (Malz, Ch.9, 22.3 Measuring Structured Credit Risk via Simulation; 22.4 Standard Tranches and Implied Credit Correlation)

MAP TIER D · #60 · 2/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #60: "Named: LO 40.i, 'just remember the impacts via the table', with one possible calculation; and the concept of implied correlation, described as 'likely to be testable.' Reported independently as 'Gaussian copula, implied correlation.' "
Shapes covered here:
S1 LO 40.i impacts table: how default probability and correlation move each tranche (Q1)
S2 the one possible calculation: reading tranche values and writedowns off the simulation grid (Q1)
S3 implied correlation backed out of a tranche price (Q1)
S4 Gaussian copula and implied correlation (Q1; copula mechanics in B3.3)

Revision notes

- **See also A9** for CR-22 a to g and m: product types, tranching, waterfall, participants and conflicts, interim cash flows, overcollateralization account, the custodian's tests and motivations. This section covers the simulation, the impacts table, sensitivities, risk factors and implied correlation.
- **Simulation procedure** (CR-22 h). Estimate parameters (each loan's default probability, physical or risk-neutral, and a pairwise copula correlation restricted to $\rho \geq 0$); draw $I = 1{,}000$ threads of $N = 100$ correlated standard normals; map each to a default time $t = -\log[1 - \Phi(z)]/\lambda$ with $\lambda = -\log(1 - \pi)$ ($\pi = 0.0225 \Rightarrow \lambda = 0.022757$); count defaults per year; pass each thread through the waterfall to get tranche losses and equity IRRs. New simulations are needed only for more threads, more loans or new correlation settings; changing recovery or collateral interest only changes the processing.
- **Impacts table, default probability** (CR-22 i, S1). For any given correlation, a higher default rate raises bond losses and lowers the equity IRR: it hurts every tranche.
- **Impacts table, correlation** (CR-22 i, S1). At low default rates the impact of correlation is relatively low. At higher default rates, higher correlation materially *raises* the equity value and *raises* the senior bond's losses: "the equity benefits from high correlation, while the senior bond is hurt by it". Malz: "correlation is more decisive for equity risk and value than default probability".
- **Mezzanine** (CR-22 i). Ambiguous: at low default rates higher correlation increases its losses (it behaves like a senior bond); at high default rates it decreases them (it behaves like equity). Table 22.3 writedowns: at $\pi = 0.0075$, 0.00% ($\rho = 0$) to 4.84% ($\rho = 0.9$); at $\pi = 0.0975$, 103.92% to 52.87%.
- **Why: clustering.** High correlation makes defaults arrive in clusters: some threads have very many, some very few. The equity distribution turns U-shaped, a "lottery ticket"; with low correlation some defaults hit almost every thread, and since equity takes the first loss, low correlation is bad for the equity. The senior is safe unless a cluster breaches its attachment point.
- **Convexity** (CR-22 i). At low correlations the equity value is positively convex in default rates ("you can't beat a dead horse"), the senior has negative convexity (losses accelerate), the mezzanine is negatively convex at low and positively convex at high default rates. At high correlations all tranches respond more nearly linearly.
- **Tranche Credit VaR** (CR-22 h, i). Equity: par minus the 10th (50th) lowest of 1,000 simulated values at 99% (95%). Bonds: the 10th (50th) highest loss minus EL. Equity VaR falls at high π and ρ because EL is already so high. Senior: 99% Credit VaR of about half of par at high correlation, virtually risk-free when defaults are uncorrelated. At $(\pi, \rho) = (0.0075, 0.30)$ the senior VaR is negative (EL above the 1% quantile loss), a simulation artifact that would disappear at 99.5% or 99.9%.
- **Default01** (CR-22 j). $-[V(\pi + 10\text{ bp}) - V(\pi - 10\text{ bp})]/20$, the change per 1 bp rise in default probability, reusing the same normal draws to avoid simulation noise. Always positive; converges to zero at very high default rates; varies most with π when correlation is low; the peak sits where pool losses approach a tranche's attachment point (equity at $\pi = 0$, mezzanine before senior). Underestimating default correlation was an

important factor in the subprime crisis. Lumpy curves show how costly simulation is, and that defaults stay "lumpy" even in a 100-loan pool.

- **Risk factors** (CR-22 k). Systematic risk (high correlation exposes even a senior bond on a well-diversified pool to large tail losses); tranche thinness (conditional on any loss, the loss is large, so 95% and 99% Credit VaR are close); granularity (lumpy pools such as CMBS and fusion deals carry concentration risk; a granular pool allows a larger senior tranche with no increase in Credit VaR); prepayment, extension and refinancing risk.
- **Standard tranches** (CR-22 l). CDX.NA.IG: equal notional CDS on 125 investment-grade names (USD 1 million each per USD 125 million); iTraxx for Europe and Asia; series initiated semiannually; a default cuts the notional by 0.8%. Standard synthetic CDO tranches: equity 0 to 3%, junior mezzanine 3 to 7%, senior mezzanine 7 to 10%, senior 10 to 15%, super senior 15 to 30%. Bespoke tranches are issued to order.
- **Implied correlation** (CR-22 l, S3). Use risk-neutral PDs from single-name CDS spreads (and risk-free discount rates); the copula correlation is the remaining unobservable input; the unique correlation at which the modelled tranche value equals the market price is the implied correlation. Tied to a tranche's attachment point: base correlation; it varies by tranche: correlation skew. Malz's example: equity priced at 5.0 with a 2% CDS-implied PD matches $\rho = 0.3$; a rise to 5.6 with no change in PD means implied correlation has risen to 0.6, an increase in the market's assessment of systematic risk.
- **As much market as credit risk** (CR-22 l). Implied credit correlation is "as much a market-risk as a credit-risk concept". Each standard tranche has risk-neutral partial spread01s to the 125 constituents rather than a default01; changes in implied correlation and spreads cause mark-to-market changes, whereas in the cash CLO they change fundamental value. Implied credit correlation is the copula correlation inferred from prices (see B3 for the six correlation concepts; D14 for compound versus base correlation).

Formulas

Hazard rate and simulated default time. $\lambda = -\log(1-\pi), \qquad t_{n,i} = -\dfrac{\log[1-\Phi(z_{n,i})]}{\lambda}$

Default01. $\text{default01} = -\dfrac{V(\pi+0.0010) - V(\pi-0.0010)}{20}$

Tranche Credit VaR (1,000 threads). Equity: par $-$ 10th lowest value; bonds: 10th highest loss $-$ EL (99%; use the 50th for 95%)

Standard tranche arithmetic. Loss per default $= \dfrac{1}{125}(1-R)$ of index notional; the notional and the spread premium shrink by $1/125 = 0.8\%$ per default

Exam traps

- **Polarity flip:** higher correlation *helps* the equity and *hurts* the senior; a rising equity price with unchanged PDs means implied correlation has *risen*.
- **Scope error, mezzanine:** it behaves like the senior at low default rates and like the equity at high ones; "always like the equity" is false.
- **Sibling swap, inputs:** implied correlation uses risk-neutral PDs from CDS spreads and risk-free rates, not physical PDs from ratings; it is as much a market-risk as a credit-risk concept.
- **Absolute claim:** a well-diversified collateral pool does not make the senior bond safe when correlation is high.
- **Intermediate result:** the equity tranche's Credit VaR can *fall* as π and ρ rise, because its expected loss is already very high.

Question D15.1

CR-22 i, l · S1, S2, S3, S4 · [Map #60]

A structured credit analyst at an Australian superannuation fund monitors the fund's equity tranche in a CLO with the same structure as the one simulated in the reading. For loans with a one-year default probability of 2.25%, the simulation grid gives (equity value in AUD million; bond writedowns in percent of tranche par):

Copula correlation ρ	0.00	0.30	0.60	0.90
Equity value	4.44	4.98	5.61	6.33
Mezzanine writedown	0.00	7.35	12.82	15.49
Senior writedown	0.00	0.52	2.14	5.05

The market price of the equity tranche has risen from AUD 4.98 million to AUD 5.61 million, while the risk-neutral default probability implied by the loans' CDS spreads has stayed at 2.25%. Which of the following statements should the analyst include in the report to the investment committee?

A. The implied correlation has fallen from 0.60 to 0.30, because a higher equity price signals that defaults are now expected to be spread more evenly across scenarios.

B. The implied correlation has risen from 0.30 to 0.60, and the mezzanine bond's expected writedown has fallen, because the mezzanine tranche behaves like the equity tranche at any default rate.

C. The implied correlation has risen from 0.30 to 0.60, reflecting a higher market assessment of systematic risk, and the senior bond's expected writedown rises from 0.52% to 2.14% of par.

D. The implied correlation has risen from 0.30 to 0.60, and because it is estimated from rating-based physical default probabilities, it is a credit risk rather than a market risk parameter.

Solution **Answer: C**

Step 1 – Back out the implied correlation

With the CDS-implied default probability fixed at 2.25%, the only free input is the copula correlation. An equity value of 4.98 matches $\rho = 0.30$; a value of 5.61 matches $\rho = 0.60$.

$\Rightarrow$ *The market now prices the equity as if defaults were more clustered: implied correlation has risen from 0.30 to 0.60.*

Step 2 – Read the other tranches at the same default probability

Senior writedown: $0.52\% \rightarrow 2.14\%$ of par (up). Mezzanine writedown: $7.35\% \rightarrow 12.82\%$ (up).

$\Rightarrow$ *At this low default rate the mezzanine sits with the senior: higher correlation hurts both bonds while it helps the equity.*

Interpretation

C is correct. Higher correlation means a low but material probability of a state with unusually many defaults, the systematic risk that reaches the senior bond even on a diversified pool, while the equity, now a "lottery ticket", gains from the higher chance of a low-default state.

Why the other options are wrong

A is incorrect. Polarity flip. Equity value increases with correlation at this default rate (4.98 at 0.30, 5.61 at 0.60), so a higher price means a higher implied correlation. Evenly spread defaults describe *low* correlation, which is bad for the equity.

B is incorrect. Scope/condition error. The mezzanine behaves like the equity only at high default rates; at $\pi = 2.25\%$ its writedown rises from 7.35% to 12.82%.

D is incorrect. Sibling swap. Implied correlation is computed from risk-neutral default probabilities taken from single-name CDS spreads, discounting at risk-free rates; Malz calls implied credit correlation "as much a market-risk as a credit-risk concept", and each standard tranche's spread01s measure a market rather than a credit risk.

LO: CR-22 i: Explain how the default probabilities and default correlations affect the credit risk in a securitization. CR-22 l: Define implied correlation and describe how it can be measured. · Reference: Malz, Ch.9, 22.3 Means of the Distributions, Table 22.3; 22.4 Implied Correlation

D16 Monetary and Fiscal Policy

Readings: CI-5 a, b, c · Monetary and Fiscal Policy: Safeguarding Stability and Trust (BIS, Annual Economic Report 2023, Section 2) (source: SchweserNotes Book 5, one rung below a GARP text)

MAP TIER D · #67 · 2/16 SOURCES TELEGRAM VARDEEZ

Specifics reported

Map #67: "Reported: 'fiscal and monetary policy.' Your ledger carries LO 104.c as introduced but unresolved: the interest-growth differential and the risks of high public debt. That is a live open item, not new coverage."
Shapes covered here:
S1 fiscal and monetary policy: how the two interact, including the tension that high debt creates (fiscal dominance) (Q1)
S2 LO 104.c: the interest-growth differential and the risks of high public debt (Q1).

Revision notes

The specifics first: high public debt and the fiscal-monetary tension (CI-5 c)

- **Interest-growth differential (S2).** The difference between interest rates and economic (GDP) growth. When rates are **below** growth, as before the pandemic, public finances can thrive; when rates rise **above** growth, the economy can slow and public spending is squeezed. The Book 5 summary defines it with policy rates; the BIS original frames it as the interest rate paid on public debt minus growth (outside-source).
- **Three primary challenges.** (1) Public finances are strained by record-high public debt, pressure to maintain spending, higher interest rates and slowing growth prospects; (2) high public debt can work against monetary tightening when fighting inflation; (3) the level of public debt introduces financial stability risks.
- **Spending pressure.** Cutting spending is hard: rising costs, inflation-linked wages and transfers, age-related spending expected to rise by 4% to 5% of GDP over 20 years, the costly global push for sustainability, and higher defense spending as geopolitical tensions rise. Inflation makes fiscal balances look better than reality (nominal revenues respond quickly). Post-GFC high debt has lowered sovereign ratings, more sharply in AEs.
- **Amplifiers of policy rates on public borrowing.** Post-GFC central bank purchases reached **25% to 50%** of outstanding government debt in major AEs; this **shortened** the effective maturity of public debt, reduced borrowing costs and **lowered remittances** from central banks to governments. Debt service is estimated to grow to 6% of GDP, worse if medium-term growth disappoints (slower labor force growth, EME slowdown). History shows higher inflation leads to higher public debt.
- **Countercyclical effects.** Large deficits support demand but raise inflation; the normal response is higher policy rates, which weaken the economy against the aim of fiscal policy, more acutely the longer inflation persists. In EMEs fiscal stimulus feeds inflation more, through exchange-rate depreciation.
- **Fiscal dominance (S1). Monetary** policy is constrained by fiscally imposed constraints: political spending pressure places a floor on fiscal flexibility, and rate rises could spark a sovereign debt crisis. Remedy: strong institutional safeguards for central bank independence so fiscal authorities cannot "hijack" monetary policy.
- **Financial stability channel.** Sovereign debt is a store of value, collateral, the most liquid asset and a pricing benchmark; the government is the ultimate backstop (the central bank as lender and market maker of last resort), so banks enjoy **higher** ratings with it. Banks rely on the state for liquidity and on sovereign bonds as collateral: a downgrade at one can spiral into the other, **most pronounced in EMEs**. High debt makes sharp yield moves more likely; long-term yields rise more in high-debt countries when policy tightens; governments locked in long-term debt at ultra-low rates, and long bonds are the most rate-sensitive.
- **Policy implications.** Monetary policy should stay focused on price stability and tolerate inflation below target while watching for overheating; fiscal policy should be as conservative as possible to keep headroom; relying on inflation to erode debt is counterproductive. Central banks should minimize balance sheets: smaller footprint and no crowding out of private lenders, less budget reliance on central bank transfers, room to expand in a future crisis. Prefer high-quality, growth-enhancing spending (infrastructure, education) over transfers; keep central banks independent; use guardrails such as fiscal rules or oversight panels.

Channels and the region of stability (CI-5 a)

- **Two channels. Fiscal**: a direct impact through government spending, taxes, the production of some services, and transfer payments. **Monetary**: the money supply via open market operations and policy rates, spilling into bond yields, borrowing costs, asset prices and exchange rates. They overlap: higher rates raise government borrowing costs and slow tax revenue; central bank profits flow to the government; too much fiscal stimulus forces a monetary response.
- **Region of stability.** The joint policy stances that deliver stable inflation and stable growth. Its boundaries vary by country, cannot be captured in a single metric, change in shape, size and locus (gradually, e.g. demographics, or rapidly, e.g. a commodity spike or pandemic), and can be altered by regulation and FX intervention. Keep a **margin of safety**: do not push policy to the outer limit.

Breaching the boundaries (CI-5 b)

- **AEs, phase 1 (1960s to mid-1980s).** Belief in fine-tuning; very high inflation fought with very high policy rates; a surge in public debt and a shift to prioritize price stability.
- **AEs, phase 2 (mid-1980s to the pandemic).** Financial liberalization and globalization (cheap EME labor) lowered inflation; boundary testing moved to **risk-taking and financial imbalances**; inflation-induced recessions gave way to longer-lasting **financial recessions**. Rates were cut in downturns and not raised in recoveries, enabling massive debt build-up. The pandemic stimulus then produced elevated inflation on top of record debt and high asset prices.
- **Why policy kept drifting. Loss of traction**: rate changes have less effect at low rates (50 or 75 bp moves needed), and fiscal stimulus weakens when debt is high. **Self-reinforcing interaction**: lower rates relax fiscal constraints and encourage debt; high debt makes monetary normalization harder.
- **EMEs.** Liberalization brought financial cycles and crises (Southern Cone early 1980s, Tequila 1994, Asia late 1990s); Latin America had hyperinflation with defaults. FX dependency means: a **smaller** region of stability; breaches bring capital outflows and depreciation; AE (especially US) policy spills over; hence the shift to flexible exchange rates and inflation targeting (Latin America tightened earlier and harder in the pandemic). See also D30 for how weak fiscal buffers raise sovereign risk premiums after geopolitical shocks.

Exam traps

- **Role misassignment on fiscal dominance:** it is *monetary* policy that is constrained by fiscal policy, not the other way round.
- **Polarity flip on the differential:** rates *below* growth help public finances; rates *above* growth strain them.
- **Polarity flip on central bank purchases:** QE *shortened* the maturity of public debt and *lowered* remittances to governments; limiting the balance sheet does the opposite.
- **Sibling swap AEs vs EMEs:** the region of stability is smaller in EMEs; the bank-sovereign loop is most pronounced in EMEs; EMEs moved to flexible exchange rates and inflation targeting.
- **Polarity flip on traction:** with high debt, fiscal stimulus is *less* powerful and monetary policy needs *more* extreme moves.
- **Absolute claim:** "inflating away the debt" is counterproductive; the government backstop *raises* bank ratings, not lowers them.

Question D16.1

CI-5 c · S1, S2 · [Map #67]

A sovereign risk analyst at a Brazilian asset manager is preparing a briefing for the investment committee on the BIS analysis of high public debt. Policy rates in several advanced economies have recently risen above nominal GDP growth after a long period below it, while government debt remains near record highs. Which of the following statements should the analyst include in the briefing?

A. With policy rates now above GDP growth, the interest-growth differential works against public finances, and high debt that makes tightening harder to sustain raises the risk of fiscal dominance.

B. Central bank purchases of 25% to 50% of government debt after the GFC lengthened the maturity of public debt, which raised central bank remittances to governments.

C. Fiscal dominance describes fiscal policy being constrained by the central bank, because high policy rates force

governments to cut spending to protect price stability.

D. The feedback loop between banks and their sovereign is most pronounced in advanced economies, because their larger debt stocks make sharp yield moves more likely.

Solution **Answer: A**

Step 1 – Read the interest-growth differential

The differential is the difference between interest rates and GDP growth. Below growth (pre-pandemic) public finances can thrive; above growth the economy can slow and public spending comes under pressure.

⇒ *The stem's move from rates below growth to rates above growth is exactly the shift that strains public finances.*

Step 2 – Link high debt to the policy tension

High debt works against monetary tightening: the tightening needed to fight persistent inflation raises the cost of government borrowing and slows growth. Fiscal dominance exists when monetary policy is constrained by fiscally imposed constraints (spending pressure; the risk that rate rises spark a sovereign debt crisis).

Interpretation

A is correct. The briefing should flag that the committee's AE sovereign exposures now face both a worse differential and a higher risk that central banks are pressed to go easy; the BIS remedy is fiscal restraint and protected central bank independence.

Why the other options are wrong

B is incorrect. Polarity flip: central bank monetization *shortened* the maturity of public debt and *lowered* remittances from central banks to governments.

C is incorrect. Role misassignment: fiscal dominance is *monetary* policy constrained by fiscal policy; the stated mechanism reverses the actors.

D is incorrect. Sibling swap: the costly feedback loop between banks and governments is most pronounced in EMEs, not AEs. (High debt does raise the risk of sharp yield moves, but that is a separate point.)

LO: CI-5 c: Describe the risks that global economies face as a result of high public debt levels, including the potential for these high debt levels, in combination with other factors, to drive tension between fiscal policy and monetary policy. · Reference: BIS, Annual Economic Report 2023, Section 2, via SchweserNotes Book 5, Reading 104, Module 104.2

D17 2023 Bank Failures and Credit Suisse (not in 2026 LOs: notes only)

Readings: Not in the 2026 LOs: evidence comes from an older Current Issues reading (map: R103, Global Financial Stability Report; the 2026 CI-4 is geopolitical risk, see D30). Related 2026 text: LTR-1 b · Liquidity Risk; LTR-2 b · Liquidity and Leverage; LTR-3 a · Early Warning Indicators; LTR-9 c · Liquidity Stress Testing; LTR-17 c · ALM and Duration Techniques; ORR-6 d · Risk Reporting; ORR-22 g · CoCos; IM-9 b; CI-1 c; CI-2 b; CI-8 b, c (IM and CI sources: SchweserNotes Book 5, one rung below a GARP text)

MAP TIER D · #69 · 2/16 SOURCES TELEGRAM VARDEEZ NOTES ONLY

Specifics reported

Map #69: "Reported: '2023 bank failure'; 'Credit Suisse'. Treat these as case studies for case↔lesson matching: the SVB duration-mismatch and uninsured-deposit-run mechanism is a different failure to the Credit Suisse confidence-and-governance collapse, and swapping the lessons is the obvious distractor."
Shapes covered here:
S1 SVB and the March 2023 US failures: duration mismatch plus an uninsured-deposit run
S2 Credit Suisse: a confidence-and-governance collapse, resolved by takeover and an AT1 write-down
S3 case↔lesson matching against the liquidity cases in the 2026 readings.

Revision notes

Status

- **Not examinable by LO in 2026.** The 2023 failures were covered by an older GFSR reading; the 2026 GFSR reading (CI-4) is about geopolitical risk (see D30). The *mechanisms*, however, are tested through the 2026 readings in the srcline, and those links are what this page is for. Event facts below are outside-source.

S1: Silicon Valley Bank and the US regional banks (outside-source unless stated)

- **SVB in brief.** Deposits of a concentrated technology and venture-capital client base surged in 2020 to 2021 and were invested heavily in long-duration Treasuries and agency MBS at low yields. As rates rose in 2022 the securities carried large unrealized losses. On 8 March 2023 SVB announced a loss on selling securities and a capital raise; depositors, most of them **uninsured**, withdrew about USD 42 billion on 9 March; the bank was closed on 10 March. Signature Bank followed on 12 March; First Republic failed on 1 May and was sold to JPMorgan Chase. US authorities guaranteed all deposits at SVB and Signature and the Fed opened the Bank Term Funding Program, lending against securities valued at par.
- **The duration mismatch (LTR-17 c).** With a **positive** leverage-adjusted duration gap, a rise in rates lowers asset values by more than liabilities, so **net worth falls**; the larger the gap, the more sensitive net worth is to rates. See A16 for the calculation.
- **The run (LTR-2 b).** Malz: no asset-liability management system can protect a fractional-reserve bank against a general loss of confidence in its ability to pay out depositors; while it transforms liquidity and maturity and owes liabilities repayable at par and on demand, no degree of liquidity protects it completely against a run.
- **Hull's causes of funding problems (LTR-1 b).** (1) Liquidity stresses in the economy (flight to quality); (2) overly aggressive funding decisions, using short-term instruments to fund long-term needs; (3) poor financial performance leading to a lack of confidence, loss of deposits and difficulty rolling funding. SVB combines (2) and (3); Northern Rock (2007), which relied on wholesale funding, is the GARP case for the same family.
- **Deposit behavior is the key assumption (LTR-9 c).** For most institutions deposit runoff is the most significant threat to liquidity and the most important behavioral dynamic to model; stress tests built on simplistic deposit assumptions will most likely fail. Stress factors include accelerated demand-deposit withdrawals and early withdrawal of term deposits.
- **Early warning (LTR-3 a).** An idiosyncratic deposit run may follow disclosure of poor performance or a systemic failure; EWIs should capture internally driven stress before it becomes public. Market signals such as widening debt and CDS spreads and a falling share price relative to peers are strong warnings.

- **Speed of a digital run (CI-1 c, CI-2 b).** SVB's run was coordinated through social media and executed by mobile banking (outside-source). The 2026 CI readings note that deepfakes and misinformation could erode confidence and trigger market selloffs or deposit runs (CI-1), and that supervisors may use GenAI to model bank runs by simulating social media interactions (CI-2).

S2: Credit Suisse (outside-source unless stated)

- **A slow loss of confidence.** Years of scandals and losses: the Greensill supply-chain funds and Archegos (both 2021), repeated leadership changes, heavy client outflows in late 2022, and material weaknesses in financial reporting disclosed in March 2023. When its largest shareholder ruled out further capital (15 March 2023), the Swiss National Bank provided a liquidity line, and on 19 March the government brokered a takeover by UBS. About CHF 16 billion of **AT1** bonds were written down to zero while shareholders received UBS shares. This was a confidence and governance failure, not a duration mismatch.
- **Archegos (IM-9 b).** The family office's total return swaps created hedge-fund-like leverage; when it collapsed in 2021, Credit Suisse lost about USD 5 billion and other banks about USD 5 billion more, prompting banks to require lower hedge fund leverage. Banks face three risks from hedge funds: revenue, credit (collateral can fall faster than expected) and counterparty (CDS and OTC derivatives).
- **Risk communication (ORR-6 d).** GARP's Box 6.5 uses Credit Suisse's 2021 annual report as a reporting example: "Our risk management procedures and policies may not be fully effective in mitigating our risk exposures in all market environments or against all types of risk, which can result in unexpected, material losses in the future." The risk section ran: key risk developments, risk oversight and governance, risk appetite framework, risk coverage and management, risk portfolio analysis.
- **CoCos and the write-down (ORR-22 g).** CoCos increase a bank's equity in distress either by converting to equity or by being **written down**, on triggers in the indenture and **not at the holder's option**; common triggers are CET1/RWA below a threshold or the regulator declaring the bank nonviable; they count as Additional Tier 1 if the trigger is 5.125% or higher, Tier 2 otherwise. Holders get little upside but bear losses close to equity's. The Credit Suisse AT1 write-down is the live example (event outside-source). See D25.
- **Cyber-driven runs (CI-8 b, c).** Loss of customer confidence after a cyber event can produce deposit runs; liquidity buffers help in theory and agent-based models are needed to assess the effect; **circuit breakers** (a temporary pause on normal rules) can limit a run after a major cyberattack, though the bank must show the attack was serious enough to count as force majeure.

S3: case↔lesson matching

- **One lesson per case.** SVB: interest rate risk in the banking book plus concentrated, uninsured, flighty deposits. Credit Suisse: cumulative governance, conduct and risk-management failures eroding confidence and franchise, ending in a forced merger and an AT1 write-down. Northern Rock (LTR-1 b): reliance on wholesale funding, run after news of emergency central bank support. Metallgesellschaft (LTR-1 b): cashflow mismatch from rolling short-dated futures to hedge long-dated commitments. LTCM: positive feedback trading in illiquid positions. Bear Stearns and Lehman (LTR-8, LTR-13 d): dealer-bank runs by repo creditors, prime brokerage clients and OTC counterparties, not insured depositors.

Formulas

Leverage-adjusted duration gap (LTR-17 c): $DGap = D_A - \frac{L}{A} D_L$

Change in net worth: $\Delta NW \approx -\left[D_A\, A - D_L\, L\right] \frac{\Delta r}{1+r}$; positive gap and rising rates: net worth falls

CoCo capital class (ORR-22 g): trigger CET1/RWA $\geq 5.125\%$: Additional Tier 1; otherwise Tier 2

Likely distractors

- **Corrupted:** "Credit Suisse failed mainly because long-dated bonds funded by uninsured deposits lost value when rates rose." **True version:** that is SVB's mechanism; Credit Suisse was a confidence and governance collapse after years of scandals and losses (case swap; outside-source).
- **Corrupted:** "SVB's failure was driven by credit losses on its loan book to technology firms." **True version:** rate-driven losses on high-quality securities plus a run by uninsured depositors (outside-source).
- **Corrupted:** "A bank with a negative leverage-adjusted duration gap loses net worth when interest rates rise."

True version: with a *positive* gap, rising rates cut asset values by more than liabilities and net worth falls (LTR-17 c).

- **Corrupted:** "Ample liquidity buffers can fully insulate a fractional-reserve bank against a depositor run." **True version:** no ALM system or degree of liquidity can protect it completely while it owes liabilities at par on demand (LTR-2 b).
- **Corrupted:** "CoCos convert into equity at the option of the holder when the bank's share price rises above a threshold." **True version:** that describes a traditional convertible; CoCos convert or are written down on distress triggers, not at the holder's option (ORR-22 g).
- **Corrupted:** "Classifying securities as held to maturity removes their interest rate risk." **True version:** the accounting label keeps unrealized losses out of reported capital, but the economic loss remains and is realized if a run forces a sale (outside-source).

D18 Inflation and Market Stress Across Bond Maturities

Readings: MR-13 a, b, c, d · Expectations, Risk Premium, Convexity, and the Shape of the Term Structure (Tuckman and Serrat, Fixed Income Securities 4e, Ch.8); MR-15 d · The Art of Term Structure Models: Volatility and Distribution (3e, Ch.10); IM-2 b · Factors (Ang, Asset Management, Ch.7)

MAP TIER D · #72 · 2/16 SOURCES TELEGRAM

Specifics reported

Map #72: "Reported: 'impact of inflation, market stress on long term / short term bond'; 'climate risk and inflation.' The differential effect across the maturity spectrum, and the risk-premium and convexity contributions to term-structure shape, are the examinable mechanics."
Shapes covered here:
S1 higher expected inflation as higher expected short rates: effect on forward rates and bond values grows with duration (Q1)
S2 market stress as a higher risk premium and higher rate volatility: premium term grows with duration, convexity term pulls the other way (Q1)
S3 a climate-related shock framed as an inflation and stress scenario for short versus long bonds (Q1).

Revision notes

- **Three determinants** (MR-13): the shape of the term structure is set by expectations of future short rates, the risk premium investors demand for bearing interest rate risk, and convexity, whose effect comes from rate volatility: $f(T) = r_0 + D\,E[dr]/dt + \lambda D - \frac{1}{2}C\sigma^2$.
- **Expectations term and inflation** (MR-13 a, S1): "the higher the instantaneous rate, the higher the forward rate; the more rates are expected to increase, the higher the forward rate; and the greater the corresponding duration, the greater the effect of expected rate changes on the forward rate." Long-run expectations can be anchored by macroeconomics: book example 4% = 1% long-run real rate + 3% long-run inflation. Higher expected inflation therefore works through this term, most strongly at long durations.
- **Granularity** (MR-13 a): short-horizon expectations can be very specific (policy meetings, tax dates, issuance calendar, quarter-end balance sheets); long-horizon expectations are smooth, which is why inflation anchors matter most at the long end.
- **Risk premium term** (MR-13 d, S2): risk-averse investors demand $E[dP/P] = r_0\,dt + \lambda D\,dt$: a premium proportional to duration. λ "may depend on time and on the level of rates, but not on the characteristics of any individual bond". The forward rate rises by λD: "the greater the corresponding interest rate risk and the greater the risk premium, the greater the forward rate". Book: $r = 1\%$, $D = 5$, $\lambda = 10$ bp gives 1.5% expected return.
- **Sharpe ratio view:** $SR = \lambda/\sigma$; 10 bp per year of duration over 100 bp rate volatility is a Sharpe ratio of 10%.
- **Convexity term under stress** (MR-13 b, c, S2): higher rate volatility lowers forward rates by $\frac{1}{2}C\sigma^2$, and C grows with term, so the effect is concentrated at the long end (book tree: forwards 9.0000%, 8.8532%, 8.5590% with flat 9% expectations). Volatility pushes long forwards *down*; the risk premium pushes them *up*. See also D12.
- **Return decomposition and maturity** (MR-13 d): $dP/P = f(T)dt - D\,dr + \frac{1}{2}C\sigma^2 dt$. A rise in rates costs return in proportion to duration, so for the same rate move long bonds lose more than short bonds; volatility adds return in proportion to convexity.
- **Expectations cannot be separated from risk premium** (MR-13): a drift of a number of basis points has the same pricing effect as a risk premium of that number per year of duration. "The term structure of interest rates cannot, on its own, be used to separate expectations of rate changes from risk premium"; only the risk-neutral process matters for pricing.
- **Two hypotheses** (MR-13): pure expectations ($\lambda = 0$; forwards are realised in the no-change scenario) versus pure risk premium ($E[dr/dt] = 0$; the term structure stays unchanged in the no-change scenario). Reality can lie between.

- **Inflation and volatility** (MR-15 d): "Periods of high inflation and high short-term interest rates are inherently unstable and, as a result, the basis-point volatility of the short rate tends to be high", which motivates CIR and lognormal models. See also D1.
- **Inflation and mean reversion** (MR-14 footnote): hyperinflation is not consistent with mean reversion and destroys a currency and its rates: "interest rates mean revert until they don't".
- **Inflation as a macro factor** (IM-2 b): high inflation is bad for both stocks and bonds. Government, investment grade and high yield bonds average 8.6%, 8.8% and 9.2% in low-inflation periods versus 5.4%, 5.3% and 6.0% in high-inflation periods: fixed payments lose real value. Part of the long-run bond risk premium compensates for doing badly when inflation is high. Government bonds do relatively well in low-growth periods (10.0% versus 3.9%) but are more volatile in recessions (15.5% versus 9.3%). See also B4.
- **Climate risk** (outside-source): no 2026 MR or IM reading in the provided material treats climate risk and inflation directly. In an exam stem, treat a climate shock through the mechanics above: its effect on expected inflation (expectations term), on uncertainty (volatility and convexity) and on required compensation (risk premium).

Formulas

Forward rate decomposition (eq. 13.20). $f(T) = \underbrace{r_0 + D\,\frac{E[dr]}{dt}}_{\text{expectations}} + \underbrace{\lambda D}_{\text{risk premium}} \underbrace{- \tfrac{1}{2}C\sigma^2}_{\text{convexity}}$

Expected return, risk-averse (eq. 13.18). $E[dP/P] = r_0\,dt + \lambda D\,dt; \qquad SR = \lambda/\sigma$ (eq. 13.19)

Instantaneous return (eq. 13.15). $dP/P = f(T)\,dt - D\,dr + \tfrac{1}{2}C\sigma^2 dt$

Exam traps

- **Polarity of the convexity term:** higher volatility *lowers* long forward rates (and supports long bond prices); it does not raise them.
- **Scope of the risk premium:** λ is common to all bonds, but the premium in a forward rate or expected return is λD: larger at long maturities, not equal across the curve.
- **Absolute claim:** forward rates alone cannot split a move into expected inflation and risk premium.
- **Sign in the return equation:** rate rises enter with $-D$, volatility with $+\tfrac{1}{2}C\sigma^2$.
- **Sibling swap with IM-2:** government bonds do relatively *well* in low-growth periods but *badly* when inflation is high; swapping the two macro factors reverses the answer.

Question D18.1

MR-13 a, b, c, d · S1, S2, S3 · [Map #72]

A treasury analyst at a UK bank is preparing a report for the asset-liability committee on how a stress scenario would affect the bank's short-dated and long-dated gilt holdings. In the scenario, a climate-related supply shock raises expected future short-term rates through higher expected inflation, while market stress raises interest rate volatility and the risk premium investors demand per unit of duration. Using the decomposition of forward rates into expectations, risk premium and convexity, which of the following statements should the analyst include in the report?

A. The rise in rate volatility raises long-term forward rates through the convexity term, so long-dated gilts cheapen relative to short-dated gilts for that reason.

B. The higher risk premium raises all forward rates by the same number of basis points, because the premium per unit of duration does not depend on any individual bond.

C. Observed forward rates allow the analyst to fully separate the part of the move due to higher expected inflation from the part due to a higher risk premium.

D. Higher expected short-term rates and a higher risk premium each raise forward rates in proportion to duration, so long-dated gilts lose more value than short-dated gilts.

Solution **Answer: D**

Step 1 – Expectations term

$$\text{expectations part of } f(T): \quad r_0 + D\,\frac{E[dr]}{dt}$$

⇒ *Higher expected inflation raises expected short rates; the effect on the forward rate is scaled by the duration of the zero of that term, so it is largest at the long end.*

Step 2 – Risk premium term

$$\text{risk premium part of } f(T): \quad \lambda D$$

⇒ *A higher λ raises every forward rate by λ times its duration: again largest at the long end.*

Step 3 – Convexity term

$$\text{convexity part of } f(T): \quad -\tfrac{1}{2}C\sigma^2$$

⇒ *Higher volatility pushes long forwards down, partly offsetting Steps 1 and 2 for long bonds; it does not add to their losses.*

Step 4 – Effect on bond values

From $dP/P = f(T)dt - D\,dr + \frac{1}{2}C\sigma^2 dt$, a rise in rates reduces value in proportion to duration, so long-dated gilts lose more than short-dated gilts. IM-2 b reaches the same direction from the asset-pricing side: high inflation is bad for bonds because their payments are fixed.

Why the other options are wrong

A is incorrect. Polarity flip. The convexity term $-\frac{1}{2}C\sigma^2$ lowers long forward rates when volatility rises; volatility supports long bond prices rather than cheapening them.

B is incorrect. Scope/condition error. λ is common to all bonds, but the premium enters as λD; dropping the duration condition makes the effect look parallel.

C is incorrect. Absolute claim. The book states the term structure cannot, on its own, separate expectations of rate changes from risk premium; only their combined risk-neutral drift is observable.

LO: MR-13 a: Explain the role of interest rate expectations in determining the shape of the term structure. MR-13 b: Apply a risk-neutral interest rate tree to assess the effect of volatility on the shape of the term structure. MR-13 c: Estimate the convexity effect using Jensen's inequality. MR-13 d: Identify the components into which the return on a bond can be decomposed, and calculate the expected return on a bond for a risk-averse investor. · Reference: Tuckman and Serrat 4e, Ch.8, 13.1 to 13.3, eqs. 13.15 to 13.20; Ang, Ch.7 (IM-2 b), Table 2.1

D19 LDA for Operational Risk

Readings: ORR-4 f · Risk Measurement and Assessment (GARP ORR Ch.4, 4.4 Operational Risk Modeling and Capital: Loss Distribution Approach; Frequency Modeling; Severity Modeling; Limitations and Constraints; Using Internal and External Loss Data; Capital Modeling for Operational Risk); ORR-4 a · Operational Loss Data; ORR-6 c · Risk Reporting (Asymmetry of Operational Risk Event Data)

MAP TIER D · #74 · 2/16 SOURCES VARDEEZ

Specifics reported

Map #74: "Reported: 'LDA approach'; and 'calculate z score (LDA)', which suggests a discriminant-analysis framing rather than the loss-distribution approach: these are two different things sharing an acronym and the recall is ambiguous. Cover both: loss frequency × severity convolution for operational capital, and linear discriminant analysis for scoring."
Shapes covered here:
S1 loss frequency × severity convolution into an aggregated annual loss distribution (Q1)
S2 operational risk capital as the 99.9th percentile of the aggregated distribution, with aggregation across units of measure (Q1; the discriminant-analysis reading of 'LDA', with the Altman Z-score, is tested in A10.2)

Revision notes

The specific: frequency × severity, then the 99.9th percentile (S1, S2)

- **Key principle.** The LDA is an old actuarial technique that decomposes loss data into **frequency** (how often events occur) and **severity** (how much they cost). Decomposition doubles the observational data points, valuable because early operational risk data were scarce. Frachot, Georges and Roncalli (Crédit Lyonnais, 2001) applied it to operational risk. Statistical modelling grew more in continental Europe; scenario-based modelling and stress testing more in the United States.
- **Convolution (S1).** Frequency and severity distributions are estimated **independently**, then convoluted into an aggregated loss distribution. The most common method is **Monte Carlo** (a million or more random draws of frequency and severity); Fast Fourier transform and Panjer recursion are equation-based alternatives needing more coding and mathematics but less computer time.
- **Capital (S2).** Economic capital is determined by extrapolating the **99.9th percentile** of the estimated loss distribution at a **one-year** horizon; the 99.9th percentile of the aggregated distribution is the stand-alone operational risk capital when only the LDA is used. Under the AMA, banks held capital for all operational losses up to 99.9% at one year ("VaR at 99.9% = Regulatory Capital").
- **Frequency: Poisson.** Discrete count of events per period (typically a year). One parameter λ, equal to both mean and variance (the intensity rate). Observations i.i.d. **Additive**: $\text{Pois}(\lambda_X) + \text{Pois}(\lambda_Y) = \text{Pois}(\lambda_X + \lambda_Y)$. In a 2009 BIS study, nine of ten AMA firms used Poisson; the rest used the negative binomial.
- **Severity: lognormal and heavier.** Continuous, asymmetric and heavy-tailed: many small losses and a few very large ones. Lognormal: $\ln X$ is normal with location μ and scale σ; positive values only; mode $e^{\mu-\sigma^2}$. Heavier tails increasingly used: Weibull and the generalized Pareto distribution (GPD).

Limitations and units of measure (ORR-4 f)

- **Two strong assumptions.** (1) Frequency and severity are independent, a strong condition given the contrast between the frequency of small and large events. (2) Within each risk class (unit of measure, UoM) losses are i.i.d., with no implicit correlation and one data-generating mechanism; tested by clustering homogeneous losses and checking for no autocorrelation.
- **UoM trade-off.** Good UoMs are homogeneous sets driven by the same factors (external fraud by business line; damage to physical assets for all lines; internal fraud by business entity; processing errors by business line). Finer segmentation: more homogeneous data, but less data, more uncertainty and harder aggregation. UoMs are aggregated with explicit correlations or **copulas** (which can capture tail dependence); the dependency structure strongly drives capital and attracts validators and regulators.
- **The tail.** LDA alone does not capture the extreme skewness of operational losses: the tail is completed with extreme value theory (block maxima or peaks over threshold), scenario data, or both. EVT is valid only if one mechanism generates all losses, which is unlikely for operational data (see A17).

Data inputs and the regulatory context (ORR-4 a, f)

- **Internal and external data.** Internal loss data are the backbone of LDA and best for the **body** (frequent, moderate losses); rules are needed for near misses, accidental gains and rapidly recovered losses (conservative firms include them in frequency and severity). External data supplement the tail: public databases (full context, scarcer) and consortia such as ORX (anonymous, abundant). Mixing requires **scaling** (size, inflation), a **cut-off mix** (severity level where external data enter) and **filtering** rules that avoid cherry-picking.
- **AMA and its four inputs.** Internal loss data (ILD), external loss data (ELD), scenario data (SD), business environment and internal control factors (BEICF): the first two are past observations, the last two prospective. AMA approval took about three years and around 30 criteria. A Tier-1 European bank calibrated body and tail with two distributions and traced every loss and scenario to the capital number.
- **After AMA.** With the new Standardized Approach (effective 1 January 2023) there is no formal requirement to model operational risk capital; modelling effort moves to Pillar 2 (ICAAP), stress testing and operational resilience. Smaller firms often hold capital at least equal to the largest scenario and to clusters of two to four scenarios that could occur together; Tier-1 firms aggregate scenarios with variance-covariance matrices or copulas. Economic capital and ICAAP largely share a purpose.
- **Why the tail dominates (ORR-6 c).** ORX 2014 to 2019: losses above EUR 10 million were 0.4% of occurrences but 66.6% of severity; losses of EUR 20,000 to 100,000 were 57.9% of occurrences but 4.4% of severity. "No averages in operational risk."
- **The other LDA.** Linear discriminant analysis (Altman's Z-score) classifies borrowers into groups from weighted ratios; it is a credit scoring tool, unrelated to capital. See also A10.

Formulas

Poisson frequency: $P(N=k)=\dfrac{\lambda^k e^{-\lambda}}{k!}$, $E[N]=\mathrm{Var}[N]=\lambda$, $\mathrm{Pois}(\lambda_X)+\mathrm{Pois}(\lambda_Y)=\mathrm{Pois}(\lambda_X+\lambda_Y)$

Lognormal severity: $f_X(x;\mu,\sigma)=\dfrac{1}{x\sigma\sqrt{2\pi}}\exp\!\left(-\dfrac{(\ln x-\mu)^2}{2\sigma^2}\right)$, $\mathrm{mode}=e^{\mu-\sigma^2}$

Aggregated annual loss: $S=\sum_{i=1}^{N}X_i$, frequency and severity independent; $P(S\le s)=\sum_{n\ge 0}P(N=n)\,P(X_1+\cdots+X_n\le s)$

Expected annual loss (compound Poisson, outside-source): $E[S]=\lambda\,E[X]$

Capital: OpRisk capital $=q_{99.9\%}(S)$ at a one-year horizon (stand-alone, LDA only)

Exam traps

- **Acronym swap:** "LDA" for scoring is linear discriminant analysis (Z-score, A10); "LDA" for capital is the loss distribution approach.
- **Dropped term:** convolving only zero- and one-event years cuts the tail off at the largest single loss; years with two or more events must be counted when the frequency is high enough to reach the 99.9% tail.
- **Sibling swap with credit capital:** the reading equates operational risk capital with the 99.9th percentile of the aggregated loss distribution; do not subtract expected loss as in the IRB unexpected-loss formula, and do not use 99%.
- **Aggregation error:** adding stand-alone UoM percentiles assumes perfect dependence; independent Poisson UoMs with the same severity merge exactly into one Poisson UoM, and dependence in general is modelled with copulas.
- **Scope/condition:** Poisson has one parameter (mean = variance); frequency and severity independence and i.i.d. within a UoM are assumptions, not facts; the 99.9% one-year standard was the AMA rule, and under the 2023 SA modelling is no longer required for Pillar 1.

Question D19.1

ORR-4 f · S1, S2 · [Map #74]

An operational risk modeller at a Swiss private bank is calibrating the loss distribution approach (LDA) for the bank's ICAAP economic capital. Two units of measure (UoMs) cover external fraud, one in private banking and one in asset management. The scenario-based calibration gives:

- External fraud, private banking: Poisson frequency, mean 0.05 events per year
- External fraud, asset management: Poisson frequency, mean 0.03 events per year
- Severity for both UoMs (same scenario set): CHF 40 million with probability 0.75 and CHF 160 million with probability 0.25

Validation has confirmed that the two UoMs are independent of each other, that frequency and severity are independent, and that the severity data are homogeneous, so the modeller merges the two into a single UoM. Run stand-alone, each UoM has a 99.9th percentile annual loss of CHF 160 million. Which of the following gives the expected annual loss and the economic capital of the merged UoM, using the confidence level and horizon applied in the reading?

A. Expected annual loss CHF 5.6 million; economic capital CHF 200 million

B. Expected annual loss CHF 5.6 million; economic capital CHF 160 million

C. Expected annual loss CHF 5.6 million; economic capital CHF 320 million

D. Expected annual loss CHF 8.0 million; economic capital CHF 200 million

Solution **Answer: A**

Step 1 – Merge the frequencies (Poisson additivity)

$\lambda = \lambda_X + \lambda_Y = 0.05 + 0.03 = 0.08$ events per year.

$\Rightarrow$ *Independent Poisson UoMs with the same severity combine exactly into one Poisson UoM with intensity 0.08: on average one external fraud event every 12.5 years across both business lines.*

Step 2 – Expected annual loss

$E[X] = 0.75 \times 40 + 0.25 \times 160 = 30 + 40 =$ CHF 70 million per event.

$E[S] = \lambda E[X] = 0.08 \times 70 =$ CHF 5.6 million (compound Poisson mean, outside-source).

$\Rightarrow$ *The yearly loss budget for this risk class: small next to the tail, as the ORX asymmetry statistics predict.*

Step 3 – Poisson probabilities of the annual event count

$P(N = k) = \lambda^k e^{-\lambda}/k!$: $P(0) = e^{-0.08} = 0.923116$; $P(1) = 0.08 \times 0.923116 = 0.073849$; $P(2) = 0.08^2/2 \times 0.923116 = 0.002954$; $P(3) = 0.000079$.

Step 4 – Convolve frequency and severity

One event: CHF 40m with $0.073849 \times 0.75 = 0.055387$; CHF 160m with $0.073849 \times 0.25 = 0.018462$.

Two events: CHF 80m with $0.002954 \times 0.75^2 = 0.001662$; CHF 200m with $0.002954 \times 2 \times 0.75 \times 0.25 = 0.001108$; CHF 320m with $0.002954 \times 0.25^2 = 0.000185$.

Three events add about 0.000033 at CHF 120m and smaller amounts higher up.

Cumulative: $P(S \le 0) = 0.923116$; $P(S \le 40) = 0.978503$; $P(S \le 80) = 0.980165$; $P(S \le 120) = 0.980198$; $P(S \le 160) = 0.998661$; $P(S \le 200) = 0.999769$.

$\Rightarrow$ *The aggregated distribution has a large mass at zero, then lumps at each sum of severities.*

Step 5 – Read the 99.9th percentile

$P(S \le 160) = 0.998661 < 0.999$ and $P(S \le 200) = 0.999769 \ge 0.999$, so $q_{99.9\%}(S) =$ CHF 200 million.

$\Rightarrow$ *The capital is set by a year with one large and one small fraud (160 + 40): once the two UoMs are merged, the 0.11% probability of that combination matters at the 99.9% level.*

Interpretation

The bank holds CHF 200 million of ICAAP economic capital for external fraud against an expected annual loss of CHF 5.6 million: the gap shows why operational risk capital is driven by rare, severe events. Merging also gives a lower figure than adding the two stand-alone results, because the UoMs are independent.

Why the other options are wrong

B is incorrect. Dropped term: convolving only zero- and one-event years caps the loss at the largest single severity, CHF 160 million; it ignores the two-event years that carry the merged distribution past 99.9%. (Using the 99th percentile, also CHF 160 million, is the same answer from the wrong confidence level.)

C is incorrect. Aggregation error: adds the stand-alone 99.9th percentiles, $160 + 160 = 320$, which implicitly assumes perfect dependence between the UoMs, contradicting the validated independence.

D is incorrect. Wrong severity mean: uses the unweighted average of the two severity points, $(40+160)/2 = 100$, so $0.08 \times 100 = 8.0$; the probabilities 0.75 and 0.25 must weight the mean.

LO: ORR-4 f: Describe approaches used to determine the level of operational risk capital for economic capital purposes, including their application and limitations. · Reference: GARP, Operational Risk and Resilience, Ch.4, 4.4 Loss Distribution Approach for Operational Risk (Key Principles, Frequency Modeling, Severity Modeling, Interrelationships and Correlations of Operational Risk Events); Capital Modeling for Operational Risk

D20 Credit Model Comparison: Vasicek, CreditRisk+, CreditMetrics, KMV

Readings: CR-10 d, e · Credit Value at Risk (Hull, RMFI 6e, Ch.19, 10.2 to 10.4); CR-5 f · Introduction to Credit Risk Modeling and Assessment (Doumpos et al., Ch.1, 5.6.3); CR-12 g · Credit Risk (Hull, OFOD 11e, Ch.24, 24.3)

MAP TIER D · #41 · 2/16 SOURCES VARDEEZ NOTES ONLY

Specifics reported

Map #41: "A four-way comparison built for the four-statement audit format. Know which is actuarial (CreditRisk+), which is ratings-migration based (CreditMetrics), which underpins the Basel IRB formula (Vasicek), and which is equity-market based (KMV/Merton family)."
Shapes covered here:
S1 CreditRisk+ as the actuarial, default-mode model (notes; distractors 1, 5)
S2 CreditMetrics as the ratings-migration, mark-to-market model (notes; distractor 2)
S3 Vasicek as the model under the Basel IRB formula (notes; distractors 3, 6)
S4 Moody's KMV as the equity-market based Merton implementation (notes; distractor 4)
S5 four-way comparison on loss recognition and correlation modelling (comparison table)

Revision notes

- **Vasicek's model** (CR-10 e, S3). A one-factor Gaussian copula model of time to default that gives high percentiles of the portfolio *default rate*: WCDR(T, X) from PD and a single credit correlation parameter ρ. For one loan the Xth loss percentile is WCDR $\times$ EAD $\times$ LGD.
- **Vasicek and Basel IRB** (CR-10 d, S3). Gordy (2003): for a large portfolio of small loans in a one-factor world, the Xth loss percentile is approximately $\sum_i$ WCDR$_i$ $\times$ EAD$_i$ $\times$ LGD$_i$. Regulatory capital for the banking book is set from this with $T = 1$ year and $X = 99.9\%$, sometimes multiplied by a maturity adjustment because a loan lasting more than a year may deteriorate without defaulting. See A31 for the IRB numbers.
- **Estimating** ρ (CR-10 d). Structural models show ρ for two companies is roughly their asset-return correlation, approximately their equity-return correlation; use the average equity correlation of the obligors, or of similar listed companies for private ones.
- **Vasicek's weakness** (CR-10 e). It "incorporates very little tail correlation"; replacing the Gaussian copula with another copula can remedy this.
- **CreditRisk+** (CR-10 e, S1). Credit Suisse Financial Products, 1997: "analytic approximations that are well established in the insurance industry" (the actuarial approach). With n loans and default probability q, the binomial count of defaults is approximated by a Poisson with mean qn when q is small and n large; it still holds approximately when PDs differ, provided all are small and q is their average.
- **Default-rate uncertainty creates correlation** (CR-10 e). If qn is gamma-distributed (mean μ, standard deviation σ), the Poisson becomes a negative binomial. Hull Table 10.4 ($\mu = 4$): the 99.9th percentile of the number of defaults is 11 at $\sigma = 0$ and 98 at $\sigma = 10$. Without uncertainty there is no default correlation and the loss distribution is fairly symmetrical; with it, positively skewed.
- **CreditRisk+ by Monte Carlo** (CR-10 e). Sample an overall default rate (e.g. from Table 9.6), link each category's rate to it by regression, sample defaults and an LGD per default, total the loss, repeat, read VaR. Refinement: default rates are autocorrelated year to year, so model next year's rate on this year's.
- **CreditRisk+ in Doumpos** (CR-5 f). In contrast to KMV, it does not use the capital structure of the company. Losses occur only in the event of bankruptcy; the default frequency is a continuous random variable; defaults are small-probability and independent, so the Poisson (mean equal to variance) fits; it calculates default events, not ratings; its advantage is the minimum data required.
- **CreditMetrics** (CR-10 e, S2). JPMorgan, 1997, built on a rating transition matrix (internal or agency) so it captures *downgrades as well as defaults*. One-year credit VaR by Monte Carlo: simulate every counterparty's end-of-year rating; if not default, revalue all transactions at the horizon using a term structure of credit spreads per rating; if default, loss = EAD $\times$ $(1 - R)$, with R sampled from an empirical distribution. Upgrades give negative losses.

- **CreditMetrics correlation** (CR-10 e; CR-12 g). Rating changes are correlated with a Gaussian copula whose correlation is typically the equity-return correlation (one-factor CAPM structure for consistency). Thresholds come from N^{-1} of cumulative transition probabilities: an A-rated firm defaults if $x_A > N^{-1}(0.9995) = 3.2905$; a B-rated firm if $x_B > N^{-1}(0.9619) = 1.7744$.
- **CreditMetrics for derivatives** (CR-10 e). With no default in year 1, the loss on a trial is $\sum_{i\geq j}(1-R)(q_i^* - q_i)v_i$: the change in expected default cost from year one onward, using the horizon spread curve's probabilities q_i^*.
- **CreditMetrics in Doumpos** (CR-5 f). A "mark-to-market model rather than a default mode model". Inputs: issuer rating, transition matrix, recovery rates, yield spreads. Horizon usually one year (up to 10). Criticised for homogeneous rating grades and fixed interest rates. Output: the distribution of the change in a loan's value; each loan can carry a different capital requirement. "Not a rating tool, nor a risk pricing model." Three stages: exposure profile, value volatility from upgrades, downgrades and defaults, then correlations to aggregate.
- **Same losses, different timing** (CR-10 e). On the same assumptions CreditMetrics and CreditRisk+ should predict the same long-run loss distribution. A loan downgraded A to BBB, then to B, then defaulting: CreditRisk+ books nothing until the year-3 default; CreditMetrics books revaluation losses in years 1, 2 and 3, which should add up to the same total.
- **Moody's KMV** (CR-5 f, S4). The best-known implementation of Merton's structural model: asset value and volatility are backed out of equity market data. Default point = short-term debt $+\frac{1}{2}$ long-term debt; distance to default is mapped to an EDF through an *empirical* distribution of historical default frequencies, not the normal. See A5.
- **Structural versus reduced form** (CR-5 f). Structural: default is endogenous, linked to assets and debt (Merton, KMV). Reduced form: default is a random event driven by an exogenous Poisson jump process, calibrated to bond and credit-derivative prices (Jarrow and Turnbull; Duffie and Singleton). Evidence is mixed for structural models; reduced form approaches were found superior in most cases.
- **Hull's summary** (CR-10 e). Vasicek and CreditRisk+ estimate losses from defaults only; the impact of downgrades is not considered (except through the maturity adjustment in the regulatory use of Vasicek). CreditMetrics differs by counting defaults and downgrades, using a Gaussian copula with a transition matrix.

Model	Approach	Loss recognised	Default correlation
Vasicek	One-factor Gaussian copula; basis of Basel IRB	Defaults only (maturity adjustment in IRB)	Single copula ρ, from equity correlations
CreditRisk+	Actuarial (insurance), Poisson / negative binomial; no capital structure	Defaults only	Uncertainty in the default rate
CreditMetrics	Rating transition matrix, Monte Carlo revaluation; mark-to-market	Defaults and downgrades (upgrades as gains)	Gaussian copula on rating changes, equity correlation
Moody's KMV	Merton structural model on equity-market data; empirical DD to EDF	Default (EDF at the horizon)	Not covered in CR-5

Formulas

Vasicek (Hull 10.1). $\text{WCDR}(T,X) = N\left(\dfrac{N^{-1}(\text{PD}) + \sqrt{\rho}\,N^{-1}(X)}{\sqrt{1-\rho}}\right)$; Xth loss percentile $\approx \sum_i \text{WCDR}_i \times \text{EAD}_i \times \text{LGD}_i$

CreditRisk+ (Hull 10.3, 10.4). $P(m) = \dfrac{e^{-qn}(qn)^m}{m!}$; with gamma uncertainty $P(m) = p^m(1-p)^\alpha \dfrac{\Gamma(m+\alpha)}{\Gamma(m+1)\Gamma(\alpha)}$, $\alpha = \mu^2/\sigma^2, p = \sigma^2/(\mu+\sigma^2)$

CreditMetrics thresholds. New rating k if $N^{-1}\left(\sum_{j<k} p_j\right) < x \leq N^{-1}\left(\sum_{j\leq k} p_j\right)$, ratings ordered from best to default

Moody's KMV default point. $\text{DP} = \text{STD} + 0.5 \times \text{LTD}$

Likely distractors

1. **Corrupted:** "CreditRisk+ estimates default probabilities from the firm's capital structure and asset volatility." (sibling swap with KMV)

   True: CreditRisk+ does not use the capital structure; it is an actuarial default-count model. Using the capital structure is the KMV/Merton approach.

2. **Corrupted:** "CreditMetrics is a default-mode model: a credit loss is recognised only when the counterparty defaults." (sibling swap with CreditRisk+)

   True: CreditMetrics is a mark-to-market model that counts losses from downgrades as well as defaults, and upgrades as negative losses.

3. **Corrupted:** "Vasicek's model captures losses from both downgrades and defaults, which is why regulators use it for IRB capital." (sibling swap with CreditMetrics)

   True: Vasicek's model estimates losses from defaults only; downgrade risk enters the regulatory implementation only through the maturity adjustment.

4. **Corrupted:** "Moody's KMV maps distance to default into a default probability with the standard normal distribution, using total debt as the default point." (scope error on both inputs)

   True: KMV uses an empirical distribution of historical default frequencies, and the default point is short-term debt plus half of long-term debt.

5. **Corrupted:** "In CreditRisk+, uncertainty about the default rate leaves the 99.9% VaR unchanged, because the expected number of defaults is the same." (absolute claim)

   True: Default-rate uncertainty creates default correlation and fattens the tail: with $\mu = 4$ the 99.9th percentile rises from 11 defaults ($\sigma = 0$) to 98 ($\sigma = 10$).

6. **Corrupted:** "Because it uses a Gaussian copula, Vasicek's model generates strong tail dependence between defaults." (polarity flip)

   True: Vasicek's model incorporates very little tail correlation; another copula is needed to remedy this.

D21 Investor Protection, MiFID and the Volcker Rule

Readings: ORR-14 a, b · Case Study: Investor Protection and Compliance Risks in Investment Activities (GARP ORR Ch.14); ORR-12 a · Guidance on Managing Outsourcing Risk (12.4 D, Incentive Compensation Review); ORR-5 d · Risk Mitigation (rule-based mistakes); ORR-3 d · Risk Identification (Basel CPBP categories)

MAP TIER D · #52 · 2/16 SOURCES VARDEEZ NOTES ONLY

Specifics reported

Map #52: "Suitability and appropriateness obligations, best execution, conflicts-of-interest management, and the proprietary-trading prohibition with its market-making and hedging exemptions. Scope/condition errors on the exemptions are the natural distractor."
Shapes covered here:
S1 suitability and appropriateness obligations
S2 best execution, with the Deutsche Bank Securities case
S3 conflicts-of-interest management under MiFID II and MiFIR
S4 the Volcker Rule's proprietary-trading prohibition and its market-making and hedging exemptions, with scope/-condition errors

Revision notes

Context (ORR-14 a)

- **Investor protection.** Legislation and regulation "aimed at ensuring that investors are fully informed about their investments when they enter into financial transactions." It also protects investors from misrepresentation by intermediaries, establishes accountability for fraud or insider trading, and supports cross-jurisdictional cooperation between regulators and law enforcement. Some of the largest settlements and penalties paid by banks and investment firms come from investor protection violations.
- **Where it sits in the taxonomy.** Compliance with financial laws usually falls in Basel event type 4, **Clients, Products & Business Practices** (CPBP), Level 2 categories 4.1 **Suitability, Disclosure & Fiduciary** and 4.2 **Improper Business or Market Practices** (see A22 for the Level 3 examples: aggressive sales, account churning, market manipulation, insider trading on the firm's account).
- **Link to AML.** Compliance requirements include strict KYC filings so that money flowing into financial products is not of doubtful origin; investor protection and AML legislation are therefore connected (see A29).

MiFID, MiFID II and MiFIR (S1, S2, S3)

- **MiFID.** Markets in Financial Instruments Directive (2004/39/EC), promulgated 2004, applicable across the EU since November 2007, in force 2007 to 2018. It prescribes conduct of business and organisational requirements for investment firms and authorisation requirements for regulated markets, defines regulatory reporting and trade transparency obligations to prevent market abuse, and sets rules for admitting instruments to trading.
- **MiFID II and MiFIR.** Adopted June 2014 after the crisis: a revised directive (MiFID II) and a regulation (MiFIR). Besides stronger investor protection, they add public disclosure of trading data and transaction reporting to regulators and supervisors. Book list of what MiFIR addresses: remuneration of traders and advisors, and conflicts of interest; fair and non-misleading communication to customers; definition and independence of investment advice; sales process and product governance; **best execution** of client trades; dealings with eligible counterparties.
- **Directive versus regulation.** EU regulations apply directly in all member states. Directives must be turned into national law, and each country may be *more* stringent, but not less, than the directive.
- **Suitability and appropriateness (S1).** In the GARP text, suitability appears as Basel category 4.1 and in the SR 13-19 example of an inappropriate incentive: variable fees that direct customers to higher-margin products without considering whether they suit the customer. Under MiFID II (outside-source): **suitability** applies to investment advice and portfolio management and requires information on the client's knowledge and experience, financial situation (including ability to bear losses) and investment objectives (including risk

tolerance); **appropriateness** applies to non-advised services in complex products and checks only knowledge and experience; execution-only dealing in non-complex products at the client's initiative needs neither.

- **Best execution (S2).** FINRA Rule 5310: seek the most favourable terms reasonably available for a customer's order; criteria include price improvement and speed of execution. The MiFID II equivalent requires all sufficient steps to obtain the best possible result considering price, costs, speed, likelihood of execution and settlement, size and nature of the order (outside-source).
- **Conflicts of interest (S3).** A specific driver of compliance risk is the conflict of interest of traders who trade both for clients' books and for the firm's own book; another is the information asymmetry between retail investors and the banks and asset managers that sell to them. MiFIR targets remuneration structures and the independence of advice; regulators watch every incentive structure that can lead to poor conduct (ORR-5: mis-selling driven by aggressive commercial incentives is a **rule-based mistake**, cured by changing the rule).

Dodd-Frank and the Volcker Rule (S4)

- **Investor Protection Act.** A component of the Dodd-Frank Wall Street Reform and Consumer Protection Act (the book dates it 2009; it was signed in July 2010, outside-source), a response to the 2007 crisis. It set up a committee to consult the SEC on new products, fee structures and trading strategies, and increased whistleblower protections.
- **Other Dodd-Frank elements.** OTC derivatives trading regulation delegated to the SEC and the CFTC; firms buying and selling derivatives must use clearinghouses; federal oversight for financial stability, with a financial stability oversight council and a reorganisation of financial regulation; the **Consumer Financial Protection Bureau**, an *independent* regulator for consumer finance (mortgages, student loans, credit cards).
- **Volcker Rule.** Contained in Dodd-Frank. It "intends to prevent commercial banks from engaging in speculative activities and proprietary trading for profit" and limits banks' investments in hedge funds and private equity funds.
- **The exemptions (outside-source).** Permitted activities include underwriting; **market making-related** activity, provided inventory is designed not to exceed the reasonably expected near-term demand of clients, customers or counterparties; **risk-mitigating hedging** of specific, identifiable risks of the bank's positions; trading in government obligations; and trading on behalf of customers. Ownership of covered funds (hedge and private equity funds) is limited, broadly to 3% of any one fund and 3% of Tier 1 capital in aggregate. A desk that holds positions beyond customer demand, or a "hedge" not tied to identifiable risks, falls back into prohibited proprietary trading.

Compliance risk management and the cases (ORR-14 b)

- **Fraud or error.** An *intentional* compliance breach is a form of internal fraud (unauthorised activity), so most of the internal fraud control framework applies. An *unintentional* breach is usually human error or faulty policies; controls against fraud also prevent errors, except deterrent controls. The difference between error and fraud is intention.
- **Drivers and controls.** General internal-fraud drivers: poor engagement, dissatisfaction or stress, insufficient resources, weak ethics culture. Specific drivers: information asymmetry, trader conflicts, volatility spikes that raise transaction volumes (more errors, easier to hide abnormal trades or insider trading). Controls: monitoring and supervision of employees and trades, strong middle and back office, policies and procedures, training, ethics culture, and consistent onboarding, screening and training of employees and third parties.
- **Fines and UBS.** US fines for investor protection violations: nearly USD 82 billion in 2000 to 2022 over 6,612 penalty records. Largest: UBS, USD 11.15 billion (July 2008): buy back USD 11 billion of securities plus USD 150 million of penalties for misrepresenting **auction rate securities** as safe, cash-equivalent products when they faced increasing liquidity risk.
- **JP Morgan spoofing.** Spoofing: a false impression of demand by rapidly entering and cancelling orders. CFTC fine of USD 920 million (September 2020) for spoofing in precious metals and US Treasury markets, over at least eight years and hundreds of thousands of spoof orders. Lesson: practices once accepted as normal became violations as investor protection regulation evolved.
- **Deutsche Bank Securities (S2).** FINRA fine of USD 2 million: from 2014 to 2018 customer market orders were first routed through its smart order router's "SuperX ping" unless customers opted out, causing execution delays and lower fill rates; the firm did not change the arrangement or test price improvement from routing directly to exchanges, and disclosed trading rebates only vaguely, with no sign they were passed to customers. Settled without admitting or denying. Regulators set punitive fines that more than offset the benefits, as a deterrent and a signal to peers.

Likely distractors

- "The Volcker Rule bars banks from all trading in securities for their own account, including market making for clients and hedging of their own positions."

  True version: it targets speculative proprietary trading for profit and limits hedge fund and private equity investments; market making within reasonably expected near-term customer demand and risk-mitigating hedging are permitted (exemptions outside-source). *Scope/condition error.*

- "A desk qualifies for the market-making exemption whenever it quotes two-way prices, whatever the size of the inventory it holds."

  True version: the exemption depends on inventory being designed not to exceed reasonably expected near-term demand of customers; inventory beyond that is proprietary trading (outside-source). *Scope/condition error: drops the condition.*

- "MiFID II is an EU regulation that applies directly and uniformly in every member state, so no country may set stricter rules."

  True version: MiFID II is a directive (MiFIR is the regulation); directives are transposed nationally and each country may be more stringent, but not less. *Sibling swap (directive versus regulation).*

- "Best execution is satisfied by routing every order to the venue that pays the broker the largest rebate, provided the rebate is mentioned in the firm's reports."

  True version: best execution means seeking the most favourable terms reasonably available for the customer (price improvement, speed of execution); Deutsche Bank Securities was fined for indirect routing, untested price improvement and vague rebate disclosure. *Polarity flip (firm's benefit for client's benefit).*

- "An unintentional compliance breach caused by a faulty policy is classified as internal fraud, since it is an unauthorised activity."

  True version: only an *intentional* breach is a form of internal fraud; unintentional breaches stem from human error or faulty policies. *Sibling swap (error versus fraud).*

- "Dodd-Frank placed consumer finance oversight in a new bureau inside the SEC, and left OTC derivatives trading to be regulated by the Federal Reserve."

  True version: the CFPB is an independent regulator; regulation of OTC derivatives trading is delegated to the SEC and the CFTC, with mandatory use of clearinghouses. *Role misassignment.*

D22 Operational Risk Identification, Mitigation and Reporting

Readings: ORR-3 a, b, c, d · Risk Identification (GARP ORR Ch.3); ORR-4 a, b, c, d, e, f, g · Risk Measurement and Assessment (Ch.4); ORR-5 a, b, c, d, e, f, g · Risk Mitigation (Ch.5); ORR-6 a, b, c, d · Risk Reporting (Ch.6)

MAP TIER D · #75 · 2/16 SOURCES VARDEEZ NOTES ONLY

Specifics reported

Map #75: "The control sibling set, preventive, detective, corrective, directive, is a named GARP corruption operator and is worth securing even though the readings are deprioritised. RCSA, KRIs and loss-data collection are the identification and assessment tools."
Shapes covered here:
S1 the control sibling set: preventive, detective, corrective, directive, with key controls, automation and testing
S2 risk and control self-assessment (RCSA): inherent and residual risk, scales, heatmap, ownership of outcomes
S3 key risk indicators, against KPIs and KCIs
S4 loss-data collection: regulatory criteria, fields, dates, boundary events, completeness

Revision notes

The specifics: controls, RCSA, KRIs, loss data (S1 to S4)

- **The control sibling set (S1; ORR-5 b).** IIA classification. **Preventive**: reduce the *likelihood* of an incident, act before the event on its causes (segregation of duties, the best-known fraud control; verification of records; signature reconciliation; limitation of access; levels of authorisation; KYC/CDD). **Detective**: alert when an incident occurs, speed resolution and limit impact (exception reports, smoke and intrusion detectors, card-fraud alerts); a detective control becomes preventive when it detects a *cause*, as a file reconciliation run before execution. **Corrective**: lessen the *impact* without changing likelihood (back-ups, IT redundancies, continuity plans, crisis communication; seatbelts and airbags). **Directive**: the prescriptions for executing a process (policies, procedures, training, guidance, governance structure, roles, supervision, handover procedures); often missing from taxonomies but effective when comprehensive.
- **Key controls and automation (ORR-5 b, c).** A **key** (primary) control is sufficient on its own to prevent the risk materialising (it can be corrective, as a cloud back-up); a non-key (secondary) control only complements. Automation raises reliability but turns human-error risk into **technology and model risk**: high-likelihood, low-impact manual errors become low-likelihood, high-impact systemic breakdowns (the "water mattress"). Failure modes: false positives and negatives, controls disabled by downtime, back-ups failing silently on a full server. Knight Capital lost nearly USD 460 million in 45 minutes from repurposed code, weak version control, insufficient testing, unprioritised alerts and no one empowered to stop trading.
- **Control design and testing (ORR-5 c).** Weak designs: **optimistic** (tick-box sign-offs needing exceptional ability or motivation), **collective** (four-eyes checks dilute accountability; better with manager and subordinate or different departments), **more of the same** (adding failed designs). Four tests in rising scrutiny: self-certification or inquiry (secondary or low-risk controls only), examination (documentary evidence; suits automated controls), observation (real time; key controls), **reperformance** (strongest: mystery shopping, bogus transactions, model replication). Effectiveness depends on tester independence, frequency proportionate to risk, scope and sample. Testing was historically internal audit's; the first line increasingly tests key controls; the ORM function does not usually test but needs assurance.
- **RCSA (S2; ORR-4 b).** Evaluates likelihood and impact of **inherent** risk (before controls) and **residual** risk (after controls); typically annual, sometimes quarterly; workshops facilitated by the second line or questionnaires; a risk assessment unit may run it. Mature firms require evidence of control testing and back-test against incidents, which shows participants **underestimate likelihood and overestimate impact**. Variants: RCA (assessment must be evidenced) and RRSA (residual only). Purpose: awareness and prioritisation, not precise measurement. Results are signed off by business-line managers; action plans are **owned by the first line** and tracked by the second or third line. Mitigation need not mean more controls: reduce volumes, limit access, redesign controls, insure.
- **Scales and heatmap (ORR-4 b).** Impact types: financial, regulatory, customer, reputation, often service continuity; four- or five-point scales, financial impact expressed relative to revenue or operating income so the scale fits units of any size. Likelihood as a percentage or "once in x years": a 1-in-10-year event has

a **10% chance of occurring in the next year**; RCSA horizons are one year at most. Heatmap colours encode appetite: green within, yellow approaching, orange above (action plan or escalated acceptance), red far beyond (urgent plan). Never multiply ratings: a frequent low-impact risk (1×4) is not a remote extreme one (4×1).

- **KRIs, KPIs, KCIs (S3; ORR-4 c).** A **KRI** signals a rise or fall in the level of a risk, in likelihood (transactions per staff member, falling satisfaction scores, higher sales targets for fraud) or in impact (key-person knowledge, data sensitivity on a server, top-10 client concentration). A **KPI** measures performance (uptime, error rates, NPS, share within SLA). A **KCI** measures control effectiveness in design or performance (missed due diligence, lack of segregation of duties, errors missed in four-eyes checks, incomplete ID files). They overlap: a failing control is a KPI, a KRI and a KCI at once (overdue confirmations); "a KPI turning amber becomes a KRI". Thresholds express appetite; Tier-1 banks reclassify an issue unresolved after about three months as a KRI.
- **Loss data: regulatory criteria (S4; ORR-4 a).** BCBS lists eight criteria under the SA; main ones: **10 years** of data; minimum threshold **EUR 20,000**; mapping to Basel event types; dates of occurrence and recovery; independent review of comprehensiveness and accuracy. Firms often use lower thresholds (commonly 1,000 or 5,000); the choice must be justified, not cherry-picking. Standard fields (ID, place, event and cause types, failed controls, four dates, gross loss, recoveries, **net loss = gross loss minus recovery**, action plans, links); use drop-down labels, not free text, in key fields (NLP now mines old free text).
- **Loss data: what to record (ORR-4 a).** Regulators require only direct financial losses; good practice adds non-financial impacts, accidental gains and near misses. **Direct** losses come from the event (remediation, compensation, fines); **indirect** losses from its consequences (attrition, morale, scrutiny, premiums), often rated 1 to 4 and monetised only in mature firms. **Grouped losses** with a common cause are one event (a regulatory requirement). Four dates: occurrence, discovery, reporting, accounting; occurrence to discovery shows visibility, discovery to reporting shows diligence; material incidents reported within days. Losses above USD 100 million at G-SIBs most often took three to five years from reporting to settlement.
- **Boundary events and completeness (ORR-4 a).** Credit-related operational losses already in credit RWAs are *excluded*; those not in credit RWAs are included; market-related operational losses are *treated as operational risk*. Completeness is checked against the general ledger (misses indirect effects and forgone revenue), P1/P2 IT logs, litigation provisions, complaints and compensation, press, back-office logs. Operational data are firm-specific, heavy-tailed, scarce, can stay hidden for years, and are complex to interpret.

Identification and assessment beyond the specifics (ORR-3, ORR-4 d to g)

- **Top-down and bottom-up (ORR-3 a).** Top-down: senior leadership, key risks to strategic objectives, two to four times a year; tools are exposures and vulnerabilities, the risk wheel (chains of cause and effect), and horizon scanning (WEF, risk.net, ORX reports, PESTLE, emerging risk committees). Bottom-up: local staff and processes; internal and external loss analysis ("could this happen to us?"), near misses ("lessons for free", reported only in a no-blame culture), RCSA, process mapping. The two are consolidated, not reconciled. Interview the longest-serving staff and recent hires (the "amazement report").
- **Extreme risk identification (ORR-3 b).** Scenario analysis identifies low-probability, high-severity events; regulators require repeatable, documented, evidence-based results. Preparation pack: external and internal losses, near misses, RCSA, KRIs, audit issues, concentrations. Senior participants, ideally an external expert, to counter **myopia** and the excessive focus on **external** causes (most large losses had internal causes). Generation (silent voting), selection (20 to 30 down to about 15 for a mid-size firm; 50 is common at international banks), and comparison with ORX or ORIC lists only afterwards. Taxonomies and Basel Levels 1 to 3 (ORR-3 c, d): see A22.
- **Factor models (ORR-4 d, e). Fault tree analysis**: AND conditions (all must occur) and OR conditions; with independence the joint probability is the product, the theoretical *minimum*; dependence calls for conditional (Bayesian) probabilities; multiply by the units of exposure, not by impact. **FAIR**: asset, threat community, threat type, effect; distributions fed to Monte Carlo. **Swiss cheese** (Reason's cumulative act model): loss only when holes in all layers align, so control *independence* matters as much as each control's reliability. **Bowtie**: event in the centre, causes and preventive barriers on the left ("5-why"), impacts with detective and corrective barriers on the right ("5-then-what"). Capital modelling (ORR-4 f): see D19 and A17.
- **Operational resilience (ORR-4 g).** BCM works by process and RTO; resilience by **important business service** and **impact tolerance**, the maximum tolerable disruption (FCA). Steps: identify IBS; set impact tolerances; map end to end; design severe but plausible scenarios; learn where tolerances are exceeded;

communication plans; annual self-assessment signed off by the board. Business impact analysis feeds tolerances; single points of failure (key people, suppliers, systems) are resilience KRIs.

Mitigation (ORR-5 a, d, e, f, g)

- **Risk responses (ORR-5 a).** ISO 31000 four Ts: **tolerate** (low inherent or well-controlled residual risk), **treat** (controls and action plans), **transfer** (insurance, outsourcing: accountability not transferable, see D2), **terminate** (remove the exposure and its revenues; last resort). External operational risks cannot be prevented, only reduced; internal ones (people, processes, systems) are far more preventable through controls. Fees partly compensate the bank for the operational risk it bears for clients; operational risk cannot be fully eliminated, hence capital.
- **Human error and process design (ORR-5 d).** Prevention through design; human error is a symptom of poor processes. **Slips** (involuntary: fix the environment and pace), **rule-based mistakes** (flawed rules or incentives, as mis-selling: change the rule), **knowledge-based mistakes** (new situations: directive controls, training, escalation), **violations** (deliberate: supervisory controls and culture). Lean Six Sigma removes waste ("DOWNTIME") and variation through DMAIC; quality improvement uses PDSA.
- **New initiatives and M&A (ORR-5 e); limiting impact (ORR-5 f).** NPAP/NIRAP: a business case with objective, alternatives, expected benefits, commercial aspects and risks; ORM involvement scaled to the project's risk rating (workshops before kick-off, monitoring during, debrief at closure); acquired firms bring "skeletons in the closet" and integration risks. Contingency planning and BCM (senior commitment, BIA, RTO, regular testing); crisis qualities: speed, competence, transparency; a technical and a communication team under a senior manager; phases: crisis, emergency response, recovery (**RPO**: data lost since the last back-up; **RTO**: tolerable downtime), restoration.
- **Transfer and reputation (ORR-5 g).** Insurance suits risks that are predictable (priceable) and transferable; fair premiums roughly equal expected loss, so large firms self-insure small losses (captives) and buy cover for tail events (cyber, business interruption, class actions). Under the SA, insurance *recoveries* reduce gross to net losses; the insurer's capacity, willingness and payment delay leave residual and liquidity risk. Reputational damage cannot really be insured or outsourced: prevention plus the **three Rs** (regret, reason, remedy), rapid response, transparency and stakeholder differentiation; reputation capital is a buffer that supports resilience (Capital One 2019).

Reporting (ORR-6 a to d)

- **Audiences (ORR-6 a).** Board risk committee (effectiveness of the framework, appetite KRIs, large events, emerging risks); audit committee (third-line assurance on internal control); ExCo (execution of the ORM framework: events, action plans, culture and use-test indicators); central ORM function and operational risk committee (aggregated, cross-cutting view); business-line managers and risk champions (their own KRIs, plans and events, benchmarked). The "reporting cake": tier 1 all you need to know to *monitor*, tier 2 to *act*, tier 3 to *decide*; high risks, near misses and defective key controls escalate unaltered.
- **Content and challenges (ORR-6 b, c).** Seven components: top-10 risks and outlook; heatmap and register; risk appetite metrics; KRIs and issues; incidents and near misses (SA tracks losses above EUR 20,000 and EUR 100,000); action plans ("zero objective" for overdue plans and audit points: discipline indicators); emerging risks. More mature firms report more forward-looking content. Rules of valuable reporting: value exceeds cost, known use, influence on decisions. Asymmetry (ORX: losses above EUR 10 million are 0.4% of events and 66.6% of value), so isolate large events, "no averages" (use median and quartiles). Qualitative ratings are not additive: aggregate by conversion to money, by categorisation ("candle" bar charts), or worst-case (prudent but over-alarming). Combined assurance coordinates the three lines.
- **External reporting (ORR-6 d).** Pillar 3 (updated 2018): qualitative description of ORM; ten years of historical losses by accounting date in prescribed templates; the business indicator and its subcomponents, with narrative. "Absence of evidence is evidence of absence": minutes and reports prove governance. Notify regulators of events meeting materiality, reputation, resilience or stability criteria; be "open and honest". Annual reports should look transparent and competent. UK firms had until March 2025 to map and test important business services within impact tolerances.

Formulas

Fault tree, AND conditions, independent: $P(\text{scenario}) = P_1 \times P_2 \times \cdots \times P_n$ (theoretical minimum); e.g. three independent controls each failing 10%: $0.1^3 = 1/1{,}000$

Fault tree, OR conditions, independent (outside-source): $P = 1 - \prod_i (1 - P_i)$
Conditional (Bayes): $P(A \mid B) = P(A \cap B)/P(B)$
Exposure scaling: expected occurrences $\approx$ probability per unit $\times$ units of exposure; e.g. 10^{-6} per employee $\times$ 100,000 employees $= 0.1$, at least a 10% likelihood
External frequency by similarity: one bank in 100 over five years: $1/(100 \times 5) = 1/500$ per bank-year
Likelihood scale: "1 in 10 years" $=$ 10% probability within the next year
Net loss: gross loss $-$ recoveries (insurance and other)

Likely distractors

- "Daily data back-ups and continuity plans are preventive controls, because they prevent the loss of data and service."

  True version: they are *corrective* controls: they lessen the impact when an event occurs and do not change its likelihood. *Sibling swap.*

- "A detective control can never act as a preventive control, since by definition it operates after an error has been recorded."

  True version: a detective control becomes preventive when it detects a cause; a file reconciliation run before a transaction is executed prevents the error. *Absolute claim.*

- "Replacing manual checks with an automated control lowers both the likelihood and the potential impact of control failures."

  True version: automation converts high-likelihood, low-impact manual errors into low-likelihood, high-impact systemic breakdowns, and adds technology and model risk. *Polarity flip.*

- "RCSA action plans for risks above appetite should be owned by the second line, since the risk function facilitated the workshop."

  True version: action plans are owned by the first line (business-line managers), who sign off the RCSA results; the second or third line tracks them. *Role misassignment.*

- "A 1-in-10-year event is expected about once over a ten-year horizon, so a ten-year horizon is the appropriate time frame for the RCSA."

  True version: it means a 10% chance of occurring within the next year; RCSA horizons are one year at most. *Scope/condition error.*

- "Operational losses connected to market risk, such as a booking error on a trading position, are excluded from the operational loss dataset and left in market risk."

  True version: market-related operational losses are treated as operational risk for capital; it is credit-related operational losses already captured in credit RWAs that are excluded. *Sibling swap (market versus credit boundary).*

D23 Sovereign Default Risk and CDS

Readings: CR-8 a, b, c, d, e, f, g · Country Risk: Determinants, Measures, and Implications (Damodaran, Country Risk 2022 Edition, Sovereign Default Risk: history, consequences, 8.1 Measuring Sovereign Default Risk, Market Interest Rates, Credit Default Swaps)

MAP TIER D · #56 · 1/16 SOURCES VARDEEZ

Specifics reported

Map #56: "The sheet's specific guidance: 'same as Level 1, but the last LO is very important on CDS.' Sovereign CDS as a measure of country default risk is therefore the one item to actually secure."
Shapes covered here:
S1 sovereign CDS spreads as a measure of country default risk, compared with ratings and bond default spreads (CR-8 g) (Q1).

Revision notes

- **The specific: sovereign CDS as a default measure** (CR-8 g; S1). Assuming away counterparty and liquidity risk, CDS prices "should provide us with updated measures of default risk in the reference entity", adjusting to current information where ratings are updated infrequently (Greece 2006 to 2010: ratings stagnant until late 2009 while CDS and default spreads moved monthly).
- **What the evidence says about CDS**: (1) changes in CDS spreads *lead* changes in sovereign bond yields and in sovereign ratings; (2) evidence is emerging that CDS spread changes are better predictors of sovereign default than *ratings*; (3) sovereign CDS spreads rise with economic policy uncertainty, together with currency depreciation; (4) CDS prices cluster (emerging market cluster 1: internal correlation 0.516 versus 0.210 external).
- **Limits of CDS**: counterparty and liquidity risk move prices for reasons unrelated to default (much of the late-2008 surge followed Lehman's failure), and the narrow market makes single names prone to liquidity squeezes. Damodaran's verdict: CDS changes provide information, "albeit noisy", but "there is little to indicate that it is superior to market default spreads (obtained from government bonds)".
- **How a sovereign CDS works** (CR-8 g): the buyer pays the spread on the notional; on a credit event (failure to pay, restructuring, bankruptcy) the seller pays par for the bond (physical) or par minus market price (cash). Colombia example: USD 10 million at 250 bps costs USD 250,000 a year. Two gaps in the protection: no credit event means no payment even if the bond price collapses, and the guarantee is only as good as the seller. Sovereign CDS are a small slice of a market dominated by corporates, then banks; few players. See also A15 and D14 for CDS mechanics.
- **Sovereign default spread** (CR-8 b, g): yield on a government's foreign-currency (e.g. US dollar) bond minus the riskless rate in that currency (Brazil, July 2022: $6.19\% - 3.02\% = 3.17\%$). It needs a default-free security in the same currency; local currency bonds of different governments cannot be compared because rate differences reflect expected inflation. Advantages over ratings: more granular (Colombia and Indonesia both Baa2, spreads 2.56% and 1.36%) and dynamic (Brazil 3.18% and Venezuela 3.09% in December 2005, 1.32% and 10.26% by December 2009). Costs: more volatile, moved by liquidity and investor demand.
- **Spreads versus ratings**: default spreads correlate with ratings and with actual default; the bond market *leads* the agencies (spreads widen before downgrades and narrow before upgrades); a rating change is still an informational event with a price impact, so ratings are not useless.
- **Foreign versus local currency defaults** (CR-8 c): foreign currency default is understandable (the borrower cannot print the currency). Local currency defaults happen because of the gold standard (pre-1971 limits on printing), a shared currency (Greece 2015 could not print euros), and a deliberate trade-off against debasement and inflation, especially where companies have foreign currency debt funding local assets. Russia's USD 39 billion ruble default and Brazil's USD 62 billion in 1990 are the large cases; countries increasingly default on both at once.
- **History**: defaults bunch in eight periods since 1820; Latin America is the epicentre (58 of 81 government defaults in 1820 to 1919; Honduras in default 79% of 115 years); governments have defaulted more on bank debt than on bonds; recent shift to local currency, non-bank and domestic-law defaults, with faster

restructurings.

- **Consequences of default** (CR-8 d): reputation loss, capital market turmoil, falling real output, political instability. Evidence: real GDP down 0.5% to 2%, mostly in year one; ratings one to two notches lower and borrowing costs 0.5% to 1% higher, fading over time; bilateral trade down 8% for up to 15 years; banking crisis probability 14% (11 points higher); after sharp devaluations, a 45% higher chance of replacing the leader and 64% for the finance chief. Total repudiation is rare; most defaults end in exchange or restructuring.
- **Drivers of sovereign default risk** (CR-8 b, e): degree of indebtedness (debt to GDP: Venezuela above 300%, but Japan, the US and France are high too, so it is incomplete), pension and social service commitments (ageing raises risk), revenues and tax base, stability of revenues (diversified economies and sales or VAT systems are more stable than income tax systems), political risk (autocracies, central bank independence), implicit backing (EU membership).
- **Sovereign ratings** (CR-8 e): cover creditworthiness to *private* creditors, not official lenders; S&P rates probability of default, Moody's probability and severity (recovery). Default = failure to pay on the due date or a restructuring. S&P's factors: political risk, economic structure, growth prospects, fiscal flexibility, government debt burden, offshore and contingent liabilities, monetary flexibility, external liquidity, external debt burden. A committee of five to ten analysts votes. Local and foreign currency ratings: *notch-up* (foreign currency anchor) or *notch-down* (local currency anchor); the gap depends on monetary policy independence. Investment grade sovereigns default far less often than speculative grade.
- **Shortcomings of ratings** (CR-8 f): upward bias, herd behaviour, too little too late, vicious cycle (overreacting in a crisis), and ratings failures (Bhatia: S&P cut Korea from AA- to B+, ten notches, in about two months in 1997). Causes: information from the governments themselves, thin analyst resources (four to five sovereigns each in 2003), revenue bias via sub-sovereign ratings, other business lines.
- **Sources of country risk** (CR-8 a): the GARP excerpt opens with sovereign default rather than the general taxonomy. Damodaran's broader framework adds risk from a country's stage in the economic life cycle, political risk (continuity, corruption, violence, expropriation), legal risk (property rights and enforcement) and economic structure (concentration in one commodity or sector) (outside-source).

Formulas

Sovereign default spread. Spread $= y_{\text{sovereign, USD}} - y_{\text{US Treasury}}$ (same currency, same maturity; Brazil: $6.19\% - 3.02\% = 3.17\%$)
CDS premium. Annual payment = notional $\times$ CDS spread (USD 10 million $\times$ 2.5% = USD 250,000)

Exam traps

- **Lead-lag polarity:** markets lead ratings: CDS changes lead bond yields and ratings, and bond spreads move before rating changes, not after.
- **Absolute claim:** CDS spreads are timely but *noisy*; there is "little to indicate" they beat bond default spreads. "Superior to all other measures" is false; "better predictor than ratings" is the supported claim.
- **Scope error:** market default spreads need a default-free bond in the same currency; comparing local currency yields across countries measures inflation as well as default.
- **Ratings are not useless:** a rating change still has a price impact.
- **Local versus foreign currency:** local currency ratings are usually at least as high as foreign currency ratings, but not always (India 2010: Ba2 local, Baa3 foreign).

Question D23.1

CR-8 g, b · S1 · [Map #56]

A sovereign risk analyst at a Singapore asset manager is presenting to the investment committee on how to monitor default risk in a portfolio of emerging market government bonds. The analyst compares agency ratings, default spreads on US dollar sovereign bonds, and sovereign CDS spreads. Which of the following statements is correct for the analyst to make?

A. Changes in sovereign CDS spreads tend to lead changes in bond yields and ratings, but a spike may reflect counterparty and liquidity risk rather than higher default risk.

B. Sovereign CDS spreads update continuously, and the evidence shows them to be clearly superior to default spreads on government bonds as a measure of default risk.

C. Default spreads on sovereign bonds tend to widen only after a rating downgrade, so the bond market confirms agency assessments rather than anticipating them.

D. Default spreads can be measured from local currency government bonds by comparing their yields across countries, without needing a default-free bond in each currency.

Solution **Answer: A**

Step 1 – What the evidence says about sovereign CDS

Damodaran: "changes in CDS spreads lead changes in the sovereign bond yields and in sovereign ratings". But "the exposure to counterparty and liquidity risk, endemic to the CDS market, can cause changes in CDS prices that have little to do with default risk": much of the late-2008 surge followed Lehman's failure, and the narrowness of the market adds liquidity squeezes.

⇒ *A is correct: CDS spreads are the most timely signal, and their known noise has a named source.*

Interpretation

The committee should use all three measures together: ratings for stability and their information events, bond default spreads for granular and dynamic pricing in a currency with a default-free benchmark, and CDS spreads as the earliest (but noisy) warning, discounting moves that coincide with dealer stress or thin trading.

Why the other options are wrong

B is incorrect. Absolute claim: CDS changes "provide information, albeit noisy", but "there is little to indicate that it is superior to market default spreads (obtained from government bonds)". The supported claim is only that CDS changes may predict default better than ratings.

C is incorrect. Polarity flip on the lead-lag: "the sovereign bond market leads ratings agencies, with default spreads usually climbing ahead of a rating downgrade and dropping before an upgrade".

D is incorrect. Scope/condition error: a market default spread needs "a default free security in the currency in which the bonds are issued". Local currency yields across countries "cannot be compared to each other, since the differences in rates can be due to differences in expected inflation".

LO: CR-8 g: Compare the use of credit ratings, market-based credit default spreads, and CDS spreads in predicting default. CR-8 b: Evaluate the methods for measuring country risk and discuss the limitations of using those methods. · Reference: Damodaran, Country Risk 2022, Sovereign Default Risk: Market Interest Rates (The Sovereign Default Spread; The Spread as a Predictor of Default) and Credit Default Swaps (CDS and Default Risk)

D24 Central Clearing and the CCP Loss Waterfall

Readings: CR-18 a, b, c, d, e, f, g, h, i, j · Central Clearing (Gregory, The xVA Challenge 4e, Ch.8: 18.1 Evolution of Central Clearing; 18.2 Mechanics of Central Clearing; 18.3 CCP Risk Management, The Loss Waterfall; 18.4 Initial Margin and Default Funds; 18.5 Impact of Central Clearing)

MAP TIER D · #59 · 1/16 SOURCES VARDEEZ

Specifics reported

Map #59: "The sheet's carve-out is explicit: 'know the loss waterfall.' Initial margin, default fund contribution of the defaulter, CCP equity tranche, mutualised default fund, assessments, in that order. Sequence errors are the distractor."
Shapes covered here:
S1 the order of the loss waterfall, with sequence errors as distractors (Q1).

Revision notes

- **The specific: the loss waterfall** (CR-18 f, g; S1). Gregory's typical waterfall (Figure 18.9), in order: (1) initial margin of the defaulter; (2) default fund contribution of the defaulter; (3) CCP *skin-in-the-game* (its own equity); (4) default fund of the non-defaulting members; (5) rights of assessment and/or other loss allocation methods; (6) remaining CCP capital; (7) liquidity support, or the CCP fails. Steps 1 and 2 are *defaulter pays*; from step 4 onward it is *survivors pay*.
- **Skin in the game**: placed after the defaulter's resources and before the mutualised fund to give the CCP "some incentive to avoid losses beyond the defaulter-pays resources". It is not in all rule books: in 2014 the Korea Exchange mutualised a USD 4.3 million loss to members' default fund without itself losing anything, because its rules put its own loss after the roughly USD 200 million default fund.
- **Beyond the default fund** (CR-18 g): *rights of assessment* are an unfunded obligation to top up a depleted fund, typically invoked if a significant fraction (e.g. 25%) is used, and generally *capped* because unlimited calls create moral hazard and hit members at the worst time. *Variation margin gains haircutting* (VMGH), probably the most common alternative, cuts the gains owed to members; it mimics bilateral economics and is guaranteed to work if the auction clears at mid-market. *Tear-up* terminates opposite trades to restore a matched book; *forced allocation* makes members accept positions at CCP prices (cannot be passed to clients). Tear-up and forced allocation are potentially infinite.
- **Initial margin of survivors is not in the waterfall**: EMIR Article 45(4) bars a CCP from using the margins of non-defaulting members; initial margin haircutting has been suggested but not used. A member can still lose through the default fund "even if it never traded with the defaulted counterparty, has no net position with the CCP", or is positioned like the defaulter.
- **What a CCP is** (CR-18 a): a financial market infrastructure interposed between buyers and sellers. Historical path: direct clearing (payment of difference), clearing rings, then complete clearing (from 1891). CBOT resisted until 1925 and the LME until 1986, because clearing homogenises counterparty risk. Clearable OTC products need standardisation, low complexity, liquidity, no wrong-way risk and enough volume; CDSs are arguably the hardest because of WWR.
- **Novation** (CR-18 b): the original contract is replaced, the CCP becomes "buyer to the seller and seller to the buyer", and the original parties have no counterparty risk to one another. The CCP runs a matched book with no net market risk but has *conditional* market risk if a member defaults.
- **Netting, multilateral offset, compression** (CR-18 c): in Gregory's three-party example, total liabilities of 180 shrink and C's liability nets to 60, removing the domino path from C to A. Compression needs actual cash flows or trades to be eliminated; a CCP can offset *risk* even between near-identical trades. Compression of cleared trades still cuts margin multipliers, leverage-ratio capital, operations and auction size.
- **Margin and default funds** (CR-18 d): variation margin daily or intraday, usually cash only; initial margin covers close-out losses at 99% or more over the MPoR (five days for OTC CCPs versus a minimum of 10 days bilaterally; one or two days for exchange-traded), may be securities. The default fund is sized by stress tests to Cover 1 (largest member and affiliates), or Cover 2 for systemically important or complex CCPs (SwapClear: largest two), and allocated simply, e.g. pro rata to initial margin, updated perhaps monthly.

- **CCP risk management** (CR-18 e): membership requirements (creditworthiness, liquidity, operational ability), margin as the primary defence ("unlike a bank, which relies largely on capital"), and default management: *macro-hedging* then *auctions* by sub-portfolio, run by a default management group with seconded member traders and rehearsed in fire drills. Lehman 2008 at SwapClear: 90% of risk hedged in one week, 66,000 trades auctioned within three weeks, about a third of the initial margin used.
- **Bilateral versus central** (CR-18 h, Table 18.2): counterparty is the original party versus the CCP; all products versus standard, vanilla, liquid ones; bilateral netting versus multilateral netting; bespoke margin versus full collateralisation including initial margin; bilateral close-out versus a coordinated default process; loss absorbency cost mainly capital versus mainly initial margin and default fund.
- **Initial margin versus default fund** (CR-18 i, Table 18.3): higher initial margin and a lower default fund mean higher cost, clients paying for their own risk, easier portability and *lower* moral hazard; a larger mutualised default fund is cheaper and gives far greater loss coverage per unit contributed, but raises moral hazard and makes portability harder. Clients post margin but do not contribute to the default fund.
- **Client clearing**: *omnibus segregation* protects a client from the clearing member's house losses but exposes it to fellow-customer risk; *individual segregation* protects against both but costs more.
- **Advantages and disadvantages** (CR-18 j): transparency, multilateral offset, loss mutualisation, legal and operational efficiency, liquidity, orderly default management; against these, moral hazard, adverse selection, bifurcation of cleared and non-cleared netting sets, procyclicality, and the CCP as a single point of failure. Clearing will not kill CVA: about 85% of a bank's CVA comes from exempt corporates and governments.

Formulas

Loss reaching the survivors' default fund.

$$L_{\text{DF, survivors}} = \min\left(\max\left(L - \text{IM}_{\text{def}} - \text{DF}_{\text{def}} - \text{SITG},\ 0\right),\ \text{DF}_{\text{survivors}}\right)$$

Loss passed to assessments and beyond. $\max\left(L - \text{IM}_{\text{def}} - \text{DF}_{\text{def}} - \text{SITG} - \text{DF}_{\text{survivors}},\ 0\right)$, never met from survivors' initial margin

Exam traps

- **Sequence swap:** skin-in-the-game sits *after* the defaulter's initial margin and default fund contribution and *before* the mutualised fund. Moving it after the mutualised fund overstates the survivors' loss.
- **Defaulter's own default fund first:** the defaulter's default fund contribution is used before anyone else's, not pooled pro rata with the survivors'.
- **Role misassignment:** non-defaulting members' *initial margin* is never in the waterfall (EMIR 45(4)); their *default fund* contributions are.
- **Scope error:** mutualised losses do not depend on having traded with the defaulter.
- **Absolute claim:** rights of assessment are generally *capped*; only tear-up and forced allocation are potentially unlimited, so "the CCP can never fail" and "a member's loss is limited to its default fund contribution" are both false in general.

Question D24.1

CR-18 f, g · S1 · [Map #59]

A risk consultant is reviewing the default rules of a European CCP on behalf of a clearing member. In a fire drill, one clearing member defaults, and the macro-hedging and auction of its portfolio cost EUR 900 million. The resources in the CCP's rule book are:

- Initial margin posted by the defaulter: EUR 500 million
- Default fund contribution of the defaulter: EUR 80 million
- CCP skin-in-the-game: EUR 50 million
- Default fund contributions of the surviving members: EUR 920 million
- Initial margin posted by the surviving members: EUR 6,400 million

Which of the following statements is correct for the consultant to include in the report?

A. The surviving members' default fund contributions absorb EUR 320 million, because the CCP's skin-in-the-game ranks after the mutualised default fund.

B. The surviving members' default fund contributions absorb nothing, because the surviving members' initial margin is applied before the mutualised default fund.

C. The surviving members' default fund contributions absorb EUR 270 million, shared only among the members that held trades with the defaulter.

D. The surviving members' default fund contributions absorb EUR 270 million, and none of the surviving members' initial margin is used.

Solution **Answer: D**

Step 1 – Defaulter pays: initial margin, then its own default fund contribution

$$900 - 500 = 400, \qquad 400 - 80 = 320$$

$\Rightarrow$ *After the defaulter's own resources, EUR 320 million of loss remains.*

Step 2 – CCP skin-in-the-game

$$320 - 50 = 270$$

$\Rightarrow$ *The CCP's equity takes the next EUR 50 million, which gives it an incentive to manage defaults well.*

Step 3 – Survivors pay: mutualised default fund

$$270 \leq 920 \;\Rightarrow\; \text{survivors' default fund absorbs EUR 270 million; nothing reaches assessments}$$

$\Rightarrow$ *About 29% of the survivors' contributions are consumed, so the CCP may invoke a capped recapitalisation, but no further layer is needed to absorb this default.*

Interpretation

The order is defaulter's initial margin, defaulter's default fund contribution, CCP skin-in-the-game, survivors' default fund, then assessments and other loss allocation. The EUR 6,400 million of surviving members' initial margin is a red herring: Gregory notes that none of the loss allocation methods lets a CCP use the initial margin of non-defaulting members (EMIR Article 45(4)), which is why initial margin is "in theory" risk free to the poster. The clearing member's real exposure is its share of the EUR 270 million plus any assessment or other allocation.

Why the other options are wrong

A is incorrect. Sequence swap: moved skin-in-the-game below the mutualised fund, so the survivors absorb $900 - 500 - 80 = 320$. In Figure 18.9 skin-in-the-game ranks immediately after the defaulter-pays resources.

B is incorrect. Role misassignment: survivors' initial margin is not in the waterfall; only the defaulter's initial margin is. Survivors pay through their default fund contributions.

C is incorrect. Scope error: correct amount, but mutualisation does not depend on trading with the defaulter. A member "can suffer default losses even if it never traded with the defaulted counterparty", has no net position with the CCP, or has a position in the same direction.

LO: CR-18 f: Provide examples of a loss waterfall. CR-18 g: Explain the different methods of absorbing losses and managing the default of one or more members of a CCP. · Reference: Gregory, The xVA Challenge 4e, Ch.8, 18.3 The Loss Waterfall, Figure 18.9; 18.2 Margin and Default Funds

D25 Contingent Convertible Bonds

Readings: ORR-22 g, with d, e, h for context · Solvency, Liquidity, and Other Regulation After the Global Financial Crisis (Carey, GARP Risk Institute 2019, 22.3 and 22.4)

MAP TIER D · #70 · 1/16 SOURCES TELEGRAM

Specifics reported

Map #70: "Reported: 'Coco bonds.' Contingent convertibles: trigger mechanics, the equity-conversion vs. principal-write-down distinction, and the AT1-below-equity inversion controversy. Low evidence weight; low cost to cover."
Shapes covered here:
S1 trigger mechanics: Core Tier 1 to RWA threshold or regulator's nonviability declaration, and the 5.125% line between Additional Tier 1 and Tier 2 (Q1)
S2 equity conversion versus principal write-down: both increase equity (Q1)
S3 AT1 ranking and the inversion controversy: CoCo holders bear losses not so different from equity holders (Q1).

Revision notes

- **Traditional convertibles** (ORR-22 g): issued by non-financial firms that wished to avoid the dilution of issuing equity before performance improved; they convert *at the option of the holder* when the share price *exceeds* thresholds in the indenture.
- **CoCos as the mirror image** (ORR-22 g): they "cause a bank's equity to increase when distress occurs, as reflected by triggers written into the indenture, and not at the option of the holder". Good times do not trigger them; distress does.
- **Conversion versus write-down** (S2): "With CoCos, equity increases either because the bond converts to equity or because its value is written down." In a write-down no new shares are issued, but the liability shrinks, so equity still rises.
- **Triggers** (S1): "a common trigger is when the ratio of Core Tier 1 Capital to RWA falls below a threshold" (a mechanical, accounting trigger), "or when a bank's primary regulator declares it to be nonviable" (a discretionary trigger owned by the regulator, not the bank or the holders).
- **Capital treatment** (S1): CoCos "may be included in Additional Tier 1 Capital if the threshold is 5.125 percent or higher, and Tier 2 capital otherwise". The AT1 bar sits above the 4.5% Core Tier 1 minimum, so the bond absorbs losses while the bank is still a going concern.
- **The cost puzzle** (ORR-22 g): holders "receive little or none of the high returns received by equity holders when a bank does well, but holders bear losses not so different from those of equity holders when a bank fails", so CoCos "should be expensive for a bank to issue". Their advantage is accounting: they do not appear in the equity account until converted, so the bank can report a higher return on equity.
- **Basel III tiers** (ORR-22 d): Tier 1 Equity (Core Tier 1) is common equity, retained earnings and limited minority interest, less goodwill, intangibles, deferred tax assets and any IRB reserve shortfall. Additional Tier 1 includes unsecured, unguaranteed, non-cumulative perpetual preferred equity subordinated to depositors and subordinated debt, callable only after five years or more, plus "debt with appropriate triggers that cause conversion to equity or write-downs". Tier 2 (subordinated debt with five years or more original maturity; general loan loss reserves capped at 1.25% of standardized RWA or 0.6% of IRB RWA) absorbs losses *after failure*. Minimums: Core Tier 1 4.5%, Tier 1 6%, total capital 8% of RWA. Tier 3 was eliminated.
- **Buffers** (ORR-22 e): capital conservation buffer 2.5%; G-SIB add-on of 1, 1.5, 2, 2.5 or 3.5%; countercyclical buffer 0 to 2.5% at national discretion. Breaching a buffer is less severe than breaching a minimum; the formal restrictions are on dividends and bonuses, plus a plan to restore ratios.
- **CoCos versus bail-in** (ORR-22 h, 22.4): to recapitalize a failing G-SIB (FSB 2014: 12 key attributes; sufficient TLAC), convertible bonds can become equity, or certain wholesale debt can be bailed in (written down or converted). Conversion terms are "written into the indentures" and "often require conversion when a bank appears to be solvent"; bail-ins "are governed by national law and details are generally chosen by authorities after they have seized control of a bank".
- **Living wills and super-equivalence** (ORR-22 h): G-SIBs prepare resolution plans on funding, recapitalization and continuing operations. Switzerland, with two huge G-SIBs it could not recapitalize in the crisis,

imposes capital requirements more onerous than Basel 3; the UK ring-fences retail from wholesale operations.

- **AT1 ranking and the inversion controversy** (S3): in the usual creditor hierarchy AT1 ranks above common equity. In March 2023 the Swiss regulator ordered a write-down to zero of about CHF 16 billion of Credit Suisse AT1 bonds under their viability terms, while shareholders received UBS shares in the rescue takeover (outside-source). The in-syllabus anchor is Carey's line that holders bear losses "not so different from those of equity holders".
- **Neighbouring ratios**: leverage ratio (Core Tier 1 to Leverage Exposure $\geq 3\%$), LCR and NSFR. See also A32; the CAR built from RWA is in D3.

Thresholds

CoCo classification (ORR-22 g). Trigger threshold (Core Tier 1 / RWA) $\geq 5.125\% \Rightarrow$ Additional Tier 1; below $5.125\% \Rightarrow$ Tier 2.

Basel III minimums (ORR-22 d). $\dfrac{\text{Core Tier 1}}{\text{RWA}} \geq 4.5\%$, $\dfrac{\text{Core Tier 1} + \text{AT1}}{\text{RWA}} \geq 6\%$, $\dfrac{\text{Tier 1} + \text{Tier 2}}{\text{RWA}} \geq 8\%$

Buffers (ORR-22 e). CCB 2.5%; G-SIB 1 to 3.5%; CCyB 0 to 2.5%. **Leverage (ORR-22 f).** Core Tier 1 / Leverage Exposure $\geq 3\%$.

Exam traps

- **Role misassignment:** conversion is triggered by the indenture (capital ratio) or by the primary regulator (non-viability), never at the holder's option. The holder's option belongs to the traditional non-financial convertible.
- **Polarity:** traditional convertibles convert when the share price *rises*; CoCos convert or write down in *distress*.
- **Sibling swap, write-down versus conversion:** both mechanisms increase equity; a write-down is not "equity-neutral".
- **Scope error on the 5.125% line:** a Core Tier 1 trigger alone does not make a CoCo AT1; the threshold must be 5.125% or higher, otherwise Tier 2.
- **Absolute claim on ranking:** CoCo holders are not shielded until shareholders are wiped out; they bear equity-like losses, and the 2023 Credit Suisse write-down hit AT1 while shareholders kept some value (outside-source).
- **Cost direction:** economically CoCos should be *expensive*; the issuer's advantage is the higher reported ROE, not a cheaper cost of capital.

Question D25.1

ORR-22 g; ORR-22 d · S1, S2, S3 · [Map #70]

A capital management analyst at a Japanese bank holding company is preparing a briefing for the asset-liability committee on a proposed issue of perpetual contingent convertible bonds. The draft term sheet states:

- Loss absorption: permanent write-down of principal; the bonds never convert into shares
- Trigger: the ratio of Core Tier 1 capital to RWA falls below 7%, or the bank's primary regulator declares the bank nonviable
- Coupons: non-cumulative

Which of the following statements is correct for the analyst to include in the briefing?

A. Because the bonds are written down rather than converted into shares, a trigger event would leave the bank's equity unchanged and would only reduce its future interest expense.

B. Because the trigger is defined on Core Tier 1 capital, the bonds would qualify as Additional Tier 1 capital whatever threshold were chosen, including the 4.5% Core Tier 1 minimum.

C. A trigger event would increase the bank's equity through the write-down, and with a 7% threshold the bonds can be included in Additional Tier 1 capital rather than Tier 2 capital.

D. Because the bonds are debt until a trigger event, their holders cannot bear any loss until the bank's common shareholders have been completely wiped out.

Solution **Answer: C**

Why C is correct

Carey: "With CoCos, equity increases either because the bond converts to equity or because its value is written down." Writing down the principal removes a liability with no matching asset loss, so equity rises. On classification, "CoCos may be included in Additional Tier 1 Capital if the threshold is 5.125 percent or higher, and Tier 2 capital otherwise"; 7% clears the bar, so the issue counts toward the 6% Tier 1 minimum.

⇒ *The write-down design absorbs losses while the bank is still a going concern, which is exactly what the higher AT1 threshold is meant to ensure.*

Why the other options are wrong

A is incorrect. Sibling swap. It treats the write-down as if only conversion created equity. Both mechanisms increase equity; the write-down simply does it without issuing shares.

B is incorrect. Scope/condition error. It drops the threshold condition: a Core Tier 1 trigger is necessary but not sufficient. A trigger at 4.5%, below 5.125%, would leave the bonds in Tier 2.

D is incorrect. Absolute claim. Carey says holders "bear losses not so different from those of equity holders when a bank fails"; a principal write-down can wipe out the holders while shareholders retain value, as with the March 2023 Credit Suisse AT1 write-down (outside-source).

LO: ORR-22 g: Describe the mechanics of contingent convertible bonds (CoCos) and explain the motivations for banks to issue them. ORR-22 d: Define in the context of Basel III and calculate where appropriate: Tier 1 capital and its components; Tier 2 capital and its components; required Tier 1 equity capital, total Tier 1 capital, and total capital. · Reference: Carey, "Solvency, Liquidity, and Other Regulation After the Global Financial Crisis", GARP Risk Institute (2019), 22.3 The Definition of Capital and 22.4 CoCos

D26 Contingency Funding Planning

Readings: LTR-11 a-c · Contingency Funding Planning (Venkat and Baird, Liquidity Risk Management: A Practitioner's Perspective, Ch.7)

MAP TIER D · #62 · 1/16 SOURCES VARDEEZ NOTES ONLY

Specifics reported

Map #62: "Governance of the CFP, trigger design and escalation, the menu of contingent funding sources and their reliability under stress, and testing requirements. Treat as thin-evidence-but-high-rating."
Shapes covered here: S1 governance of the CFP: who designs, invokes, oversees and approves (notes: governance and roles); S2 trigger design and escalation: EWIs, liquidity health measures, escalation levels (notes: monitoring and escalation); S3 the menu of contingent actions and their reliability under stress, including signalling effects (notes: contingent actions, market signals); S4 testing requirements and operational readiness (notes: testing).

Revision notes

- **What a CFP is** (LTR-11 a): a CFP "serves as a logical connection to its companion, the liquidity stress testing framework, by linking the stress test results and other related information as inputs to the CFP governance, menu of contingent liquidity actions, and decision framework." Low-impact, high-probability events are handled in business-as-usual (BAU) funding and liquidity management; the CFP addresses the *high-impact, low-probability* end.
- **Relationship with stress testing** (LTR-11 a): CFP scenarios should be aligned to the liquidity stress testing scenarios and to recovery and resolution plans. The stress tests cover systemic and idiosyncratic risks, market (asset) and funding liquidity, over short-term and prolonged periods; the CFP "in turn should provide a tactical mechanism for escalating a developing crisis to management's attention and ensuring actionable responses are available." The CFP may add its own scenarios outside the limit structure: the Fedwire intraday debit cap is exceeded, specific counterparties fail, or Federal Home Loan Bank funding becomes unavailable. See also A26 (LTR-9 stress test design).
- **Supervisory backdrop** (LTR-11 b): the 2008 EU report on 84 institutions; the Fed's Enhanced Prudential Standards require a CFP for US banking organisations with USD 50 billion or more of assets, with policies on governance, roles, liquidity measures, triggers, contingent actions and communication, tailored to the institution and regularly tested. CLAR and daily liquidity reporting link to its measures; the CFP bridges BAU liquidity management and recovery and resolution planning. Smaller institutions often embed their CFP in business continuity plans; larger ones formalise it, often with several CFPs by subsidiary or legal entity.
- **Five design considerations** (LTR-11 b): (I) aligned to business and risk profiles, with the risk appetite statement tied in through quantifiable EWIs, limits and escalation levels, and refreshed as the business changes (leaders build it into strategic planning); (II) integrated with broader risk management (ERM, capital management, business continuity and crisis management) and explicitly linked to the liquidity stress test through the limit structure and escalation levels; (III) operational and actionable but a flexible playbook, balancing recommended actions with management discretion; (IV) inclusive of stakeholder groups (ALCO, risk and capital committee, investment committee, business units, finance, treasury, risk, operations and technology); (V) supported by a communication plan.
- **Five key components** (LTR-11 c): (1) governance and oversight; (2) scenarios and liquidity gap analysis; (3) contingent actions; (4) monitoring and escalation; (5) data and reporting.
- **Roles** (LTR-11 c, S1): *corporate treasury* monitors the funding and liquidity profile in BAU; "the treasurer, in consultation with the CFO and others, may invoke the CFP and convene the liquidity crisis team (LCT)". The *LCT* (C-level executives, heads of segments, geographies and legal entities) is the central point of contact, monitors continuously, recommends CFP actions, and "is responsible for designing the CFP and submitting it to the senior management group for review and approval". The *management committee* oversees the LCT during a crisis and consults the board. The *board* is actively engaged as advisor and counsel, especially on strategic actions such as large asset or subsidiary sales.
- **Communication** (LTR-11 c): messaging is centrally managed ("everyone should be working off the same page"), but each relationship stays with its existing owner: business units with clients and counterparties;

corporate treasury with regulators, rating agencies and clearing banks; investor relations with investors, analysts and media; legal and compliance with regulators and supervisors. Clear, timely communication sustains confidence; loss of confidence "can quickly spiral downward when rumors start to take over".

- **Policies and procedures** (LTR-11 c): document the CFP and align it with risk management, business continuity and recovery planning. Illustrative outline: purpose, related policies, governance, roles, review and approval, periodic review, stress testing and scenarios, methodology (scenario design, inputs, key assumptions, data control, model validation), monitoring and escalation (liquidity gap analysis, contingent actions), reporting frequency and briefing decks.
- **Testing** (LTR-11 c, S4): periodically test targeted elements for relevance and execution effectiveness. Business divestitures, large asset sales and Federal Reserve borrowing "may be impractical or unavailable for testing"; debt issuance, brokered deposit issuance and limited securities sales from the investment portfolio may be appropriate to test. Also test governance, escalation, communication, coordination and reporting; frequent simulation exercises improve response times and expose gaps a purely theoretical design would miss.
- **Contingent actions menu** (LTR-11 c, S3): chosen in light of the size, timing and pattern of the shortfall, the expected relief and the ability to execute. Examples: maintain committed lines of credit; tighten underwriting and dial back lending; raise deposit pricing; roll off (not reinvest) maturing securities; shift short-term to longer-term funding; brokered CDs or direct-to-consumer deposits; securitise retail assets; pledge assets at the discount window; sell liquid assets; draw securitisation conduits; issue subordinated debt; slow asset growth; at-call loans; sell consumer loans or businesses; raise equity; cut capital distributions and discretionary spending.
- **Reliability under stress** (LTR-11 c, S3): availability depends on whether the stress is systemic or idiosyncratic and how severe. Impediments: securitisation markets shut; repo access restricted by solvency concerns, downgrades or reputation; downgrade-driven collateral and margin calls; predatory margin practices; higher custodian cash deposits; higher funding costs; deposit runoff; collapse of interbank and wholesale funding; counterparties unwilling to roll. The CFP should document mitigants for each.
- **Signalling risk** (LTR-11 c, S3): early defensive steps (curtailing business, tightening credit, limiting risky counterparties) may "send inadvertent signals" that reduce the availability of later actions. Hedge funds may move prime brokerage elsewhere; debt holders may force buybacks (agreeing drains liquidity, refusing questions viability). Accessing a sovereign or emergency facility may make lenders "immediately withdraw existing credit"; selling businesses or suspending reverse repo can read as distress, so such actions belong to later stages.
- **Early warning indicators** (LTR-11 c, S2): market and business measures (external and internal) plus liquidity health measures (internal), chosen together so the institution can tell idiosyncratic from systemic stress. Macro: repo spreads, haircut trends, CDS spreads. Industry: sector profitability, rating actions, capital adequacy, peer analysis. Institution-specific: stock price drop, rapid asset growth funded by volatile liabilities, negative publicity, falling asset quality or earnings, rating downgrades, wider debt and CDS spreads, higher collateral demands, counterparties refusing unsecured or longer-term deals, early withdrawal requests, debt buyback calls, FX volatility. EWIs are most useful before or at the onset of a crisis. See also D7 (LTR-3).
- **Liquidity health measures** (LTR-11 c, S2): projected net funding requirements to unused funding capacity; non-core funding to long-term assets; overnight borrowings to total assets; short-term liabilities to total assets; funding source concentration; funding maturity profile; used to total borrowing capacity; liquid assets to volatile liabilities; unpledged eligible collateral to total assets; loans to commitments; for advanced firms, daily liquidity reporting, LCR and NSFR. Their importance "is related to their associated limits".
- **Escalation levels** (LTR-11 c, S2): no set number; three to five is common practice. Level 1: elevated monitoring, triggered for example by stress test results showing liquidity falling below risk appetite targets or a shift in market perception; LCT convened, communication plan executed. Level 2: noticeable market or idiosyncratic events hurting the business; closer monitoring of peers and counterparties, curtailing activity and new credit, strategic sales of less liquid assets, evaluating larger actions. Level 3: dramatic steps, focus "primarily on survival", resembling recovery and resolution. Movement up *or* down "should be explicitly considered and approved by the LCT"; Level 1 actions may need only ALCO or the Treasurer, Levels 2 and 3 involve senior management and/or the board. Expert judgement is not to be understated.
- **Data and reporting** (LTR-11 c): many institutions would benefit from more frequent liquidity reporting beyond treasury and funding desks (senior management, ALCO, risk committees, board), with context and

qualitative guidance; reports should capture intraday positions, contingent liabilities and capacity usage in funding sources.

- **Institution type and structure** (LTR-11 b, c): depositories focus on deposit runoff; wholesale-funded firms on credit lines and funding markets; insurers fund from policy liabilities and emphasise cash flow matching. Revolver lenders must model drawdowns; prime brokers and dealers face client withdrawals and collateral calls; wholesale borrowers face higher margins and haircuts. One document rarely covers everything, so firms run a *portfolio of CFPs* by business, product, geography and FX, and legal entity (ring-fenced entities under another regulator have their own; some build a currency CFP).
- **Liquidity and capital** (LTR-11 b): historically siloed; an integrated approach brings common EWIs and a joint view of liquidity and capital stress responses, aligned with recovery and resolution and ERM. Effective CFP attributes: strong management involvement, alignment with capital and risk frameworks, a wide range of scenarios, a documented action plan, and a communication plan. The CFP "is a playbook" that needs regular updating.

Likely distractors

- **Corrupted:** The CFP is designed mainly for frequent, low-impact funding shortfalls, while severe, low-probability events are left to business-as-usual liquidity management.

  True: BAU manages the low-impact, high-probability events; the CFP addresses the high-impact, low-probability ones. (Polarity flip.)

- **Corrupted:** Once an early warning indicator breaches its limit, the institution moves automatically to the next escalation level without further approval.

  True: movement between levels, up or down, "should be explicitly considered and approved by the LCT", and management's expert judgement matters. (Scope/condition error.)

- **Corrupted:** A well-designed institution should cover all businesses, geographies and legal entities in a single CFP document.

  True: a single document is unlikely to cover them clearly; institutions run a portfolio of CFPs (business, product, geography and FX, legal entity, ring-fenced entities). (Absolute claim.)

- **Corrupted:** Investor relations should manage the institution's communications with rating agencies and clearing banks during a liquidity crisis.

  True: corporate treasury owns regulators and supervisors, rating agencies and clearing banks; investor relations owns investors, analysts and media. (Role misassignment.)

- **Corrupted:** Drawing early on an emergency central bank facility is a low-cost first response, because it signals that official support is in place.

  True: accessing a sovereign or emergency facility may lead lenders to withdraw existing credit at once; such actions may be necessary later but early signals of weakness "may actually precipitate a liquidity crunch". (Polarity flip on the signalling effect.)

- **Corrupted:** Periodic CFP testing should include business divestitures and Federal Reserve borrowing, since these are the actions most likely to be needed in a severe crisis.

  True: those actions "may be impractical or unavailable for testing"; debt issuance, brokered deposits and limited portfolio sales may be tested, along with governance, escalation and communication. (Scope/condition error.)

D27 Intraday Liquidity and Dealer Bank Failure Mechanics

Readings: LTR-6 a-c · Intraday Liquidity Risk Management (Venkat and Baird, Ch.4); LTR-8 a-c · The Failure Mechanics of Dealer Banks (Duffie, Journal of Economic Perspectives 24:1, 2010); LTR-1 b · Liquidity Risk (Hull, RMFI 5e, Ch.24): Northern Rock, Ashanti Goldfields and Metallgesellschaft

MAP TIER D · #65 · 1/16 SOURCES VARDEEZ NOTES ONLY

Specifics reported

Map #65: "Intraday liquidity monitoring tools and the payment-system angle; for dealer banks, the runs on repo, on prime-brokerage balances, and on OTC derivatives counterparties. Case↔lesson matching against Northern Rock is the likely format."
Shapes covered here: S1 intraday liquidity monitoring tools: measures of flows versus measures of risk levels (notes: measurement bullets); S2 the payment-system angle: uses and sources of intraday liquidity, daylight overdraft policy, FMU defences and tools (notes: LTR-6 bullets); S3 dealer-bank runs on repo, on prime-brokerage balances and by OTC derivatives counterparties, ending with the clearing bank (notes: LTR-8 bullets); S4 case↔lesson matching against Northern Rock (notes: case table, LTR-1 b).

Revision notes

- **Why daylight overdrafts exist** (LTR-6 a): in the high-rate Volcker era banks ran balances down to reserve and correspondent minimums and went negative intraday. Central banks worried about (1) inadvertently lending to a member that fails during the day and (2) overdrafts acting as a "shadow" money supply (peak overdrafts near 10% of M1 then, 0.7% in 2013). The Fed's 1985 policy introduced the *net debit cap* (self-assessed, board-approved); fees followed: 24 bp above a deductible of 10% of capital (1994), 36 bp (1995), and in 2008 50 bp for unsecured overdrafts, zero for fully collateralised ones, no deductible, and a 150 bp penalty for institutions without discount window access. Even with about USD 1.5 trillion at the Fed at day end, overdrafts arise from market conventions (fed funds delivered at once but returned first thing next morning; FMU settlement windows at day end) and from intraday credit to clients (payroll before an expected incoming wire; custody clients buying before a maturity settles).
- **Uses of intraday liquidity** (LTR-6 a, S2): *outgoing wire transfers* ("typically the largest use"; payments are throttled near the debit cap); *settlements at PCS systems* (mostly once a day, late afternoon, a use or a source); *funding of nostro accounts* (managed to a target average monthly balance as compensation to the correspondent, a use or a source); *collateral pledging* (OTC trading, public deposits, margin calls); *asset purchases and funding* (securities, client loans, fixed assets).
- **Sources of intraday liquidity** (LTR-6 a, S2): *cash balances* at the central bank and correspondents; *incoming funds flow* ("the largest source of intraday funding in periods of normal market function", real-time LVPS or batch); *intraday credit* (the Fed's net debit cap is an unsecured committed line that charges interest; the Bank of England requires collateral of the highest-quality government securities but charges no interest; interbank daylight lines are generally uncommitted and free; FMUs let positions build until settlement); *liquid assets* (low volatility, predictable); *overnight borrowings* (fed funds, Libor, Eurodollar, weighing excess liquidity cost against breaching a daylight limit); *other term funding* (FHLB, term repo; an incremental, not consistent, source). The treasurer's three complications: variable cash flows, market forces (unexpected margin calls, central bank desks) and no single real-time data source.
- **Governance** (LTR-6 b): leaders manage intraday risk actively rather than as a cost of doing business, putting settlement and systemic risk in the risk taxonomy and risk appetite. Three lines of defence: *Treasury* (first line) manages intraday and end-of-day funding positions; *Corporate Risk Management* (second line, where expertise is emphasised) oversees policy, monitors all funding desks, aggregates reporting and gives an independent view; *Internal Audit* (third line) assesses adherence. Committees include treasury, operations, IT and lines of business; risk self-assessments cover settlement risk of new products. Monitoring has two perspectives: intraday credit the bank *extends* to clients and intraday credit the bank *uses*; the second is harder (FMU and correspondent data gaps, aggregation).
- **Measures for understanding flows** (LTR-6 c, S1): total payments (sent and received by client type and for the bank, net settlement position at any time, trends); other cash transactions and settlement positions at every FMU (the Fed tracks collateral at one-minute intervals); time-sensitive obligations (amounts and

deadlines); total intraday credit lines extended to clients (mostly uncommitted and undisclosed); total bank intraday credit lines available and their usage (as expected in resolution planning).

- **Measures for monitoring risk levels** (LTR-6 c, S1): *daily maximum intraday liquidity usage* (largest net negative balance relative to the committed or uncommitted line, peak and average tracked monthly, per account and on a consolidated basis where liquidity moves freely); *intraday credit relative to Tier 1 capital* (total and unsecured, available and used; available unsecured credit to Tier 1 is "a rough measure of the inadvertent systemic risk"); *client intraday credit usage* (client peak overdraft relative to its line; flags clients to re-price or change); *payment throughput* (share of outgoing payments by time of day, FMU deadlines, efficiency of FMUs' use of daylight credit).
- **Intraday stress testing** (LTR-6 c): BCBS (April 2013) suggests at least four scenario categories: own financial stress, counterparty stress, customer stress (a correspondent's customer bank) and market-wide credit or liquidity stress; also reverse stress tests and events such as natural disasters. Benefits include the management discussion, playbooks and a data aggregation capability.
- **FMU lines of defence** (LTR-6 c, S2): (1) membership criteria; tiering (indirect members routed through direct ones) concentrates risk: CHAPS had 18 direct members in 2010 against 866 at Target2 and over 7,000 at Fedwire; (2) monitoring participants (financial statements, intraday margin cycles, cutting or removing the net debit cap, pre-funding, extra collateral, letters of credit); (3) risk mutualisation (prefunded guarantee funds; central bank underwriting, as the Fed does for Fedwire to give immediate finality). Tools: net debit caps, collateral, liquidity savings mechanisms (bilateral and multilateral netting, throughput requirements, prioritisation, intra-system lending), staggered settlement windows, contingent backup lines.
- **Dealer bank lines of business** (LTR-8 a): securities dealing, underwriting and trading (plus proprietary trading); repo intermediation through clearing accounts at JPMorgan Chase and BNY Mellon (tri-party repo USD 2.5 trillion in 2007); OTC derivatives run as a matched book under ISDA master agreements (about USD 600 trillion notional; dealer exposures of USD 25,372 billion fall to USD 3,744 billion after netting; over 80% of 2008 collateral was cash, with collateral demanded doubling to about USD 4 trillion; USD 23 trillion of USD 28 trillion CDS notional was dealer-to-dealer); prime brokerage to hedge funds (mostly Morgan Stanley, Goldman Sachs, Bear Stearns); asset management including internal hedge funds; off-balance-sheet SPEs and SIVs. Risks include market, credit and counterparty risk, reputational support of internal funds and SIVs (Bear USD 3.2 billion, HSBC USD 35 billion, Citigroup USD 49 billion), and diseconomies of scope in governance.
- **Run 1: short-term secured creditors** (LTR-8 b, S3): dealers financed inventories with mostly overnight repo at average haircuts under 2%; Bear and Lehman had leverage above 30. Creditors "may have little or no incentive to renew" when solvency is doubted: they can reinvest with another dealer and avoid having to sell collateral (money market funds under Rule 2a-7) or litigation. A refusal en masse forces fire sales, which lower the value of unsold collateral and repo capacity: a "death spiral" that can spread to other banks. Investment-grade corporate bond haircuts rose from under 5% to about 20% after Lehman. Mitigants: credit lines, a buffer of cash and liquid securities, laddered liability maturities; lender-of-last-resort financing; insured deposits are less runnable, but Rule 23A bars using them to fund broker-dealer affiliates.
- **Run 2: prime brokerage clients** (LTR-8 b, S3): in the UK client cash and securities are commingled with the broker's assets ("equivalent to uninsured deposits"); US Rule 15c3-3 pools free credit balances. Example: cash of Hedge Fund A USD 150 million, margin loan to B USD 100 million, USD 50 million in reserve; if A leaves, the broker must find USD 100 million. Brokers re-hypothecate up to USD 140 of client assets per USD 100 of margin loan (equity advance rate 50%). Even *without* a run, clients can keep demanding margin loans the weakened broker cannot refinance. Re-pledgeable client collateral fell 69% at Morgan Stanley (USD 832 billion to USD 294 billion), 51% at Merrill Lynch, 30% at Goldman Sachs; Lehman's London clients could not reclaim re-hypothecated assets.
- **Run 3: derivatives counterparties** (LTR-8 b, S3): counterparties cut exposure by borrowing from the dealer, entering trades that make the dealer pay cash, harvesting in-the-money positions, or novating to another dealer (dealers began refusing Bear Stearns novations, spreading alarm). Collateral-on-downgrade triggers drain cash (Morgan Stanley: USD 498.3 million for one notch, a further USD 1,456.2 million for two; the proximate cause of AIG's bailout). Counterparty exposures to Citibank fell from USD 126 billion to USD 81 billion while those to JPMorgan grew. Most OTC derivatives are "qualifying financial contracts" exempt from the automatic stay, so close-out at replacement cost (for example 0.2% bid-offer on USD 30 trillion notional, about USD 60 billion) drains the estate with priority, encouraging others to run first.
- **Final step: loss of cash settlement privileges** (LTR-8 b, S3): clearing banks extend daylight overdrafts against securities in the clearing account (intraday peak around 10 a.m., "several hundred billion dollars").

When liquidity is in doubt the clearing bank uses its *right of offset* to stop payments that would take the account negative; JPMorgan's refusal to process Lehman's wires left Lehman unable to settle, and it filed. Duffie's composite "Alpha Bank" runs the whole sequence: failed equity raise, client bailouts, prime brokerage flight, counterparties' exposure cuts and refused novations, repo non-renewal, then the clearing bank.

- **Policy measures** (LTR-8 c): higher capital (higher for uncleared derivatives), supervisory councils, special resolution powers, securitisation retention, trade repositories. Lender-of-last-resort facilities and TARP-style capital injections limited fire sales but are costly and raise moral hazard. For repo: a tri-party repo utility with less discretion in rolling dealers' repos (Bernanke), central bank insurance of tri-party repo, or an "emergency bank" funded by repo participants with discount window access. For derivatives: central clearing (multilateral netting; only standard contracts, so not AIG's customised CDS; CCP strength and implicit government backing are crucial). For recapitalisation: distress-contingent convertible debt and mandatory rights offerings triggered by leverage or liquidity thresholds.
- **Northern Rock** (LTR-1 b, S4): a top-five UK mortgage lender funding much of its book with short-term debt instruments and securitisation (Granite). After August 2007 it could not replace maturing funding, although the FSA judged it solvent. It sought Bank of England support on 12 September; the BBC broke the news on 13 September and the first British bank run in 150 years followed (about GBP 2 billion withdrawn by 17 September), ending only when the government guaranteed all deposits. Emergency borrowing at a penalty rate reached GBP 25 billion by February 2008; it was nationalised. Lessons: liquidity is not solvency; wholesale funding disappears in stress; news of central bank borrowing can itself trigger a run. See also A32.
- **Ashanti Goldfields and Metallgesellschaft** (LTR-1 b, S4): both hedged illiquid long-term positions with margined contracts. Ashanti sold gold forward; when 15 European central banks limited gold sales (September 1999) gold jumped over 25%, Ashanti could not meet margin calls and had to restructure. MG sold 5- to 10-year fixed-price oil supply contracts and hedged with rolled short-dated futures; oil fell, margin calls mounted, management closed the hedges for a USD 1.33 billion loss. Lesson: not to avoid futures and forwards, but to ensure access to funding for the cash flow mismatch.
- **Case↔lesson matching** (S4): Northern Rock: short-term wholesale funding and a retail run triggered by news of central bank support. Bear Stearns: repo lenders refused to roll even agency collateral, prime brokerage clients withdrew cash and unencumbered securities, dealers refused novations. Lehman: tri-party clearing bank demanded collateral and invoked its right of offset; London clients lost re-hypothecated assets. AIG: collateral-on-downgrade triggers. Ashanti and Metallgesellschaft: margin calls on hedges of illiquid assets. See also D6 (repo cases in Tuckman and Serrat).

Formulas and numbers

Daily maximum intraday liquidity usage. $\dfrac{\text{largest net negative balance in the day}}{\text{committed or uncommitted intraday credit line}}$ (track peak and average)

Intraday credit relative to Tier 1 capital. $\dfrac{\text{intraday credit (total or unsecured; available or used)}}{\text{Tier 1 capital}}$

Client intraday credit usage. $\dfrac{\text{client's peak daily intraday overdraft}}{\text{client's intraday credit line}}$; **payment throughput:** $\dfrac{\text{outgoing payments made by time } t}{\text{total outgoing payments}}$

Fed daylight overdraft fees. 24 bp (1994) → 36 bp (1995) → 50 bp unsecured, 0 bp fully collateralised, 150 bp penalty without discount window access (2008)

Repo haircut. 10% haircut: USD 100 million of securities secures a USD 90 million loan

Collateral netting. A's exposure to B is 100 and B's exposure to A is 80 under one master agreement, fully collateralised: B posts $100 - 80 = 20$ with A

Re-hypothecation cap (US). up to USD 140 of client assets per USD 100 of margin loan; equity advance rate 50%

Close-out friction. termination cost ≈ bid-offer × notional, e.g. $0.2\% \times 30$ trillion $= 60$ billion

Likely distractors

- **Corrupted:** Outgoing wire transfers are typically a bank's largest source of intraday liquidity, while incoming payment flows are its largest use.

  True: outgoing wires are typically the largest *use*; incoming funds flow is the largest *source* in normal markets.

(Polarity flip.)

- **Corrupted:** The Federal Reserve requires intraday overdrafts to be collateralised and charges no interest, while the Bank of England offers an unsecured committed line and charges for its use.

  True: the Fed's net debit cap is an unsecured committed line that charges interest; the Bank of England requires top-quality government collateral and charges no interest. (Sibling swap.)

- **Corrupted:** Internal Audit, as the second line of defence, oversees funding policy and aggregates intraday liquidity reporting across the bank's funding desks.

  True: that is Corporate Risk Management (second line); Treasury is the first line, Internal Audit the third, assessing adherence to intraday policies. (Role misassignment.)

- **Corrupted:** Because they hold collateral after a haircut, repo creditors have a strong incentive to keep rolling overnight repo to a dealer whose solvency is in doubt.

  True: they "may have little or no incentive to renew": they can reinvest with other dealers and avoid forced collateral sales or litigation. (Polarity flip.)

- **Corrupted:** Because most OTC derivatives are subject to the automatic stay at bankruptcy, derivatives counterparties have little reason to cut their exposure to a weakening dealer.

  True: most are qualifying financial contracts *exempt* from the stay; close-out drains the estate with priority, which raises others' incentive to run first. (Polarity flip.)

- **Corrupted:** Northern Rock failed when its clearing bank exercised its right of offset and stopped processing its payment instructions.

  True: that is the Lehman (and Duffie's Alpha Bank) endgame; Northern Rock's lesson is reliance on short-term wholesale funding and a retail run triggered by news of Bank of England support. (Case cross-mapping.)

D28 Portfolio Construction and the No-Trade Region

Readings: IM-4 a, b, c, d, e, f, g, h, i · Portfolio Construction (Grinold and Kahn, Active Portfolio Management 2e, Ch.14; RMIM Ch.4, 4.1 to 4.10)

MAP TIER D · #66 · 1/16 SOURCES VARDEEZ NOTES ONLY

Specifics reported

Map #66: "Alpha refinement and neutralisation, transaction costs, and the optimal no-trade region: the band [−sell cost, +purchase cost] around the target. The four construction techniques (screens, stratification, linear programming, quadratic programming) form a ready-made four-statement audit."
Shapes covered here:
S1 alpha refinement: scaling, trimming, and neutralisation (benchmark, cash, risk-factor, industry)
S2 transaction costs: amortisation over the holding period
S3 the optimal no-trade region for MCVA and for alpha, and Leland's band around a target allocation
S4 the four techniques as a four-statement audit: screens, stratification, LP, QP

Revision notes

- **Inputs** (IM-4 a): current portfolio, alphas, covariance estimates, transactions cost estimates and an active risk aversion. Only the current portfolio is measured with near certainty; alphas are "often unreasonable and subject to hidden biases"; covariances and costs are noisy; even risk aversion is uncertain. The standard objective is active return minus an active risk penalty.
- **Refining instead of constraining** (IM-4 b): any construction process, however sophisticated, can be replaced by refining the alphas and then running a simple unconstrained mean/variance optimisation (modified alphas $\alpha' = 2\lambda'_A V h_{PA}$). In the Major Market Index example, no short sales and a 5% cap above benchmark pulled the alphas toward zero: their standard deviation fell from 2.00% to 0.57%. Constraints are, in part, safeguards against poor research.
- **Scale the alphas** (S1): $\alpha =$ volatility $\times$ IC $\times$ score, with score mean 0 and standard deviation 1, so the scale is Std$\{\alpha\} =$ volatility $\times$ IC (IC 0.05 and residual risk 30% give 1.5%). Rescale if the inputs do not have this scale. Alternative: match the implied IR to the manager's ex ante IR. Scaling and trimming are applied as the final step.
- **Trim outliers** (S1): examine alphas larger than about three times the scale; set those based on questionable data to zero and pull genuine ones in to three times scale. The extreme version forces alphas into a normal distribution, which uses only the ranking information.
- **Neutralisation** (IM-4 c): removing biases or undesirable bets from alphas; decided *a priori*, not by trying all possibilities. Benchmark-neutral: the benchmark has zero alpha, so the optimal portfolio has a beta of 1 and avoids benchmark timing (subtract $\beta_n \alpha_B$; in the example α_B was only 1.6 bp). Cash-neutral: no active cash position (both are possible at once). Risk-factor-neutral: neutralise against factors the manager cannot forecast. Industry-neutral: subtract the cap-weighted industry average alpha. Neutralisation is not uniquely defined.
- **Transaction costs** (IM-4 d, S2): the third "rubber chicken". They turn a one-dimensional trade-off (one knob: active risk aversion) into a two-dimensional one, so the scale of the alphas must now be precise. Costs occur at a point in time while alpha accrues over the horizon: the annualised transactions cost is the round-trip cost divided by the holding period in years. Example: two 4% annual-alpha stocks, costs USD 0.75 per trade on USD 100: a 6-month holding costs 3% a year, a 24-month holding 0.75% a year.
- **Practical details** (IM-4 e): choose the active risk aversion from the IR and the desired active risk, $\lambda_A =$ IR$/(2\psi_P)$: IR 0.5 and 5% active risk give 0.05 (check percents versus decimals). A higher aversion to *specific* risk reduces bets on single stocks (the biggest losers) and reduces dispersion across multiple portfolios. Alpha coverage: stocks forecast but not in the benchmark get benchmark weight zero; benchmark stocks without forecasts get $\alpha = 0$ after the covered alphas are recentred by their value-weighted average.
- **Portfolio revisions** (IM-4 f): revise whenever new information arrives, if the trade-off is specified correctly. A manager who underestimates costs and revises daily churns the portfolio; a crude but effective cure is to revise less often. Shorter-horizon alphas are noisier, but the costs stay the same whether reacting to signal or noise.

- **MCVA** (S3): the marginal contribution to value added of stock n, $\text{MCVA}_n = \alpha_n - 2\lambda_A \psi_P \,\text{MCAR}_n$, measures how risk-adjusted alpha changes as the holding rises against cash; MCAR is the marginal contribution to active risk.
- **No-trade region** (IM-4 g, S3): the current portfolio is optimal while $-\text{SC}_n \le \text{MCVA}_n \le \text{PC}_n$. Book example: PC 0.50%, SC 0.75%; an MCVA of 0.80% gives a net benefit from buying of 0.30%; an MCVA of -1.30% gives a net benefit from selling of 0.55%. Bandwidth = PC + SC = 1.25%. Just after a purchase the stock sits at the upper end, so alpha must fall by 1.25% before a sale is considered.
- **Leland and information horizons** (S3): around an optimal 60/40 allocation with linear costs there is a no-trade region; outside it, trade back to the *boundary*, not the target, which cuts turnover and costs roughly in half with effectively no change in risk. The region's size depends on costs, risk aversion and the expected return and risk of stocks and bonds. After-cost IR falls with one-way cost and with shorter signal half-lives: we pay costs, and we "lose by intimidation".
- **Screens** (IM-4 h, S4): rank by alpha, take the top N, equal- or cap-weight; buy/hold/sell lists regulate turnover. Strengths: simple, clear cause and effect, easy to computerise, robust (depends solely on ranking, so wild alphas do not alter the result). Weaknesses: ignores all alpha information except rankings, no protection against biases (no utilities if all utilities rank low), fragmentary risk control.
- **Stratification** (S4): "glorified screening". Split stocks into exclusive categories (e.g. 10 sectors × 3 sizes = 30), screen within each, weight categories at benchmark weights. Adds robustness against alpha biases *across* categories; still ignores information, cannot slightly over- and underweight categories, and risk control fails if important dimensions are excluded.
- **Linear programming** (S4): "space-age stratification". Characterises stocks along risk dimensions (industry, size, volatility, beta) that need not be exclusive; can include explicit costs, turnover limits and position limits; maximises alpha less costs while staying close to the benchmark. Uses all alpha information, but has difficulty producing a prespecified number of stocks, and risk-control dimensions must not work against the alphas.
- **Quadratic programming** (S4): "the ultimate": explicitly handles alpha, risk and transaction costs and includes LP as a special case. Weakness: far more inputs (500 stocks need 500 volatilities and 124,750 correlations), and the optimiser exploits noise in covariances. True market vol 17%: estimates within 1% barely hurt, errors above 3% matter, and an underestimate to 12% gives negative value added. Lesson: get good covariance estimates rather than abandon QP.
- **The test** (Muller): identical alphas with IC 0.1 on 500 stocks, so the target IR is $0.1\sqrt{500} = 2.24$. QP had the highest and most consistent ex post IR (average 1.88, standard deviation 0.40); a stratified portfolio had the single highest (2.82) but no consistency; a screen had a negative IR in one period; none reached 2.24, partly because of constraints. Alternatives to mean/variance (semivariance, downside risk, shortfall probability) add effort without improving results (Kahn and Stefek; Grinold 1999). See also A18 for the IR.
- **Dispersion** (IM-4 i): maximum minus minimum return across separate accounts running the same strategy. Client-driven dispersion (client constraints) is beyond the manager's control; dispersion from differing betas and factor exposures (betas 0.9 to 1.1 with a 35% market return gives 7%) should be controlled; dispersion from holding different assets is partly optimal because of transaction costs. More portfolios mean more dispersion for a given tracking error.
- **Managing dispersion:** if alphas and risk are constant, dispersion never disappears (a transaction cost barrier); with changing alphas the accounts converge (five portfolios converged over roughly 5 years, fastest with the lowest costs). Example bound: $\lambda_A = 0.10$, round-trip costs 2%, 10% turnover give a tracking error of 1%. Dual-benchmark optimisation reduces dispersion only by giving up return.

Formulas

Value added and figure of merit. $\text{VA} = \alpha_P - \lambda_A \psi_P^2$; construction maximises VA − TC
Alpha structure and scale. α = volatility × IC × score, $\text{Std}\{\alpha\}$ = volatility × IC; trim at about 3× scale
Benchmark-neutral alpha. $\alpha'_n = \alpha_n - \beta_n \alpha_B$

Alpha coverage (eqs. 4.5, 4.6). $H\{N_1\} = \sum_{n\in N_1} h_{B,n}$, $\alpha\{N_1\} = \dfrac{\sum_{n\in N_1} h_{B,n}\alpha_n}{H\{N_1\}}$; $\alpha'_n = \alpha_n - \alpha\{N_1\}$ $(n \in N_1)$, $\alpha'_n = 0$ $(n \in N_0)$

Annualised transactions cost. $\dfrac{\text{round-trip cost}}{\text{holding period in years}}$

Active risk aversion (eq. 4.3). $\lambda_A = \dfrac{\text{IR}}{2\,\psi_P}$ (ψ_P in percent)
MCVA (eq. 4.7). $\text{MCVA}_n = \alpha_n - 2\lambda_A \psi_P\,\text{MCAR}_n$
No-trade region (eqs. 4.8, 4.9). $-\text{SC}_n \le \text{MCVA}_n \le \text{PC}_n \iff 2\lambda_A\psi_P\text{MCAR}_n - \text{SC}_n \le \alpha_n \le 2\lambda_A\psi_P\text{MCAR}_n + \text{PC}_n$; width $= \text{PC}_n + \text{SC}_n$
Fundamental law (target IR). $\text{IR} = \text{IC}\sqrt{\text{BR}}$ ($0.1\sqrt{500} = 2.24$)
Expected dispersion (eq. 4.13). $E\{r_{PA,\max} - r_{PA,\min}\} = 2\,\Phi^{-1}\{(\tfrac{1}{2})^{1/N}\}\,\psi$

Likely distractors

- **Corrupted:** "With purchase cost 0.50% and sales cost 0.75%, the no-trade band for a stock's MCVA is 0.50% wide, since only the purchase cost must be recovered."

  True: the band runs from $-$SC to $+$PC, so its width is $0.50\% + 0.75\% = 1.25\%$; after a purchase, alpha must fall by 1.25% before a sale is considered.

- **Corrupted:** "When a stock/bond mix drifts outside Leland's no-trade region, the optimal strategy is to trade back to the 60/40 target."

  True: trade back to the *boundary* of the region; this cuts turnover and transactions costs roughly in half with effectively no change in risk.

- **Corrupted:** "Screens are sensitive to wild alpha estimates because they use the full magnitude of each alpha."

  True: screens depend solely on ranking, so wild positive or negative alphas do not alter the result; their weakness is that they ignore all information except rankings and control risk only fragmentarily.

- **Corrupted:** "Linear programming, like stratification, requires the risk dimensions to partition the stocks into mutually exclusive categories."

  True: exclusive categories are the stratification method; LP characterises stocks along several non-exclusive risk dimensions (industry, size, volatility, beta).

- **Corrupted:** "Quadratic programming needs fewer inputs than the other techniques, so it is the least exposed to estimation error."

  True: QP needs far more inputs (124,750 correlations for 500 stocks) and the optimiser exploits noise in covariances; an underestimate of market volatility to 12% against a true 17% turns value added negative.

- **Corrupted:** "The annualised transactions cost equals the round-trip cost multiplied by the holding period in years, so a 6-month holding halves the annual cost."

  True: divide by the holding period: a 6-month holding doubles the annual cost and a 24-month holding halves it.

D29 Crypto, Tokenisation and Digital Resilience

Readings: CI-6 a, b, c, d · Regulating the Crypto Ecosystem: The Case of Unbacked Crypto Assets (IMF Fintech Note, 2022); CI-7 a, b, c, d, e · Tokenization and Financial Market Inefficiencies (IMF Fintech Note, 2025); CI-8 a, b, c · Digital Resilience and Financial Stability: The Quest for Policy Tools in the Financial Sector (Martinez, Banco de España, 2023) (source: SchweserNotes Book 5, Readings 105 to 107, one rung below a GARP text)

MAP TIER D · #68 · 1/16 SOURCES VARDEEZ NOTES ONLY

Specifics reported

Map #68: "Because Current Issues rotates every year, the absence of recall evidence carries almost no information. Treat the provider ratings as the only signal available and cover all eight readings: they are 8 questions of the 80."
Shapes covered here:
S1 CI-6: crypto asset categories, ecosystem entities and their regulation, national approaches, BCBS treatment, Bali Fintech Agenda
S2 CI-7: tokenisation, trust, sharedness and programmability, lifecycle frictions, externalities, retail and broker effects, ledger models and trust failures
S3 CI-8: cyber versus ICT risk, cloud concentration, contagion to financial risk, macroprudential tools and their challenges

Revision notes

- **Crypto assets and DLT** (CI-6 a): digital assets that use cryptography and rely on distributed ledger technology (DLT), a sequenced, standardised, cryptographically secured record on a typically decentralised network, distributed simultaneously to all participants. Blockchain is the best-known DLT; records are generally immutable.
- **Unbacked crypto and the six categories** (CI-6 a): unbacked crypto assets' value is set by supply and demand; they are not money (not generally used as medium of exchange, store of value or unit of account), are highly volatile with low intrinsic value, and are mostly speculative. No single international classification exists; six broad categories: unbacked crypto assets, NFTs, security tokens, utility tokens, stablecoins, CBDCs.
- **Issuers** (CI-6 b): decentralised issuance gives anonymity, liquidity and broad access but limits disclosure and monitoring. Pre-mining and ICOs expose participants to price manipulation and pump-and-dump schemes. Regulatory response: white papers, notification, suitability tests for retail users, fair marketing, consumer education.
- **Exchanges and wallets** (CI-6 b): exchanges give direct retail access, trade from their own inventory and offer custody, which exposes them to operational and cyber risk; IOSCO's framework stresses governance, regulated access, controls, market integrity, transparency, AML/CFT, product offering rules, conflicts of interest and off-chain transactions. Wallets hold private keys: custodial wallets reduce loss risk but add cyber risk; hot (online) wallets are faster but hackable; cold (offline) wallets resist cyber risk but can be physically lost. Regulation: segregation and safekeeping, cyber resilience, record keeping, AML/CFT, prudential requirements, cyber insurance, wind-down arrangements, conflicts of interest.
- **Validators, technology and institutions** (CI-6 b): private blockchains reduce interoperability and can fragment and concentrate; public blockchains avoid single points of failure but carry user-behaviour risks. Regulation aims to be technology neutral. Banks' direct exposures call for capital deductions and higher risk weights; indirect exposures come through derivatives, loans and services. A crypto provider that is "systemic" may be designated an FMI, but users can switch exchange or go on-chain, so close supervision is still needed.
- **National approaches and bans** (CI-6 c): EU (stablecoins, licensing), Switzerland (progressive; ledger-based securities), UK (security tokens first, broader later), Albania (licensing; legal for trading since 2020), Nigeria (not legal tender; banks barred), Japan (bespoke provider rules; stablecoins since 2020). Bypassing via unregulated exchanges and VPNs is common (Egypt, Nigeria; mining in China despite the 2021 ban). The IMF and BCBS judge a blanket ban ineffective; targeted restrictions (crypto derivatives in Japan and the UK, crypto as payment) can work in the short term.
- **BCBS treatment** (CI-6 c): banks' crypto holdings split into lower-risk assets (tokenised traditional assets

with risk weights similar to the underlying; stablecoins possibly lower) and higher-risk assets (unbacked crypto assets), with a proposed 1,250% risk weight applied to the maximum of long and short positions. (The BCBS standard calls these Group 1 and Group 2: outside-source.)

- **Bali Fintech Agenda** (CI-6 d): IMF and World Bank framework; legal certainty to build investor confidence. Considerations: monitoring (web searches, horizon scanning, innovation offices), prioritising resources, scope (all key-function entities; heavier where unbacked crypto and stablecoins are supported by centralised entities; user education where crypto is niche), domestic collaboration (who owns AML/CFT), continuous risk assessment, with cross-border cooperation.
- **Tokens and tokenisation** (CI-7 a): tokens are assets or representations of assets on a digital ledger, a digital certificate of ownership; tokenisation is a collection of tools, not a single technology. Native tokens exist only on the ledger (Bitcoin, Ether, digital art); non-native tokens represent off-ledger assets (equities, bonds, real estate); assets can also migrate between ledgers, destroyed on the source and reissued.
- **Trust, sharedness, programmability** (CI-7 a): trust (assets cannot move without consent; reliable execution), sharedness (accessible to all relevant parties, ownership transferable and updated on the ledger), programmability (smart contracts: immutable code that automates instructions, e.g. coupon-paying bonds, qualified-investor restrictions).
- **Lifecycle frictions** (CI-7 b): information asymmetries, search frictions, transaction costs, counterparty risk, across issuance, trading, servicing and redemption; intermediaries (investment banks, CSDs, clearinghouses, CCPs) reduce them for fees. Tokenisation: issuance (no separate record-keeping or custody; embedded data, e.g. BNP Paribas 2022 bond with ESG data), trading (simultaneous or instantaneous settlement; fewer search frictions in OTC markets), servicing (automated coupons and dividends). It cannot fully remove counterparty risk between initiation and execution: a CCP or locked-up collateral is needed, at an opportunity cost.
- **Externalities, internalities, market power** (CI-7 b, c): shock transmission rises with speed, leverage (cheaper debt issuance), deposit migration to wholesale funding, rehypothecation and automated smart-contract trading (flash crashes). Infrastructure investment matters more as users grow (single point of failure). Network externalities raise liquidity, but non-interoperable ledgers fragment liquidity and raise search frictions and costs. Knowledge externalities from open code can weaken incentives to innovate.
- **Retail and brokers** (CI-7 d): direct retail trading amplifies internalities (trading poorly understood complex assets to avoid advisory fees, under-diversification). Instant broker switching and no clearinghouse reduce brokers' market power; but privately owned ledgers create new market power (monopoly rents, monetised data, free access then fees once dominant).
- **Ledger models and trust failures** (CI-7 e): sharedness via single ledger, common ledger (escrowed non-native assets with escrow certificates) or compatible ledgers (standardised technology, interoperable). Trust via single-operator systems (operator credibility and legal protection; risk even higher than permissioned), permissionless DLT (attack on consensus, e.g. more than 50% of PoW computing power; fee manipulation, front running, sandwich attacks; hard to enforce compliance) or permissioned DLT (collusion among authorised participants, checked by reputation and regulation).
- **Cyber versus ICT risk** (CI-8 a): cyber risk is loss of confidentiality, integrity and availability (CIA) from a cyberattack; ICT risk is disruption to information and communication systems with similar effects that does not stem from a cyberattack. Some institutions rank cyber risk above market risk; loss data are too scarce for reliable estimates, which has hindered cyber insurance.
- **Cloud computing** (CI-8 a): adoption has *decreased* cyber risk but *increased* ICT risk, and moves some but not all cyber risk to providers; cyber and ICT risk are now integrated with possible systemic effects. All major banks use multiclouds (less than half of asset managers and insurers), at the cost of weaker stewardship. About 80% of cloud customers had an outage in three years; about 20% of those were significant.
- **From cyber to financial risk** (CI-8 b): the key question is how technology risks lead to financial risks (illiquidity, excessive leverage, loss of trust). Severity is lower for confidentiality and higher for integrity and availability. Contagion is non-linear, so agent-based models (still in their infancy) are proposed, also as a basis for cyber stress tests. A disrupted bank may see no withdrawals while depositors run on other banks. Institutions must be reliable enough to recover quickly, which need not mean 100% reliability.
- **Policy measures** (CI-8 c): certification and standardisation as a starting point; consolidation of cybersecurity bodies; forward-looking, principles-based operational resilience rules (better for single entities than for systemic risk); ESRB proposal for EU-wide incident reporting; DORA covers ICT third-party (cloud) risk and penetration testing.

- **Macroprudential tools** (CI-8 c): circuit breakers, a temporary pause on normal rules after a major attack, complementing deposit controls (challenge: proving the attack is force majeure); cooperative arrangements, from tick-the-box to judgment-based standards, ICT buffers such as Sheltered Harbor (customers of an attacked co-owner bank transact at another co-owner bank); structural measures for systemically important technological institutions (SITIs), multicloud or private clouds, "forced diversification" (penalties raise enforceability issues). For the other CI readings see A14 (CI-1, CI-2), A28 (CI-3), D30 (CI-4) and D16 (CI-5).

Likely distractors

- **Corrupted:** "Banks' heavy adoption of cloud computing has reduced both their cyber risk and their ICT risk."

  True: cloud adoption has decreased cyber risk but increased ICT risk, and has merged the two with potential systemic implications.

- **Corrupted:** "A systemic cyber shock is most severe for data confidentiality and least severe for data availability."

  True: severity is lower for confidentiality and higher for integrity and for the availability of data and systems.

- **Corrupted:** "Sheltered Harbor, which lets customers of an attacked co-owner bank transact at another co-owner bank, is an example of a circuit breaker, a long-term substitute for deposit controls."

  True: Sheltered Harbor is a cooperative arrangement (an ICT buffer); circuit breakers are a *temporary* pause that complements deposit controls.

- **Corrupted:** "A permissioned DLT is most exposed to an attacker acquiring more than 50% of the network's computing power."

  True: that is the consensus attack on a permissionless proof-of-work ledger; a permissioned DLT fails if authorised participants collude to alter the ledger.

- **Corrupted:** "Fragmented, non-interoperable tokenised ledgers mainly amplify the transmission of financial shocks."

  True: fragmentation reduces network externalities and liquidity and raises search frictions and transaction costs; shock amplification comes from speed, leverage and interconnectedness.

- **Corrupted:** "The IMF and BCBS favour a blanket ban on crypto activities, and the BCBS assigns tokenised traditional assets a 1,250% risk weight."

  True: a blanket ban is judged ineffective (targeted restrictions are preferred); the 1,250% weight is proposed for higher-risk unbacked crypto assets, while tokenised traditional assets get risk weights similar to the underlying.

D30 Geopolitical Risk and Financial Stability

Readings: CI-4 a, b, c, d, e, f · Global Financial Stability Report, Chapter 2 (IMF, April 2025) (source: SchweserNotes Book 5, one rung below a GARP text)

2026 LO · NOT IN MAP

Specifics reported

Not in the Priority Map: CI-4 is examinable in 2026 and no map entry carries its specifics. The map files the current CI-4 LOs (103.a to 103.e) under #19 Climate Risk and #69 2023 Bank Failures, but the specifics reported there come from older GFSR chapters (see B5 and D17).
Shapes covered here: S1 channels from geopolitical risk to asset prices and financial stability, and policy responses (Q1); S2 effects on banks and nonbank financial institutions: valuations, lending, fund flows (Q1); S3 pricing in equity and option markets (GPR betas, downside and tail-risk premiums) (Q2); S4 sovereign risk premiums, and commodity importers vs exporters (Q2).

Revision notes

Channels and policy (CI-4 a)

- **Geopolitical risk.** Adverse effects on financial markets, trade relations and growth from major conflicts: wars, terrorism, and trade disputes involving tariffs or sanctions. Even the *threat* of rising tension can depress asset prices and redirect capital flows.
- **Two channels (S1).** The **economic channel**: disruptions to trade, capital flows, supply chains and production capacity (Russia 2022: stock and sovereign bond prices fell, oil and grain rose; sovereign debt burdens can rise with military spending or lost revenue). The **market sentiment channel**: uncertainty, risk aversion and lower confidence, which can depress valuations and raise volatility even without direct conflict.
- **Policy responses.** Exposed entities should assess, measure and manage the exposures. Countries, especially EMs, strengthen resilience through better regulatory frameworks, preserved fiscal and monetary flexibility and deeper financial markets. Financial institutions hold adequate capital *and* liquidity buffers; capital buffers alone are not sufficient.

Who wins and who loses (CI-4 b, c)

- **Countries.** Those directly involved or closely linked suffer most (physical damage, sanctions, trade and investment disruption). **Commodity exporters**, especially energy producers, often **benefit** from higher prices; **importers** face currency depreciation and wider sovereign spreads (European importers after 2022).
- **Sectors and asset classes.** Defense and energy gain; banking, retail and utilities decline. Oil and gold rise; equities fall, especially in EMs; safe-haven sovereign bonds appreciate (lower yields); currencies of commodity importers weaken. Event studies: stocks fall about 3% immediately after major events, up to 9% for severe ones. Wars cause deeper and longer effects than sanctions or trade restrictions; the 1973 oil embargo and the 1990 invasion of Kuwait caused sustained declines.
- **Stock market responses.** Average reactions are modest: about -0.3% after a country-specific shock; large regional shocks (at least two standard deviations above normal) up to **7x** larger and more persistent; global shocks up to **20x**, persisting up to six months. Domestic events: about -1% within a month, about -5% in EMs with military conflict. The VIX and tail risk spike.
- **Cross-border effects.** Through trade and financial linkages: countries close to the conflict can see declines up to 2.5x larger; firms with revenue exposure, subsidiaries or major shareholders in affected countries fall too, more so in EMs.
- **Russia 2022.** Defense and energy firms with revenue exposure to Russia and Ukraine fell, while the same sectors rose elsewhere; European markets fell more than other AEs (energy imports, links); firms with subsidiaries there lost about 2.5% within a week. Aggregate exposure to Russia has since declined, but risen for some Asia-Pacific and European firms.
- **US-China trade tensions (2018 to 2024).** Chinese equities fell about 4% on average after tariff announcements (nearly 8% after the May 2019 US announcement); US firms about 1.3% on average, 1.6% to 1.8% after China's August 2019 retaliation. The study covers no 2025 tariff events.

Pricing geopolitical risk (CI-4 d, e)

- **Sovereign risk premiums (S4).** Measured by sovereign CDS spreads, which widen after conflicts on fiscal and debt-sustainability worries: about **180 bps** in EMs within a month vs about **40 bps** in AEs. Long yields rise in vulnerable EMs; safe-haven yields (US, UK, Germany, Switzerland, Japan) fall, more so the bigger the event. Conflicts hitting key trading partners also raise premiums. Below-median reserve adequacy: about +100 bps; below-median institutional quality: about +120 bps. High debt-to-GDP, thin fiscal and reserve buffers and weak institutions all amplify. See also D16 for high public debt.
- **Sign of the risk premium (S3).** Assets that **lose** value when geopolitical risk rises carry **positive** risk premiums; assets that **gain** carry low or **negative** premiums, accepted for their hedging value.
- **GPR betas.** Sensitivity of equity returns to geopolitical shocks: **energy and defense positive**, consumer-oriented sectors negative. The energy-versus-other-sectors GPR beta premium moved from negative to positive after the 2022 invasion: investors came to prefer hedging stocks.
- **Option markets.** Premiums for downside protection (puts) and tail protection (deep out-of-the-money puts) rise sharply in major shocks, **tail premiums the most**. After the 2022 invasion, put premiums rose for Russian firms and for European firms with exposure; the 2018 to 2019 tariff weeks raised premiums in the US and China but did not spill over meaningfully elsewhere.

Banks and nonbanks (CI-4 f)

- **Market, credit and liquidity risks rise (S2).** Volatility triggers selloffs that erode asset and collateral values and tighten lending; capital outflows amplify volatility in less liquid markets; sanctions, capital controls and cyberattacks compound the stress. Lending suffers through weaker borrower credit quality: less cross-border lending, lower asset values and bank equity, higher rollover risk on foreign debt.
- **Exposures.** As of 2024, markets with elevated geopolitical risk were about 8% of cross-border bank claims, 10% of cross-border bank liabilities and 13% of equity fund assets. Banks have **reduced** direct exposure to Russia and Ukraine since the 2014 annexation of Crimea; investment funds cut direct exposure by about 60% after 2022.
- **Fund flows.** Bond funds with 10% exposure to conflict-affected countries saw a 2.3% drop in flows and a 1% fall in returns (equity funds less). Funds with 10% direct exposure to Russia and Ukraine lost about 6% in a week and 8% over six months after the invasion; the 2018 to 2024 tariff announcements caused no meaningful fund losses. Redemption pressure and forced sales, not inflows, follow major events. Remedy: buffers, no concentrated exposures, better risk management, and regulation that accounts for geopolitical risk.

Numbers to know

Stock reaction, country-specific shock	about -0.3%; regional up to 7x; global up to 20x (up to six months)
Domestic events / EMs in military conflict	about -1% / about -5% within a month
Sovereign CDS, one month after major conflicts	EMs about +180 bps; AEs about +40 bps
Weak reserves / weak institutions	about +100 bps / about +120 bps
US-China tariffs	China about -4% (nearly -8% May 2019); US about -1.3%
Exposure to high-GPR markets (2024)	8% bank claims, 10% bank liabilities, 13% equity fund assets
Funds, 10% Russia/Ukraine exposure	about -6% in one week, -8% over six months

Exam traps

- **Sibling swap with CI-5:** the two channels are *economic* and *market sentiment*, not fiscal and monetary.
- **Role/sibling swap, importers vs exporters:** exporters (energy producers) gain from price spikes; importers suffer depreciation and wider spreads.
- **Polarity flip on GPR betas and premiums:** energy and defense have *positive* GPR betas (they gain), so they act as hedges with low or negative risk premiums; consumer sectors have negative GPR betas.
- **Scope error across cases:** Russia 2022 hit funds and European options; US-China tariffs moved US and Chinese markets but did not spill over meaningfully and caused no meaningful fund losses.
- **Polarity flip on exposures:** banks *reduced* direct exposure to Russia and Ukraine after 2014; funds cut theirs

by about 60% after 2022.

- **Absolute claim:** capital buffers alone, or cutting only direct exposure, do not protect against indirect contagion; tail-protection premiums rise *more* than ordinary downside premiums.

Question D30.1 CI-4 a, f · S1, S2 · [self-gen]

The CRO of a German bank is preparing a board briefing on the IMF's analysis of geopolitical risk and financial stability after a surge in military tensions in a region where the bank has EUR 3.1 billion of cross-border claims. Which of the following statements should the CRO include in the briefing?

A. Geopolitical risk moves asset prices through an economic channel and a market sentiment channel, and falling asset and collateral values combined with capital outflows can tighten bank lending.

B. Geopolitical risk moves asset prices through a fiscal channel and a monetary channel, so the main policy response the IMF recommends is that central banks cut rates as soon as conflicts begin.

C. Because US-China tariff announcements caused no meaningful fund losses, funds with 10% direct exposure to Russia and Ukraine also avoided material losses after the 2022 invasion.

D. Since the 2014 annexation of Crimea, banks have steadily increased their direct exposure to Russia and Ukraine, raising their vulnerability when the 2022 invasion began.

Solution **Answer: A**

Step 1 – Name the channels

The economic channel works through disruptions to trade, capital flows and production capacity; the market sentiment channel through uncertainty, risk aversion and lower investor confidence.

Step 2 – Trace the effect to banks

Higher volatility triggers selloffs that erode asset and collateral values, leading to tighter lending; capital outflows amplify volatility in less liquid markets; weaker borrowers reduce cross-border lending and raise rollover risk on foreign debt.

⇒ *For this bank, the regional claims are exposed through both credit quality and collateral values, and funding can come under rollover pressure.*

Interpretation

A is correct. The board should hear that adequate capital *and* liquidity buffers, limits on concentrated exposures and stronger risk management are the IMF's prescriptions; cutting direct exposure alone does not address indirect contagion.

Why the other options are wrong

B is incorrect. Sibling swap (case cross-mapping): fiscal and monetary channels belong to CI-5 (Monetary and Fiscal Policy); CI-4's channels are economic and market sentiment, and the reading's policy advice is resilience (regulatory frameworks, fiscal and monetary flexibility, deeper markets, buffers), not rate cuts.

C is incorrect. Scope/condition error: extends the tariff result to a war. Funds with 10% direct exposure lost about 6% within a week and 8% over six months after the invasion; only the tariff announcements caused no meaningful fund losses.

D is incorrect. Polarity flip: banks *reduced* their direct exposure to Russia and Ukraine over the decade after 2014.

LO: CI-4 a: Identify and explain the key channels through which geopolitical risk influences asset prices and financial stability and discuss policy measures to address potential consequences. · CI-4 f: Describe the implications of geopolitical risk exposure for the resilience of banks and nonbank financial institutions, including effects on equity valuations, lending, and investment fund flows. · Reference: IMF, Global Financial Stability Report, April 2025, Ch.2, via SchweserNotes Book 5, Reading 103, Modules 103.1 and 103.2

Question D30.2 CI-4 b, d, e · S3, S4 · [self-gen]

A portfolio manager at an Australian multi-asset fund is reviewing how markets priced geopolitical risk in past conflicts before setting AUD 400 million of hedges for a potential new regional conflict. Which of the following conclusions from the IMF analysis is correct?

A. Energy and defense stocks typically have negative GPR betas, because consumer demand and private investment weaken when geopolitical tensions rise.

B. Within a month of major conflicts, sovereign CDS spreads widened about 180 bps in emerging markets versus about 40 bps in advanced economies, while safe-haven yields fell.

C. During the 2018 to 2019 tariff announcements, put premiums rose in the United States and China and spilled over strongly to most other global equity markets.

D. Commodity importers typically gain from geopolitical shocks through higher prices, while exporters face currency depreciation and wider sovereign spreads.

Solution **Answer: B**

Step 1 – Sovereign risk premiums (CI-4 d)

Sovereign CDS spreads rise sharply after military conflicts as investors worry about fiscal stability and debt sustainability: about 180 bps in EMs within one month vs about 40 bps in AEs. Safe-haven yields (US, UK, Germany, Switzerland, Japan) fall, more so the bigger the event.

⇒ *EM sovereign exposure needs the larger hedge; safe-haven duration acts as a hedge.*

Step 2 – Equity and option pricing (CI-4 e)

Energy and defense have positive GPR betas (they gain when tensions rise) and act as hedges, so they carry low or negative risk premiums; consumer sectors have negative GPR betas. Put premiums rise, tail-risk premiums the most; the tariff-week increases stayed in the US and China.

Step 3 – Importers and exporters (CI-4 b)

Exporters (especially energy producers) often benefit from higher prices; importers face currency depreciation and wider sovereign spreads.

Interpretation

B is correct. The manager should size sovereign hedges toward vulnerable, importing EMs, use safe-haven duration and tail protection, and treat energy and defense equity as partial hedges.

Why the other options are wrong

A is incorrect. Sibling swap: attributes the consumer-oriented sectors' negative GPR beta (weaker demand) to energy and defense, whose GPR betas are positive.

C is incorrect. Scope/condition error: the tariff-related rise in option premiums was confined to the United States and China and did not spill over meaningfully to other global markets.

D is incorrect. Role swap between importers and exporters: exporters gain from higher prices; importers suffer depreciation and wider spreads.

LO: CI-4 d: Evaluate the effects of geopolitical shocks on sovereign risk premiums. · CI-4 e: Explain how investors price geopolitical risk in equity and option markets, including changes in downside risk and tail-risk premiums. · CI-4 b: Analyze how global geopolitical risk events affect countries, sectors, and asset classes differently, including their varying impacts on commodity importers and exporters. · Reference: IMF, Global Financial Stability Report, April 2025, Ch.2, via SchweserNotes Book 5, Reading 103, Modules 103.1 and 103.2

Answer key

B1.1	**C**	B1.2	**C**	B2.1	**B**	B2.2	**D**	B2.3	**D**	B3.1	**D**
B3.2	**D**	B3.3	**D**	B4.1	**C**	B4.2	**C**	C1.1	**B**	C1.2	**C**
C2.1	**C**	C2.2	**B**	C2.3	**D**	C3.1	**A**	C3.2	**D**	C4.1	**C**
C4.2	**B**	C4.3	**A**	C5.1	**B**	C5.2	**B**	C6.1	**A**	C6.2	**D**
C7.1	**A**	C8.1	**D**	C8.2	**B**	C9.1	**A**	C9.2	**D**	D1.1	**A**
D1.2	**B**	D2.1	**D**	D2.2	**D**	D3.1	**D**	D3.2	**D**	D4.1	**D**
D4.2	**D**	D5.1	**B**	D5.2	**A**	D6.1	**D**	D6.2	**A**	D7.1	**C**
D7.2	**D**	D8.1	**A**	D8.2	**B**	D9.1	**A**	D9.2	**C**	D10.1	**C**
D10.2	**A**	D11.1	**A**	D11.2	**B**	D12.1	**C**	D13.1	**A**	D14.1	**D**
D15.1	**C**	D16.1	**A**	D18.1	**D**	D19.1	**A**	D23.1	**A**	D24.1	**D**
D25.1	**C**	D30.1	**A**	D30.2	**B**						

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